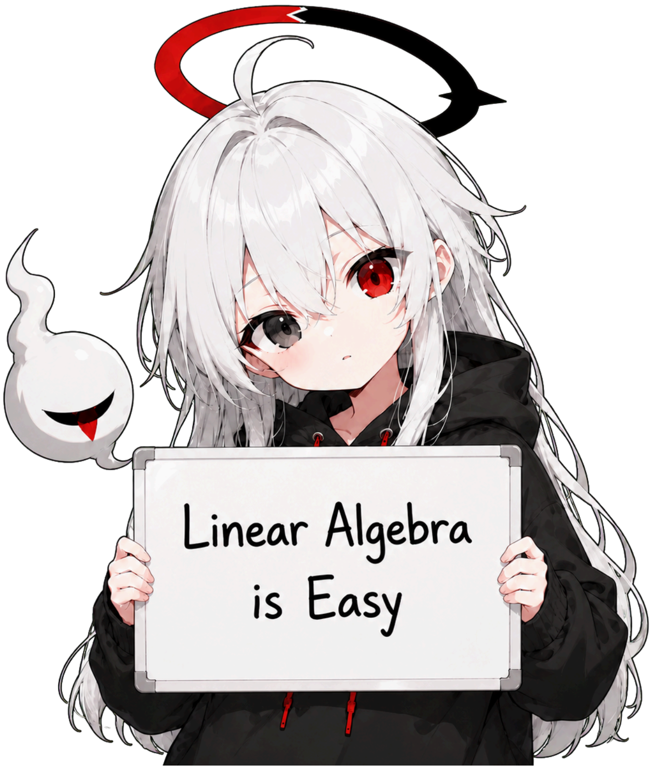
Linear Algebra F*cked Up

Higher Algebra



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This is my personal notes on Higher Algebra.

This work is done along the help of OpenAI's GPT 5.6 sol (codex), in assist of template design and (around 50% of) content generated, under my own supervision.

Most of the content is based on the great work of *Jacob Lurie, Higher Algebra*, which is available at https://www.math.ias.edu/~lurie/papers/HA.pdf

0 Preliminaries

0.1 Pointed Simplicial Sets

0.1.1 Definition

Definition 0.1.1 (Pointed Simplicial Sets). The category of *pointed simplicial sets* is defined to be

$$\mathbf{sSet}_* \coloneqq \mathbf{sSet}_{\Delta^0/} = (\Delta^0 \downarrow \mathbf{sSet})$$

where Δ^0 is the obviously the terminal object of **sSet**.

Remark. Since for every n, there is a unique

$$\Delta^n \to \Delta^0$$

the base point of a pointed simplicial set $\Delta^0 \underset{0}{\overset{x}{\to}} x$ induces a constant n-simplex

$$(x_0)_n \in x_n, \quad \Delta^n \to \Delta^0 \to x$$

Geometrically, it is a n-simplex that is "collapsed" to the base point, with all its vertices identified with the base point.

0.1.2 Simplicial n-loops

Just like what we did in topology, we can define some "loops" in simplicial sets.

Definition 0.1.2 (Simplicial n-Loop). A *simplicial n-loop* of a pointed simplicial set (x, x_0) is a simplex $\sigma \in x_n$, where

$$d_i\sigma = (x_0)_{n-1}, \quad i = 0, \ldots, n$$

which means that the border of σ degenerate to the base point.

Remark. Equivalently, the restriction of $\sigma : \Delta^n \to x$ to $\partial\Delta^n$ is the constant map at x_0.

$$\begin{tikzcd}
\partial\Delta^n \arrow[r] \arrow[d] & \Delta^0 \arrow[d, "x_0"] \\
\Delta^n \arrow[r, "\sigma"'] & x
\end{tikzcd}$$

Because the border was collapsed to a point, σ factors uniquely through the quotient

$$\Delta^n/\partial\Delta^n \to x$$

where the notation $\Delta^n/\partial\Delta^n$ is somehow called the *n-sphere* S^n, defined by above. Hence we say "simplicial n-loop $\iff$ pointed simplicial map $S^n \to x$"

0.2 Wedge Products and Smash Products

0.2.1 Wedge Products

Definition 0.2.3 (Wedge Product). Suppose $\Delta^0 \to x$ and $\Delta^0 \to y$ are two pointed simplicial sets, the *wedge product* of x and y is defined to be

$$x \vee y := x \sqcup_{\Delta^0} y$$

basically connecting the two base points together.

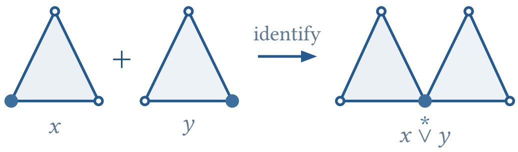

Figure 1: The wedge product identifies the marked base vertices of two pointed simplicial sets.

Remark. The wedge product is functorial in both variables, it is a bifunctor of 1-category

$$(- \vee -) : \mathbf{sSet}_* \times \mathbf{sSet}_* \to \mathbf{sSet}_*$$

also on the category of animae, this is also a ∞-bifunctor

$$(- \vee -) : \mathbf{Ani}_* \times \mathbf{Ani}_* \to \mathbf{Ani}_*$$

0.2.2 Smash Products

Definition 0.2.4 (Smash Product). Suppose $\Delta^0 \to x$ and $\Delta^0 \to y$ are two pointed simplicial sets, the *smash product* of x and y is defined to be

$$x \wedge y := \frac{x \times y}{x \vee y}$$

basically collapsing the wedge product to a point.

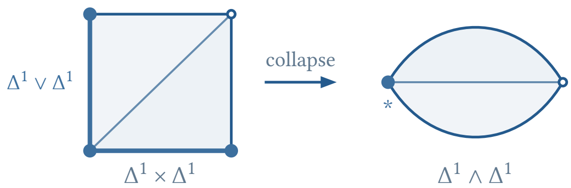

Figure 2: A minimal model of the smash product: the highlighted wedge in the triangulated product is collapsed to the marked vertex.

0.2.3 Some Theories on Wedge and Smash

Before discussing the formal properties, it is useful to keep the following analogy in mind: pointed spaces behave much like modules, with the base point playing the role of zero, the wedge product the role of direct sum, and the smash product the role of tensor product. This is a guide to intuition rather than an identification of the two settings.

R-modules	pointed spaces
0	$*$
$m \oplus n$	$x \vee y$
$m \otimes_R n$	$x \wedge y$
$\mathrm{Hom}_R(m, n)$	$\mathrm{Map}_*(x, y)$

All the equivalences below are natural in the pointed simplicial sets involved. They can equally be read in the ∞-category $\mathbf{Ani}_*$ of pointed animae.

Proposition 0.2.1 (Wedge Product Laws). The wedge product is symmetric and associative, and the one-point object $*$ is its unit:

$$x \vee y \simeq y \vee x$$
$$(x \vee y) \vee z \simeq x \vee (y \vee z)$$
$$x \vee * \simeq x$$

These equivalences express the fact that $x \vee y$ is the coproduct of x and y in the pointed category.

Proposition 0.2.2 (Smash Product Laws). Let $S^0 = \Delta^0 \sqcup \Delta^0$, with one summand chosen as the base point. The smash product is symmetric and associative, its unit is S^0, and the one-point object is absorbing:

$$x \wedge y \simeq y \wedge x$$
$$(x \wedge y) \wedge z \simeq x \wedge (y \wedge z)$$
$$x \wedge S^0 \simeq x$$
$$x \wedge * \simeq *$$

Proposition 0.2.3 (Distributivity and Tensor-Hom Adjunction). Smash product distributes over wedge product:

$$x \wedge (y \vee z) \simeq (x \wedge y) \vee (x \wedge z).$$

Moreover, pointed mapping spaces satisfy the natural adjunction

$$\mathrm{Map}_*(x \wedge y, z) \simeq \mathrm{Map}_*(x, \mathrm{Map}_*(y, z)).$$

Thus $x \wedge (-)$ preserves colimits, which explains the distributive law.

Proof. Write x_0, y_0, and z_0 for the chosen base points. By definition, the quotient map

$$q : x \times y \to x \wedge y$$

collapses $x \vee y = (x \times \{y_0\}) \cup (\{x_0\} \times y)$ to the base point. Hence a pointed map $g : x \wedge y \to z$ is the same thing as a map $f : x \times y \to z$ satisfying

$$f(x, y_0) = z_0 = f(x_0, y)$$

for every x and y.

Currying f gives a map

$$x \to \mathrm{Map}(y, z), \quad x \mapsto (y \mapsto f(x, y))$$

The condition $f(x, y_0) = z_0$ says that every map $y \to z$ in its image is pointed, so this map factors through $\mathrm{Map}_*(y, z)$. The condition $f(x_0, y) = z_0$ says that x_0 is sent to the constant base-point map, so the curried map $x \to \mathrm{Map}_*(y, z)$ is itself pointed. This construction is reversible and natural. Applied degreewise, it gives the equivalence of pointed mapping spaces

$$\mathrm{Map}_*(x \wedge y, z) \simeq \mathrm{Map}_*(x, \mathrm{Map}_*(y, z))$$

By symmetry, $x \wedge (-)$ is therefore left adjoint to $\mathrm{Map}_*(x, -)$ and preserves all colimits. Since $y \vee z$ is the coproduct of y and z in the pointed category, preservation of this coproduct gives

$$x \wedge (y \vee z) \simeq (x \wedge y) \vee (x \wedge z)$$

as claimed. □

In other words, $(\mathbf{sSet}_*, \wedge, S^0)$ is a closed symmetric monoidal category, with internal Hom object $\mathrm{Map}_*(-, -)$. Likewise, $(\mathbf{Ani}_*, \wedge, S^0)$ is a closed symmetric monoidal ∞-category.

Remark (Spheres and Suspension). For pointed spheres one obtains

$$S^m \wedge S^n \simeq S^{m+n}$$

the proof is obvious and direct.

0.3 Suspension and Looping

0.3.1 Suspension

Intuitively, the suspension of a pointed simplicial set is obtained by taking the product with a simplicial interval and collapsing the two ends to points. Formally, we have

Definition 0.3.5 (Suspension). The *suspension* of a pointed simplicial set x is defined to be

$$\Sigma x := S^1 \wedge x$$

0.3.2 Looping

According to the "n-loop = pointed map from n-sphere" principle, we have

Definition 0.3.6 (Loop). The *loop* of a pointed simplicial set x is defined to be

$$\Omega x := \mathrm{Map}_*(S^1, x)$$

0.3.3 Some theories on Suspension and Looping

Theorem 0.3.1 (Suspension-Loop Adjunction). There is a natural equivalence of pointed mapping spaces

$$\mathrm{Map}_*(\Sigma x, y) \simeq \mathrm{Map}_*(x, \Omega y)$$

for every pointed simplicial sets x and y. In other words, there is an adjunction

$$\Sigma : \mathbf{sSet}_* \rightleftarrows \mathbf{sSet}_* : \Omega$$

Proof. Immediately by definition and by Proposition 0.2.3. □

0.3.4 Module Analogy

Let R be a commutative ring. For a fixed R-module ℓ, the ordinary tensor–Hom adjunction is

$$\mathrm{Hom}_R(\ell \otimes_R m, n) \simeq \mathrm{Hom}_R(m, \mathrm{Hom}_R(\ell, n)).$$

Thus the pointed-space dictionary extends from individual operations to adjoint functors:

$$\begin{aligned} k \wedge (-) \quad &\leftrightarrow \quad \ell \otimes_R (-) \\ \mathrm{Map}_*(k, -) \quad &\leftrightarrow \quad \mathrm{Hom}_R(\ell, -). \end{aligned}$$

Taking $k = S^1$ on the space side gives $\Sigma \dashv \Omega$. The closer algebraic analogue appears in the derived category $D(R)$, where the shifted unit $R[1]$ satisfies

$$\begin{aligned} R[1] \otimes^{\mathrm{L}}_R m &\simeq m[1] \\ \mathrm{RHom}_R(R[1], n) &\simeq n[-1]. \end{aligned}$$

Hence the shift $[1]$ is left adjoint to $[-1]$, and the two are inverse equivalences. This last feature belongs to the stable setting: after pointed spaces are stabilized to spectra, Σ and Ω likewise become inverse equivalences; in $\mathbf{Ani}_*$ they are adjoint but not inverse in general. We can even claim ambitiously

> “Stable homotopy theory = Linear algebra over the sphere spectrum $\mathbb{S}$.”

We will explore this analogy in further chapters.

0.4 Homotopy Groups

The simplicial n-loops introduced earlier are individual representatives. To obtain an invariant of the homotopy type of x, we identify representatives that are homotopic through pointed maps.

Definition 0.4.7 (Homotopy Groups). Let (x, x_0) be a pointed anima, represented for example by a pointed Kan complex. For every $n \geq 0$, define

$$\pi_n(x, x_0) := \pi_0 \mathrm{Map}_*(S^n, x) = [S^n, x]_*,$$

where $[S^n, x]_*$ denotes the set of pointed homotopy classes of maps. In terms of iterated loop spaces, the same definition is

$$\pi_n(x, x_0) = \pi_0(\Omega^n x),$$

where $\Omega^0 x = x$ and the base point of $\Omega^n x$ is the constant loop at x_0.

Remark (Why a Kan Replacement Is Needed). If x is an arbitrary pointed simplicial set, the expression above means the *derived* pointed mapping space. Concretely, choose a pointed Kan fibrant replacement $x \to x^{\mathrm{fib}}$ and set

$$\pi_n(x, x_0) := \pi_0 \mathrm{Map}_*\left(S^n, x^{\mathrm{fib}}\right).$$

This makes π_n invariant under weak equivalence. When x is already Kan, no replacement is necessary.

Remark (Simplicial Description). For a pointed Kan complex (x, x_0), a map $S^n = \Delta^n / \partial\Delta^n \to x$ is exactly an n-simplex $\sigma \in x_n$ satisfying

$$d_i \sigma = (x_0)_{n-1}, \quad 0 \le i \le n.$$

Therefore $\pi_n(x, x_0)$ is the set of such simplicial n-loops modulo pointed simplicial homotopy relative to $\partial\Delta^n$. The quotient by homotopy is essential: the raw set of n-loops is not itself a homotopy invariant.

Proposition 0.4.4 (Group Structure). For $n \ge 1$, $\pi_n(x, x_0)$ is a group; for $n \ge 2$, it is abelian. The product of two classes $[f]$ and $[g]$ is represented by the composite

$$S^n \xrightarrow{\text{pinch}} S^n \vee S^n \xrightarrow{f \vee g} x.$$

Equivalently, this is concatenation of components in the n-fold loop space $\Omega^n x$. The identity is the constant map at x_0.

Proof sketch. The loop space $\Omega^n x$ has concatenation, so its connected components form a group when $n \ge 1$. For $n \ge 2$, the two loop directions give two compatible multiplications; the Eckmann–Hilton argument makes the product commutative. □

0.4.1 The Whitehead Theorem

Theorem 0.4.2 (Whitehead Theorem). Suppose $f : x \to y$ is a map of animae. Suppose

$$\pi_0(x) \xrightarrow{\sim} \pi_0(y), \quad \pi_n(x, x_0) \xrightarrow{\sim} \pi_n(y, f(x_0))$$

holds for every base point x_0. Then f is a categorical equivalence.

Part I

Stability

1 Stable ∞-Categories

1.1 The Motivation of the Subject

Ordinary category theory remembers a *set* of maps between two objects. This is often enough in classical algebra, but it is too small for homotopy theory and derived mathematics. There, two maps may be connected by a homotopy, two homotopies may themselves be homotopic, and these higher relations affect later constructions. Higher Algebra keeps this information instead of collapsing it.

Localization. When quasi-isomorphisms or weak equivalences are inverted, an ordinary localization remembers only homotopy classes of maps. An ∞-categorical localization retains the whole mapping space. Its truncation is

$$\mathrm{Hom}_{hC}(x, y) = \pi_0 \mathrm{Map}_C(x, y)$$

Thus the ordinary derived category is useful, but it is only a shadow of the enhanced object.

Coherence. "Associative up to homotopy" is not enough by itself. The chosen homotopies must agree with one another, their agreements must also agree, and so on. $\mathbb{A}_\infty$- and $\mathbb{E}_n$-algebras, encoded by ∞-operads, package this infinite system of compatibility conditions.

Stability. Stable ∞-categories provide functorial fibers, cofibers, shifts, and mapping spectra. They unify derived categories and spectra while retaining the information that a triangulated homotopy category forgets.

> **Remark (Worldview).** Higher Algebra replaces "unique by equality" with "a contractible space of coherent choices." Its guiding principle is simple: keep enough structure so that the next construction is invariant, functorial, and composable.

In short, ∞-categories retain higher morphisms, stable ∞-categories handle derived and exact phenomena, and ∞-operads describe algebraic operations up to every required coherence. These tools are used because the mathematics is already higher, not because abstraction is an end in itself.

1.2 First Definitions: Stability

1.2.1 Pointed ∞-Categories

> **Definition 1.2.1 (Zero Object, Pointed ∞-Category).** Let C be an ∞-category, a *zero object* is an object which is both initial and terminal. We will say that C is *pointed* if it admits a zero object. In this case, we will denote the zero object by $0 \in C$.

Obviously, if C is pointed, then the zero object is unique up to a contractible space of choices. The condition is equals to:

Proposition 1.2.1. Let C be an ∞-category, C is pointed iff:

1. C admits initial and terminal objects $0, 1$.
2. Exists $0 \to 1$.

1.2.2 Triangle, Fiber and Cofiber

Definition 1.2.2 (Triangle, (co)Fiber Sequence). Let C be a pointed ∞-category, a *triangle* in C is a diagram $\Delta^1 \times \Delta^1 \to C$ depicted as

$$\begin{tikzcd} x \arrow[r, "f"] \arrow[d] & y \arrow[d, "g"] \\ 0 \arrow[r] & z \end{tikzcd}$$

where 0 is the zero object. We say that a triangle in C is a *fiber sequence* if it is a pullback square, and a *cofiber sequence* if it is a pushout square.

A triangle is basically a composite of two maps

$$x \xrightarrow{f} y \xrightarrow{g} z, \quad gf \simeq 0$$

if this triangle is a pullback, then $x \simeq y \times_z 0$, where $y \times_z 0$ is exactly the fiber of g, hence $x \simeq \mathrm{fib}(g)$. Similarly, if this triangle is a pushout, then $z \simeq y \sqcup_x 0$, where $y \sqcup_x 0$ is exactly the cofiber of f, hence $z \simeq \mathrm{cofib}(f)$.

Remark. Let C be a pointed ∞-category, a triangle in C consists of the following data:

1. A pair of morphisms $f : x \to y$ and $g : y \to z$ in C.
2. A 2-simplex in C corresponding to a diagram

$$\begin{tikzcd} & y \arrow[dr, "g"] \arrow[d, phantom, "///"] & \\ x \arrow[ur, "f"] \arrow[rr, "h"'] & {} & z \end{tikzcd}$$

in C, which witnesses h as a composite of f and g.
3. A 2-simplex in C corresponding to a diagram

$$\begin{tikzcd} & 0 \arrow[dr, "0"] \arrow[d, phantom, "///"] & \\ x \arrow[ur, "0"] \arrow[rr, "h"'] & {} & z \end{tikzcd}$$

in C, which may view as a nullhomotopy of h.

By the notion of triangle, we can define the notion of fiber and cofiber in a pointed ∞-category.

Definition 1.2.3 ((co)Fiber). Let C be a pointed ∞-category containing a morphism $g : x \to y$, a *fiber* of g is a fiber sequence

$$\begin{tikzcd} w \arrow[r] \arrow[d] & x \arrow[d, "g"] \\ 0 \arrow[r] & y \end{tikzcd}$$

Dually, a *cofiber* of g is a cofiber sequence

$$\begin{tikzcd} x \arrow[r, "g"] \arrow[d] & y \arrow[d] \\ 0 \arrow[r] & z \end{tikzcd}$$

We write $w = \mathrm{fib}(g)$ and $z = \mathrm{cofib}(g)$.

Proposition 1.2.2. Fix a pointed ∞-category C and a morphism $f : x \to y$, if a cofiber of f exists, then it is unique up to an equivalence. Precisely, consider the full subcategory

$$E \subset \mathbf{Fun}(\Delta^1 \times \Delta^1, C)$$

spanned by the cofiber sequences, let $\theta : E \to \mathbf{Fun}(\Delta^1, C)$ be the forgetful functor, which send a diagram

$$\begin{tikzcd} x \arrow[r, "g"] \arrow[d] & y \arrow[d] \\ 0 \arrow[r] & z \end{tikzcd}$$

to the morphism $g : x \to y$. Then θ is a Kan fibration. If every morphism in C admits a cofiber, then θ is a trivial Kan fibration.

Proposition 1.2.3. The functor

$$\mathrm{cofib} : \mathbf{Fun}(\Delta^1, C) \to C$$

is the left adjoint of the follwing functor:

$$C \simeq \mathbf{Fun}(\Delta^0, C) \to \mathbf{Fun}(\Delta^1, C)$$

by sending every object x to the morphism $0 \to x$. Hence cofib preserves colimits.

1.2.3 Stability

Definition 1.2.4 (Stable ∞-Category). An ∞-category C is *stable* if it satisfies the following conditions:

1. C is pointed.
2. Every morphism in C admits a fiber and a cofiber.
3. A triangle in C is a fiber sequence if and only if it is a cofiber sequence.

Proposition 1.2.4. If C is a stable ∞-category, then C^{op} is also stable.

1.2.4 Closure Properties

Proposition 1.2.5. Let C be a stable ∞-category and k a simplicial set. Then the ∞-category $\mathbf{Fun}(k, C)$ is stable.

Proposition 1.2.6. Let C be a small stable ∞-category, and let κ be a regular cardinal, then the ∞-category $\mathrm{Ind}_\kappa(C)$ is stable.

1.3 Homotopy Category of a Stable ∞-Category

1.3.1 Intro

Let m be a module over a commutative ring R. It admits a resolution

$$\dots \to p_2 \to p_1 \to p_0 \to m \to 0$$

by projective R-modules p_i. There are generally many different choices, but they are always *quasi-isomorphic* to each other: that is, a chain map between two resolutions that induces a isomorphism on homology. It's often convenient to work in the *derived category* $\mathrm{D}(R)$ of the ring R, the category obtained from the category of chain complexes of R-modules by formally inverting quasi-isomorphisms.

The derived category $\mathrm{D}(R)$ of a commutative ring R is usually not an abelian category. A morphism $f : x' \to x$ in $\mathrm{D}(R)$ usually does not have a kernel in $\mathrm{D}(R)$. Instead, one can associate to f its cofiber (*mapping cone*) x'', which is well-defined up to noncanonical isomorphism.

Verdier introduced the notion of a *triangulated category* to axiomatize the properties of the derived category by formating the mapping cones, we will soon review the theory of triangulated categories and then show that the homotopy category of a stable ∞-category is triangulated.

1.3.2 Basic Homological Algebra

Definition 1.3.5 (Additive Category, Abelian Category). Let A be a category, we say that A is *additive* if it satisfies the following conditions:

1. A admits a zero object 0.
2. A admits finite products and coproducts.

For every pair of objects $x, y \in A$, a *zero morphism* is a map factors through $x \to 0 \to y$. It is unique, and denoted by 0.

3. For every pair x, y the map $x \sqcup y \to x \times y$ described by

$$\begin{pmatrix} \mathrm{id}_x & 0 \\ 0 & \mathrm{id}_y \end{pmatrix}$$

   is an isomorphism, and denote the inverse by $\varphi_{x,y}$.

We can now define the *sum* of two morphisms $f, g : x \to y$ to be the morphism $f + g : x \to y$ by the composition

$$x \xrightarrow{\Delta_x} x \times x \xrightarrow{f \times g} y \times y \xrightarrow{\varphi_{y,y}} y \sqcup y \xrightarrow{\nabla_y} y$$

where Δ_x is the diagonal map and ∇_y is the codiagonal map. This construction makes $\mathrm{Hom}_A(x, y)$ into a commutative monoid with identity 0.

4. For every x, y the commutative monoid $\mathrm{Hom}_A(x, y)$ is an abelian group. i.e. every morphism $f : x \to y$ admits an additive inverse $-f : x \to y$ such that $f + (-f) = 0$.

An additive category A is *abelian*, if every morphism admits a kernel and a cokernel, and every monomorphism is a kernel of its cokernel, and the canonical map

$$\mathrm{coker}(\ker(f) \to x) \to \ker(y \to \mathrm{coker}(f))$$

is an isomorphism for every morphism $f : x \to y$ in A.

Definition 1.3.6 (Triangulated Category). A *triangulated category* consists of the following data:

1. An additive category A.
2. A translation functor $A \to A$ which is an equivalence of categories, by $x \mapsto x[1]$.
3. A collection of *distinguished triangles*

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

These data are required to satisfy the following axioms:

1. Every morphism $f : x \to y$ in A can be extended to distinguished triangle in A.
2. The collection of distinguished triangles is stable under isomorphisms.
3. Given an object $x \in A$, the diagram

$$x \xrightarrow{\mathrm{id}_x} x \longrightarrow 0 \longrightarrow x[1]$$

   is a distinguished triangle.
4. A diagram

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

   is a distinguished triangle if and only if the diagram

$$y \xrightarrow{g} z \xrightarrow{h} x[1] \xrightarrow{-f[1]} y[1]$$

   is a distinguished triangle.
5. Given a commutative diagram

$$
\begin{tikzcd}
x \arrow[r] \arrow[d, "f"'] & y \arrow[r] \arrow[d] & z \arrow[r] \arrow[d, dashed] & x[1] \arrow[d, "f[1]"] \\
x' \arrow[r] & y' \arrow[r] & z' \arrow[r] & x'[1]
\end{tikzcd}
$$

where the rows are distinguished triangles, then there exists a morphism $z \to z'$ making the whole diagram commutative.

6. (*Octahedral Axiom*) uppose given three distinguished triangles

$$
x \xrightarrow{f} y \xrightarrow{u} y/x \xrightarrow{d} x[1], \quad y \xrightarrow{g} z \xrightarrow{v} z/y \xrightarrow{d'} y[1], \quad x \xrightarrow{g\circ f} z \xrightarrow{w} z/x \xrightarrow{d'''} x[1]
$$

in A, there exists a fourth distinguished triangle

$$
y/x \xrightarrow{\varphi} z/x \xrightarrow{\psi} z/y \xrightarrow{\theta} y/x[1]
$$

such that the diagram

$$
\begin{tikzcd}
 & & z \arrow[drr, "w"] \arrow[ddrr, "v"'] & & \\
x \arrow[urr, "g \circ f"] \arrow[d, "f"'] & & & & z/x \arrow[llll, "{d'', +1}"] \arrow[d, "\psi"] \\
y \arrow[uurr, "g"] \arrow[drr, "u"'] & & & & z/y \arrow[llll, "{d', +1}"] \arrow[dll, "{\theta, +1}"] \\
 & & y/x \arrow[uull, "{d, +1}"] \arrow[uurr, "\varphi"'] & &
\end{tikzcd}
$$

commutes.

1.3.3 Homotopy Category of a Stable ∞-Category

Let C be a stable ∞-category. The entire construction rests on two facts: pushout squares are pullback squares, and the space of choices of a limit or colimit is contractible. The first produces the usual homological operations; the second makes them coherent.

1.3.3.1 Suspension and Additivity

Definition 1.3.7 (Suspension and Loop Objects). For $x \in C$, define Σx and Ωx by a pushout square and a pullback square, respectively:

$$
\begin{tikzcd}
x \arrow[r] \arrow[d] & 0 \arrow[d] \\
0 \arrow[r] & \Sigma x
\end{tikzcd}
\qquad
\begin{tikzcd}
\Omega x \arrow[r] \arrow[d] & 0 \arrow[d] \\
0 \arrow[r] & x
\end{tikzcd}
$$

The left square is a pushout and the right square is a pullback. We write $x[1] = \Sigma x$ and $x[-1] = \Omega x$

Proposition 1.3.7 (Suspension Is an Equivalence). The functors $\Sigma, \Omega : C \to C$ are inverse equivalences. Hence $[1] : hC \to hC$ is an equivalence.

Proof. The square defining Σx is also a pullback, so $\Omega\Sigma x \simeq x$. Dually, $\Sigma\Omega x \simeq x$. These equivalences are natural because the relevant spaces of choices are contractible. □

Proposition 1.3.8 (Additivity of the Homotopy Category). The homotopy category hC is additive.

Proof. The zero map points every $\mathrm{Map}_C(x, y)$, and suspension gives

$$\mathrm{Map}_C(\Sigma x, y) \simeq \Omega \mathrm{Map}_C(x, y)$$

Therefore

$$\mathrm{Hom}_{hC}(x, y) \simeq \pi_2 \mathrm{Map}_C(x[-2], y)$$

The right side is an abelian group, and naturality makes composition bilinear. Thus hC is preadditive.

The zero object remains a zero object in hC. For $x, y \in C$, put

$$p := \mathrm{cofib}\left(x[-1] \xrightarrow{0} y\right)$$

The pushout universal property and the suspension equivalence give

$$\mathrm{Map}_C(p, t) \simeq \mathrm{Map}_C(x, t) \times \mathrm{Map}_C(y, t)$$

Hence p is a coproduct. A finite coproduct in a preadditive category is a biproduct: its projections are determined by the matrices $(\mathrm{id}, 0)$ and $(0, \mathrm{id})$. Thus hC is additive. □

1.3.3.2 Distinguished Triangles

For a morphism $f : x \to y$, choose a coherent diagram in C

$$\begin{tikzcd}
x \arrow[r, "f"] \arrow[d] & y \arrow[r] \arrow[d] & 0 \arrow[d] \\
0 \arrow[r] & C_f \arrow[r, "\delta_f"] & \Sigma x
\end{tikzcd}$$

in which both small squares are pushouts. Then $C_f \simeq \mathrm{cofib}\,(f)$, the outer rectangle gives Σx, and δ_f is the connecting morphism.

Definition 1.3.8 (Distinguished Triangle in hC). A triangle in hC

$$x \xrightarrow{f} y \xrightarrow{u} z \xrightarrow{\delta} x[1]$$

is *distinguished* if it is isomorphic to a cofiber triangle

$$x \xrightarrow{f} y \to C_f \xrightarrow[f]{\delta} \Sigma x$$

arising from a coherent two-pushout diagram as above.

Restriction from coherent two-pushout diagrams to the arrow f is a trivial Kan fibration. Consequently its fibers are contractible: the triangle is independent of all choices up to isomorphism in hC, and coherent squares of arrows lift to morphisms of cofiber triangles.

Lemma 1.3.1 (The Sign from Transposing a Suspension Square). Interchanging the two zero directions in a pushout square defining Σx induces $-\mathrm{id}_{\Sigma x}$ in hC.

Proof. Under the natural identification

$$\mathrm{Hom}_{hC}(\Sigma x, y) \simeq \pi_1 \mathrm{Map}_C(x, y)$$

transposition is loop reversal, hence multiplication by -1. Yoneda gives the result. □

Theorem 1.3.1 (Triangulated Homotopy Category). Let C be a stable ∞-category. With translation $x[1] = \Sigma x$ and the distinguished triangles defined above, hC is a triangulated category.

Proof. We verify Verdier's axioms.

TR1: existence, invariance, and the identity triangle. Every arrow $f : x \to y$ extends to

$$x \xrightarrow{f} y \longrightarrow C_f \xrightarrow[f]{\delta} x[1]$$

by two pushouts. Isomorphism invariance is part of the definition, and $C_{\mathrm{id}_x} \simeq 0$ gives

$$x \xrightarrow{\mathrm{id}_x} x \longrightarrow 0 \longrightarrow x[1]$$

TR2: rotation. The right pushout square says that $x[1]$ is the cofiber of $y \to C_f$. One further pushout gives

$$y \xrightarrow{u} C_f \xrightarrow[f]{\delta} x[1] \xrightarrow{-f[1]} y[1]$$

The minus sign is precisely the sign of the preceding lemma. Since shifts preserve pushouts, desuspension and repeated rotation prove the converse.

TR3: morphisms of triangles. A commutative square in hC is a coherent square in C after choosing representatives and a homotopy. The trivial Kan fibration of cofiber extensions lifts it to a morphism of the two cofiber triangles. Its third component $C_f \to C_{f'}$ is the required map; uniqueness is neither asserted nor needed.

TR4: the octahedral axiom. For $x \xrightarrow{f} y \xrightarrow{g} z$, choose coherent cofiber models. Pushout pasting gives

$$\begin{tikzcd} x \arrow[r, "f"] \arrow[d] & y \arrow[r, "g"] \arrow[d] & z \arrow[d] \\ 0 \arrow[r] & C_f \arrow[r, "\varphi"] & C_{gf} \end{tikzcd}$$

The left square and outer rectangle are pushouts, hence so is the right square. Pushing it out along $C_f \to 0$ identifies its cofiber with $C_g = z \sqcup_y 0$ and produces the fourth distinguished triangle

$$C_f \xrightarrow{\varphi} C_{gf} \xrightarrow{\psi} C_g \xrightarrow{\theta} C_{f[1]}$$

All four triangles are faces of one coherent pushout diagram. Its remaining faces give exactly the commutative octahedron, including the shifted signs fixed by TR2. Thus TR4 holds and hC is triangulated. □

Corollary 1.3.1 (The Stable–Triangulated Dictionary). Passing from C to hC carries the basic stable constructions to the usual triangulated ones:

stable ∞-category	triangulated category
zero object	zero object
Σ, Ω	shifts $[1], [-1]$
finite product = finite coproduct	biproduct
cofiber/fiber sequence	distinguished triangle
exact functor	triangulated functor

Thus every formal theorem about triangulated categories applies to hC. Exact functors descend to triangulated functors, and mapping spaces give the usual long exact sequences. The enhancement C retains more: its cofibers and comparison maps are coherently functorial, unlike cones in hC.

1.4 Exact Functors

1.4.1 The Definition

Definition 1.4.9 (Exact Functor). A functor $F : C \to D$ between stable ∞-categories is *exact* if it preserves zero objects and sends cofiber sequences in C to cofiber sequences in D. Equivalently, F preserves fiber sequences.

We have the following equivalent characterizations of exact functors.

Proposition 1.4.9. Let $F : C \to D$ be a functor between stable ∞-categories, then the following are equivalent:

1. F is exact.

2. F preserves finite limits.
3. F preserves finite colimits.

Proof sketch. Suppose first that F is exact. It preserves suspensions, desuspensions, and finite biproducts. Moreover, the pushout of a span $y \xleftarrow{f} x \xrightarrow{g} z$ is the cofiber of

$$x \xrightarrow{(f,-g)} y + z$$

Hence F preserves pushouts. Since it also preserves the zero object, it preserves all finite colimits. Dually, or by expressing a pullback as the fiber of $y + z \to w$, it preserves all finite limits.

Conversely, a functor preserving finite colimits preserves the initial object and pushout squares, hence the zero object and cofiber sequences. A functor preserving finite limits preserves the terminal object and pullback squares; in a stable ∞-category these are again the zero object and cofiber sequences. Thus either condition implies exactness. □

1.4.2 Ext Functor

Definition 1.4.10 (Ext Functor). We write

$$\mathrm{Ext}^n_C(x, y) \coloneqq \mathrm{Hom}_{hC}(x, y[n])$$

2 Homological Algebra Reviewed

2.1 t-Structures

2.1.1 Definition of t-Structures on Triangulated Categories

Recall that we say a full sub ∞-category $C' \subset C$ is a *localization* if the inclusion functor has a left adjoint. We will introduce a class of localizations called t-*structures* which are particularly important in homological algebra.

Definition 2.1.1 (t-Structure on Triangulated Category). Let D be a triangulated category. A t-*structure* on D is a pair of full subcategories $\left(D_{\geq 0}, D_{\leq 0}\right)$ stable under isomorphisms, such that the following conditions hold:

1. For $x \in D_{\geq 0}$ and $y \in D_{\leq 0}$, we have $\mathrm{Hom}_D(x, y[-1]) = 0$.
2. We have inclusions $D_{\geq 0}[1] \subset D_{\geq 0}$ and $D_{\leq 0}[-1] \subset D_{\leq 0}$.
3. For any $x \in D$, there exists a fiber sequence $x' \to x \to x''$ (i.e. $x' = \mathrm{fib}\,(x \to x'')$) where $x' \in D_{\geq 0}$ and $x'' \in D_{\leq -1}$.

Remark. We denote $D_{\geq n}$ for $D_{\geq 0}[n]$ and $D_{\leq n}$ for $D_{\leq 0}[n]$ for any integer n.

Remark. In this definition, either of the full subcategories $D_{\geq 0}$ and $D_{\leq 0}$ determines the other. For example, an object $y \in D$ belongs to $D_{\leq -1}$ if and only if $\mathrm{Hom}_D(x, y)$ vanishes for all $x \in D_{\geq 0}$.

2.1.2 Definition of t-Structures on Stable ∞-Categories

Definition 2.1.2 (t-Structure on Stable ∞-Category). Let C be a stable ∞-category, a t-*structure* on C is a t-structure hC, if C is equipped with a t-structure, we let $C_{\geq n}$ and $C_{\leq n}$ denote the full subcategories of C spanned by those objects which belong to $(hC)_{\geq n}$ and $(hC)_{\leq n}$, respectively.

Proposition 2.1.1. Let C be a stable ∞-category with a t-structure, for each $n \in \mathbb{Z}$, the full subcategory $C_{\leq n}$ is a localization of C.

Proof. Since shifts are equivalences and carry $C_{\leq n}$ to translates of $C_{\leq -1}$, it is enough to treat $n = -1$.

Recall the criterion for a full subcategory to be a localization. The inclusion $i : D \to C$ has a left adjoint if, for every $x \in C$, there is a map $\eta_x : x \to Lx$ with $Lx \in D$ such that composition with η_x induces an equivalence

$$\mathrm{Map}_C(Lx, y) \to \mathrm{Map}_C(x, y)$$

for every $y \in D$. Indeed, this is exactly the universal property characterizing Lx as the reflection of x into D; these universal objects assemble into a functor $L : C \to D$ left adjoint to i.

Apply the third axiom of the t-structure to x. It gives a fiber, hence also cofiber, sequence

$$x' \to x \underset{x}{\overset{\eta}{\to}} x''$$

with $x' \in C_{\geq 0}$ and $x'' \in C_{\leq -1}$. We claim that x'' has the required universal property. Fix $y \in C_{\leq -1}$. Applying $\mathrm{Map}_C(-, y)$ gives a fiber sequence

$$\mathrm{Map}_C(x'', y) \to \mathrm{Map}_C(x, y) \to \mathrm{Map}_C(x', y)$$

It therefore remains to show that $\mathrm{Map}_C(x', y)$ is contractible.

For $k \in \mathbb{Z}$, write

$$\mathrm{Ext}^k_C(a, b) := \mathrm{Hom}_{hC}(a, b[k]) \simeq \mathrm{Hom}_{hC}(a[-k], b)$$

If $k \leq 0$, then $-k \geq 0$, so the shift axiom gives $x'[-k] \in C_{\geq 0}$. The orthogonality axiom and $y \in C_{\leq -1}$ now imply

$$\mathrm{Ext}^k_C(x', y) \simeq \mathrm{Hom}_{hC}(x'[-k], y) \simeq 0$$

For every $m \geq 0$, stability identifies the homotopy groups of the mapping space with these groups:

$$\pi_m \mathrm{Map}_C(x', y) \simeq \mathrm{Ext}^{-m}_C(x', y) \simeq 0$$

Mapping spaces are Kan complexes. The case $m = 0$ says that this mapping space is connected, and the higher cases say that all its homotopy groups vanish; the Whitehead criterion therefore makes it contractible. Hence $\mathrm{Map}_C(x'', y) \to \mathrm{Map}_C(x, y)$ is an equivalence for every $y \in C_{\leq -1}$, so $x \mapsto x''$ defines the left adjoint $\tau_{\leq -1} : C \to C_{\leq -1}$. Shifting gives the localization $\tau_{\leq n} : C \to C_{\leq n}$ for every $n \in \mathbb{Z}$. □

Corollary 2.1.1. Each $C_{\leq n}$ is closed under all limits which exist in C. Dually, each $C_{\geq n}$ is closed under all colimits which exist in C.

2.2 Truncation Functors

2.2.1 Definition of Truncation Functors

Indeed, local objects are characterized by equivalences of mapping spaces, and such equivalences are preserved under limits. The second statement is dual.

Remark (Notation). Write

$$\tau_{\leq n} : C \to C_{\leq n}, \quad \tau_{\geq n} : C \to C_{\geq n}$$

for the left and right adjoints to the respective inclusions. Every $x \in C$ then fits naturally into a fiber–cofiber sequence

$$\tau_{\geq n}x \to x \to \tau_{\leq n-1}x$$

Both truncation functors preserve every $C_{\leq m}$ and every $C_{\geq m}$. For example, $\tau_{\leq n}$ either acts as the identity on $C_{\leq m}$ or lands in the smaller subcategory $C_{\leq n}$. The assertion for $\tau_{\geq n}$ follows from the displayed fiber sequence and closure under limits. The remaining statements are dual.

Remark (Warning). The symbol $\tau_{\leq n}$ here denotes truncation for a t-structure, not the truncation of an arbitrary ∞-category by homotopy dimension. A nonzero object of a stable ∞-category is never truncated in the latter sense: for arbitrarily large r, its identity gives a nonzero class in $\pi_r \mathrm{Map}_C(x[-r], x)$.

The notions agree after restricting the source to $C_{\geq 0}$. For $k \geq -1$, an object x belongs to $C_{\leq k}$ exactly when $\mathrm{Map}_C(y, x)$ is k-truncated for every $y \in C_{\geq 0}$.

2.2.2 Commuting Truncations and the Heart

For $m, n \in \mathbb{Z}$, the universal properties give a natural transformation

$$\theta : \tau_{\leq m} \circ \tau_{\geq n} \to \tau_{\geq n} \circ \tau_{\leq m}$$

Proposition 2.2.2. The transformation θ is an equivalence. If $m < n$, both sides are zero; if $m \geq n$, they define the truncation of x to the interval $[n, m]$.

Proof sketch. Both sides lie in $C_{\geq n} \cap C_{\leq m}$. When $m < n$, orthogonality makes this intersection zero. When $m \geq n$, apply $\mathrm{Ext}^0_C(-, y)$ for $y \in C_{\geq n} \cap C_{\leq m}$ to the two truncation sequences. Orthogonality identifies the outer maps, and the five lemma identifies the middle map induced by θ. Yoneda then shows that θ is an equivalence. □

Definition 2.2.3 (Heart). The *heart* of the t-structure is

$$C^{\heartsuit} := C_{\geq 0} \cap C_{\leq 0}$$

Define the homology-object functors by

$$\pi_0 := \tau_{\leq 0} \circ \tau_{\geq 0} \simeq \tau_{\geq 0} \circ \tau_{\leq 0}, \quad \pi_n(x) := \pi_0(x[-n])$$

For $x, y \in C^{\heartsuit}$ and $r > 0$, orthogonality gives

$$\pi_r \mathrm{Map}_C(x, y) \simeq \mathrm{Ext}^{-r}_C(x, y) \simeq 0$$

Thus mapping spaces in $C^{\heartsuit}$ are discrete, so this ∞-category is the nerve of its homotopy category. The category $hC^{\heartsuit}$ is abelian; kernels and cokernels are obtained by applying π_0 to fibers and cofibers. Our indexing is homological: the object above is denoted $\pi_n(x)$ rather than $\mathrm{H}_n(x)$.

2.2.3 General Localizations

Not every localization of a stable ∞-category comes from a t-structure. Given a collection S of morphisms, the localization $S^{-1}C$ is the full subcategory of S-*local* objects. An object x is S-local when, for every $f : y' \to y$ in S, composition with f induces an equivalence

$$\mathrm{Map}_C(y,x) \to \mathrm{Map}_C(y',x)$$

To compare this with a t-structure, complete f to a cofiber sequence

$$y' \to y \to y''$$

Applying $\mathrm{Ext}^i_C(-,x)$ gives

$$\ldots \to \mathrm{Ext}^i_C(y'',x) \to \mathrm{Ext}^i_C(y,x) \to \mathrm{Ext}^i_C(y',x) \to \mathrm{Ext}^{i+1}_C(y'',x) \to \ldots$$

Consequently, x is f-local exactly when

$$\mathrm{Ext}^i_C(y,x) \to \mathrm{Ext}^i_C(y',x)$$

is an isomorphism for every $i \le 0$. In particular, locality implies $\mathrm{Ext}^i_C(y'',x) = 0$ for $i \le 0$, while vanishing for $i \le 1$ is sufficient for locality. If y' is zero, the exact criterion reduces simply to $\mathrm{Ext}^i_C(y,x) = 0$ for every $i \le 0$.

2.3 Some More Propositions

2.3.1 Quasisaturated Localizations

Definition 2.3.4 (Quasisaturated Collection). Let C admit pushouts. A collection S of morphisms is *quasisaturated* if

1. every equivalence belongs to S;
2. S satisfies two-out-of-three for every composable pair;
3. a pushout of a morphism in S again belongs to S.

Intersections of quasisaturated collections are quasisaturated. Hence every collection of maps has a smallest quasisaturated closure, called the collection it *generates*.

Definition 2.3.5 (Closed under Extensions). A full subcategory $E \subset C$ is *closed under extensions* if, for every fiber–cofiber sequence $x \to y \to z$, the conditions $x, z \in E$ imply $y \in E$.

Let $L : C \to C$ be an idempotent localization, let C_L be its essential image, and let S_L be the collection of maps f for which $L(f)$ is an equivalence. The collection S_L is quasisaturated.

Proposition 2.3.3 (t-Structure Localizations). The following conditions are equivalent:

1. S_L is generated by a collection of maps of the form $0 \to a$;
2. S_L is generated by all $0 \to a$ such that $La \simeq 0$;
3. the local subcategory C_L is closed under extensions;
4. for $a \in C$ and $b \in C_L$, the map $\mathrm{Ext}^1_C(La,b) \to \mathrm{Ext}^1_C(a,b)$ is injective;
5. $C_{\ge 0} := \{a \mid La \simeq 0\}$ and $C_{\le -1} := \{a \mid La \simeq a\}$ determine a t-structure on C.

Proof sketch. The implication (1)→(2) follows by enlarging the chosen generators to all L-acyclic objects. For (2)→(3), mapping an acyclic generator into a fiber sequence shows that an extension of local objects is local.

If (3) holds, an extension of La by a local object has local middle term; consequently an extension class which becomes split after pullback along $a \to La$ was already split. This is (4).

Under (4), the fiber sequence

$$\mathrm{fib}\,(a \to La) \to a \to La$$

gives the required truncation sequence. Adjunction gives orthogonality, and the injectivity in (4) supplies the remaining shifted orthogonality. Hence the two classes in (5) form a t-structure. Finally, if (5) holds, the cofiber of every map inverted by L is in $C_{\geq 0}$; pushouts and two-out-of-three show that the maps $0 \to a$ with $a \in C_{\geq 0}$ generate all of S_L. □

Thus the localizations arising as negative truncations are exactly those generated, up to quasi-saturation, by killing objects.

2.3.2 Boundedness and Completion

Definition 2.3.6 (Bounded Objects). Let C^+ consist of the objects lying in some $C_{\geq n}$, and let C^- consist of those lying in some $C_{\leq n}$. Put

$$C^b := C^+ \cap C^-$$

The t-structure is *left bounded*, *right bounded*, or *bounded* when $C = C^+$, $C = C^-$, or $C = C^b$, respectively.

Definition 2.3.7 (Left Completion). The *left completion* $\hat{C}$ is the ∞-category of compatible Postnikov towers $(x_n)_{n\in\mathbb{Z}}$ with

$$x_n \in C_{\leq n}, \quad \tau_{\leq n} x_{n+1} \simeq x_n$$

Equivalently,

$$\hat{C} \simeq \lim_n C_{\leq n}$$

The canonical functor is

$$C \to \hat{C}, \quad x \mapsto (\tau_{\leq n} x)_n$$

We call C *left complete* when this functor is an equivalence.

Proposition 2.3.4 (Basic Properties of Completion). The left completion has the following properties:

1. $\hat{C}$ is stable and inherits a t-structure;
2. the functor $C \to \hat{C}$ is exact and induces equivalences

$$C_{\leq n} \simeq \hat{C}_{\leq n}$$

   for every n;
3. it induces an equivalence $C^+ \simeq \hat{C}^+$ on left bounded objects.

Proof sketch. Limits and cofibers of compatible towers are formed levelwise, with the shift reindexing the tower, so $\hat{C}$ is stable. The two halves of its t-structure are detected levelwise. A tower bounded above stabilizes at the corresponding truncation, which proves (2); shifting proves (3). □

In particular, left completion changes only objects which are unbounded below. Passing to the left bounded part and taking left completion are inverse constructions. Right completion and right completeness are defined dually.

> **Proposition 2.3.5 (Criterion for Left Completeness).** Suppose that C admits countable products and that $C_{\geq 0}$ is closed under them. Then the following are equivalent:
>
> 1. C is left complete;
> 2. the only object in $\bigcap_{n\in\mathbb{Z}} C_{\geq n}$ is the zero object.

Proof sketch. Every countable tower has a limit, computed by the fiber sequence

$$\lim_n x_n \to \prod_n x_n \xrightarrow{\mathrm{id}-\mathrm{shift}} \prod_n x_n$$

Hence a compatible Postnikov tower has a limit in C. For every x, the cofiber of the completion map

$$x \to \lim_n \tau_{\leq n} x$$

belongs to every $C_{\geq n}$. Condition (2) therefore makes this map an equivalence. Conversely, if x belongs to every $C_{\geq n}$, then all its truncations vanish; left completeness forces $x \simeq 0$. □

2.4 Filtered Objects and Spectral Sequences

Let C be a stable ∞-category with a t-structure and heart $A \simeq C^{\heartsuit}$. A *filtered object* is a functor $x : N(\mathbb{Z}) \to C$, depicted as

$$\ldots \to x_{-1} \xrightarrow{f^0} x_0 \xrightarrow{f^1} x_1 \to \ldots$$

Its pth associated graded object is

$$\mathrm{gr}^p x := \mathrm{cofib}\left(x_{p-1} \to x_p\right)$$

We now construct a spectral sequence in A beginning with

$$E_1^{p,q} \simeq \pi_{p+q}\mathrm{gr}^p x$$

2.4.1 Interval Cofibers

> **Definition 2.4.8 (Complex Associated to a Filtration).** Adjoin a least element $-\infty$ to $\mathbb{Z}$. The filtered object x extends to an interval diagram with values $x_{i,j}$ for $-\infty \leq i \leq j$, defined by
>
> $$x_{-\infty,j} := x_j, \quad x_{i,j} := \mathrm{cofib}\left(x_i \to x_j\right)$$

Thus $x_{i,i} \simeq 0$, and for $i \le j \le k$ there is a canonical fiber–cofiber sequence

$$x_{i,j} \to x_{i,k} \to x_{j,k} \to x_{i,j}[1]$$

Equivalently, the square with vertices $x_{i,j}, x_{i,k}, 0, x_{j,k}$ is a pushout. Conversely, every interval diagram satisfying these two properties is determined by the filtration $x_{-\infty,j}$. More precisely, restriction to the $-\infty$th row is an equivalence of ∞-categories. Hence all interval cofibers and connecting maps above are functorial and independent of choices up to a contractible space.

Remark (Underlying Chain Complex). Set

$$c_p := x_{p-1,p}[-p]$$

The connecting maps of adjacent intervals give

$$\dots \to c_1 \to c_0 \to c_{-1} \to \dots$$

in hC. Two consecutive maps compose to zero because they are adjacent boundary maps in a coherent pushout diagram. Thus every filtered object has a canonical chain complex of graded pieces in its triangulated homotopy category.

2.4.2 The Spectral Sequence

Proposition 2.4.6 (Spectral Sequence of a Filtered Object). For $r \ge 1$, define an object of the heart by

$$E_r^{p,q} := \operatorname{im}\big(\pi_{p+q}x_{p-r,p} \to \pi_{p+q}x_{p-1,p+r-1}\big)$$

The connecting maps of the interval sequences induce differentials

$$d_r : E_r^{p,q} \to E_r^{p-r,q+r-1}$$

They satisfy $d_r^2 = 0$, and there are natural isomorphisms

$$E_{r+1}^{p,q} \simeq \ker\big(d_r : E_r^{p,q} \to E_r^{p-r,q+r-1}\big) / \operatorname{im}\big(d_r : E_r^{p+r,q-r+1} \to E_r^{p,q}\big)$$

In particular, (E_r, d_r) is a spectral sequence in A, natural in the filtered object x.

Proof sketch. For $r = 1$, both terms in the defining image are $x_{p-1,p}$, giving the announced E_1-page. For general r, the differential is induced by the boundary map between the two relevant interval cofibers. The pushout identities imply that two successive boundary maps compose to zero. Exactness in the heart then identifies the image defining E_{r+1} with the kernel of d_r modulo the image of the incoming d_r. □

The differential has bidegree $(-r, r-1)$ and lowers total degree by one. On the first page one may equivalently write

$$E_1^{p,q} \simeq \pi_q c_p$$

so d_1 is simply the homology differential obtained from the underlying chain complex of graded pieces.

> **Remark.** If C is the derived ∞-category of an abelian category and x is a filtered chain complex, this construction recovers the classical spectral sequence of a filtered complex. The interval-diagram formulation is useful because it makes every higher differential canonical and functorial.

2.4.3 Convergence

> **Definition 2.4.9 (Compatibility with Sequential Colimits).** The t-structure on C is *compatible with sequential colimits* if C admits colimits indexed by $\mathbb{N}$ and $C_{\leq 0}$ is closed under them. Under this assumption the functors $\pi_n : C \to A$ preserve sequential colimits, and sequential colimits in the heart are exact.

> **Proposition 2.4.7 (Convergence for a Bounded-Below Filtration).** Suppose the t-structure is compatible with sequential colimits and $x_p \simeq 0$ for all sufficiently negative p. Put
>
> $$x_\infty := \operatorname*{colim}_p x_p, \quad a_n := \pi_n x_\infty$$
>
> Then the spectral sequence converges to $\pi_* x_\infty$. Explicitly, for fixed (p, q) the differentials eventually vanish, and a_n has the exhaustive increasing filtration
>
> $$F^p a_n := \operatorname{im}\big(\pi_n x_p \to a_n\big)$$
>
> whose associated graded pieces are
>
> $$E_\infty^{p,q} \simeq F^p a_{p+q} / F^{p-1} a_{p+q}$$

Proof sketch. The lower bound makes the incoming and outgoing intervals defining d_r eventually constant or zero, so every bidegree stabilizes. Since π_n commutes with the sequential colimit, the images of $\pi_n x_p \to \pi_n x_\infty$ form an exhaustive filtration. Applying the long exact sequence to

$$x_{p-1} \to x_p \to \operatorname{gr}^p x$$

identifies the stable page with the indicated subquotient. □

2.5 The Dold–Kan Correspondence

The classical Dold–Kan correspondence says that a simplicial object in an abelian category is exactly a chain complex concentrated in nonnegative degrees. Its stable ∞-categorical form replaces chain complexes by nonnegative filtrations; this retains all higher coherences and connects directly with the preceding spectral sequence.

2.5.1 Normalization and Denormalization

Let A be an abelian category and let $x : \Delta^{\mathrm{op}} \to A$ be a simplicial object, with face maps d_i and degeneracy maps s_i. Its *normalized chain complex* Nx is defined by

$$N_n x := \cap_{i=1}^n \ker(d_i : x_n \to x_{n-1}), \quad \partial := d_0 : N_n x \to N_{n-1} x$$

The simplicial identities give $\partial^2 = 0$. If

$$D_n x := \sum_{i=0}^{n-1} \mathrm{im}(s_i)$$

is the degenerate subobject, then there is a canonical decomposition

$$x_n \simeq N_n x \oplus D_n x$$

More generally, in an additive category the normalized piece is the image of the idempotent

$$p_n := (1 - s_{n-1} d_n) \ldots (1 - s_0 d_1)$$

whenever this idempotent splits.

Definition 2.5.10 (The Dold–Kan Functor). Let $c = (c_n, \partial)$ be a chain complex in nonnegative degrees. The *Dold–Kan functor* constructs a simplicial object $\mathrm{DK}\,(c)$ whose degree-n object is

$$\mathrm{DK}(c)_n \simeq \oplus_{\eta:[n]\twoheadrightarrow[k]}\, c_k$$

where the sum runs over all order-preserving surjections $\eta : [n] \twoheadrightarrow [k]$. For $\alpha : [m] \to [n]$, factor

$$\eta \circ \alpha : [m] \twoheadrightarrow [\ell] \hookrightarrow [k]$$

into a surjection followed by an injection. On the η-summand, the induced structure map is the identity if the injection is the identity, is $\partial : c_k \to c_{k-1}$ if it omits only 0, and is zero otherwise. This rule defines all faces and degeneracies; the equality $\partial^2 = 0$ gives the simplicial identities.

Thus the copies indexed by nonidentity surjections freely supply exactly the degenerate simplices missing from a chain complex. For example, a complex concentrated in degree zero gives a constant simplicial object, while a complex with a in degree one and zero differential has $\mathrm{DK}\,(a[1])_n \simeq a^{\oplus n}$.

Theorem 2.5.1 (Classical Dold–Kan). For every additive category A, the functor

$$\mathrm{DK} : \mathrm{Ch}_{\geq 0}(A) \to \mathbf{Fun}\,(\Delta^{\mathrm{op}}, A)$$

is fully faithful. It is an equivalence when A is idempotent complete. In particular, for every abelian category there are inverse equivalences

$$\mathrm{DK} : \mathrm{Ch}_{\geq 0}(A) \overset{\sim}{\rightleftarrows} \mathbf{Fun}\,(\Delta^{\mathrm{op}}, A) : N$$

Proof sketch. The simplicial identities produce a normalization idempotent on x_n. Its image is $N_n x$, and splitting the complementary degeneracy idempotents yields the canonical decomposition

$$x_n \simeq \oplus_{\eta:[n]\twoheadrightarrow[k]}\, N_k x$$

This identifies $N\,\mathrm{DK}\,(c)$ with c and $\mathrm{DK}\,(Nx)$ with x. Additivity is enough for full faithfulness; idempotent completeness is exactly what guarantees the required splittings and hence essential surjectivity. In an abelian category the normalization can equivalently be formed by the displayed intersections of kernels. □

Corollary 2.5.2. If x is a simplicial abelian group, then its underlying simplicial set is a Kan complex and

$$\pi_n(x) \simeq \mathrm{H}_n(Nx)$$

Thus simplicial homotopy groups become ordinary homology groups under normalization.

2.5.2 Stable Dold–Kan

For a stable ∞-category C, write

$$\mathrm{Fil}_{\geq 0}(C) := \mathbf{Fun}\,(\mathrm{N}_\bullet(\mathbb{N}), C), \quad sC := \mathbf{Fun}\left(\mathrm{N}_\bullet(\Delta)^{\mathrm{op}}, C\right)$$

Objects of the first category are sequences $d_0 \to d_1 \to \ldots$, while objects of the second are simplicial objects of C.

Theorem 2.5.2 (Stable Dold–Kan). Every stable ∞-category C admits inverse equivalences

$$\mathrm{DK} : \mathrm{Fil}_{\geq 0}(C) \overset{\sim}{\rightleftarrows} sC : N^{\mathrm{st}}$$

No t-structure, countable colimits, or idempotent-completeness assumption is required.

The stable normalization N^{st} sends a simplicial object x to its skeletal filtration

$$d_n := |\mathrm{sk}_n x|$$

where $|\mathrm{sk}_n x|$ is the finite geometric realization of the n-skeleton. If $\ell_n x$ denotes the latching object, stability gives

$$\mathrm{cofib}\,(d_{n-1} \to d_n)[-n] \simeq \mathrm{cofib}\,(\ell_n x \to x_n)$$

The right-hand side is the stable normalized object in degree n. Hence the underlying chain complex of this filtration is the normalized chain complex of x in hC. The equivalence itself contains the higher null-homotopies and compatibilities which that chain complex alone forgets.

Remark (Relation with the Spectral Sequence). Suppose C has a t-structure with heart A. Applying π_q degreewise to x gives a simplicial object of A, and the first page of the spectral sequence of its skeletal filtration is

$$E_1^{p,q} \simeq N_p\left(\pi_q x\right)$$

Its first differential is the normalized differential. If the geometric realization exists and the convergence hypotheses above hold, then

$$E_1^{p,q} \Longrightarrow \pi_{p+q}|x|$$

In this sense the stable Dold–Kan equivalence is the structural source of the spectral sequence attached to a simplicial object.

3 Derived ∞-Categories

3.1 Nerves of dg-Categories

3.1.1 Introduction of the Goal

Let A be an additive category. The chain complexes in A can be organized into an ∞-category C, described informally as follows:

1. The objects of C are chain complexes in A.
2. Given $x_*, y_* \in C$, the morphisms $x_* \to y_*$ are chain maps.
3. Given $f, g : x_* \to y_*$, a 2-morphism $f \to g$ is represented by a chain homotopy, that is, maps $h_n : x_n \to y_{n+1}$ satisfying
$$d_y \circ h_n + h_{n-1} \circ d_x = f_n - g_n$$
4. …

To make this precise, we could proceed in several steps:

1. Let $\mathbf{Ch}(A)$ be the ordinary category of chain complexes in A.
2. For $x_*, y_* \in \mathbf{Ch}(A)$, form the mapping chain complex
$$\underline{\mathrm{Hom}}_A(x_*, y_*)_m := \prod_{n\in\mathbb{Z}} \mathrm{Hom}_A(x_n, y_{n+m})$$
with differential
$$(\partial f)_n := d_y f_n - (-1)^m f_{n-1} d_x$$
Its degree-zero cycles are chain maps, its degree-zero boundaries are null-homotopic maps, and
$$\mathrm{H}_m(\underline{\mathrm{Hom}}_A(x_*, y_*)) \simeq \mathrm{Hom}_{h\mathbf{Ch}(A)}(x_*, y_*[-m])$$
Composition is a chain map, so $\mathbf{Ch}(A)$ is enriched over $\mathbf{Ch}(\mathbf{Ab})$; in other words, it is a dg-category.
3. The connective truncation
$$\tau_{\geq 0} : \mathbf{Ch}(\mathbf{Ab}) \to \mathbf{Ch}_{\geq 0}(\mathbf{Ab})$$
is right-lax monoidal. Concretely, it comes with coherent maps
$$\tau_{\geq 0} m \otimes \tau_{\geq 0} n \to \tau_{\geq 0}(m \otimes n)$$
Applying it to every mapping complex preserves units and composition. Thus $\mathbf{Ch}(A)$ is enriched over nonnegatively graded chain complexes.
4. By the Dold–Kan correspondence from the preceding chapter,
$$\mathrm{DK} : \mathbf{Ch}_{\geq 0}(\mathbf{Ab}) \to \mathbf{Fun}\left(\Delta^{\mathrm{op}}, \mathbf{Ab}\right)$$

is an equivalence. The Alexander–Whitney construction equips DK with a right-lax monoidal structure, so it again transports enrichment. The resulting simplicial abelian group of morphisms is

$$\mathrm{Map}_\Delta(x_*, y_*) \coloneqq \mathrm{DK}\left(\tau_{\geq 0}\underline{\mathrm{Hom}}_A(x_*, y_*)\right)$$

5. Forgetting the abelian-group structure degreewise turns these mapping objects into simplicial sets. Their bilinear composition gives a simplicial category, denoted $\mathbf{Ch}_\Delta(A)$. Every simplicial abelian group is a Kan complex, so every mapping simplicial set of $\mathbf{Ch}_\Delta(A)$ is Kan; hence this simplicial category is fibrant.
6. Finally, take the homotopy coherent nerve

$$C_{\mathrm{dg}}(A) \coloneqq N_{\mathrm{hc}}(\mathbf{Ch}_\Delta(A))$$

Since $\mathbf{Ch}_\Delta(A)$ is fibrant, $C_{\mathrm{dg}}(A)$ is an ∞-category. It has the same objects as $\mathbf{Ch}(A)$, while its mapping spaces retain chain maps, chain homotopies, and all higher coherent homotopies.

This procedure is indeed too complicated, so we will introduce some new concepts to simplify it.

3.1.2 What is a dg(differential graded)-category?

Definition 3.1.1 (dg-Category). Let k be a commutative ring. A *dg-category* over k consists of the following data:

1. A collection $\{x, y, \ldots\}$ whose elements are called *objects* of C.
2. For every pair of $x, y \in C$, a *chain complex* of k-modules

$$\ldots \to \mathrm{Map}_C(x, y)_1 \to \mathrm{Map}_C(x, y)_0 \to \mathrm{Map}_C(x, y)_{-1} \to \ldots$$

which we denote by $\mathrm{Map}_C(x, y)_*$.

3. For every triple $x, y, z \in C$ a composition map

$$\mathrm{Map}_C(y, z)_* \otimes_k \mathrm{Map}_C(x, y)_* \to \mathrm{Map}_C(x, z)_*$$

which we can identify with a collection of k-bilinear maps

$$\circ : \mathrm{Map}_C(y, z)_p \times \mathrm{Map}_C(x, y)_q \to \mathrm{Map}_C(x, z)_{p+q}$$

satisfying the *Leibniz rule*

$$d(g \circ f) = dg \circ f + (-1)^p g \circ df$$

4. For each $x \in C$, an *identity morphism* $\mathrm{id}_x \in \mathrm{Map}_C(x, x)_0$ such that

$$g \circ \mathrm{id}_x = g, \quad \mathrm{id}_x \circ f = f$$

forall $f \in \mathrm{Map}_C(y, x)_p$ and $g \in \mathrm{Map}_C(x, y)_q$.

The composition law is required to be associative in the following sense: for every triple $f \in \mathrm{Map}_C(w, x)_p$, $g \in \mathrm{Map}_C(x, y)_q$ and $h \in \mathrm{Map}_C(y, z)_r$, we have

$$h \circ (g \circ f) = (h \circ g) \circ f$$

In the case $k = \mathbb{Z}$, we refer to C as a *dg-category* without further qualification.

We now introduce a construction called the dg-nerve.

Definition 3.1.2 (dg-Nerve). Let C be a dg-category, we associate to C to a simplicial set $\mathrm{N}^{\mathrm{dg}}_\bullet(C)$ called the *dg-nerve*. For each $n \geq 0$, we define

$$\mathrm{N}^{\mathrm{dg}}_\bullet(C)_n \simeq \mathrm{Hom}_{\mathbf{sSet}}\left(\Delta^n, \mathrm{N}^{\mathrm{dg}}_\bullet(C)\right)$$

to be the set of all ordered pairs $\left(\{x_i\}_{0\leq i\leq n}, \{f_I\}\right)$ where:

1. For $0 \leq i \leq n$, x_i denotes an object of the dg-category C.
2. For every subset $I = \{i_- < i_m < i_{m-1} < \ldots < i_1 < i_+\} \subset [n]$ with $m \geq 0$, f_I is an element of the abelian group $\mathrm{Map}_C\left(x_{i_-}, x_{i_+}\right)_m$, satisfying
$$d f_I = \sum_{1\leq r\leq m} (-1)^r \left(f_{I-\{i_r\}} - f_{i_r<\ldots<i_1<i_+} \circ f_{i_-<i_m<\ldots<i_r}\right)$$

If $\alpha : [m] \to [n]$ is a nondecreasing function, then the induced map $\mathrm{N}^{\mathrm{dg}}_\bullet(C)_n \to \mathrm{N}^{\mathrm{dg}}_\bullet(C)_m$ is given by

$$\left(\{x_i\}_{0\leq i\leq n}, \{f_I\}\right) \mapsto \left(\left\{x_{\alpha(j)}\right\}_{0\leq j\leq m}, \{g_J\}\right)$$

where

$$g_J = \begin{cases} f_{\alpha(J)} & \text{if } \alpha|_J \text{ is injective} \\ \mathrm{id}_{x_i} & \text{if } J = \{j, j'\} \text{ with } \alpha(j) = \alpha(j') = i \\ 0 & \text{otherwise} \end{cases}$$

Example. Let C be a dg-category, then:

1. A 0-simplex of $\mathrm{N}^{\mathrm{dg}}_\bullet(C)$ is an object of C.
2. A 1-simplex of $\mathrm{N}^{\mathrm{dg}}_\bullet(C)$ is a morphism of C, that is, a pair of objects $x, y \in C$ together with an element $f \in \mathrm{Map}_C(x, y)_0$ satisfying $df = 0$.
3. A 2-simplex of $\mathrm{N}^{\mathrm{dg}}_\bullet(C)$ is a triple of objects $x, y, z \in C$ together with triple of morphisms
$$f \in \mathrm{Map}_C(x, y)_0, \quad g \in \mathrm{Map}_C(y, z)_0, \quad h \in \mathrm{Map}_C(x, z)_0$$
satisfying $df = dg = dh = 0$, together with an element $k \in \mathrm{Map}_C(x, z)_1$ satisfying $dk = (g \circ f) - h$.

Remark. Let C be a dg-category and let C_0 denote its underlying ordinary category. Then $\mathrm{N}_\bullet(C_0)$ is isomorphic to the simplicial subset of $\mathrm{N}^{\mathrm{dg}}_\bullet(C)$ whose n-simplices are the pairs $\left(\{x_i\}_{0\leq i\leq n}, \{f_I\}\right)$ with $f_I = 0$ whenever I has more than two elements. In particular, $\mathrm{N}_\bullet(C_0) \to \mathrm{N}^{\mathrm{dg}}_\bullet(C)$ is bijective on n-simplices for $n \leq 1$.

Proposition 3.1.1 (The dg-Nerve is an ∞-Category). If C is a dg-category, then the simplicial set $\mathrm{N}^{\mathrm{dg}}_\bullet(C)$ is an ∞-category.

Proof. It is enough to fill every inner horn. Fix $0 < j < n$ and a map

$$\varphi_0 : \Lambda^n_j \to \mathrm{N}^{\mathrm{dg}}_\bullet(C)$$

Unwinding the definition, the horn specifies objects $x_0, \dots, x_n$ and elements f_I satisfying the dg-nerve relation for every nonempty subset $I \subseteq [n]$, except for

$$I = [n], \quad I = [n] - \{j\}$$

Indeed, these are precisely the two nondegenerate simplices which do not lie in the jth horn. Define the missing elements by

$$f_{[n]-\{j\}} := \sum_{0<p<n} (-1)^{p-j} f_{\{p,p+1,\dots,n\}} \circ f_{\{0,\dots,p\}} - \sum_{0<p<n, p \neq j} (-1)^{p-j} f_{[n]-\{p\}}$$

and set

$$f_{[n]} := 0$$

We verify that these choices satisfy the two missing relations. With $f_{[n]} = 0$, the relation indexed by $[n]$ has zero left-hand side; solving its right-hand side for the only unknown term $f_{[n]-\{j\}}$ gives exactly the displayed formula.

It remains to check the relation indexed by $[n] - \{j\}$. For any subset I, let R_I be the left-hand side minus the right-hand side of its dg-nerve relation. Expanding $dR_{[n]}$ with $d^2 = 0$ and the Leibniz rule, the terms obtained by deleting two vertices cancel in pairs, and the threefold composites cancel by associativity. Since $R_I = 0$ for every proper face already contained in the horn, the result reduces to

$$dR_{[n]} = (-1)^j R_{[n]-\{j\}}$$

Our definition gives $R_{[n]} = 0$, so its differential vanishes and hence $R_{[n]-\{j\}} = 0$. We have therefore obtained an n-simplex extending φ_0. Every inner horn admits a filler, so $\mathrm{N}^{\mathrm{dg}}_\bullet(C)$ is an ∞ -category. □

Remark (The dg-Nerve Need Not Be Stable). The preceding proposition asserts only that $\mathrm{N}^{\mathrm{dg}}_\bullet(C)$ is an ∞-category. It need not be stable. For example, let C have one object x with

$$\mathrm{Map}_C(x, x)_* = k[0]$$

for a nonzero ring k. Then $\mathrm{Map}_{\mathrm{N}^{\mathrm{dg}}_\bullet(C)}(x, x)$ is the discrete space underlying k, hence is not contractible. Since x is the only object, a zero object would have to be x; therefore $\mathrm{N}^{\mathrm{dg}}_\bullet(C)$ is not even pointed and cannot be stable.

Stability requires additional structure on C. A standard sufficient hypothesis is that C be *pretriangulated*: it contains a zero object and is closed under shifts and mapping cones in a way compatible with its dg enrichment. Under this hypothesis $\mathrm{N}^{\mathrm{dg}}_\bullet(C)$ is stable. In particular, the dg-category $\mathbf{Ch}(A)$ of chain complexes is pretriangulated, which is why $\mathbf{Ch}_\infty(A) = \mathrm{N}^{\mathrm{dg}}_\bullet(\mathbf{Ch}(A))$ is stable in the next section.

3.1.3 Mapping Spaces and the Coherent Nerve

For a dg-category C, let hC denote the ordinary category with the same objects and morphism sets

$$\mathrm{Hom}_{hC}(x, y) := \mathrm{H}_0\big(\mathrm{Map}_C(x, y)_*\big)$$

Proposition 3.1.2 (Homotopy Category and Mapping Spaces). There is a canonical isomorphism

$$hC \simeq h\mathrm{N}_\bullet^{\mathrm{dg}}(C)$$

For every $x, y \in C$, the right mapping space of the dg-nerve is naturally isomorphic to

$$\mathrm{Map}^R_{\mathrm{N}_\bullet^{\mathrm{dg}}(C)}(x, y) \simeq \mathrm{DK}\left(\tau_{\geq 0}\mathrm{Map}_C(x, y)_*\right)$$

Consequently, for every $m \geq 0$,

$$\pi_m \mathrm{Map}_{\mathrm{N}_\bullet^{\mathrm{dg}}(C)}(x, y) \simeq \mathrm{H}_m\big(\mathrm{Map}_C(x, y)_*\big)$$

Proof sketch. The canonical map from the nerve of the closed degree-zero morphisms to $\mathrm{N}_\bullet^{\mathrm{dg}}(C)$ is the identity on objects and morphisms. A 2-simplex shows that two cycles $f, g \in \mathrm{Map}_C(x, y)_0$ represent the same morphism exactly when $f - g = dh$ for some $h \in \mathrm{Map}_C(x, y)_1$. This gives the first isomorphism.

An n-simplex of the right mapping space has fixed initial and terminal vertices x, y. Unwinding the dg-nerve relation identifies its remaining elements, after the standard change of signs, with an n-simplex of the Dold–Kan object of $\tau_{\geq 0}\mathrm{Map}_C(x, y)_*$. The last assertion is then the identity $\pi_m \, \mathrm{DK}\,(c) \simeq \mathrm{H}_m(c)$. □

The same formula constructs a simplicial category C_Δ. Its objects are those of C, and

$$\mathrm{Map}_{C_\Delta}(x, y) := \mathrm{DK}\left(\tau_{\geq 0}\mathrm{Map}_C(x, y)_*\right)$$

Composition is induced by the right-lax monoidal Alexander–Whitney map. Every mapping object is a simplicial abelian group and hence a Kan complex, so C_Δ is a fibrant simplicial category.

Proposition 3.1.3 (Comparison of the Two Nerves). There is a natural functor

$$\theta : N_{\mathrm{hc}}(C_\Delta) \to \mathrm{N}_\bullet^{\mathrm{dg}}(C)$$

which is an equivalence of ∞-categories.

Proof sketch. The functor is the identity on objects and is obtained by applying the Alexander–Whitney formula to coherent simplices. On every pair x, y, the preceding proposition identifies the induced map of right mapping spaces with the identity of $\mathrm{DK}\left(\tau_{\geq 0}\mathrm{Map}_C(x, y)_*\right)$. Thus θ is fully faithful and essentially surjective, hence an equivalence. □

This comparison explains why the direct formula for $\mathrm{N}_\bullet^{\mathrm{dg}}(C)$ and the longer construction by simplicial enrichment encode the same ∞-category.

3.1.4 Functoriality and the Model Structure

Definition 3.1.3 (dg-Functor). Let C, D be dg-categories over a commutative ring k. A dg-functor $F : C \to D$ assigns an object $F(x)$ to every $x \in C$ and a chain map

$$F_{x,y} : \mathrm{Map}_C(x, y)_* \to \mathrm{Map}_D(F(x), F(y))_*$$

to every pair x, y, compatibly with identities and composition. It induces functors $hF : hC \to hD$ and $\mathrm{N}^{\mathrm{dg}}_\bullet(F) : \mathrm{N}^{\mathrm{dg}}_\bullet(C) \to \mathrm{N}^{\mathrm{dg}}_\bullet(D)$.

Theorem 3.1.1 (Model Structure on dg-Categories). The category $\mathbf{dgCat}_k$ of small dg-categories over k admits a combinatorial model structure characterized as follows.

1. A dg-functor $F : C \to D$ is a weak equivalence, also called a *quasi-equivalence*, exactly when hF is an equivalence and every map

   $$\mathrm{Map}_C(x, y)_* \to \mathrm{Map}_D(F(x), F(y))_*$$

   is a quasi-isomorphism.
2. It is a fibration exactly when these maps of complexes are degreewise surjective and hF is an isofibration: every isomorphism $F(x) \to y$ in hD lifts to an isomorphism $x \to x'$ in hC.

Proposition 3.1.4 (The dg-Nerve is Right Quillen). Let L_{dg} denote the left adjoint of the dg-nerve. There is a Quillen adjunction

$$L_{\mathrm{dg}} : \mathbf{sSet}_{\mathrm{Joyal}} \rightleftarrows \mathbf{Cat}_{\mathrm{dg},k} : \mathrm{N}^{\mathrm{dg}}_\bullet$$

Thus L_{dg} is left Quillen and $\mathrm{N}^{\mathrm{dg}}_\bullet$ is right Quillen. The right adjoint sends a dg-category to its dg-nerve, while the left adjoint freely generates a dg-category from a simplicial set.

In particular, a quasi-equivalence of dg-categories induces an equivalence of ∞-categories, and a fibration induces an isofibration of dg-nerves.

The left adjoint is characterized by natural bijections

$$\mathrm{Hom}_{\mathbf{Cat}_{\mathrm{dg},k}}\big(L_{\mathrm{dg}}(\kappa), C\big) \simeq \mathrm{Hom}_{\mathbf{sSet}}\big(\kappa, \mathrm{N}^{\mathrm{dg}}_\bullet(C)\big)$$

for every simplicial set κ and dg-category C. Equivalently, it is the left Kan extension of its values on standard simplices:

$$L_{\mathrm{dg}}(\kappa) \simeq \operatorname*{colim}_{\Delta^n \to \kappa} L_{\mathrm{dg}}(\Delta^n)$$

Here $L_{\mathrm{dg}}(\Delta^n)$ is the universal dg-category whose dg-functors to C are the n-simplices of $\mathrm{N}^{\mathrm{dg}}_\bullet(C)$. This universal description is all that is needed for the Quillen adjunction; an explicit model can be obtained by applying normalized chains to simplicial rigidification.

Proof sketch. The dg-nerve preserves small limits and filtered colimits, so it admits a left adjoint. Suppose $F : C \to D$ is a fibration. In a lifting problem for an inner horn, the target filler determines the missing element in a mapping complex of D. Degreewise surjectivity lifts this element to C; the same formula used in the inner-horn proof then supplies the remaining face. Thus $\mathrm{N}^{\mathrm{dg}}_\bullet(F)$ is an inner fibration. The isofibration condition on hF lifts equivalences and makes it an isofibration.

If F is also a weak equivalence, the mapping-space formula above turns each quasi-isomorphism of mapping complexes into a weak homotopy equivalence, while the equivalence hF gives essential surjectivity. Hence $\mathrm{N}_\bullet^{\mathrm{dg}}(F)$ is a categorical equivalence. Therefore $\mathrm{N}_\bullet^{\mathrm{dg}}$ preserves fibrations and trivial fibrations and is right Quillen. □

Remark (Summary of the Construction). The preceding discussion can be organized into the following chain of ideas.

1. A dg-category stores morphisms from x to y in the chain complex $\mathrm{Map}_C(x, y)_*$, rather than in a set.
2. Connective truncation followed by Dold–Kan turns this complex into the mapping space of the dg-nerve:

$$\mathrm{Map}_{\mathrm{N}_\bullet^{\mathrm{dg}}(C)}(x, y) \simeq \mathrm{DK}\left(\tau_{\geq 0}\mathrm{Map}_C(x, y)_*\right)$$

   Consequently,

$$\pi_m \mathrm{Map}_{\mathrm{N}_\bullet^{\mathrm{dg}}(C)}(x, y) \simeq \mathrm{H}_m\big(\mathrm{Map}_C(x, y)_*\big)$$

   Thus the homology of a mapping complex becomes the homotopy of a mapping space. In degree zero this recovers $h\mathrm{N}_\bullet^{\mathrm{dg}}(C) \simeq hC$.
3. The direct dg-nerve and the homotopy coherent nerve of the associated simplicial category are equivalent. They are two models of the same ∞-category: the former is explicit, while the latter explains the simplicial enrichment behind it.
4. The model structure on dg-categories is designed so that a quasi-equivalence is exactly a quasi-isomorphism on every mapping complex together with an equivalence on homotopy categories. The mapping-space formula therefore shows that the dg-nerve sends every quasi-equivalence to an equivalence of ∞-categories.
5. These facts are compatible with the Quillen adjunction

$$L_{\mathrm{dg}} : \mathbf{sSet}_{\mathrm{Joyal}} \rightleftarrows \mathbf{Cat}_{\mathrm{dg},k} : \mathrm{N}_\bullet^{\mathrm{dg}}$$

   Hence the dg-nerve is the bridge from differential graded algebra to ∞-categories: chain homotopies and their higher compatibilities are retained as higher simplices. This statement gives only a Quillen *adjunction*. It does not assert that the two model categories are Quillen equivalent.

3.2 Derived ∞-Categories

The dg-nerve remembers chain homotopies and all their higher coherences, but it does not yet identify quasi-isomorphic complexes. The derived ∞-category is obtained by performing exactly this final localization.

3.2.1 Definition and Universal Property

Let A be an abelian category. Regard $\mathbf{Ch}(A)$ as the dg-category whose mapping complexes were defined above, and put

$$\mathbf{Ch}_\infty(A) \coloneqq \mathrm{N}^{\mathrm{dg}}_\bullet(\mathbf{Ch}(A))$$

Its homotopy category is the ordinary homotopy category $K(A)$ of chain complexes. Let W be the collection of quasi-isomorphisms, that is, chain maps $f : x \to y$ for which every map

$$\mathrm{H}_n(f) : \mathrm{H}_n(x) \to \mathrm{H}_n(y)$$

is an isomorphism.

Definition 3.2.4 (Derived ∞-Category). The *derived* ∞*-category* of A is the ∞-categorical localization

$$\mathbf{D}(A) \coloneqq \mathbf{Ch}_\infty(A)\left[W^{-1}\right]$$

It has the same objects as $\mathbf{Ch}_\infty(A)$, but every quasi-isomorphism is made into an equivalence. We write

$$q : \mathbf{Ch}_\infty(A) \to \mathbf{D}(A)$$

for the localization functor.

The definition is characterized by the universal property

$$\mathbf{Fun}\,(\mathbf{D}(A), C) \simeq \mathbf{Fun}_W(\mathbf{Ch}_\infty(A), C)$$

where the right-hand side is the full subcategory of functors which carry every quasi-isomorphism to an equivalence. Thus a construction on complexes descends to $\mathbf{D}(A)$ precisely when it is invariant under quasi-isomorphisms; the descent and all its coherences are then unique up to a contractible space of choices.

Proposition 3.2.5 (Stable Verdier Quotient). The ∞-category $\mathbf{Ch}_\infty(A)$ is stable. Let $\mathrm{Ac}(A)$ be its full stable subcategory of acyclic complexes. Then

$$\mathbf{D}(A) \simeq \mathbf{Ch}_\infty(A)/\ \mathrm{Ac}(A)$$

is a stable Verdier quotient, and q is exact. Moreover,

$$h\mathbf{D}(A) \simeq K(A)/\ \mathrm{Ac}(A)$$

is the classical triangulated derived category of A.

Proof sketch. Shifts and cofibers in $\mathbf{Ch}_\infty(A)$ are represented by shifts and mapping cones of complexes. A map f is a quasi-isomorphism exactly when $\mathrm{cofib}(f)$ is acyclic. Localizing at W is therefore the same as killing the stable subcategory $\mathrm{Ac}(A)$. A quotient of a stable ∞-category by a stable subcategory is stable, and passage to the homotopy category gives the usual Verdier quotient. □

Consequently, a cofiber sequence in $\mathbf{D}(A)$ is represented by a mapping-cone sequence

$$x \xrightarrow{f} y \to \mathrm{Cone}\,(f) \to x[1]$$

and becomes a distinguished triangle in $h\mathbf{D}(A)$. The ∞-category retains the functorial cofiber and all higher coherences which its triangulated homotopy category forgets.

3.2.2 Homology, the t-Structure, and Ext

Proposition 3.2.6 (Standard t-Structure). The derived ∞-category has a canonical t-structure defined by

$$\mathbf{D}(A)_{\geq 0} := \{x : \mathrm{H}_n(x) = 0 \ \text{ for } \ n < 0\}$$

$$\mathbf{D}(A)_{\leq 0} := \{x : \mathrm{H}_n(x) = 0 \ \text{ for } \ n > 0\}$$

Its truncation functors are induced by the good truncations of complexes, and its homology-object functors are the ordinary homology functors

$$\pi_n(x) \simeq \mathrm{H}_n(x)$$

The inclusion of objects concentrated in degree zero induces an equivalence of abelian categories

$$A \simeq \mathbf{D}(A)^{\heartsuit}$$

Proof sketch. Good truncations give a functorial fiber–cofiber sequence

$$\tau_{\geq 0} x \to x \to \tau_{\leq -1} x$$

and the two terms have the required homology vanishing. If $x \in \mathbf{D}(A)_{\geq 0}$ and $y \in \mathbf{D}(A)_{\leq -1}$, every map $x \to y$ vanishes by truncation, giving orthogonality. A complex with homology only in degree zero is quasi-isomorphic to that homology object placed in degree zero, which identifies the heart with A. □

The usual bounded variants are recovered directly from this t-structure:

$$\mathbf{D}^{+}(A) := \cup_{m \in \mathbb{Z}} \mathbf{D}(A)_{\geq m}, \quad \mathbf{D}^{-}(A) := \cup_{n \in \mathbb{Z}} \mathbf{D}(A)_{\leq n}$$

$$\mathbf{D}^{b}(A) := \cup_{m \leq n} \left(\mathbf{D}(A)_{\geq m} \cap \mathbf{D}(A)_{\leq n}\right)$$

For $x, y \in \mathbf{D}(A)$ and $m \geq 0$, stability gives

$$\pi_m \mathrm{Map}_{\mathbf{D}(A)}(x, y) \simeq \mathrm{Hom}_{h\,\mathbf{D}(A)}(x[m], y)$$

For $a, b \in A \simeq \mathbf{D}(A)^{\heartsuit}$, define

$$\mathrm{Ext}^n_A(a, b) := \mathrm{Hom}_{h\mathbf{D}(A)}(a, b[n]) \quad (n \geq 0)$$

Equivalently,

$$\mathrm{Ext}^n_A(a, b) \simeq \pi_0 \mathrm{Map}_{\mathbf{D}(A)}(a, b[n])$$

This recovers the classical Yoneda Ext groups. Notice the direction of the shift: with our homological indexing, $\pi_m \mathrm{Map}_{\mathbf{D}(A)}(a, b) \simeq \mathrm{Hom}_{h\,\mathbf{D}(A)}(a, b[-m])$. Thus positive Ext is seen by shifting the target before taking the mapping space.

3.2.3 Resolutions and Derived Functors

The localization definition is conceptual, while resolutions make it computable.

Definition 3.2.5 (K-Projective and K-Injective Complexes). A complex p is *K-projective* if

$$\mathrm{Map}_{\mathbf{Ch}_{\infty}(A)}(p, a) \simeq *$$

for every acyclic complex a. Dually, a complex i is K-*injective* if

$$\mathrm{Map}_{\mathbf{Ch}_\infty(A)}(a,i) \simeq *$$

for every acyclic a.

Proposition 3.2.7 (Computing the Localization). If every complex x admits a quasi-isomorphism $p \to x$ with p K-projective, then the full dg-subcategory of K-projective complexes presents $\mathbf{D}(A)$. Dually, if every x admits a quasi-isomorphism $x \to i$ with i K-injective, then the K-injective complexes present $\mathbf{D}(A)$.

With either kind of replacement, derived mapping spaces can be computed by

$$\mathrm{Map}_{\mathbf{D}(A)}(x,y) \simeq \mathrm{DK}\big(\tau_{\geq 0}\underline{\mathrm{Hom}}_A(p,y)\big)$$

$$\mathrm{Map}_{\mathbf{D}(A)}(x,y) \simeq \mathrm{DK}\big(\tau_{\geq 0}\underline{\mathrm{Hom}}_A(x,i)\big)$$

Proof sketch. A quasi-isomorphism between K-projective complexes is a chain-homotopy equivalence, and the same holds for K-injectives. Moreover, mapping from a K-projective complex or into a K-injective complex already sends quasi-isomorphisms to equivalences. Hence the displayed mapping objects satisfy the universal property of the localization. □

If A is a Grothendieck abelian category, every complex admits a K-injective resolution; in particular, $\mathbf{D}(A)$ is a presentable stable ∞-category. For module categories one may also use K-projective resolutions.

Let $F : A \to B$ be additive and apply it degreewise to complexes. If F is exact, it preserves quasi-isomorphisms and therefore induces an exact functor

$$\mathbf{D}(F) : \mathbf{D}(A) \to \mathbf{D}(B)$$

For a nonexact functor, its total left and right derived functors, when they exist, are the Kan extensions through q. In terms of adapted resolutions, they are computed by

$$\mathrm{L}F(x) \simeq F(p), \quad \mathrm{R}F(x) \simeq F(i)$$

The point of a resolution is precisely that the right-hand side no longer depends on the chosen representative of x.

Example (Modules over a Ring). For a ring R, the derived ∞-category is

$$\mathbf{D}(R) \coloneqq \mathbf{D}(\mathbf{Mod}_R)$$

If $p \to m$ is a K-projective resolution and n is another complex, then

$$m \otimes^{\mathrm{L}}_R n \simeq p \otimes_R n, \quad \mathrm{RHom}_R(m,n) \simeq \underline{\mathrm{Hom}}_R(p,n)$$

When m, n are modules placed in degree zero, our homological convention gives

$$\mathrm{H}_r\big(m \otimes^{\mathrm{L}}_R n\big) \simeq \mathrm{Tor}^R_r(m,n), \quad \mathrm{H}_{-r}(\mathrm{RHom}_R(m,n)) \simeq \mathrm{Ext}^r_R(m,n)$$

Remark (Core Idea). The construction has two logically separate stages:

$$\mathbf{Ch}(A) \xrightarrow{\text{dg-nerve}} \mathbf{Ch}_\infty(A) \xrightarrow{\text{invert quasi-isomorphisms}} \mathbf{D}(A)$$

The first stage promotes chain homotopies to coherent higher morphisms. The second discards the choice of resolution by identifying complexes with the same derived information. Stability then organizes cones into cofiber sequences, the standard t-structure recovers homology and the original abelian category, and resolutions provide concrete formulas for mapping spaces and derived functors.

4 Spectra

4.1 Why We Need Spectra?

4.1.1 "Linear" Matters

Since the title of this book is actually "Linear Algebra", we have to fix one thing: since the homotopy theory of pointed spaces is not "linear". For pointed spaces X, Y, consider

$$[X, Y]_* := \pi_0 \mathrm{Map}_*(X, Y)$$

to be the homotopy classes of pointed maps, it is only a set, not a group. Things changes if X is a suspension

$$X = \Sigma X'$$

then, through

$$\mathrm{Map}_*(\Sigma X', Y) \simeq \mathrm{Map}_*(X', \Omega Y)$$

We have

$$[\Sigma X', Y]_* = \pi_1 \mathrm{Map}_*(X', Y)$$

naturally has a group structure. And if we suspension it again by

$$X = \Sigma^2 X''$$

then we have

$$\left[\Sigma^2 X'', Y\right]_* = \pi_2 \mathrm{Map}_*(X'', Y)$$

which is an abelian group. In fact, by suspending the space enough times, making the homotopy theory more and more "linear".

By this idea, we would like to consider

$$[X, Y]_* \to [\Sigma X, \Sigma Y]_* \to \left[\Sigma^2 X, \Sigma^2 Y\right]_* \to \ldots$$

and take the limit

$$[X, Y]^s = \varinjlim_n [\Sigma^n X, \Sigma^n Y]$$

called the *stable homotopy classes of maps* from X to Y.For fine spaces, this forms an abelian group.

4.1.2 Inverting Suspension

There is a fatal problem exists in the suspension of pointed animae, which is that the suspension functor

$$\Sigma : \mathbf{Ani}_* \to \mathbf{Ani}_*$$

is not invertible, there does not exist a functor

$$\Sigma^{-1} : \mathbf{Ani}_* \to \mathbf{Ani}_*$$

corresponding to the inverse of suspension (i.e. we want the ∞-category to be stable). Stable homotopy theory wants to invert the suspension functor, and this is the main reason why we need *spectra*, which is a world where suspension forms an equivalence.

$$\Sigma : \mathbf{Sp} \xrightarrow{\sim} \mathbf{Sp}$$

just like the $[1]$ functor on the derived category of an abelian category.

4.2 Brown Representability Theorem

From now on, we will build a relation between cohomology theory and the theory of spectra, we will start with the Brown representability theorem, which is a very important theorem in stable homotopy theory.

4.2.1 Generalized Cohomology Theories

Definition 4.2.1 (Reduced Generalized Cohomology Theory). A *reduced generalized cohomology theory* on pointed CW complexes is a family of contravariant functors

$$E^q : (h\mathbf{CW}_*)^{\mathrm{op}} \to \mathbf{Ab} \quad (q \in \mathbb{Z})$$

with the following properties.

1. *Reducedness.* $E^q(*) = 0$ for every q.
2. *Suspension.* There are natural isomorphisms
$$E^q(x) \simeq E^{q+1}(\Sigma x)$$
3. *Exactness.* Every cofiber sequence
$$a \to x \to x/a \to \Sigma a$$
induces a natural long exact sequence
$$\dots \to E^q(x/a) \to E^q(x) \to E^q(a) \to E^{q+1}(x/a) \to \dots$$
4. *Wedge axiom.* For every family $\{x_\alpha\}$,
$$E^q\left(\bigvee_\alpha x_\alpha\right) \simeq \prod_\alpha E^q(x_\alpha)$$

The adjective *generalized* means that no dimension axiom is imposed.

Thus E^q is the degree-q cohomology functor, not the qth power of a single functor. Exactness implies the Mayer–Vietoris gluing property, while the wedge axiom is already one of the Brown axioms. Consequently, after forgetting the abelian-group structure, every E^q is a Brown functor.

Example. Reduced singular cohomology $\tilde{\mathrm{H}}^q(-;G)$ is the basic example. Its dimension axiom says

$$\tilde{\mathrm{H}}^0(S^0;G) \simeq G, \quad \tilde{\mathrm{H}}^q(S^0;G) \simeq 0 \quad (q \neq 0)$$

Generalized theories such as topological K-theory need not satisfy this axiom.

Remark (Homology versus Cohomology). A reduced generalized *homology* theory is instead a covariant family

$$E_q : h\mathbf{CW}_* \to \mathbf{Ab}$$

which sends wedges to direct sums and cofiber sequences to long exact sequences. Brown representability below directly concerns the contravariant cohomology functors E^q. Homology theories acquire their representing description after passing to spectra.

4.2.2 Classical Version

Let $h\mathbf{CW}_*$ denote the homotopy category of pointed CW complexes.

Definition 4.2.2 (Brown Functor). A contravariant functor

$$F : (h\mathbf{CW}_*)^{\mathrm{op}} \to \mathbf{Set}_*$$

is a *Brown functor* if it satisfies:

1. *Wedge axiom.*

$$F\left(\bigvee_\alpha x_\alpha\right) \simeq \prod_\alpha F(x_\alpha)$$

2. *Mayer–Vietoris axiom.* If $x = a \cup b$ is a union of pointed subcomplexes whose interiors cover x, then

$$F(x) \to F(a) \times_{F(a\cap b)} F(b)$$

   is surjective.

Every representable functor $[-, y]_*$ has these properties. The wedge axiom is the coproduct universal property, while Mayer–Vietoris follows from the homotopy extension property for inclusions of CW subcomplexes.

Theorem 4.2.1 (Classical Brown Representability). Every Brown functor is representable. More precisely, there exist a pointed CW complex y and a universal class $u \in F(y)$ such that

$$T_u : [x, y]_* \to F(x), \quad [f] \mapsto f^*u$$

is a natural bijection for every pointed CW complex x. The representing space y is unique up to pointed homotopy equivalence.

Proof sketch. The representing pair (y, u) is built by cells. First wedge onto y enough spheres so that every class in $F(S^n)$ is the pullback of u; this forces surjectivity on spheres. Next attach cells along every map $S^n \to y$ which pulls u back to the basepoint; the Mayer–Vietoris axiom extends u across these cells and kills the kernel. Repeating this in every dimension and taking a mapping telescope gives

$$[S^n, y]_* \simeq F(S^n)$$

for all n.

A relative version of the same cell construction extends a map $a \to y$ from a CW subcomplex $a \subset x$ whenever its pullback of u agrees with a prescribed class in $F(x)$. Taking $a = *$ proves that T_u is surjective. Applying the relative statement to the two ends of a cylinder shows that two maps with the same pullback of u are homotopic, so T_u is injective. Whitehead's theorem upgrades the weak equivalences arising in the construction to homotopy equivalences, and Yoneda gives uniqueness of y. □

> **Remark (From Cohomology Theories to Spectra).** Let $\{E^q\}_{q\in\mathbb{Z}}$ be a reduced generalized cohomology theory on pointed CW complexes. Each functor E^q satisfies the wedge and Mayer–Vietoris axioms, so Brown representability gives pointed CW complexes e_q with
>
> $$E^q(x) \simeq \left[x, e_q\right]_*$$
>
> The suspension isomorphism and the suspension–loop adjunction give
>
> $$\left[x, e_q\right]_* \simeq \left[\Sigma x, e_{q+1}\right]_* \simeq \left[x, \Omega e_{q+1}\right]_*$$
>
> Yoneda therefore yields equivalences
>
> $$e_q \simeq \Omega e_{q+1}$$
>
> Hence the representing spaces assemble into an Ω-spectrum. This is the classical reason that spectra represent generalized cohomology theories.

4.2.3 Representability in an ∞-Category

Let C be an ∞-category. A space-valued presheaf

$$P : C^{\mathrm{op}} \to \mathbf{Ani}$$

is *representable* if there is an object $y \in C$ and a natural equivalence

$$P(x) \simeq \mathrm{Map}_C(x, y)$$

for every $x \in C$. By the ∞-categorical Yoneda lemma, choosing such an equivalence is the same as choosing a *universal element* $u \in P(y)$ for which evaluation induces the displayed equivalences. The representing object is unique up to equivalence.

Passing to connected components gives the ordinary representable functor

$$\mathrm{Hom}_{hC}(-, y) \simeq \pi_0\mathrm{Map}_C(-, y)$$

on the homotopy category. Accordingly, a functor $F : (hC)^{\mathrm{op}} \to \mathbf{Set}$ is representable when there are $y \in C$ and $u \in F(y)$ such that

$$\mathrm{Hom}_{hC}(x, y) \to F(x), \quad [f] \mapsto f^*u$$

is a bijection for every x. The theorem below concerns this Set-valued notion: it reconstructs the representing object from the behavior of F on colimits in the underlying ∞-category.

For an ordinary presentable category D, the adjoint functor theorem says that $F : D^{\mathrm{op}} \to \mathbf{Set}$ is representable exactly when it carries colimits in D to limits of sets. The difficulty here is that hC

need not possess the colimits which exist in C. Brown representability replaces that unavailable criterion by conditions imposed directly on coproducts and pushouts in C.

Remark (Basic Principle). A representable contravariant functor turns colimits into limits, since

$$\mathrm{Map}_C\Big(\operatorname*{colim}_i x_i, y\Big) \simeq \lim_i \mathrm{Map}_C(x_i, y)$$

After applying π_0, arbitrary coproducts still become products. For a pushout, connected components retain the existence, but not necessarily the uniqueness, of a gluing; this explains the surjectivity condition in Brown representability.

4.2.4 Cogroup Objects and Compact Generators

Definition 4.2.3 (Cogroup Object). Let D be a category with finite coproducts. A *cogroup object* is an object $s \in D$ equipped with a comultiplication

$$\Delta : s \to s \sqcup s$$

together with a counit and coinverse, such that for every $y \in D$ the induced operation

$$\mathrm{Hom}_D(s, y) \times \mathrm{Hom}_D(s, y) \simeq \mathrm{Hom}_D(s \sqcup s, y) \xrightarrow{\Delta^*} \mathrm{Hom}_D(s, y)$$

is a group law. Equivalently, $\mathrm{Hom}_D(s, -)$ naturally takes values in groups.

The guiding example is a suspension. If C admits finite colimits and $x \to \emptyset$ is a map to its initial object, then

$$\Sigma x := \emptyset \sqcup_x \emptyset$$

is a cogroup object of hC. Indeed, $\mathrm{Hom}_{hC}(\Sigma x, y)$ is a fundamental group of a mapping space, and its group operation is loop concatenation. In a stable ∞-category, hC is additive, so every object is a cogroup object; the comultiplication is the diagonal $s \to s + s$ and the coinverse is $-\,\mathrm{id}_s$.

Definition 4.2.4 (Compact Generators). An object $s \in C$ is *compact* if $\mathrm{Map}_C(s, -)$ preserves filtered colimits. A set $\{s_\alpha\}_{\alpha \in A}$ *generates* C *under small colimits* if the smallest full subcategory containing every s_α and closed under equivalences and small colimits is all of C.

Compactness lets maps out of s_α detect a colimit at a finite stage; generation means that the functors $\mathrm{Map}_C(s_\alpha, -)$ jointly detect equivalences. These properties abstract the roles played by spheres and CW cells in the classical proof.

4.2.5 General Brown Representability

Theorem 4.2.2 (Brown Representability). Let C be a presentable ∞-category containing a set $\{s_\alpha\}_{\alpha \in A}$ such that:

1. every s_α is a cogroup object of hC;
2. every s_α is compact;
3. the objects s_α generate C under small colimits.

Then a functor $F : (hC)^{\mathrm{op}} \to \mathbf{Set}$ is representable if and only if it satisfies the following conditions.

1. For every collection $\{c_\beta\}$, the canonical map
$$F\left(\sqcup_\beta c_\beta\right) \to \prod_\beta F\left(c_\beta\right)$$
is a bijection.
2. For every pushout $d' \simeq c' \sqcup_c d$, the canonical map
$$F(d') \to F(c') \times_{F(c)} F(d)$$
is surjective.

Proof sketch. Necessity follows from the mapping-space formula above. For sufficiency, begin with a pair (x, u) and coproduct copies of the generators until
$$\mathrm{Hom}_{hC}(s_\alpha, x) \to F(s_\alpha)$$
is surjective for every α. Use pushouts to kill its kernels, and iterate. Cogroup structures make these kernels groups, compactness lets maps from s_α factor through a finite stage, and generation shows that the resulting universal pair represents F on every object. This is exactly the classical argument *add generators, then kill relations*, with compact cogroup generators replacing spheres. □

Corollary 4.2.1. Every compactly generated presentable stable ∞-category satisfies the hypotheses of Brown representability, because every object of its homotopy category is a cogroup object. In particular, a contravariant functor on its homotopy category is representable exactly when it takes coproducts to products and pushouts to weak pullbacks of sets.

For connected pointed spaces, the compact cogroup generator is S^1; its suspensions detect all higher homotopy groups. Thus the generalized theorem recovers the classical Brown theorem, while its stable case is the form that will be used for spectra.

Remark (From Cohomology to Homotopy). Brown representability turns a reduced generalized cohomology theory into homotopy-theoretic data. If e_q represents E^q, then
$$E^q(x) \simeq \left[x, e_q\right]_*, \quad E^q(S^m) \simeq \pi_m\left(e_q\right)$$
Thus cohomology classes are homotopy classes of maps, and the coefficient groups of the theory are homotopy groups of the representing spaces. Moreover, Yoneda identifies a natural cohomology operation $E^q \to E'^r$ with a homotopy class of maps $e_q \to e'_r$.

The suspension axiom organizes the spaces e_q into an Ω-spectrum, so a whole cohomology theory becomes a single stable homotopy type. Brown's theorem directly concerns contravariant cohomology theories. Covariant homology theories enter after stabilization: a representing spectrum e defines homology by the homotopy groups of $e \wedge \Sigma^\infty x$.

4.3 Spectrum Objects

Spectrum objects are the basic device for extracting stable information from an unstable ∞-category. The key input is *excision*: a linear functor should turn a gluing problem into a pullback problem.

4.3.1 Reduced Excisive Functors

The classical model is singular homology. The simplicial abelian group $\mathbb{Z}\mathrm{Sing}(x)_\bullet$ is a Kan complex and

$$\mathrm{H}_n(x;\mathbb{Z}) \simeq \pi_n\big(\mathbb{Z}\mathrm{Sing}(x)_\bullet\big) \quad (n \geq 0)$$

For an open cover $x = u \cup v$, the Mayer–Vietoris sequence is obtained by applying homotopy groups to the homotopy pullback square associated with u, v, $u \cap v$, and x. This is the prototype for the following definition.

Definition 4.3.5 (Reduced Excisive Functor). Let C admit pushouts and let D admit pullbacks. A functor $F : C \to D$ is *excisive* if it carries every pushout square in C to a pullback square in D.

If C and D have final objects, then F is *reduced* if it carries a final object of C to a final object of D. We write

$$\mathbf{Exc}_*(C, D) \subseteq \mathbf{Fun}(C, D)$$

for the full subcategory of reduced excisive functors.

Remark (Excision Is Exactness). If C and D are stable, a functor $F : C \to D$ is reduced and excisive if and only if it is exact. Thus excision is the unstable formulation of the familiar requirement that a linear functor preserve exact triangles.

Equivalently, F is exact precisely when it preserves zero objects and the canonical map

$$\Sigma_D F(x) \to F(\Sigma_C x)$$

is an equivalence for every x.

There is a useful one-object test for excision. Let C be pointed with finite colimits, let D be pointed with finite limits, and let $F : C \to D$ be reduced. Applying F to the suspension square of x produces a canonical map

$$\eta_x : F(x) \to \Omega_D F(\Sigma_C x)$$

Proposition 4.3.1 (Suspension Test for Excision). The functor F is excisive if and only if η_x is an equivalence for every $x \in C$.

Proof sketch. If F is excisive, it sends the suspension pushout of x to the pullback defining $\Omega_D F(\Sigma_C x)$. Conversely, compare an arbitrary pushout square with the four suspension squares of its vertices. The maps η_x identify the resulting comparison map with an equivalence, so the image square is a pullback. □

4.3.2 The Definition of a Spectrum Object

Let $\mathbf{Ani}_*^{\mathrm{fin}}$ be the ∞-category of finite pointed animae. It is generated by S^0 under finite colimits.

Definition 4.3.6 (Spectrum Object). Let C be an ∞-category with finite limits. A *spectrum object* of C is a reduced excisive functor

$$x : \mathbf{Ani}_*^{\mathrm{fin}} \to C$$

The full ∞-category of spectrum objects is

$$\mathbf{Sp}(C) := \mathbf{Exc}_*\left(\mathbf{Ani}_*^{\mathrm{fin}}, C\right)$$

Remark (Why Does This Definition Produce Spectra?). At first sight, a spectrum object is merely a functor in $\mathbf{Fun}\left(\mathbf{Ani}_*^{\mathrm{fin}}, C\right)$. The reduced and excisive conditions force this functor to contain an infinite sequence of deloopings.

For every $n \geq 0$, the suspension square

$$\begin{tikzcd} S^n \arrow[r] \arrow[d] & * \arrow[d] \\ * \arrow[r] & \Sigma S^n \simeq S^{n+1} \end{tikzcd}$$

is a pushout in $\mathbf{Ani}_*^{\mathrm{fin}}$. Applying x gives

$$\begin{tikzcd} x(S^n) \arrow[r] \arrow[d] & x(*) \arrow[d] \\ x(*) \arrow[r] & x\left(S^{n+1}\right) \end{tikzcd}$$

Reducedness says $x(*)$ is terminal, while excision says that the second square is a pullback. Therefore

$$x(S^n) \simeq * \times_{x(S^{n+1})} * \simeq \Omega x\left(S^{n+1}\right)$$

Setting $x_n := x(S^n)$ gives coherent equivalences

$$x_n \simeq \Omega x_{n+1}$$

Hence one reduced excisive functor automatically contains the infinite delooping data

$$x_0 \simeq \Omega x_1, \quad x_1 \simeq \Omega x_2, \quad x_2 \simeq \Omega x_3, \quad \ldots$$

The functor

$$\Omega^\infty : \mathbf{Sp}(C) \to C, \quad x \mapsto x_0$$

is evaluation at S^0. More precisely, if C_* denotes the ∞-category of pointed objects of C, then

$$\mathbf{Sp}(C) \simeq \lim\left(\ldots \xrightarrow{\Omega} C_* \xrightarrow{\Omega} C_*\right)$$

When C is already pointed, this becomes the tower displayed at the beginning of this section.

Remark (Finite versus Compact). The category $\mathbf{Ani}_*^{\mathrm{fin}}$ need not contain every compact pointed anima. The full subcategory of compact objects of $\mathbf{Ani}_*$ is its idempotent completion. The finite category is the one used above because of its universal property under finite colimits.

4.3.3 Classical Sequential Spectra

The classical point-set model makes the tower of deloopings explicit.

Definition 4.3.7 (Sequential Spectrum). A *sequential prespectrum* is a sequence of pointed spaces $(x_0, x_1, \ldots)$ equipped with structure maps

$$\sigma_n : \Sigma x_n \to x_{n+1}$$

Equivalently, by adjunction, it has maps

$$\tilde{\sigma}_n : x_n \to \Omega x_{n+1}$$

It is an Ω-*spectrum* if every $\tilde{\sigma}_n$ is a weak homotopy equivalence.

For $k \in \mathbb{Z}$, the stable homotopy group of a prespectrum x is

$$\pi_k^s(x) := \varinjlim_n \pi_{k+n}(x_n)$$

where n is taken sufficiently large that $k + n \geq 0$. A map is a *stable equivalence* if it induces an isomorphism on every π_k^s. The classical stable homotopy category is obtained by inverting these maps. Every sufficiently well-behaved prespectrum admits a spectrification $x \to qx$ to an Ω-spectrum with the same stable homotopy groups.

Example (Basic Spectra). For a pointed space x, its suspension prespectrum has levels

$$(\Sigma^\infty x)_n := \Sigma^n x$$

After spectrification this defines the suspension spectrum $\Sigma^\infty x \in \mathbf{Sp}$. In particular,

$$\mathbb{S} := \Sigma^\infty S^0$$

is the sphere spectrum, and its homotopy groups are the stable stems

$$\pi_k(\mathbb{S}) \simeq \varinjlim_n \pi_{k+n}(S^n)$$

If a is an abelian group, the Eilenberg–Mac Lane spectrum $\mathrm{H}a$ has spaces $K(a, n)$ and structure equivalences

$$K(a, n) \simeq \Omega K(a, n + 1)$$

It is characterized by $\pi_0(\mathrm{H}a) \simeq a$ and $\pi_k(\mathrm{H}a) \simeq 0$ for $k \neq 0$. Ordinary reduced cohomology is then represented by

$$\tilde{\mathrm{H}}^q(x; a) \simeq [\Sigma^\infty x, \Sigma^q\, \mathrm{H}a]_{\mathbf{Sp}}$$

Remark (Classical versus ∞-Categorical Spectra). The two descriptions encode the same stable homotopy theory.

classical model	∞-categorical model
spaces x_n and maps $\Sigma x_n \to x_{n+1}$	a functor $\mathbf{Ani}_*^{\mathrm{fin}} \to \mathbf{Ani}$
Ω-spectrum condition	reduced excision
invert stable equivalences	equivalences already encode localization
chosen point-set structure maps	all homotopy coherences retained

More precisely, localizing any standard model category of classical spectra at its stable equivalences presents the ∞-category $\mathbf{Sp} \simeq \mathbf{Sp}(\mathbf{Ani})$. Thus an ∞-categorical spectrum should be compared with the stable homotopy type of a classical spectrum, not with one chosen point-set presentation.

4.3.4 Stability

Theorem 4.3.3 (Stability Criterion). For a pointed ∞-category C, the following conditions are equivalent.

1. The ∞-category C is stable.
2. It admits finite colimits and $\Sigma_C : C \to C$ is an equivalence.
3. It admits finite limits and $\Omega_C : C \to C$ is an equivalence.

Proof sketch. In a stable ∞-category, pushouts and pullbacks agree, so suspension and looping are inverse equivalences. Conversely, if looping is invertible, every pullback can be delooped into a pushout; hence pullbacks and pushouts agree. The suspension version is dual. □

Theorem 4.3.4 (Stability of Excisive Functors). Let C be pointed and admit finite colimits, and let D admit finite limits. Then $\mathbf{Exc}_*(C, D)$ is a stable ∞-category. In particular, $\mathbf{Sp}(D)$ is stable.

Proof sketch. Reduced excisive functors are closed under finite limits, which are computed pointwise. Their loop functor is also pointwise, and the suspension test identifies it with an equivalence. The stability criterion therefore applies. Taking $C = \mathbf{Ani}_*^{\mathrm{fin}}$ gives the assertion for spectrum objects. □

Corollary 4.3.2 (Recognizing Stability by Spectrum Objects). Let C admit finite limits. Then C is stable if and only if evaluation at S^0 is an equivalence

$$\Omega_C^\infty : \mathbf{Sp}(C) \xrightarrow{\sim} C$$

In this case, we denote its inverse by

$$\Sigma_C^\infty : C \xrightarrow{\sim} \mathbf{Sp}(C)$$

Thus

$$\Omega^\infty_C \Sigma^\infty_C \simeq \mathrm{id}_C, \quad \Sigma^\infty_C \Omega^\infty_C \simeq \mathrm{id}_{\mathbf{Sp}(C)}$$

For $c \in C$, the spectrum $\Sigma^\infty_C c$ is determined levelwise by

$$(\Sigma^\infty_C c)_n \simeq \Sigma^n_C c$$

Proof sketch. If C is stable, then Ω_C is an equivalence, so the tower

$$\ldots \xrightarrow{\Omega_C} C \xrightarrow{\Omega_C} C$$

is essentially constant. Its limit is therefore C, and evaluation at the zeroth level is an equivalence. Conversely, $\mathbf{Sp}(C)$ is stable; hence an equivalence $\mathbf{Sp}(C) \simeq C$ transports the stable structure to C. □

Remark. In general, Σ^∞ denotes the suspension-spectrum or stabilization functor and is left adjoint to Ω^∞ when this adjoint exists. It is only when C is already stable that this adjunction becomes an adjoint equivalence and Σ^∞_C is literally the inverse of Ω^∞_C.

4.3.5 The Universal Property of Stabilization

Theorem 4.3.5 (Universal Property). Let C be pointed and admit finite colimits, and let D admit finite limits. Composition with evaluation at S^0 induces an equivalence

$$\mathbf{Exc}_*(C, \mathbf{Sp}(D)) \xrightarrow{\sim} \mathbf{Exc}_*(C, D)$$

Hence every reduced excisive functor $C \to D$ has an essentially unique refinement whose values are coherent spectrum objects. Taking $C = \mathbf{Ani}^{\mathrm{fin}}_*$ shows in particular that stabilization is idempotent

$$\mathbf{Sp}(\mathbf{Sp}(D)) \simeq \mathbf{Sp}(D)$$

Remark (Formal Properties). Spectrum objects and their limits are computed pointwise. Consequently, for every simplicial set K there is a natural equivalence

$$\mathbf{Sp}(\mathbf{Fun}(K, C)) \simeq \mathbf{Fun}(K, \mathbf{Sp}(C))$$

If C is presentable, the reduced and excisive conditions define an accessible localization of a functor category. Hence $\mathbf{Sp}(C)$ is again presentable, as well as stable.

Remark (Homology Theories). A reduced spectrum-valued homology theory is, at its core, a reduced excisive functor

$$E : \mathbf{Ani}^{\mathrm{fin}}_* \to \mathbf{Sp}$$

and its graded groups are recovered by

$$E_n(x) \coloneqq \pi_n E(x)$$

Excision produces Mayer–Vietoris long exact sequences, while preservation of arbitrary wedges supplies the usual wedge axiom when the theory is extended from finite pointed animae to all pointed animae. Thus the spectrum packages all degrees and all connecting maps of a homology theory into one object.

4.4 ∞-Category of Spectra

4.4.1 Spectrum

The most important example of a stable ∞-category is the category of spectra itself. The preceding construction turns the classical sequence of spaces and deloopings into the following intrinsic definition.

Definition 4.4.8 (The ∞-Category of Spectra). A *spectrum* is a spectrum object of the ∞-category of animae. We write

$$\mathbf{Sp} := \mathbf{Sp}(\mathbf{Ani}) \simeq \mathbf{Sp}(\mathbf{Ani}_*)$$

for the ∞-category of spectra. Equivalently,

$$\mathbf{Sp} \simeq \mathbf{Exc}_*\left(\mathbf{Ani}_*^{\mathrm{fin}}, \mathbf{Ani}\right)$$

Proposition 4.4.2 (Equivalent Sequential Description). Equivalently, a spectrum is a sequence of pointed animae

$$x = (x_0, x_1, x_2, \ldots)$$

together with coherent equivalences

$$\varepsilon_n : x_n \xrightarrow{\sim} \Omega x_{n+1} \quad (n \geq 0)$$

A morphism $f : x \to y$ is a sequence of pointed maps $f_n : x_n \to y_n$ compatible, up to coherent homotopy, with the equivalences ε_n. In particular,

$$\mathbf{Sp} \simeq \lim\left(\ldots \xrightarrow{\Omega} \mathbf{Ani}_* \xrightarrow{\Omega} \mathbf{Ani}_*\right)$$

and the infinite-loop-space functor is evaluation at the zeroth term

$$\Omega^\infty x \simeq x_0$$

Remark. This is the intrinsic ∞-categorical version of a classical Ω-spectrum. The word *coherent* is essential: besides the equivalences $x_n \simeq \Omega x_{n+1}$, the ∞-categorical limit keeps all higher compatibilities automatically. Thus one may safely remember a spectrum as the infinite delooping sequence

$$x_0 \simeq \Omega x_1, \quad x_1 \simeq \Omega x_2, \quad x_2 \simeq \Omega x_3, \quad \ldots$$

Definition 4.4.9 (Sphere Spectrum). The *sphere spectrum* is the suspension spectrum of the pointed 0-sphere

$$\mathbb{S} := \Sigma^\infty S^0 \in \mathbf{Sp}$$

In the classical sequential model it is represented by the suspension prespectrum

$$S^0, S^1, S^2, \ldots$$

with the canonical structure maps $\Sigma S^n \simeq S^{n+1}$, followed by spectrification. Its homotopy groups are the stable stems

$$\pi_k(\mathbb{S}) \simeq \varinjlim_n \pi_{k+n}(S^n)$$

Remark (The Stable Homotopy Analogue of $\mathbb{Z}$). The sphere spectrum plays the role of the integers in stable homotopy theory. The analogy is structural:

- $\mathbb{Z}$ is the tensor unit in **Ab**, while $\mathbb{S}$ is the smash-product unit in **Sp**

$$\mathbb{Z} \otimes a \simeq a, \quad \mathbb{S} \wedge x \simeq x$$

- $\mathbb{Z}$ is the free rank-one generator of **Ab**, while $\mathbb{S}$ is a compact generator of **Sp**; maps from its shifts recover every homotopy group

$$\mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^n \mathbb{S}, x) \simeq \pi_n(x)$$

- Their endomorphisms in degree zero agree

$$\mathrm{End}_{h\mathbf{Sp}}(\mathbb{S}) \simeq \pi_0(\mathbb{S}) \simeq \mathbb{Z}$$

However, $\mathbb{S}$ is not the Eilenberg–Mac Lane spectrum $\mathrm{H}\mathbb{Z}$. The latter has only one nonzero homotopy group, whereas the higher stable stems $\pi_n(\mathbb{S})$ retain genuinely homotopical information. Thus $\mathbb{S}$ is the homotopically enriched analogue of $\mathbb{Z}$, not merely $\mathbb{Z}$ regarded as a spectrum.

Remark (Relation with the Classical Category). The homotopy category $h\mathbf{Sp}$ is the classical stable homotopy category. In particular, its morphisms are stable homotopy classes of maps. The various point-set models of spectra give different presentations, but after inverting stable equivalences they all present the same ∞-category **Sp**.

Remark (A Spectrum as a Homology Theory). Let $e \in \mathbf{Sp}$. Under the definition above, e is a reduced excisive functor

$$e : \mathbf{Ani}_*^{\mathrm{fin}} \to \mathbf{Ani}$$

For a map $a \to x$ of finite animae, form the pointed quotient

$$x/a := x \sqcup_a *$$

The relative groups associated with e are

$$e_n(x, a) := \pi_n(e(x/a)) \quad (n \geq 0)$$

where the base point of $e(x/a)$ is induced by $e(*) \to e(x/a)$. Negative degrees are recovered by shifting the spectrum e.

Reducedness gives $e_n(*) = 0$. Excision turns homotopy pushout squares of finite pointed animae into homotopy pullback squares, so applying homotopy groups produces excision, Mayer–Vietoris sequences, and the connecting maps of a generalized homology theory. Thus a spectrum is not merely a sequence of spaces: it coherently packages all degrees of a homology theory.

Corollary 4.4.3. The ∞-category $\mathbf{Sp}$ is stable. Its suspension and loop functors are inverse equivalences

$$\Sigma : \mathbf{Sp} \overset{\sim}{\leftrightarrow} \mathbf{Sp} : \Omega$$

The stable ∞-category $\mathbf{Sp}$ also carries a canonical Postnikov t-structure, organized by the vanishing of its homotopy groups. This will relate spectra to ordinary abelian groups through the heart of $\mathbf{Sp}$.

4.4.2 The Postnikov t-Structure

For $x \in \mathbf{Sp}$ and $n \in \mathbb{Z}$, define its nth homotopy group by

$$\pi_n(x) := \mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^n \mathbb{S}, x)$$

The infinite loop-space structure makes this an abelian group in every degree, including degrees 0 and 1.

Proposition 4.4.3 (Postnikov t-Structure on Spectra). The full subcategories

$$\mathbf{Sp}_{\geq 0} := \{x \in \mathbf{Sp} : \pi_n(x) = 0 \ \text{ for } \ n < 0\}$$

and

$$\mathbf{Sp}_{\leq 0} := \{x \in \mathbf{Sp} : \pi_n(x) = 0 \ \text{ for } \ n > 0\}$$

determine an accessible t-structure on $\mathbf{Sp}$. It is both left and right complete.

If a spectrum x is represented by compatible spaces $x_n \simeq \Omega x_{n+1}$, then

$$x \in \mathbf{Sp}_{\leq m} \Leftrightarrow x_n \ \text{ is } \ (n+m)\text{-truncated for every } \ n \geq 0$$

Thus the truncations of a spectrum are ordinary Postnikov truncations, performed compatibly at every level.

Corollary 4.4.4 (The Heart of Spectra). The heart of the Postnikov t-structure is canonically equivalent to the category of abelian groups

$$\mathbf{Sp}^{\heartsuit} \simeq \mathbf{Ab}$$

Under this equivalence, an abelian group a corresponds to the Eilenberg–Mac Lane spectrum $\mathrm{H}a$. Its nth space is $K(a, n)$, and its only nonzero homotopy group is

$$\pi_0(\mathrm{H}a) \simeq a$$

Proof sketch. A spectrum in the heart has each level x_n both $(n-1)$-connected and n-truncated, hence $x_n \simeq K(a,n)$ for one abelian group a. Conversely, the spaces $K(a,n)$ form an Ω-spectrum. Completeness follows because a spectrum is reconstructed from its compatible Postnikov truncations; a spectrum whose homotopy groups all vanish is therefore zero. □

Remark (The General Construction). More generally, if C is presentable, then $\mathbf{Sp}(C)$ has an accessible t-structure whose negative part is characterized by

$$x \in \mathbf{Sp}(C)_{\leq -1} \Leftrightarrow \Omega^\infty x \text{ is final in } C$$

Equivalently, its connective part is generated under colimits and extensions by the suspension spectra of objects of C.

4.4.3 Homotopy Groups and Compact Generation

Corollary 4.4.5 (Stable Whitehead Theorem). A map $f : x \to y$ of spectra is an equivalence if and only if

$$\pi_n(f) : \pi_n(x) \to \pi_n(y)$$

is an isomorphism for every $n \in \mathbb{Z}$.

Proof sketch. Apply the long exact sequence of homotopy groups to $\mathrm{fib}(f)$. If every $\pi_n(f)$ is an isomorphism, then all homotopy groups of $\mathrm{fib}(f)$ vanish. Completeness of the Postnikov t-structure forces $\mathrm{fib}(f) \simeq 0$. □

Proposition 4.4.4 (Compact Generation). The sphere spectrum $\mathbb{S}$ is a compact generator of $\mathbf{Sp}$ in the stable sense: its shifts $\{\Sigma^n \mathbb{S}\}_{n\in\mathbb{Z}}$ jointly detect equivalences. Consequently, $\mathbf{Sp}$ is compactly generated, and its compact objects are precisely the retracts of finite cell spectra.

Proof sketch. Maps from the shifts of $\mathbb{S}$ compute the homotopy groups

$$\mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^n \mathbb{S}, x) \simeq \pi_n(x)$$

so generation follows from the stable Whitehead theorem. Compactness is the fact that homotopy groups commute with filtered colimits. The compact objects are then obtained from $\mathbb{S}$ by finite cofibers, shifts, and retracts. □

Remark. The proposition above is basically telling that the entire $\mathbf{Sp}$ is generated by $\mathbb{S}$ via suspensions, colimits and retracts.

Remark (Colimits and Infinite Loop Spaces). The functors $\pi_n : \mathbf{Sp} \to \mathbf{Ab}$ preserve products and filtered colimits. The infinite-loop-space functor has the stronger connective statement

$$\Omega^\infty : \mathbf{Sp}_{\geq 0} \to \mathbf{Ani}$$

preserves sifted colimits. Outside the connective part, Ω^∞ is a right adjoint and should not be expected to preserve arbitrary colimits.

4.5 Presentable Stable ∞-Categories

Presentability is especially simple in the stable setting: finite colimits are already built into stability, so arbitrary coproducts control all small colimits.

4.5.1 Coproducts Control Colimits

Proposition 4.5.5 (Colimits in a Stable ∞-Category). Let C be stable.

1. The category C admits all small colimits if and only if it admits all small coproducts.
2. If C and D admit small colimits, an exact functor $F : C \to D$ preserves all small colimits if and only if it preserves small coproducts.
3. If these conditions hold, an object $x \in C$ is compact if and only if every map

$$x \to \sqcup_{\alpha \in A} y_\alpha$$

factors, up to homotopy, through a finite subcoproduct.

Proof sketch. In a stable category, finite colimits are finite limits and are therefore already available. General colimits can be assembled from coproducts and cofibers, proving the first two statements. For the last, write a coproduct as the filtered colimit of its finite subcoproducts and apply the definition of compactness. □

Thus, for exact functors, the apparently global problem of preserving every colimit reduces to checking only coproducts.

4.5.2 Presentability and Generators

Recall that an object $g \in C$ *generates* a stable ∞-category if

$$\pi_0 \mathrm{Map}_C(g, x) = * \Rightarrow x \simeq 0$$

Equivalently, all shifts of g jointly detect equivalences. A κ-compact object is one whose mapping-space functor preserves κ-filtered colimits.

Theorem 4.5.6 (Presentability Criterion). A stable ∞-category C is presentable if and only if:

1. C admits small coproducts;
2. its homotopy category hC is locally small;
3. for some regular cardinal κ, it has a κ-compact generator g.

Proof sketch. A presentable category has a small family of compact objects; their coproduct gives one κ-compact generator for sufficiently large κ. Conversely, close the shifts of g under κ-small colimits. This gives a small stable subcategory whose κ-Ind-completion maps fully faithfully to C. Since g detects zero objects, this map is an equivalence. □

> Remark. This criterion is visible entirely in hC. It is therefore the stable ∞-categorical counterpart of compact-generation criteria for triangulated categories.

4.5.3 Stabilization and Its Presentable Universal Property

> **Theorem 4.5.7 (Presentability of Stabilization).** If C is presentable, then $\mathbf{Sp}(C)$ is presentable and
>
> $$\Omega^\infty : \mathbf{Sp}(C) \to C$$
>
> admits a left adjoint $\Sigma^\infty_+ : C \to \mathbf{Sp}(C)$. If C is pointed, we write this functor as Σ^∞.

The important point is not merely the existence of this adjunction, but its universal property.

> **Theorem 4.5.8 (Free Stable Presentable Category).** Let C and D be presentable and suppose that D is stable. Precomposition with Σ^∞_+ induces an equivalence
>
> $$\mathbf{Fun}^{\mathrm{L}}(\mathbf{Sp}(C), D) \xrightarrow{\sim} \mathbf{Fun}^{\mathrm{L}}(C, D)$$
>
> where $\mathbf{Fun}^{\mathrm{L}}$ denotes the ∞-category of colimit-preserving functors.

Taking $C = \mathbf{Ani}$ gives the fundamental characterization

$$\mathbf{Fun}^{\mathrm{L}}(\mathbf{Sp}, D) \xrightarrow{\sim} D, \quad F \mapsto F(\mathbb{S})$$

Hence $\mathbf{Sp}$ is the stable presentable ∞-category freely generated under colimits by one object, the sphere spectrum. This is the precise categorical sense in which $\mathbb{S}$ behaves like the free rank-one object $\mathbb{Z}$.

> Remark. If $G : D \to \mathbf{Sp}(C)$ is exact, then G admits a left adjoint exactly when $\Omega^\infty G : D \to C$ does. Stabilization therefore does not create a new obstruction to adjointness.

4.5.4 Stable Localizations and Accessible t-Structures

> **Proposition 4.5.6 (Stable Localizations).** Let C be stable and let $C' \subseteq C$ be a reflective full subcategory with localization $L : C \to C'$. Then C' is stable if and only if L is left exact.

Since a localization is a left adjoint, it already preserves colimits. Thus left exactness is precisely what makes it exact and allows stability to pass to the essential image.

> **Theorem 4.5.9 (Presentations by Spectral Presheaves).** An ∞-category C is presentable and stable if and only if there exist a small ∞-category E and an accessible left-exact localization
>
> $$\mathbf{Fun}(E^{\mathrm{op}}, \mathbf{Sp}) \to C$$

This is the stable analogue of presenting an ordinary Grothendieck category by a localization of a presheaf category: spaces are replaced by spectra, and left exactness preserves the stable structure.

Proposition 4.5.7 (Generating Accessible t-Structures). Let C be presentable and stable.

1. If $U \subseteq C$ is presentable and closed under small colimits and extensions, then there is a t-structure with $C_{\geq 0} = U$.
2. Every small family $\{x_\alpha\}$ generates an accessible t-structure: its connective part is the smallest full subcategory containing all x_α and closed under extensions and small colimits.

Remark (Accessibility Test). For a t-structure on a presentable stable ∞-category, the following conditions are equivalent: $C_{\geq 0}$ is presentable, $C_{\leq 0}$ is presentable, and the truncation functors $\tau_{\geq 0}$ and $\tau_{\leq 0}$ are accessible. Any of these conditions is what we mean by an *accessible t-structure.*

4.6 Mapping Spectra

Mapping spaces in a stable ∞-category admit compatible deloopings. A single spectrum therefore packages maps in every shifted degree.

4.6.1 Construction

Let C be a locally small stable ∞-category and let $x, y \in C$. For $n \geq 0$, set

$$m_n := \mathrm{Map}_C(x, \Sigma^n y)$$

Since $\Sigma^n y \simeq \Omega\Sigma^{n+1} y$ and mapping out of x preserves limits, there are natural equivalences

$$m_n \simeq \Omega m_{n+1}$$

These equivalences, with their canonical coherences, define an Ω-spectrum.

Definition 4.6.10 (Mapping Spectrum). The *mapping spectrum* from x to y is the spectrum

$$\mathrm{Map}_C^{\mathbf{Sp}}(x, y) \in \mathbf{Sp}$$

whose nth space is $\mathrm{Map}_C(x, \Sigma^n y)$. Its infinite loop space is the original mapping space

$$\Omega^\infty \mathrm{Map}_C^{\mathbf{Sp}}(x, y) \simeq \mathrm{Map}_C(x, y)$$

4.6.2 Basic Properties

Proposition 4.6.8. For $x, y \in C$ and $a, b, n \in \mathbb{Z}$, the mapping spectrum satisfies:

1. *Homotopy groups recover shifted maps.*

$$\pi_n \mathrm{Map}_C^{\mathbf{Sp}}(x, y) \simeq \mathrm{Hom}_{hC}(\Sigma^n x, y) \simeq \mathrm{Hom}_{hC}(x, \Sigma^{-n} y)$$

2. *Shifts become suspensions of spectra.*

$$\mathrm{Map}_C^{\mathbf{Sp}}\left(\Sigma^a x, \Sigma^b y\right) \simeq \Sigma^{b-a}\mathrm{Map}_C^{\mathbf{Sp}}(x, y)$$

3. *Exactness.* The functors

$$\mathrm{Map}_C^{\mathbf{Sp}}(x, -) : C \to \mathbf{Sp}, \quad \mathrm{Map}_C^{\mathbf{Sp}}(-, y) : C^{\mathrm{op}} \to \mathbf{Sp}$$

   are exact.

4. *Detection of equivalences.* A map $f : y \to z$ is an equivalence if and only if $\mathrm{Map}_C^{\mathbf{Sp}}(x, f)$ is an equivalence for every $x \in C$.

For example, a cofiber sequence $x \to y \to z$ induces fiber sequences

$$\mathrm{Map}_C^{\mathbf{Sp}}(w, x) \to \mathrm{Map}_C^{\mathbf{Sp}}(w, y) \to \mathrm{Map}_C^{\mathbf{Sp}}(w, z)$$

and

$$\mathrm{Map}_C^{\mathbf{Sp}}(z, w) \to \mathrm{Map}_C^{\mathbf{Sp}}(y, w) \to \mathrm{Map}_C^{\mathbf{Sp}}(x, w)$$

Taking homotopy groups recovers the familiar long exact sequences of Ext-groups. In the notation of Chapter 2,

$$\pi_n \mathrm{Map}_C^{\mathbf{Sp}}(x, y) \simeq \mathrm{Ext}_C^{-n}(x, y)$$

Remark (Spectral Enrichment). Composition of maps refines canonically to maps of spectra

$$\mathrm{Map}_C^{\mathbf{Sp}}(y, z) \wedge \mathrm{Map}_C^{\mathbf{Sp}}(x, y) \to \mathrm{Map}_C^{\mathbf{Sp}}(x, z)$$

together with unit maps

$$\mathbb{S} \to \mathrm{Map}_C^{\mathbf{Sp}}(x, x)$$

These maps are coherently associative and unital. In other words, every stable ∞-category is canonically enriched over spectra. Ordinary mapping spaces are obtained by applying Ω^∞ and therefore remember only the infinite-loop-space face of this richer object.

Remark (The Presentable Case). If C is presentable and stable, it is canonically tensored over $\mathbf{Sp}$. For $e \in \mathbf{Sp}$ and $x \in C$, there is an object $e \otimes x \in C$ characterized by the tensor–Hom adjunction

$$\mathrm{Map}_C^{\mathbf{Sp}}(e \otimes x, y) \simeq \mathrm{Map}_{\mathbf{Sp}}^{\mathbf{Sp}}\left(e, \mathrm{Map}_C^{\mathbf{Sp}}(x, y)\right)$$

The functor $e \mapsto e \otimes x$ preserves colimits and sends $\mathbb{S}$ to x. For $C = \mathbf{Sp}$, the mapping spectrum is the usual function spectrum $F(x, y)$, and

$$\mathrm{Map}_{\mathbf{Sp}}^{\mathbf{Sp}}(\mathbb{S}, y) \simeq y$$

4.7 Cohomology and Homology Theories

Fix a spectrum $E \in \mathbf{Sp}$ and a pointed anima $x \in \mathbf{Ani}_*$. In formulas below, we use the abbreviations

$$E \wedge x := E \wedge \Sigma^\infty x, \quad \mathrm{Map}^{\mathbf{Sp}}(x, E) := \mathrm{Map}^{\mathbf{Sp}}_{\mathbf{Sp}}(\Sigma^\infty x, E)$$

Thus the space x is first stabilized; smash products and mapping spectra are then formed inside $\mathbf{Sp}$.

4.7.1 Cohomology First

Definition 4.7.11 (Reduced Generalized Cohomology Theory). A *reduced generalized cohomology theory* on pointed animae is a family of contravariant functors

$$E^n : (h\mathbf{Ani}_*)^{\mathrm{op}} \to \mathbf{Ab} \quad (n \in \mathbb{Z})$$

satisfying the following axioms.

1. *Reducedness.*

$$E^n(*) = 0$$

2. *Suspension.* There are natural isomorphisms

$$E^n(\Sigma x) \simeq E^{n-1}(x)$$

3. *Exactness.* Every cofiber sequence

$$a \to x \to x/a \to \Sigma a$$

   induces a natural long exact sequence

$$\dots \to E^n(x/a) \to E^n(x) \to E^n(a) \to E^{n+1}(x/a) \to \dots$$

4. *Wedge axiom.* For every family $\{x_\alpha\}$,

$$E^n\left(\bigvee_\alpha x_\alpha\right) \simeq \prod_\alpha E^n(x_\alpha)$$

The word *generalized* means that no dimension axiom is required.

Definition 4.7.12 (Representing Spectrum). Let $\{E^n\}_{n\in\mathbb{Z}}$ be a reduced generalized cohomology theory. A *representing spectrum* is a spectrum $E \in \mathbf{Sp}$ together with natural isomorphisms

$$E^n(x) \simeq \mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^\infty x, \Sigma^n E)$$

for every pointed anima x and every $n \in \mathbb{Z}$.

The more intrinsic object is not the separate family of abelian-group-valued functors E^n, but the single spectrum-valued functor

$$E^\bullet : \mathbf{Sp}^{\mathrm{op}} \to \mathbf{Sp}, \quad y \mapsto \mathrm{Map}^{\mathbf{Sp}}_{\mathbf{Sp}}(y, E)$$

The original theory on pointed animae is obtained by precomposing with the suspension-spectrum functor

$$\mathbf{Ani}_* \xrightarrow{\Sigma^\infty} \mathbf{Sp} \xrightarrow{\mathrm{Map}^{\mathbf{Sp}}_{\mathbf{Sp}}(-,E)} \mathbf{Sp}$$

and then taking homotopy groups. Explicitly,

$$E^n(x) := \pi_{-n}\mathrm{Map}^{\mathbf{Sp}}(x, E) \simeq \mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^\infty x, \Sigma^n E)$$

Thus all groups $E^n(x)$ are the homotopy groups of one mapping spectrum, not unrelated invariants in different degrees.

Proposition 4.7.9 (A Spectrum Represents Cohomology). Every spectrum E defines a reduced generalized cohomology theory by the formula above. The representing spectrum is unique up to equivalence.

Proof sketch. The suspension formula follows from the shift formula for mapping spectra. Mapping into E carries cofiber sequences to fiber sequences and wedges to products. Taking homotopy groups gives exactness and the wedge axiom. Uniqueness follows from the stable Yoneda lemma. □

Remark (Brown Representability). Conversely, Brown representability says that a reduced generalized cohomology theory satisfying the usual size hypotheses is represented by a spectrum E. Thus a whole family $\{E^n\}_{n\in\mathbb{Z}}$ is encoded by one stable object rather than by unrelated functors in each degree.

4.7.2 Homology

Definition 4.7.13 (Reduced Generalized Homology Theory). A *reduced generalized homology theory* is a family of covariant functors

$$E_n : h\mathbf{Ani}_* \to \mathbf{Ab} \quad (n \in \mathbb{Z})$$

satisfying reducedness, natural suspension isomorphisms

$$E_n(\Sigma x) \simeq E_{n-1}(x)$$

exactness for cofiber sequences, and the wedge axiom

$$E_n\left(\bigvee_\alpha x_\alpha\right) \simeq \oplus_\alpha E_n(x_\alpha)$$

Proposition 4.7.10 (Homology Represented by a Spectrum). Every spectrum E defines a reduced generalized homology theory by

$$E_n(x) := \pi_n(E \wedge x)$$

A cofiber sequence $a \to x \to x/a$ induces long exact sequences

$$\dots \to E_n(a) \to E_n(x) \to E_n(x/a) \to E_{n-1}(a) \to \dots$$

Proof sketch. Smashing with E preserves colimits, so it preserves cofiber sequences and wedges. Taking homotopy groups gives the long exact sequence and turns wedges of spectra into direct sums of abelian groups. □

4.7.3 The Derived Algebra Dictionary

The analogy with derived algebra becomes literal after replacing the ground ring R by the sphere spectrum $\mathbb{S}$. A coefficient spectrum E behaves like a module, mapping spectra behave like derived Hom, and smash products behave like derived tensor products.

derived algebra	stable homotopy
$\mathrm{RHom}_R(m, n)$	$\mathrm{Map}^{\mathbf{Sp}}(x, E)$
$\mathrm{Ext}^q_R(m, n)$	$E^q(x)$
$m \otimes^{\mathrm{L}}_R n$	$x \wedge E$
$\mathrm{Tor}^R_q(m, n)$	$E_q(x)$

The two numerical theories are obtained by taking homotopy groups of the corresponding spectrum-valued constructions

$$E^q(x) = \pi_{-q}\mathrm{Map}^{\mathbf{Sp}}(x, E), \quad E_q(x) = \pi_q(x \wedge E)$$

Example (Ordinary and Stable Theories). For an abelian group a, the Eilenberg–Mac Lane spectrum $\mathrm{H}a$ recovers ordinary reduced theories

$$(\mathrm{H}a)_n(x) \simeq \tilde{\mathrm{H}}_n(x; a), \quad (\mathrm{H}a)^n(x) \simeq \tilde{\mathrm{H}}^n(x; a)$$

Taking $E = \mathbb{S}$ instead gives stable homotopy and stable cohomotopy

$$\mathbb{S}_n(x) \simeq \pi_n(\Sigma^\infty x), \quad \mathbb{S}^n(x) \simeq \mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^\infty x, \Sigma^n \mathbb{S})$$

Remark. For an unpointed anima x, first adjoin a disjoint base point and use x_+. Thus the unreduced theory is written

$$E_n(x) := \pi_n(E \wedge \Sigma^\infty_+ x)$$

4.8 The Eilenberg–Mac Lane Spectrum

4.8.1 The Construction

Let $a \in \mathbf{Ab}$. For $n \geq 0$, let $a[n]$ denote the nonnegatively graded chain complex with a in degree n and zero elsewhere. Apply the Dold–Kan functor and forget the abelian-group structure to obtain a pointed anima

$$K(a, n) := \mathrm{DK}(a[n])$$

Every simplicial abelian group is a Kan complex, and Dold–Kan identifies homotopy groups with homology groups. Hence

$$\pi_i K(a,n) \simeq \begin{cases} a \text{ if } & i = n \\ 0 \text{ if } & i \neq n \end{cases}$$

Thus $K(a,n)$ is an Eilenberg–Mac Lane space. Shifting the chain complex corresponds to delooping, giving natural equivalences

$$K(a,n) \simeq \Omega K(a,n+1)$$

These equivalences are coherent in n and therefore define an Ω-spectrum

$$K(a,0), K(a,1), K(a,2), \ldots$$

Definition 4.8.14 (Eilenberg–Mac Lane Spectrum). The spectrum determined by the sequence above is the *Eilenberg–Mac Lane spectrum* of a, denoted

$$\mathrm{H}a \in \mathbf{Sp}$$

Equivalently, $\mathrm{H}a$ is the object corresponding to a under the equivalence

$$\mathbf{Sp}^{\heartsuit} \simeq \mathbf{Ab}$$

Proposition 4.8.11. The spectrum $\mathrm{H}a$ is characterized up to equivalence by

$$\pi_i(\mathrm{H}a) \simeq \begin{cases} a & \text{if } i = 0 \\ 0 & \text{if } i \neq 0 \end{cases}$$

The construction is functorial and defines a fully faithful functor

$$\mathrm{H} : \mathbf{Ab} \to \mathbf{Sp}$$

whose essential image is the heart $\mathbf{Sp}^{\heartsuit}$.

Proof sketch. The zeroth space of $\mathrm{H}a$ is the discrete group $K(a,0)$, and the equivalences $K(a,n) \simeq \Omega K(a,n+1)$ identify the stable homotopy groups with the displayed Dold–Kan calculation. The description of the Postnikov heart then gives uniqueness, functoriality, and full faithfulness. □

Remark (Why It Represents Ordinary Theory). For every pointed anima x, the general formulas represented by a spectrum specialize to

$$\tilde{\mathrm{H}}^n(x;a) \simeq \mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^\infty x, \Sigma^n \mathrm{H}a)$$

and

$$\tilde{\mathrm{H}}_n(x;a) \simeq \pi_n(\mathrm{H}a \wedge \Sigma^\infty x)$$

Hence the Dold–Kan construction of the spaces $K(a,n)$, the spectrum $\mathrm{H}a$, and ordinary singular homology and cohomology are three levels of the same construction.

Remark. The spectrum $\mathrm{H}\mathbb{Z}$ is a commutative ring spectrum, and every $\mathrm{H}a$ is naturally an $\mathrm{H}\mathbb{Z}$-module. This is the stable-homotopical form of the ordinary $\mathbb{Z}$-module structure on an abelian group. It also clarifies the earlier distinction: $\mathrm{H}\mathbb{Z}$ encodes ordinary homological algebra, while the sphere spectrum $\mathbb{S}$ retains the higher stable stems.

Part II

Operad Theory

5 ∞-Operads

5.1 Motivations

5.1.1 From Binary Products to Many Inputs

A commutative monoid is usually presented by a binary multiplication and a unit, subject to strict identities. A symmetric monoidal category $\mathscr{C}$ has analogous data

$$1 \in \mathscr{C}, \quad \otimes : \mathscr{C} \times \mathscr{C} \to \mathscr{C}$$

but associativity, symmetry, and the unit laws now hold only through coherent isomorphisms. Trying to repeat this definition in an ∞-category would lead to isomorphisms between isomorphisms, then homotopies between those isomorphisms, and so on. Listing this infinite hierarchy explicitly is not a workable definition.

The useful change of viewpoint is to regard an n-fold tensor product as a single n-ary operation. Its meaning is captured by multimorphisms

$$\mathrm{Hom}_{\mathscr{C}}(c_1 \otimes \ldots \otimes c_n, d) \simeq \mathrm{Mul}_{\mathscr{C}}(c_1, \ldots, c_n; d)$$

For example, a linear map $u \otimes v \to w$ is the same thing as a bilinear map $u \times v \to w$. Both parenthesizations of $u \otimes v \otimes w$ therefore represent trilinear maps, which explains why the associator is canonical. Operads turn this observation into a definition: all finite-arity operations and their substitutions are recorded at once.

5.1.2 Pointed Finite Sets as Bookkeeping

Let $\mathbf{Fin}_*$ be the category of finite pointed sets, and write

$$\langle n \rangle = \{*, 1, \ldots, n\}$$

A pointed map $\alpha : \langle n \rangle \to \langle m \rangle$ may be viewed as a partially defined map from the n inputs to the m outputs: an input sent to $*$ is simply unused. Two kinds of maps will be especially important.

- The inert map $\rho^i : \langle n \rangle \to \langle 1 \rangle$ selects the ith input and sends every other input to $*$
- The active fold map $\mu_n : \langle n \rangle \to \langle 1 \rangle$ sends every non-basepoint to 1

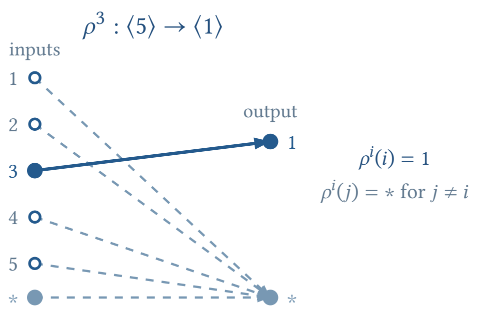

Figure 3: The inert projection ρ^i retains the ith input and sends every other input to the basepoint.

Remark (Why the Pointed Set $\langle n\rangle$ Is So Effective). The elements $1, \ldots, n$ represent input slots; the basepoint $*$ is not an additional input or output, but a place for unused data. Consequently, one pointed map

$$\alpha : \langle n\rangle \to \langle m\rangle$$

records three pieces of information at once.

- The fiber $\alpha^{-1}(j)$ specifies which inputs are grouped together to produce the jth output
- The fiber $\alpha^{-1}(*)$ records the inputs which are discarded
- If $\alpha^{-1}(j)$ is empty, the jth output is produced by a nullary operation, hence by a constant or unit map

In particular, $\langle 0\rangle = \{*\}$ represents zero inputs. The unique active map $\langle 0\rangle \to \langle 1\rangle$ therefore encodes nullary operations.

More generally, an active map sends no non-basepoint to $*$, while an inert map selects exactly one input for every output. Every pointed map factors, essentially uniquely, as

$$\langle n\rangle \xrightarrow{i} \langle k\rangle \xrightarrow{a} \langle m\rangle, \quad \alpha = a \circ i$$

with i inert and a active. The inert part selects and forgets inputs; the active part groups the remaining inputs into genuine operations. This active–inert separation is the central reason that $\mathbf{Fin}_*$ can encode units, tensor products, projections, and all finite substitutions in one small category.

Given a symmetric monoidal category $\mathscr{C}$, package its tensor product into a category $\mathscr{C}^{\otimes}$ over $\mathbf{Fin}_*$

$$p : \mathscr{C}^{\otimes} \to \mathbf{Fin}_*$$

An object over $\langle n\rangle$ is a finite list $[c_1, \ldots, c_n]$. A morphism over $\alpha : \langle n\rangle \to \langle m\rangle$ is a family of maps

$$\otimes_{i\in\alpha^{-1}(j)} c_i \to d_j \quad (1 \leq j \leq m)$$

The tensor over an empty fiber is interpreted as the unit. Composition is substitution: first tensor together the inputs belonging to each intermediate output, then tensor those blocks according to the next pointed map.

This projection contains the entire monoidal structure. Classically it has the following two properties.

1. The functor p is an opfibration, so maps in $\mathbf{Fin}_*$ admit coherent pushforward along cocartesian lifts
2. The inert projections induce equivalences

$$\mathscr{C}^{\otimes}_{\langle n\rangle} \simeq \prod_{i=1}^{n} \mathscr{C}^{\otimes}_{\langle 1\rangle} \simeq \mathscr{C}^n$$

Remark (How the Usual Structure Is Recovered). Put $\mathscr{C} = \mathscr{C}^{\otimes}_{\langle 1\rangle}$. The fiber over $\langle 0\rangle$ determines the unit, pushforward along μ_2 gives $c \otimes d$, and permutations of $\langle n\rangle$ give the symmetry. Different composites in $\mathbf{Fin}_*$ encode different parenthesizations. Their functorial compatibility supplies the associator, the unit constraints, and the Mac Lane coherence diagrams automatically.

Thus the logical process is

finite inputs → a map in $\mathbf{Fin}_*$ → cocartesian transport → tensor and all its coherences

5.1.3 The Higher-Categorical Form

The same description survives unchanged after replacing categories by ∞-categories. A symmetric monoidal ∞-category is encoded by a coCartesian fibration

$$p : \mathscr{C}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

for which the inert maps ρ^i induce the Segal equivalence

$$\mathscr{C}^{\otimes}_{\langle n\rangle} \simeq \prod_{i=1}^{n} \mathscr{C}^{\otimes}_{\langle 1\rangle}$$

The fiber over $\langle 1\rangle$ is the underlying ∞-category $\mathscr{C}$. Since composition of cocartesian transport is coherent by construction, the infinitely many higher associativity and symmetry homotopies no longer have to be specified one at a time.

The classical theory below isolates the many-input structure which will later be transferred to the ∞-categorical setting.

5.2 Operads

5.2.1 Colored Operads

An ordinary category organizes unary maps $x \to y$. A colored operad instead allows a finite collection of inputs and a single output, with every input and output carrying a type, or *color*.

Definition 5.2.1 (Colored Operad). A *colored operad* $\mathcal{O}$ consists of the following data.

1. A collection of objects, called the *colors* of $\mathcal{O}$
2. For every finite set I, colors $\{x_i\}_{i\in I}$, and a color y, a set of multimorphisms
$$\mathrm{Mul}_{\mathcal{O}}(\{x_i\}_{i\in I}; y)$$
3. For every map $\alpha : I \to J$, with $I_j = \alpha^{-1}(j)$, substitution maps
$$\prod_{j\in J} \mathrm{Mul}_{\mathcal{O}}\Big(\{x_i\}_{i\in I_j}; y_j\Big) \times \mathrm{Mul}_{\mathcal{O}}\Big(\{y_j\}_{j\in J}; z\Big) \to \mathrm{Mul}_{\mathcal{O}}\big(\{x_i\}_{i\in I}; z\big)$$
4. Identity operations
$$\mathrm{id}_x \in \mathrm{Mul}_{\mathcal{O}}(\{x\}; x)$$

Substitution is associative and the identity operations are left and right units. Reindexing the finite set I by a bijection gives the symmetric action on the inputs.

Remark (Meaning of Substitution). The map $\alpha : I \to J$ groups the original inputs into the fibers $I_j = \alpha^{-1}(j)$. Choose an inner operation for every group and one outer operation

$$\varphi_j \in \mathrm{Mul}_{\mathcal{O}}\Big(\{x_i\}_{i\in I_j}; y_j\Big), \quad \psi \in \mathrm{Mul}_{\mathcal{O}}\Big(\{y_j\}_{j\in J}; z\Big)$$

Substitution first applies each φ_j to the inputs in I_j, producing one intermediate output of color y_j, and then feeds all these outputs into ψ. The resulting operation is

$$\psi \circ \Big(\{\varphi_j\}_{j\in J}\Big) \in \mathrm{Mul}_{\mathcal{O}}(\{x_i\}_{i\in I}; z)$$

For example, take $J = \{1, 2\}$, $I_1 = \{1\}$, and $I_2 = \{2, 3, 4\}$. A unary operation g, a ternary operation h, and a binary operation f then compose as

$$(f \circ (g, h))(a_1, a_2, a_3, a_4) = f(g(a_1), h(a_2, a_3, a_4))$$

Thus operadic composition is simply simultaneous insertion into all input slots. Associativity says that a multilevel insertion gives the same result whether it is performed one level at a time or all at once.

The use of a finite indexing set, rather than a chosen ordering, is the form most compatible with $\mathbf{Fin}_*$. Every colored operad has an underlying category with

$$\mathrm{Hom}_{\mathcal{O}}(x, y) = \mathrm{Mul}_{\mathcal{O}}(\{x\}; y)$$

An ordinary category is therefore the special colored operad whose multimorphism sets are empty unless there is exactly one input.

Remark (Simplicial Variant). Replacing every set $\mathrm{Mul}_{\mathcal{O}}(\{x_i\}_{i\in I}; y)$ by a simplicial set, with simplicial substitution maps, gives a *simplicial colored operad*. We will return to this enrichment only after the classical theory has been packaged categorically.

5.2.2 Operads: The Single-Color Case

Definition 5.2.2 (Operad). An *operad* is a colored operad $\mathcal{O}$ having only one color, denoted by 1. For every $n \geq 0$, put

$$\mathcal{O}_n = \mathrm{Mul}_{\mathcal{O}}\big(\{1\}_{1\leq i\leq n}; 1\big)$$

We call $\mathcal{O}_n$ the set of n-ary operations. The operad is equivalently determined by the collection $\{\mathcal{O}_n\}_{n\geq 0}$, the actions of the symmetric groups Σ_n, the identity $\mathrm{id} \in \mathcal{O}_1$, and the substitution maps

$$\mathcal{O}_m \times \prod_{1\leq i\leq m} \mathcal{O}_{n_i} \to \mathcal{O}_{n_1+\ldots+n_m}$$

These data are unital, associative, and equivariant under the symmetric group actions.

Here $\mathcal{O}_0$, $\mathcal{O}_1$, and $\mathcal{O}_2$ are the nullary, unary, and binary operations. One may picture substitution by grafting rooted trees; associativity says that the result does not depend on the order of grafting.

Remark (Nullary Operations Are Constants). A nullary operation has no inputs. For a colored operad, an operation with output color y is an element of

$$\mathrm{Mul}_{\mathcal{O}}(\varnothing; y)$$

so it behaves as a constant of color y. In the colored operad associated to a symmetric monoidal category $\mathscr{C}$, such a constant is represented by a morphism from the unit object

$$1_{\mathscr{C}} \to y$$

Thus, in the one-colored case, the elements of $\mathcal{O}_0$ are precisely the constant operations.

5.2.3 A Basic Example

Example (A Symmetric Monoidal Category as a Colored Operad). Every symmetric monoidal category $\mathscr{C}$ determines a colored operad with colors $\mathrm{Ob}(\mathscr{C})$ and

$$\mathrm{Mul}_{\mathscr{C}}\big(\{x_i\}_{i \in I}; y\big) = \mathrm{Hom}_{\mathscr{C}}(\otimes_{i \in I} x_i, y)$$

Composition is composition of maps after tensoring. This example has an additional representability property: the functor

$$y \mapsto \mathrm{Mul}_{\mathscr{C}}\big(\{x_i\}_{i \in I}; y\big)$$

is represented by $\otimes_{i \in I} x_i$. A general colored operad supplies the multimorphism sets but need not supply an object representing them. This is the precise reason that symmetric monoidal categories form a special class of operads.

5.2.4 Packaging a Colored Operad over $\mathbf{Fin}_*$

The definition of a colored operad is asymmetric: a morphism has a finite collection of inputs but only one output. Following Higher Algebra, we correct this asymmetry by packaging the data into an ordinary category over $\mathbf{Fin}_*$.

Example (The Category $\mathcal{O}^{\otimes}$). Let $\mathcal{O}$ be a colored operad. Define a category $\mathcal{O}^{\otimes}$ as follows.

1. Its objects are finite, possibly empty, sequences of colors

$$[x_1, \ldots, x_m]$$

2. A morphism from $[x_1, \ldots, x_m]$ to $[y_1, \ldots, y_n]$ consists of a pointed map

$$\alpha : \langle m \rangle \to \langle n \rangle$$

   and, for every $1 \le j \le n$, an operation

$$\varphi_j \in \mathrm{Mul}_{\mathcal{O}}\big(\{x_i\}_{i \in \alpha^{-1}(j)}; y_j\big)$$

   Inputs sent by α to $*$ are discarded. An empty fiber $\alpha^{-1}(j)$ is allowed and is governed by a nullary operation.

3. The identity of $[x_1, \ldots, x_m]$ lies over the identity of $\langle m \rangle$ and is given by the unary operations $\{\mathrm{id}_{x_i}\}_{1 \le i \le m}$

4. Suppose $\left(\alpha, \{\varphi_j\}_{1\le j\le n}\right)$ is followed by $\left(\beta, \{\psi_k\}_{1\le k\le r}\right)$. Their composite lies over
$$\beta \circ \alpha : \langle m\rangle \to \langle r\rangle$$
and its kth component is the operadic substitution
$$\theta_k = \psi_k \circ \left(\{\varphi_j\}_{j\in\beta^{-1}(k)}\right) \in \mathrm{Mul}_{\mathcal{O}}\left(\{x_i\}_{i\in(\beta\circ\alpha)^{-1}(k)}; z_k\right)$$

The unit and associativity axioms of $\mathcal{O}$ imply the corresponding category axioms for $\mathcal{O}^{\otimes}$.

Remark (Meaning of a Morphism in $\mathcal{O}^{\otimes}$). A morphism in $\mathcal{O}^{\otimes}$ has two layers.

1. The pointed map $\alpha : \langle m\rangle \to \langle n\rangle$ is a *wiring diagram*. If $\alpha(i) = j$, the input wire x_i is routed to the jth output; if $\alpha(i) = *$, that wire is discarded
2. For every output j, the decoration
$$\varphi_j \in \mathrm{Mul}_{\mathcal{O}}\left(\{x_i\}_{i\in\alpha^{-1}(j)}; y_j\right)$$
combines exactly the wires routed to j and produces an output of color y_j

Thus a many-output morphism is not a new primitive operation: it is a family of ordinary one-output operadic operations performed in parallel.

For example, consider $\alpha : \langle 4\rangle \to \langle 2\rangle$ given by
$$\alpha(1) = \alpha(3) = 1, \quad \alpha(2) = 2, \quad \alpha(4) = *$$

A morphism over α from $[x_1, x_2, x_3, x_4]$ to $[y_1, y_2]$ is precisely a pair
$$\varphi_1 \in \mathrm{Mul}_{\mathcal{O}}(\{x_1, x_3\}; y_1), \quad \varphi_2 \in \mathrm{Mul}_{\mathcal{O}}(\{x_2\}; y_2)$$

The input x_4 plays no role. If $\mathcal{O}$ comes from a symmetric monoidal category, the same data are maps
$$x_1 \otimes x_3 \to y_1, \quad x_2 \to y_2$$

Composition now has a transparent meaning: connect the output wires of the first diagram to the input wires of the second and substitute the vertex labels. The pointed maps compose, while the operations compose by operadic substitution.

There is a canonical projection
$$\pi : \mathcal{O}^{\otimes} \to \mathbf{Fin}_*$$

On objects it sends $[x_1, \ldots, x_m]$ to $\langle m\rangle$. On morphisms it sends $\left(\alpha, \{\varphi_j\}\right)$ to α. Thus π remembers the assignment of inputs to outputs and forgets the operations decorating that assignment.

Definition 5.2.3 (Inert Morphism). A morphism $f : \langle m\rangle \to \langle n\rangle$ in $\mathbf{Fin}_*$ is *inert* if every $i \in \{1, \ldots, n\}$ has exactly one preimage under f. In particular, $\rho^i : \langle n\rangle \to \langle 1\rangle$ is inert.

Remark. Write $\langle n\rangle^{\circ} = \{1, \ldots, n\}$ for the non-basepoint part. Every inert morphism
$$f : \langle m\rangle \to \langle n\rangle$$
determines an injective map

$$\alpha : \langle n \rangle^{\circ} \to \langle m \rangle^{\circ}$$

characterized by

$$f^{-1}(\{i\}) = \{\alpha(i)\}$$

Thus an inert map chooses one input for every output; all inputs outside the image of α are sent to the basepoint. Conversely, every such injection determines a unique inert map.

Remark (What the Packaging Remembers). The fiber over $\langle 1 \rangle$ is the underlying category of $\mathcal{O}$, while the inert maps ρ^i identify the fiber over $\langle n \rangle$ with $\mathcal{O}^n$. Morphisms over the active fold μ_n recover the sets $\mathrm{Mul}_{\mathcal{O}}\big(\{x_i\}_{1 \le i \le n}; y\big)$. Consequently, both multimorphisms and their substitution law can be reconstructed from

$$\pi : \mathcal{O}^{\otimes} \to \mathbf{Fin}_*$$

5.3 Toward ∞-Operads

5.3.1 Definition of an ∞-Operad

Let

$$p : \mathcal{O}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

be a functor between ∞-categories. Write $\mathcal{O}^{\otimes}_{\langle n \rangle}$ for its fiber over $\langle n \rangle$. If x and y lie over $\langle m \rangle$ and $\langle n \rangle$, respectively, and $f : \langle m \rangle \to \langle n \rangle$, let

$$\mathrm{Map}^{f}_{\mathcal{O}^{\otimes}}(x, y)$$

denote the union of those components of $\mathrm{Map}_{\mathcal{O}^{\otimes}}(x, y)$ which lie over f.

Definition 5.3.4 (∞-Operad). An ∞-*operad* is a functor

$$p : \mathcal{O}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

satisfying the following conditions.

1. *Inert lifts.* For every inert $f : \langle m \rangle \to \langle n \rangle$ and every $x \in \mathcal{O}^{\otimes}_{\langle m \rangle}$, there is a p-coCartesian morphism

   $$x \to f_! x$$

   lying over f
2. *Decomposition of targets.* Let $x \in \mathcal{O}^{\otimes}_{\langle m \rangle}$ and $y \in \mathcal{O}^{\otimes}_{\langle n \rangle}$. Choose p-coCartesian morphisms $y \to y_i$ over $\rho^i : \langle n \rangle \to \langle 1 \rangle$. Then every $f : \langle m \rangle \to \langle n \rangle$ induces an equivalence

   $$\mathrm{Map}^{f}_{\mathcal{O}^{\otimes}}(x, y) \simeq \prod_{i=1}^{n} \mathrm{Map}^{\rho^i \circ f}_{\mathcal{O}^{\otimes}}(x, y_i)$$
3. *Existence of tuples.* For every finite collection $x_1, \ldots, x_n \in \mathcal{O}^{\otimes}_{\langle 1 \rangle}$, there is an object $x \in \mathcal{O}^{\otimes}_{\langle n \rangle}$ and p-coCartesian morphisms $x \to x_i$ over the maps ρ^i

Condition (1) says that an n-tuple can be projected to selected entries. Condition (2) says that a morphism with n outputs is exactly a collection of n independent one-output morphisms. Condition (3) says that every finite list of colors really occurs as an object over $\langle n \rangle$.

Remark (The Segal Form). Let

$$\mathcal{O} = \mathcal{O}^{\otimes}_{\langle 1 \rangle}$$

be the *underlying* ∞*-category*. Conditions (2) and (3) together say that the inert projections induce an equivalence

$$\mathcal{O}^{\otimes}_{\langle n \rangle} \simeq \mathcal{O}^{n}$$

Thus objects over $\langle n \rangle$ may be written as formal tuples $x_1 \oplus \ldots \oplus x_n$. The symbol $\oplus$ here means concatenation of colors, not a coproduct in $\mathcal{O}$.

Strictly speaking, this definition generalizes *colored* operads. To obtain the single-colored analogue, one additionally chooses an essentially surjective functor $\Delta^0 \to \mathcal{O}$.

5.3.2 Multimorphism Spaces

Let $x_1, \ldots, x_n, y \in \mathcal{O}$ and let

$$\mu_n : \langle n \rangle \to \langle 1 \rangle$$

be the active fold map. Using the Segal equivalence, choose the object $x_1 \oplus \ldots \oplus x_n$ over $\langle n \rangle$.

Definition 5.3.5 (Multimorphism Space). The space of operations with inputs $x_1, \ldots, x_n$ and output y is

$$\mathrm{Mul}_{\mathcal{O}}\big(\{x_i\}_{1 \le i \le n}; y\big) := \mathrm{Map}^{\mu_n}_{\mathcal{O}^{\otimes}}(x_1 \oplus \ldots \oplus x_n, y)$$

For $n = 1$ this is the ordinary mapping space of the underlying ∞-category. In general, the three axioms above equip these spaces with substitution maps which are associative up to coherent homotopy. This is the precise higher-categorical replacement of a classical colored operad.

5.3.3 Fundamental Examples

Definition 5.3.6 (The Commutative ∞-Operad). The *commutative* ∞*-operad* is the identity functor

$$\mathrm{N}_{\bullet}(\mathbf{Fin}_*) \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

regarded as an ∞-operad. We denote it by

$$\mathbf{Comm}^{\otimes}$$

Remark (Meaning of $\mathbf{Comm}^{\otimes}$). Its underlying ∞-category has a single color x. For every $n \ge 0$, the active fold μ_n is the unique n-ary operation, so

$$\mathrm{Mul}_{\mathbf{Comm}}\big(\{x\}_{1 \le i \le n}; x\big) \simeq \Delta^0$$

Thus $\mathbf{Comm}^{\otimes}$ is not itself a commutative algebra object. It is the *theory* of commutative algebra objects: a map of ∞-operads

$$\mathbf{Comm}^{\otimes} \to \mathscr{C}^{\otimes}$$

chooses an object of $\mathscr{C}$ together with multiplication, unit, symmetry, and all higher coherences. Moreover, $\mathbf{Comm}^{\otimes}$ is terminal among ∞-operads, since every $p : \mathcal{O}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ has a unique map to it, namely p itself. A $\mathbf{Comm}$-monoidal ∞-category is therefore exactly a symmetric monoidal ∞-category.

Definition 5.3.7 (The Trivial ∞-Operad). Let $\mathbf{Triv} \subset \mathbf{Fin}_*$ be the subcategory containing every object and only the inert morphisms. The *trivial ∞-operad* is the inclusion

$$\mathrm{N}_{\bullet}(\mathbf{Triv}) \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

We denote its source by

$$\mathbf{Triv}^{\otimes} := \mathrm{N}_{\bullet}(\mathbf{Triv})$$

Remark (Meaning of $\mathbf{Triv}^{\otimes}$). This is the minimal one-colored ∞-operad. If x denotes its unique color, then

$$\mathrm{Mul}_{\mathbf{Triv}}(\{x\}_{1\le i\le n}; x) \simeq \begin{cases} \Delta^0 & \text{if } n = 1 \\ \varnothing & \text{if } n \neq 1 \end{cases}$$

Hence it contains the required unary identity but no constants and no genuine multi-input operations. A map

$$\mathbf{Triv}^{\otimes} \to \mathcal{O}^{\otimes}$$

is determined by the image of x, so it merely chooses a color of $\mathcal{O}$. In particular, a $\mathbf{Triv}$-algebra in a monoidal ∞-category is just an object with no additional algebraic structure.

Remark (The Two Extremes). There is a canonical inclusion of ∞-operads

$$\mathbf{Triv}^{\otimes} \to \mathbf{Comm}^{\otimes}$$

The source retains only the structural inert maps; the target supplies one operation in every arity. Restriction along this inclusion sends a commutative algebra object to its underlying object. Thus $\mathbf{Triv}^{\otimes}$ and $\mathbf{Comm}^{\otimes}$ have two extremal universal roles: the former remembers only an object, while the latter specifies one coherently commutative operation in every arity.

Proposition 5.3.1 (Classical Operads Give Discrete ∞-Operads). If $\mathcal{O}$ is a classical colored operad, then

$$\mathrm{N}_{\bullet}(\mathcal{O}^{\otimes}) \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

is an ∞-operad.

Proof. Inert maps lift by selecting the corresponding colors and decorating the lift with identity operations. The definition of morphisms in $\mathcal{O}^{\otimes}$ gives the product decomposition in condition (2), while the list $[x_1, \ldots, x_n]$ supplies condition (3). All operation spaces are discrete, so no additional homotopy coherence is introduced. □

Definition 5.3.8 (Operadic Nerve). Let $\mathcal{O}$ be a simplicial colored operad and let $\mathcal{O}^{\otimes}$ be its simplicial category constructed above. Its *operadic nerve* is the homotopy coherent nerve

$$\mathrm{N}^{\otimes}_{\bullet}(\mathcal{O}) := \mathrm{N}_{\bullet}(\mathcal{O}^{\otimes})$$

together with the canonical map

$$\mathrm{N}^{\otimes}_{\bullet}(\mathcal{O}) \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

We call $\mathcal{O}$ *fibrant* if every simplicial set $\mathrm{Mul}_{\mathcal{O}}(\{x_i\}_{i \in I}; y)$ is a Kan complex.

Proposition 5.3.2 (Operadic Nerve Theorem). If $\mathcal{O}$ is a fibrant simplicial colored operad, then its operadic nerve $\mathrm{N}^{\otimes(\mathcal{O})}_{\bullet}$ is an ∞-operad. Its underlying ∞-category is the homotopy coherent nerve of the underlying simplicial category of colors.

Proof. The mapping simplicial sets of $\mathcal{O}^{\otimes}$ are coproducts of finite products of Kan complexes, hence are Kan. Its homotopy coherent nerve is therefore an ∞-category. Inert lifts are induced by identity operations, the mapping space decomposition in condition (2) is already an isomorphism of simplicial sets, and the fiber over $\langle n \rangle$ is the n-fold product of the fiber over $\langle 1 \rangle$. □

5.3.4 Reading the Definition: Objects and Operations

The definition becomes more transparent once we separate *colors*, *wiring*, and *operations*. The map

$$p : \mathcal{O}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

records only the arity and wiring of a construction; the fiberwise data record the colors and the operations decorating that wiring.

Remark (What Are the Objects?). The genuine colors form the ∞-category

$$\mathcal{O} = \mathcal{O}^{\otimes}_{\langle 1 \rangle}$$

An object $x \in \mathcal{O}$ is therefore one color. More generally, the Segal equivalence identifies

$$\mathcal{O}^{\otimes}_{\langle n \rangle} \simeq \mathcal{O}^{n}$$

so an object over $\langle n \rangle$ is merely a formal tuple $x_1 \oplus \ldots \oplus x_n$. It is not a tensor product and need not define an object of $\mathcal{O}$. The inert maps ρ^i extract its entries. Their coCartesian lifts make this extraction canonical up to a contractible space of choices.

Remark (What Are the Operations?). A morphism of the underlying ∞-category $\mathcal{O}$ is only a *unary* operation. An n-ary operation is instead a morphism in the total ∞-category $\mathcal{O}^{\otimes}$ lying over the active fold map

$$\mu_n : \langle n\rangle \to \langle 1\rangle$$

Thus its space is

$$\mathrm{Mul}_{\mathcal{O}}\big(\{x_i\}_{1\le i\le n}; y\big) = \mathrm{Map}^{\mu_n}_{\mathcal{O}^{\otimes}}(x_1 \oplus \dots \oplus x_n, y)$$

A point of this space is an operation, a path is a homotopy between operations, and its higher simplices encode higher homotopies. This is the precise difference from a classical operad, whose operation spaces are discrete sets.

Remark (How to Read a General Morphism). Consider a morphism over

$$f : \langle m\rangle \to \langle n\rangle$$

from $x_1 \oplus \dots \oplus x_m$ to $y_1 \oplus \dots \oplus y_n$. The map f is its wiring diagram. For each output j, the fiber $f^{-1}(j)$ specifies the inputs entering that output. Condition (2) says that the entire mapping space is

$$\prod_{j=1}^{n} \mathrm{Mul}_{\mathcal{O}}\big(\{x_i\}_{i\in f^{-1}(j)}; y_j\big)$$

Consequently, a general morphism is exactly one operation for every output, performed in parallel. Inputs mapped to $*$ are discarded. An empty fiber gives a nullary operation, hence a constant of the corresponding output color.

The three axioms can now be read as follows.

1. Inert lifts provide the structural operations which select and rearrange entries of a tuple
2. The mapping-space decomposition says that distinct outputs carry independent one-output operations
3. The existence axiom says that every finite tuple of colors is available

Finally, composition in the ∞-category $\mathcal{O}^{\otimes}$ substitutes the output operations of one wiring diagram into the input slots of the next. Because this is composition in an ∞-category, associativity, units, and all higher compatibilities are supplied coherently rather than imposed by an infinite list of equations. In short, an ∞-operad is a system of colors and spaces of many-input, one-output operations, together with coherently associative substitution; the map to $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ is the device which keeps the combinatorics of the inputs separate from the homotopy theory of the operations.

5.4 Active–Inert Structure

5.4.1 Active and Inert Morphisms

Definition 5.4.9 (Active Morphism). A pointed map

$$f : \langle m \rangle \to \langle n \rangle$$

in $\mathbf{Fin}_*$ is *active* if

$$f^{-1}(*) = \{*\}$$

Thus an active map uses every input. In contrast, an inert map selects exactly one input for every output and may discard the remaining inputs.

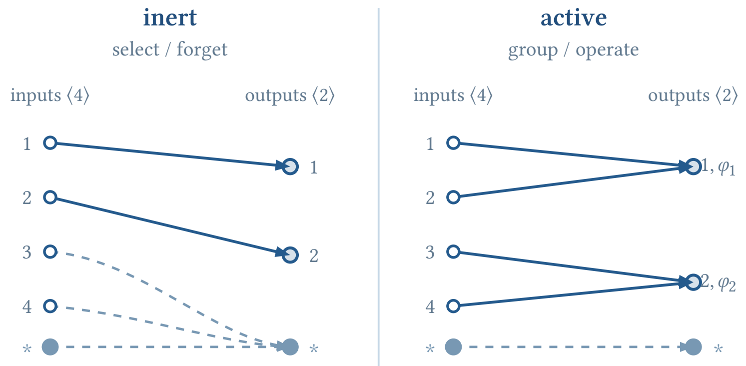

Figure 4: An inert map selects one wire for every output and may discard the others. An active map sends no ordinary input to the basepoint and groups all inputs into output operations.

Remark (How to Read the Wires). For a pointed map $f : \langle m \rangle \to \langle n \rangle$, a wire ending at j says that its input lies in $f^{-1}(j)$. These are precisely the inputs of the operation producing the jth output. A wire ending at $*$ is unused.

In the left panel,

$$i(1) = 1, \quad i(2) = 2, \quad i(3) = i(4) = *$$

so each output retains exactly one input. In the right panel,

$$a(1) = a(2) = 1, \quad a(3) = a(4) = 2$$

so all inputs are used and the two fibers label the operations

$$\varphi_1 \in \mathrm{Mul}_{\mathcal{O}}(\{x_1, x_2\}; y_1), \quad \varphi_2 \in \mathrm{Mul}_{\mathcal{O}}(\{x_3, x_4\}; y_2)$$

Proposition 5.4.3 (Active–Inert Factorization). Every morphism f in $\mathbf{Fin}_*$ factors, uniquely up to a unique isomorphism, as

$$f = a \circ i$$

where i is inert and a is active. Hence the inert and active morphisms form a factorization system on $\mathbf{Fin}_*$.

Proof. Let $S \subseteq \{1, \ldots, m\}$ be the set of inputs not sent to $*$, and choose an identification $S \simeq \{1, \ldots, k\}$. First define an inert map $i : \langle m \rangle \to \langle k \rangle$ which keeps precisely the elements of S. The restriction of f to S then defines an active map $a : \langle k \rangle \to \langle n \rangle$, and $f = a \circ i$. Another choice differs only by a permutation of $\langle k \rangle$. Finally, a square with inert left edge and active right edge has a unique filler:

each output of the inert map has one specified preimage, while the active map discards none. This gives the factorization system. □

The inert part performs bookkeeping: it selects, permutes, or forgets inputs. The active part then groups the surviving inputs into genuine many-input operations. This factorization lifts to every ∞-operad.

Definition 5.4.10 (Active and Inert Morphisms in an ∞-Operad). Let $p : \mathcal{O}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ be an ∞-operad. A morphism u of $\mathcal{O}^{\otimes}$ is

- *inert* if $p(u)$ is inert and u is p-coCartesian
- *active* if $p(u)$ is active

Proposition 5.4.4 (Lifted Active–Inert Factorization). The inert and active morphisms form a factorization system on $\mathcal{O}^{\otimes}$. In particular, every morphism factors, through a contractible space of choices, as an inert morphism followed by an active morphism.

Proof. Given $u : x \to z$, factor its image as $p(u) = a \circ i$. Choose the p-coCartesian lift $\tilde{i} : x \to y$ supplied by the inert-lift axiom. The universal property of $\tilde{i}$ produces a morphism $\tilde{a} : y \to z$ over a and a homotopy

$$u \simeq \tilde{a} \circ \tilde{i}$$

The space of such choices is contractible. Orthogonality follows from the corresponding factorization system on $\mathbf{Fin}_*$ and the mapping-space universal property of coCartesian morphisms. □

Thus inert morphisms contain only structural bookkeeping, while active morphisms contain the operadic operations and their composites.

5.5 Maps of ∞-Operads

Definition 5.5.11 (Map of ∞-Operads). A map from $p : \mathcal{O}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ to $q : \mathcal{P}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ is a functor

$$F : \mathcal{O}^{\otimes} \to \mathcal{P}^{\otimes}$$

such that $q \circ F = p$ and F preserves inert morphisms. We denote the ∞-category of such maps by

$$\mathbf{Alg}_{\mathcal{O}}(\mathcal{P})$$

where morphisms are natural transformations over $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$, which can be viewed as a full subcategory of $\mathbf{Fun}_{/\mathrm{N}_{\bullet}(\mathbf{Fin}_*)}(\mathcal{O}^{\otimes}, \mathcal{P}^{\otimes})$.

Commuting with the projection means that F preserves arities and wiring. Preservation of inert morphisms means that it also respects the decomposition of tuples into colors. Consequently, F is precisely a map on colors together with compatible maps on all multimorphism spaces which preserve identities and operadic substitution.

Proposition 5.5.5 (Detecting Maps of ∞-Operads). Let $F : \mathcal{O}^{\otimes} \to \mathcal{P}^{\otimes}$ be a functor over $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$. It is enough to check that F preserves inert morphisms lying over the elementary projections

$$\rho^i : \langle n \rangle \to \langle 1 \rangle$$

If it does, then it preserves every inert morphism and hence is a map of ∞-operads.

Proof. Under the Segal equivalences, an object over $\langle n \rangle$ is determined by its n projections. An inert morphism is likewise determined by the corresponding elementary projections and equivalences between their targets. Since the family $\{\rho^i\}_{1 \le i \le n}$ jointly detects these data, preservation of their coCartesian lifts implies preservation of every inert morphism. □

5.5.1 The ∞-Category of ∞-Operads

Maps of ∞-operads are closed under composition and contain all identity maps. Together with their coherent homotopies, they therefore form an ∞-category.

Definition 5.5.12 (The ∞-Category $\mathbf{Op}_\infty$). Let $\mathbf{Op}_\infty^{\Delta}$ be the simplicial category whose objects are small ∞-operads $\mathcal{O}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ and whose mapping Kan complexes are

$$\mathrm{Map}_{\mathbf{Op}_\infty^{\Delta}}(\mathcal{O}, \mathcal{P}) \coloneqq \mathbf{Alg}_{\mathcal{O}}(\mathcal{P})^{\simeq}$$

Here the right-hand side is the maximal Kan complex of $\mathbf{Alg}_{\mathcal{O}}(\mathcal{P})$: its vertices are maps of ∞-operads, and its paths are equivalences between such maps. Define

$$\mathbf{Op}_\infty \coloneqq \mathrm{N}_{\bullet}\big(\mathbf{Op}_\infty^{\Delta}\big)$$

and call it the *∞-category of ∞-operads.*

The final section gives a marked-simplicial-set model for $\mathbf{Op}_\infty$. Besides making the preceding construction concrete, that model proves that $\mathbf{Op}_\infty$ is presentable and therefore admits all small limits and colimits.

Remark (Mapping Category versus Mapping Space). The ∞-category $\mathbf{Alg}_{\mathcal{O}}(\mathcal{P})$ also contains noninvertible natural transformations between operad maps. The mapping object of $\mathbf{Op}_\infty$ must be a space, so only its maximal subgroupoid $\mathbf{Alg}_{\mathcal{O}}(\mathcal{P})^{\simeq}$ appears in the definition of $\mathbf{Op}_\infty$.

Proposition 5.5.6 (Detecting Equivalences of ∞-Operads). A map $F : \mathcal{O}^{\otimes} \to \mathcal{P}^{\otimes}$ is an equivalence in $\mathbf{Op}_\infty$ if and only if both of the following hold.

1. The underlying functor on colors $\mathcal{O} \to \mathcal{P}$ is essentially surjective
2. For every $x_1, \dots, x_n, y \in \mathcal{O}$, the induced map

   $$\mathrm{Mul}_{\mathcal{O}}\big(\{x_i\}_{1 \le i \le n}; y\big) \to \mathrm{Mul}_{\mathcal{P}}\big(\{F(x_i)\}_{1 \le i \le n}; F(y)\big)$$

   is a homotopy equivalence

Proof sketch. The second condition is operadic full faithfulness. The Segal decomposition reconstructs every mapping space in $\mathcal{O}^{\otimes}$ from the multimorphism spaces and the wiring maps in $\mathbf{Fin}_*$. The first condition gives essential surjectivity over $\langle 1\rangle$, and hence, by the Segal equivalence, over every $\langle n\rangle$. Thus F is an equivalence of the total ∞-categories over $\mathrm{N}_\bullet(\mathbf{Fin}_*)$. The converse follows by restricting such an equivalence to colors and to the fibers over the active fold maps. □

Remark (The Two Universal Examples). The earlier examples have the following meaning inside $\mathbf{Op}_\infty$:

$$\mathrm{Map}_{\mathbf{Op}_\infty}(\mathcal{O}, \mathbf{Comm}) \simeq \Delta^0, \quad \mathrm{Map}_{\mathbf{Op}_\infty}(\mathbf{Triv}, \mathcal{O}) \simeq \mathcal{O}^{\simeq}$$

Hence $\mathbf{Comm}^{\otimes}$ is terminal in $\mathbf{Op}_\infty$, while a map out of $\mathbf{Triv}^{\otimes}$ is exactly the choice of a color, up to equivalence.

Let's recall the definition of an isofibration, which is a special kind of inner fibration.

Definition 5.5.13 (Isofibration). A functor $F : \mathcal{C} \to \mathcal{D}$ between ∞-categories is an *isofibration* if it is an inner fibration and equivalences lift: for every $x \in \mathcal{C}$ and every equivalence $u : F(x) \to y$ in $\mathcal{D}$, there is an object $\tilde{y} \in \mathcal{C}$ and an equivalence $\tilde{u} : x \to \tilde{y}$ satisfying

$$F(\tilde{y}) = y, \quad F(\tilde{u}) = u$$

Definition 5.5.14 (Fibration of ∞-Operads). A map of ∞-operads

$$F : \mathcal{O}^{\otimes} \to \mathcal{P}^{\otimes}$$

is a *fibration of ∞-operads* if its underlying functor of ∞-categories is an isofibration.

This notion allows colors and operations to vary in families. For an inner fibration of ∞-operads, it can be detected using only inert morphisms.

Proposition 5.5.7 (Detecting Fibrations of ∞-Operads). Let $F : \mathcal{O}^{\otimes} \to \mathcal{P}^{\otimes}$ be a map of ∞-operads which is an inner fibration. Then F is a fibration of ∞-operads if and only if every inert morphism $F(x) \to y$ in $\mathcal{P}^{\otimes}$ admits an inert lift with source x. In this case, the inert morphisms of $\mathcal{O}^{\otimes}$ are exactly the F-coCartesian morphisms whose images in $\mathcal{P}^{\otimes}$ are inert.

Proof. Suppose first that inert morphisms admit inert lifts. Every equivalence in $\mathcal{P}^{\otimes}$ is inert, so an equivalence $F(x) \to y$ lifts to an inert morphism over an isomorphism of pointed finite sets. Such a lift is itself an equivalence. Hence F is an isofibration.

Conversely, suppose that F is an isofibration and let $u : F(x) \to y$ be inert. First choose in $\mathcal{O}^{\otimes}$ an inert morphism $v : x \to x'$ over the same pointed map as u. The morphisms $F(v)$ and u are two coCartesian lifts of that pointed map, so their targets are related by an equivalence $e : F(x') \to y$, together with a homotopy $u \simeq e \circ F(v)$. Lift e to an equivalence $\tilde{e} : x' \to \tilde{y}$ and use the inner-fibration property to lift the comparison 2-simplex. The resulting composite $\tilde{e} \circ v$ is an inert lift of u. The same comparison shows that the inert morphisms in $\mathcal{O}^{\otimes}$ are precisely the F-coCartesian morphisms lying over inert morphisms. □

5.5.2 $\mathcal{O}$-Monoidal ∞-Categories

Proposition 5.5.8 (Recognizing $\mathcal{O}$-Monoidal Structures). Let $p : \mathcal{O}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ be an ∞-operad and let

$$q : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$$

be a coCartesian fibration. The following conditions are equivalent.

1. The composite $p \circ q$ exhibits $\mathcal{C}^{\otimes}$ as an ∞-operad
2. Whenever $x \in \mathcal{O}^{\otimes}_{\langle n \rangle}$ corresponds to $x_1 \oplus \ldots \oplus x_n$, the inert projections induce an equivalence

$$\mathcal{C}_x \simeq \prod_{i=1}^{n} \mathcal{C}_{x_i}$$

Proof. Lift the inert projections $x \to x_i$ coCartesianly along q. Since inert maps already satisfy the Segal equivalence in $\mathcal{O}^{\otimes}$, the Segal map for $\mathcal{C}^{\otimes}$ restricts over x to

$$\mathcal{C}_x \to \prod_{i=1}^{n} \mathcal{C}_{x_i}$$

Hence the object part of the operadic Segal condition is precisely (2). The coCartesian mapping-space formula and the mapping-space decomposition in $\mathcal{O}^{\otimes}$ identify the corresponding morphism conditions. Thus the full ∞-operad axioms for $p \circ q$ are equivalent to the fiberwise product condition. □

Definition 5.5.15 ($\mathcal{O}$-Monoidal ∞-Category). When the equivalent conditions above hold, we say that q exhibits $\mathcal{C}^{\otimes}$ as an *$\mathcal{O}$-monoidal ∞-category*.

This definition turns every operation of $\mathcal{O}$ into an operation on the corresponding fibers. More precisely, if

$$f \in \mathrm{Mul}_{\mathcal{O}}(\{x_i\}_{1 \le i \le n}; y)$$

then coCartesian transport along f gives a functor, well-defined up to a contractible space of choices,

$$\otimes_f : \prod_{i=1}^{n} \mathcal{C}_{x_i} \to \mathcal{C}_y$$

Thus an $\mathcal{O}$-monoidal category assigns a category of objects to every color and a many-variable tensor functor to every operation. Operadic substitution in $\mathcal{O}$ becomes composition of these tensor functors, with all higher compatibilities inherited from coCartesian transport.

Example (Symmetric Monoidal ∞-Categories). Take the commutative ∞-operad

$$\mathbf{Comm}^{\otimes} = \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

A **Comm**-monoidal ∞-category is exactly a symmetric monoidal ∞-category

$$p : \mathscr{C}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

with

$$\mathscr{C}^{\otimes}_{\langle n\rangle} \simeq \mathscr{C}^n$$

CoCartesian transport along the active maps

$$\langle 0\rangle \to \langle 1\rangle, \quad \mu_2 : \langle 2\rangle \to \langle 1\rangle$$

produces respectively the unit object and the tensor product

$$1 \in \mathscr{C}, \quad \otimes : \mathscr{C} \times \mathscr{C} \to \mathscr{C}$$

The maps μ_n produce the n-fold tensor products, while permutations in $\mathbf{Fin}_*$ supply symmetry. Functoriality and coCartesian transport encode the associativity, unit, symmetry, and all of their higher coherences.

Remark (Symmetric Monoidal Means coCartesian). Equivalently, a symmetric monoidal ∞-category is an ∞-operad

$$p : \mathscr{C}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

whose structure map p is a coCartesian fibration. An arbitrary ∞-operad supplies coCartesian lifts only for inert maps; requiring p to be coCartesian supplies them for *every* pointed map, in particular for the active fold maps which define tensor products. The base $\mathbf{Fin}_*$ also contains all permutations, so their coCartesian transport provides the symmetry. Thus "symmetric" is encoded by the coCartesian operad over $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$, rather than by adding a braiding separately.

5.6 Algebra Objects

> " $\mathscr{O}^{\otimes}$-algebra = a realization in $\mathscr{C}^{\otimes}$ of the operations and laws prescribed by $\mathscr{O}^{\otimes}$. "

Here $\mathscr{O}^{\otimes}$ is the *syntax*: its colors specify the types of objects, its multimorphism spaces specify the available operations, and operadic substitution specifies how those operations must compose. The monoidal ∞-category $\mathscr{C}^{\otimes}$ is the *semantic environment* in which this syntax is interpreted. Concretely, a map of ∞-operads sends every color x to an object $a(x)$ and every operation

$$f \in \mathrm{Mul}_{\mathscr{O}}\big(\{x_i\}_{1\le i\le n}; y\big)$$

to a morphism

$$a(f) : a(x_1) \otimes \ldots \otimes a(x_n) \to a(y)$$

Preservation of inert morphisms respects the input slots, while preservation of composition says that substitution in $\mathscr{O}$ becomes composition in $\mathscr{C}^{\otimes}$. Thus all the equations prescribed by $\mathscr{O}$ hold in $\mathscr{C}$, coherently rather than merely on the nose.

5.6.1 Algebra Objects as Operadic Lifts

Definition 5.6.16 (Relative Algebra Objects). Let

$$p : \mathscr{C}^{\otimes} \to \mathscr{O}^{\otimes}$$

be a fibration of ∞-operads, and let $\alpha : \mathscr{P}^{\otimes} \to \mathscr{O}^{\otimes}$ be a map of ∞-operads. A *$\mathscr{P}$-algebra object of $\mathscr{C}$ relative to $\mathscr{O}$* is a map of ∞-operads $a : \mathscr{P}^{\otimes} \to \mathscr{C}^{\otimes}$ such that

$$p \circ a = \alpha$$

The full subcategory of $\mathbf{Fun}_{\mathscr{O}^{\otimes}}(\mathscr{P}^{\otimes}, \mathscr{C}^{\otimes})$ spanned by these maps is denoted by

$$\mathbf{Alg}_{\mathscr{P}/\mathscr{O}}(\mathscr{C})$$

When $\mathscr{P} = \mathscr{O}$ and $\alpha = \mathrm{id}$, we abbreviate this to $\mathbf{Alg}_{\mathscr{O}}(\mathscr{C})$. When the structural map α is understood, we also omit the base $\mathscr{O}$ from the subscript.

Thus an algebra object is a lift of the operadic theory $\mathscr{P}^{\otimes}$ through p. Preservation of inert morphisms fixes the underlying tuple of objects; the images of active morphisms supply the actual operations. Functoriality then encodes their unit, substitution, and all higher coherence laws.

Example (Commutative Algebra Objects). If $\mathscr{C}^{\otimes} \to \mathbf{Comm}^{\otimes}$ is symmetric monoidal, define

$$\mathbf{CAlg}(\mathscr{C}) := \mathbf{Alg}_{\mathbf{Comm}}(\mathscr{C})$$

An object $c \in \mathbf{CAlg}(\mathscr{C})$ consists of an underlying object of $\mathscr{C}$ together with coherent operations

$$1 \to c, \quad c^{\otimes n} \to c \quad (n \geq 1)$$

invariant under permutations and compatible with substitution. If $\mathscr{C}$ is an ordinary symmetric monoidal category, its operadic nerve recovers the nerve of the ordinary category of commutative algebra objects in $\mathscr{C}$.

5.6.2 Three Basic Operads

Proposition 5.6.9 (The Trivial Operad Classifies Objects). Let $\mathscr{C}^{\otimes} \to \mathbf{Comm}^{\otimes}$ be a symmetric monoidal ∞-category, and use the canonical inclusion $\mathbf{Triv}^{\otimes} \to \mathbf{Comm}^{\otimes}$. Evaluation at the unique color induces an equivalence

$$\mathbf{Alg}_{\mathbf{Triv}/\mathbf{Comm}}(\mathscr{C}) \simeq \mathscr{C}$$

Proof. The operad $\mathbf{Triv}^{\otimes}$ contains only the inert bookkeeping maps and the unary identity. Hence a map $\mathbf{Triv}^{\otimes} \to \mathscr{C}^{\otimes}$ is determined by the image c of its unique color. All required inert lifts are then forced, up to a contractible space of choices, by the coCartesian structure of $\mathscr{C}^{\otimes}$. □

Definition 5.6.17 (The $\mathbb{E}_0$-Operad). Let $\mathbf{Fin}_*^{\mathrm{inj}}$ be the subcategory of $\mathbf{Fin}_*$ containing every object and those pointed maps $f : \langle m\rangle \to \langle n\rangle$ for which each nonbasepoint of the target has at most one preimage. Define

$$\mathbb{E}_0^{\otimes} := \mathrm{N}_\bullet\left(\mathbf{Fin}_*^{\mathrm{inj}}\right) \to \mathrm{N}_\bullet(\mathbf{Fin}_*)$$

Remark ($\mathbb{E}_0^{\otimes}$ Contains More Than the Inert Maps). An inert map requires every nonbasepoint of the target to have *exactly one* preimage. The defining condition for $\mathbb{E}_0^{\otimes}$ is weaker: every such fiber has cardinality at most one, and may therefore be empty. Thus $\mathbf{Triv}^{\otimes}$ contains the maps which only select and rearrange inputs, whereas $\mathbb{E}_0^{\otimes}$ additionally allows an output to receive no input. Such an empty fiber represents a nullary operation, which becomes the unit map $1 \to c$ in an $\mathbb{E}_0$-algebra.

The only active operations of $\mathbb{E}_0^{\otimes}$ have arity zero or one. The nullary operation is a constant, hence becomes a unit map, while the unary operation is the identity. Consequently there are inclusions

$$\mathbf{Triv}^{\otimes} \to \mathbb{E}_0^{\otimes} \to \mathbf{Comm}^{\otimes}$$

and restriction gives forgetful functors

$$\mathbf{CAlg}(\mathscr{C}) \to \mathbf{Alg}_{\mathbb{E}_0}(\mathscr{C}) \to \mathscr{C}$$

Proposition 5.6.10 ($\mathbb{E}_0$-Algebras Are Pointed Objects). Let $\mathscr{C}^{\otimes}$ be a symmetric monoidal ∞-category with unit 1. Then $\mathbf{Alg}_{\mathbb{E}_0}(\mathscr{C})$ is equivalent to the ∞-category whose objects are pairs

$$(c, \eta : 1 \to c)$$

In particular, an $\mathbb{E}_0$-algebra is a unital object with no multiplication.

Proof. The active map $\langle 0\rangle \to \langle 1\rangle$ supplies $\eta : 1 \to c$, and $\mathbb{E}_0^{\otimes}$ has no active operations of arity at least two. The Segal condition forces all remaining values from c, while the inert maps carry no further algebraic choices. Thus restriction to the nullary operation and the unique color is fully faithful and essentially surjective. □

Remark (A Useful Hierarchy). The three operads isolate three successive amounts of structure:

$$\begin{aligned} \mathbf{Triv} &: \text{an object} \\ \mathbb{E}_0 &: \text{a map } \ 1 \to c \\ \mathbf{Comm} &: \text{a coherently commutative unital multiplication on } \ c \end{aligned}$$

This is why forgetting a commutative multiplication naturally passes first through its underlying pointed object and then through its underlying object.

5.6.3 Pointwise Algebra Objects

Proposition 5.6.11 (Algebras in Functor Categories). Let $\mathscr{C}^{\otimes} \to \mathscr{O}^{\otimes}$ be an $\mathscr{O}$-monoidal ∞-category and let k be a simplicial set. Set

$$\mathscr{D}^{\otimes} := \mathbf{Fun}\big(k, \mathscr{C}^{\otimes}\big) \times_{\mathbf{Fun}(k, \mathscr{O}^{\otimes})} \mathscr{O}^{\otimes}$$

where $\mathscr{O}^{\otimes} \to \mathbf{Fun}\big(k, \mathscr{O}^{\otimes}\big)$ sends an object to the corresponding constant diagram. Then $\mathscr{D}^{\otimes} \to \mathscr{O}^{\otimes}$ is $\mathscr{O}$-monoidal, its fiber at $x \in \mathscr{O}$ is

$$\mathscr{D}_x \simeq \mathbf{Fun}(k, \mathscr{C}_x)$$

and every operadic operation is computed pointwise. Consequently,

$$\mathbf{Alg}_{\mathscr{O}}(\mathscr{D}) \simeq \mathbf{Fun}(k, \mathbf{Alg}_{\mathscr{O}}(\mathscr{C}))$$

Proof. CoCartesian transport in a functor category is detected pointwise. Hence the Segal product decomposition and every tensor operation in $\mathscr{D}^{\otimes}$ are obtained by applying the corresponding construction in $\mathscr{C}^{\otimes}$ at each vertex of k. A section of $\mathscr{D}^{\otimes}$ preserving inert morphisms is therefore exactly a k-diagram of such sections of $\mathscr{C}^{\otimes}$. □

5.6.4 Monoidal Functors

Definition 5.6.18 ($\mathscr{O}$-Monoidal Functor). Let $p : \mathscr{C}^{\otimes} \to \mathscr{O}^{\otimes}$ and $q : \mathscr{D}^{\otimes} \to \mathscr{O}^{\otimes}$ be $\mathscr{O}$-monoidal ∞-categories. An *$\mathscr{O}$-monoidal functor* is a functor

$$F : \mathscr{C}^{\otimes} \to \mathscr{D}^{\otimes}$$

over $\mathscr{O}^{\otimes}$ which carries p-coCartesian morphisms to q-coCartesian morphisms. When $\mathscr{O} = \mathbf{Comm}$, this is a symmetric monoidal functor.

The ∞-category of such functors is denoted by

$$\mathbf{Fun}^{\otimes}_{\mathscr{O}}(\mathscr{C}, \mathscr{D})$$

Equivalently, F consists of functors $F_x : \mathscr{C}_x \to \mathscr{D}_x$ for the colors x of $\mathscr{O}$, together with coherent equivalences

$$F_y\big(\otimes_f (c_1, \dots, c_n)\big) \simeq \otimes_f \left(F_{x_1}(c_1), \dots, F_{x_n}(c_n)\right)$$

for every operation $f \in \mathrm{Mul}_{\mathscr{O}}\big(\{x_i\}_{1 \le i \le n}; y\big)$.

Proposition 5.6.12 (Fiberwise Equivalence Criterion). An $\mathscr{O}$-monoidal functor $F : \mathscr{C}^{\otimes} \to \mathscr{D}^{\otimes}$ is an equivalence if and only if, for every color $x \in \mathscr{O}$, the induced functor

$$F_x : \mathscr{C}_x \to \mathscr{D}_x$$

is an equivalence. In the one-colored case, it is therefore enough to test the underlying functor $\mathscr{C} \to \mathscr{D}$.

Proof. The Segal equivalence identifies the fiber over a tuple $x_1 \oplus \dots \oplus x_n$ with the product of the fibers over its entries. Thus equivalence on all color fibers implies equivalence on every fiber of $\mathscr{O}^{\otimes}$. Since F preserves coCartesian transport, it is a natural transformation between the diagrams classified by the two coCartesian fibrations, and such a transformation is an equivalence exactly when it is pointwise an equivalence. □

Remark. The monoidal hypothesis is essential: an arbitrary map of ∞-operads can agree on the underlying unary ∞-categories while differing on genuine multi-input operations.

5.7 ∞-Preoperads

The definition of an ∞-operad is intrinsic but not especially convenient for constructing limits, colimits, or derived mapping spaces. Marked simplicial sets provide a model in which the inert morphisms are recorded as part of the object.

5.7.1 Marked Simplicial Sets

Definition 5.7.19 (Marked Simplicial Set). A *marked simplicial set* is a pair $\overline{x} = (x, M)$, where x is a simplicial set and M is a collection of edges containing every degenerate edge. We write $x^\flat$ when only the degenerate edges are marked and $x^\sharp$ when every edge is marked.

Definition 5.7.20 (∞-Preoperad). An ∞-*preoperad* is a marked simplicial set $\overline{x} = (x, M)$ equipped with a map

$$r : x \to \mathrm{N}_\bullet(\mathbf{Fin}_*)$$

such that $r(e)$ is inert whenever $e \in M$. A morphism of ∞-preoperads is a map over $\mathrm{N}_\bullet(\mathbf{Fin}_*)$ which preserves marked edges. We denote their category by $\mathbf{POp}_\infty$.

Equivalently, $\mathbf{POp}_\infty$ is the overcategory of marked simplicial sets over $\mathrm{N}_\bullet(\mathbf{Fin}_*)$ with all inert edges marked. Cartesian product with $k^\sharp$ for a simplicial set k tensors this category over simplicial sets and gives the simplicial mapping objects $\mathrm{Map}_{\mathbf{POp}_\infty}(\overline{x}, \overline{y})$.

Definition 5.7.21 (Marked Operad Notation). If $p : \mathcal{O}^\otimes \to \mathrm{N}_\bullet(\mathbf{Fin}_*)$ is an ∞-operad, write

$$\mathcal{O}^{\otimes,\natural}$$

for the corresponding ∞-preoperad in which precisely the inert morphisms are marked.

Thus a preoperad is only raw marked data over $\mathrm{N}_\bullet(\mathbf{Fin}_*)$; the operadic horn filling and Segal conditions will instead appear as fibrancy conditions.

5.7.2 The Operadic Model Structure

Theorem 5.7.1 (∞-Operadic Model Structure). The category $\mathbf{POp}_\infty$ admits a left proper combinatorial simplicial model structure characterized by the following properties.

1. A map $\overline{x} \to \overline{y}$ is a cofibration if and only if the underlying map $x \to y$ is a monomorphism

2. It is a weak equivalence if and only if, for every ∞-operad $\mathcal{O}^{\otimes}$, the induced map

$$\mathrm{Map}_{\mathbf{POp}_\infty}(\overline{y}, \mathcal{O}^{\otimes,\natural}) \to \mathrm{Map}_{\mathbf{POp}_\infty}(\overline{x}, \mathcal{O}^{\otimes,\natural})$$

is a weak homotopy equivalence

3. The fibrant objects are precisely the marked operads $\mathcal{O}^{\otimes,\natural}$. A map between fibrant objects is a fibration exactly when its underlying map is a fibration of ∞-operads

Proof sketch. Start with marked simplicial sets over $\mathrm{N}_\bullet(\mathbf{Fin}_*)$ and localize so that the lifting conditions force inert coCartesian lifts, the decomposition of multimorphism spaces, and the Segal equivalences. The resulting fibrant objects satisfy exactly the three axioms of an ∞-operad. The weak equivalences are then detected by enriched mapping spaces into these fibrant objects. General localization results for combinatorial simplicial model categories give existence and left properness. □

Corollary 5.7.1 (A Presentation of $\mathbf{Op}_\infty$). The simplicial localization of $\mathbf{POp}_\infty$ at the operadic weak equivalences is equivalent to $\mathbf{Op}_\infty$. Consequently, $\mathbf{Op}_\infty$ is presentable and admits all small limits and colimits.

Proof sketch. Every preoperad is cofibrant, since the map from the initial object is a monomorphism. After fibrant replacement, the enriched mapping Kan complex from $\mathcal{O}^{\otimes,\natural}$ to $\mathcal{P}^{\otimes,\natural}$ is precisely $\mathbf{Alg}_{\mathcal{O}}(\mathcal{P})^{\simeq}$. These are the mapping spaces used to define $\mathbf{Op}_\infty$. Presentability follows from the combinatorial model, and small limits and colimits follow from presentability. □

5.7.3 Small Models and Unary Operads

Remark (Small Models for Triv and $\mathbb{E}_0$). There are operadic weak equivalences

$$\{\langle 1\rangle\}^\flat \to \mathbf{Triv}^{\otimes,\natural}, \quad (\Delta^1)^\flat \to \mathbb{E}_0^{\otimes,\natural}$$

where the edge of Δ^1 in the second map represents the active map $\langle 0\rangle \to \langle 1\rangle$. Thus, up to operadic weak equivalence, $\mathbf{Triv}$ is generated by one color and $\mathbb{E}_0$ by one nullary operation, or equivalently one unit arrow.

Proposition 5.7.13 (∞-Categories as Unary ∞-Operads). There is a fully faithful functor

$$\mathbf{Cat}_\infty \to \mathbf{Op}_\infty$$

which regards an ∞-category as an ∞-operad having only unary operations. Its essential image consists precisely of those $p : \mathcal{O}^{\otimes} \to \mathrm{N}_\bullet(\mathbf{Fin}_*)$ which factor through $\mathbf{Triv}^{\otimes}$.

Proof sketch. If an operad factors through $\mathbf{Triv}^{\otimes}$, it has no nullary or genuinely multi-input operations. It is therefore determined by its fiber over $\langle 1\rangle$ and the unary composition there. Maps between such operads are exactly functors between these underlying ∞-categories, which proves full faithfulness. □

Remark (Symmetric Monoidal ∞-Categories inside $\mathbf{Op}_\infty$). Symmetric monoidal ∞-categories and symmetric monoidal functors form a subcategory

$$\mathbf{Cat}_\infty^{\otimes} \subset \mathbf{Op}_\infty$$

Its objects are those ∞-operads $\mathscr{C}^{\otimes} \to \mathrm{N}_\bullet(\mathbf{Fin}_*)$ whose projection is a coCartesian fibration, and its morphisms are the operad maps which preserve the corresponding coCartesian morphisms. This subcategory is not full: an arbitrary operad map need not be symmetric monoidal.

6 ∞-Operads: Constructions

We now study how categorical constructions interact with $\mathcal{O}$-monoidal structures. Recall that a symmetric monoidal ∞-category is the special case of an ∞-operad $\mathcal{C}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_{*})$ whose structure map is a coCartesian fibration.

6.1 Subcategories of $\mathcal{O}$-Monoidal ∞-Categories

6.1.1 Tensor-Closed Subcategories

Let $p : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$ be an $\mathcal{O}$-monoidal ∞-category. If $\mathcal{D} \subset \mathcal{C}$ is a full subcategory stable under equivalence, let $\mathcal{D}^{\otimes} \subset \mathcal{C}^{\otimes}$ be the full subcategory consisting of tuples whose components all belong to $\mathcal{D}$. For a color $x \in \mathcal{O}$, write

$$\mathcal{D}_x := \mathcal{D} \cap \mathcal{C}_x$$

Proposition 6.1.1 (Inheritance by a Tensor-Closed Subcategory). Suppose that, for every operation

$$f \in \mathrm{Mul}_{\mathcal{O}}\big(\{x_i\}_{1 \le i \le n}; y\big)$$

the associated tensor functor satisfies

$$\otimes_f \left(\prod_{i=1}^{n} \mathcal{D}_{x_i} \right) \subset \mathcal{D}_y$$

Then the restriction

$$p|_{\mathcal{D}^{\otimes}} : \mathcal{D}^{\otimes} \to \mathcal{O}^{\otimes}$$

is a coCartesian fibration of ∞-operads. Hence $\mathcal{D}$ inherits an $\mathcal{O}$-monoidal structure, and the inclusion $\mathcal{D}^{\otimes} \to \mathcal{C}^{\otimes}$ is $\mathcal{O}$-monoidal.

Proof sketch. The Segal equivalence identifies the fiber of $\mathcal{D}^{\otimes}$ over a tuple with the product of the corresponding fibers $\mathcal{D}_{x_i}$. The closure hypothesis says exactly that coCartesian transport along every active map stays inside $\mathcal{D}^{\otimes}$; inert transport merely selects components and does so automatically. Thus the restricted projection is coCartesian and still satisfies the operadic Segal condition. □

Remark (The Symmetric Monoidal Criterion). If $\mathcal{C}$ is symmetric monoidal, a full replete subcategory $\mathcal{D} \subset \mathcal{C}$ inherits the restricted symmetric monoidal structure if and only if it contains the unit and is closed under the binary tensor product

$$1 \in \mathcal{D}, \quad d \otimes d' \in \mathcal{D}$$

The nullary operation detects the unit, while iteration of the binary operation gives all finite tensor products.

6.1.2 Colocalizations and Algebra Objects

Proposition 6.1.2 (Fiberwise Colocalization). In the situation above, suppose moreover that every inclusion $i_x : \mathscr{D}_x \to \mathscr{C}_x$ admits a right adjoint r_x. Then the functors $\{r_x\}_{x \in \mathcal{O}}$ assemble into a map of ∞-operads

$$r^{\otimes} : \mathscr{C}^{\otimes} \to \mathscr{D}^{\otimes}$$

which is right adjoint to the inclusion. There is a counit $i^{\otimes} \circ r^{\otimes} \to \mathrm{id}$ lying over the identity of $\mathcal{O}^{\otimes}$.

Proof sketch. Apply r_x componentwise in every Segal fiber. Closure under all tensor operations ensures that the universal colocalizing arrows are preserved by operadic transport. Their spaces of choices are contractible, so the fiberwise right adjoints assemble coherently and preserve inert morphisms. □

Corollary 6.1.1 (Colocalization of Algebra Objects). For every map $\alpha : \mathscr{P}^{\otimes} \to \mathcal{O}^{\otimes}$, the inclusion induces a fully faithful functor

$$\mathbf{Alg}_{\mathscr{P}/\mathcal{O}}(\mathscr{D}) \to \mathbf{Alg}_{\mathscr{P}/\mathcal{O}}(\mathscr{C})$$

and postcomposition with $r^{\otimes}$ is its right adjoint. Thus an algebra object colocalizes by applying r_x to its underlying objects and then using the induced operadic structure.

6.1.3 Compatible Monoidal Localizations

Remark (Recall: Localization and Colocalization). A *localization* of an ∞-category $\mathscr{C}$ consists of a full subcategory $i : \mathscr{D} \hookrightarrow \mathscr{C}$ for which the inclusion admits a left adjoint

$$\ell : \mathscr{C} \rightleftarrows \mathscr{D} : i$$

We usually regard ℓ as the endofunctor $i \circ \ell$ of $\mathscr{C}$. Its unit $\eta : \mathrm{id} \to \ell$ is idempotent: both $\ell\eta_c$ and $\eta_{\ell c}$ are equivalences. An object d is *local* if η_d is an equivalence, equivalently if d belongs to the essential image of ℓ. The adjunction is characterized by

$$\mathrm{Map}_{\mathscr{C}}(\ell c, d) \simeq \mathrm{Map}_{\mathscr{C}}(c, d)$$

for every local object d. A morphism g is an *ℓ-equivalence* if $\ell(g)$ is an equivalence; equivalently, precomposition with g induces an equivalence of mapping spaces into every local object.

Dually, a *colocalization* is a full subcategory whose inclusion admits a right adjoint r, with counit $r \to \mathrm{id}$. Thus localization reflects objects into a subcategory, while colocalization coreflects them.

Definition 6.1.1 (Compatibility with an $\mathcal{O}$-Monoidal Structure). Suppose that for every color $x \in \mathcal{O}$ we have a localization $\ell_x : \mathscr{C}_x \to \mathscr{C}_x$. A morphism g in $\mathscr{C}_x$ is an *ℓ_x-equivalence* if $\ell_{x(g)}$ is an equivalence. The family $\{\ell_x\}_{x \in \mathcal{O}}$ is *compatible with the $\mathcal{O}$-monoidal structure* if, for every operation

$$f \in \mathrm{Mul}_{\mathcal{O}}\big(\{x_i\}_{1\le i\le n}; y\big)$$

and every family of ℓ_{x_i}-equivalences g_i, the morphism

$$\otimes_f\,\big(\{g_i\}_{1\le i\le n}\big)$$

is an ℓ_y-equivalence.

Remark (One-Colored Test). For a localization $\ell : \mathscr{C} \to \mathscr{C}$ of a symmetric monoidal ∞-category, compatibility is equivalent to the elementary condition

$$g \;\text{ is an }\; \ell\text{-equivalence} \Longrightarrow g \otimes \mathrm{id}_z \;\text{ is an }\; \ell\text{-equivalence}$$

for every object z. By symmetry and iteration, this tests all variables and all arities.

Lemma 6.1.1 (Fiberwise Localization Lemma). Let $q : \mathscr{E} \to \mathscr{B}$ be a coCartesian fibration, and let ℓ_E, ℓ_B be localizations whose essential images are $\mathscr{E}'$ and $\mathscr{B}'$. Suppose that q restricts to $q' : \mathscr{E}' \to \mathscr{B}'$ and carries ℓ_E-equivalences to ℓ_B-equivalences. Then ℓ_E carries q -coCartesian morphisms to q'-coCartesian morphisms, and q' is a coCartesian fibration.

Proof sketch. The coCartesian property is detected by the usual homotopy-Cartesian square of mapping spaces. Mapping into a local object identifies the localized and unlocalized mapping spaces, while compatibility identifies their bases. The defining square therefore remains homotopy Cartesian after applying the localizations. □

Proposition 6.1.3 (Monoidal Localization). Let $\{\ell_x\}_{x\in\mathcal{O}}$ be a compatible family of localizations, and let $\mathscr{D}_x$ be the essential image of ℓ_x. Then:

1. the fiberwise functors assemble into a localization $\ell^{\otimes} : \mathscr{C}^{\otimes} \to \mathscr{D}^{\otimes}$ over $\mathcal{O}^{\otimes}$
2. the projection $\mathscr{D}^{\otimes} \to \mathcal{O}^{\otimes}$ is a coCartesian fibration of ∞-operads
3. the inclusion is a map of ∞-operads and $\ell^{\otimes}$ is an $\mathcal{O}$-monoidal functor

The induced operation on local objects is characterized by

$$\otimes_f^{\mathscr{D}}\,(d_1, \ldots, d_n) \simeq \ell_y\big(\otimes_f^{\mathscr{C}}\,(d_1, \ldots, d_n)\big)$$

Proof sketch. Compatibility says that each tensor functor descends through the localizations. The fiberwise localization lemma shows that the descended transport is coCartesian. Applying the Segal equivalence fiberwise then constructs $\mathscr{D}^{\otimes}$ and shows that $\ell^{\otimes}$ preserves the required coCartesian morphisms. The unit maps $\mathrm{id} \to \ell_x$ assemble because their spaces of extensions are contractible. □

Remark (Restriction versus Localization). If local objects are already closed under the tensor products of $\mathscr{C}$, the inherited tensor is simply the restriction considered above. In general they need not be tensor-closed: the localized category still becomes monoidal, but one must tensor in $\mathscr{C}$ and then apply ℓ.

6.1.4 Compatible t-Structures

Definition 6.1.2 (Monoidal Compatibility of t-Structures). Suppose every fiber $\mathscr{C}_x$ is stable, every operation $\otimes_f$ is exact in each variable, and each fiber carries a t-structure. The family is *compatible with the $\mathcal{O}$-monoidal structure* if

$$\otimes_f\left(\prod_{i=1}^{n}\left(\mathscr{C}_{x_i}\right)_{\geq 0}\right)\subset\left(\mathscr{C}_y\right)_{\geq 0}$$

for every operation f of $\mathcal{O}$.

Proposition 6.1.4 (Connective and Truncated Objects). For a compatible family of t-structures:

1. the connective objects $\mathscr{C}_{\geq 0}$ inherit an $\mathcal{O}$-monoidal structure
2. exactness gives the degree estimate
$$\otimes_f\left(\prod_{i=1}^{n}\left(\mathscr{C}_{x_i}\right)_{\geq m_i}\right)\subset\left(\mathscr{C}_y\right)_{\geq \sum_i m_i}$$
3. for every $k \geq 0$, the truncation localizations $\tau_{\leq k} : (\mathscr{C}_x)_{\geq 0} \to (\mathscr{C}_x)_{\geq 0}$ form a compatible family and hence the subcategories $(\mathscr{C}_x)_{\geq 0} \cap (\mathscr{C}_x)_{\leq k}$ inherit the induced $\mathcal{O}$-monoidal structure

Proof sketch. The first statement is the tensor-closed subcategory criterion. For the third, it is enough to vary one input. If $g : c \to c'$ is a $\tau_{\leq k}$-equivalence between connective objects, then its fiber belongs to degree $\geq k+1$. Exactness in that variable and the degree estimate put the fiber of $\otimes_f(g)$ in degree $\geq k+1$, so applying $\tau_{\leq k}$ makes $\otimes_f(g)$ an equivalence. This is precisely compatibility of the truncation localizations. □

Corollary 6.1.2 (The Heart Inherits the Tensor Product). Taking $k = 0$, the hearts of a compatible family of t-structures inherit the corresponding monoidal operations. In particular, the heart of a symmetric monoidal stable ∞-category with compatible t-structure is an ordinary symmetric monoidal category.

6.2 Slicing ∞-Operads

6.2.1 Relative Slice Constructions

Let $q : x \to s$ be a map of simplicial sets and let $p : s \times k \to x$ be a map over s, so that the triangle

$$\begin{tikzcd} & x \arrow[dr, "q"] & \\ s \times k \arrow[ur, "p"] \arrow[rr, "\mathrm{pr}_1"'] & & s \end{tikzcd}$$

commutes.

Definition 6.2.3 (Relative Slices). The *relative undercategory* $x_{p/} \to s$ is characterized by the following universal property: for every $u : y \to s$, maps $y \to x_{p/}$ over s are naturally identified with maps

$$F : y \times k^{\triangleright} \to x$$

over s whose restriction to $y \times k$ is $p \circ (u \times \mathrm{id}_k)$. Dually, the *relative overcategory* $x_{/p} \to s$ is defined by replacing the right cone $k^{\triangleright}$ with the left cone $k^{\triangleleft}$.

When $s = \Delta^0$, these are the usual undercategory and overcategory. More generally, slicing is computed fiberwise: for every vertex $a \in s$,

$$\left(x_{p/}\right)_a \simeq (x_a)_{p_a/}, \quad \left(x_{/p}\right)_a \simeq (x_a)_{/p_a}$$

where $x_a = x \times_s \{a\}$ and $p_a : k \to x_a$ is the induced diagram.

6.2.2 Slices of ∞-Operads

Let $q : \mathscr{C}^{\otimes} \to \mathscr{O}^{\otimes}$ be a fibration of ∞-operads and let

$$p : k \to \mathbf{Alg}_{\mathscr{O}}(\mathscr{C})$$

be a diagram of $\mathscr{O}$-algebra objects. Evaluation gives a map

$$\tilde{p} : \mathscr{O}^{\otimes} \times k \to \mathscr{C}^{\otimes}$$

over $\mathscr{O}^{\otimes}$. Define

$$\mathscr{C}^{\otimes}_{p/\mathscr{O}} := \left(\mathscr{C}^{\otimes}\right)_{\tilde{p}/}, \quad \mathscr{C}^{\otimes}_{/p/\mathscr{O}} := \left(\mathscr{C}^{\otimes}\right)_{/\tilde{p}}$$

If $k = \Delta^0$ and p selects an algebra object a, we abbreviate these to $\mathscr{C}^{\otimes}_{a/\mathscr{O}}$ and $\mathscr{C}^{\otimes}_{/a/\mathscr{O}}$. Their fibers over $\langle 1 \rangle$ are the ordinary slices $\mathscr{C}_{a/}$ and $\mathscr{C}_{/a}$, respectively.

Theorem 6.2.1 (Operadic Slice Theorem). With the notation above:

1. the induced maps

$$q' : \mathscr{C}^{\otimes}_{p/\mathscr{O}} \to \mathscr{O}^{\otimes}, \quad q'' : \mathscr{C}^{\otimes}_{/p/\mathscr{O}} \to \mathscr{O}^{\otimes}$$

   are fibrations of ∞-operads

2. a morphism in either sliced operad is inert if and only if its image in $\mathscr{C}^{\otimes}$ is inert
3. if q is a coCartesian fibration, then the map q'' is a coCartesian fibration, so the relative overcategory inherits an $\mathscr{O}$-monoidal structure
4. the map q' is also coCartesian provided every section $p(t) : \mathscr{O}^{\otimes} \to \mathscr{C}^{\otimes}$, for $t \in k_0$, preserves all coCartesian morphisms

Proof sketch. The universal property of relative slices translates inner-horn lifting in the sliced objects into inner-horn lifting for q. Fiberwise, the slices are ordinary undercategories or overcategories, and the Segal product decomposition is therefore inherited from $\mathscr{C}^{\otimes}$. The same cone description identifies inert morphisms and coCartesian transport. For the undercategory, transport must also move the chosen source diagram p; this is why the additional coCartesian-preservation hypothesis on every section $p(t)$ is necessary. □

Remark (The Asymmetry). For an $\mathcal{O}$-monoidal category, slicing *over* an algebra object is automatically $\mathcal{O}$-monoidal. Slicing *under* it generally is not: the algebra section must itself be $\mathcal{O}$-monoidal, meaning that it preserves every coCartesian morphism rather than only the inert ones required of an algebra object.

6.2.3 Slicing over a Commutative Algebra

Corollary 6.2.3 (The Monoidal Overcategory). Let $\mathscr{C}^{\otimes}$ be a symmetric monoidal ∞-category and let $a \in \mathbf{CAlg}(\mathscr{C})$. Then the ordinary slice $\mathscr{C}_{/a}$ is the underlying ∞-category of a symmetric monoidal ∞-category. For objects $u : x \to a$ and $v : y \to a$, their tensor product is the composite

$$x \otimes y \xrightarrow{u \otimes v} a \otimes a \xrightarrow{m} a$$

where $m : a \otimes a \to a$ is the multiplication. The unit is the unit map $1 \to a$.

Proof sketch. Apply the operadic slice theorem with $\mathcal{O}^{\otimes} = \mathbf{Comm}^{\otimes}$ and $k = \Delta^0$. The formula is simply coCartesian transport in the sliced operad. Associativity, the unit law, and symmetry follow coherently from the corresponding structure maps of the commutative algebra a. □

Remark (The Undercategory). The undercategory $\mathscr{C}_{a/}$ inherits the analogous symmetric monoidal structure only when the section $a : \mathbf{Comm}^{\otimes} \to \mathscr{C}^{\otimes}$ is symmetric monoidal. This is the "trivial algebra" case: its structural maps, including $1 \to a$ and $a \otimes a \to a$, are equivalences. Without this hypothesis, there is no canonical way to tensor two arrows whose common source is a.

6.3 Coproducts of ∞-Operads

The ∞-category $\mathbf{Op}_{\infty}$ is presentable, so coproducts of ∞-operads exist abstractly. Their total ∞ -categories cannot simply be joined over $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$: the arities contributed by the two factors must first be combined. The following construction makes this bookkeeping explicit.

6.3.1 Splitting a Finite Set of Inputs

For a subset $s \subset \langle n \rangle$ containing the basepoint, write $[s]$ for the pointed finite set obtained from s by its induced order.

Definition 6.3.4 (The Category Sub). An object of $\mathbf{Sub}$ is a triple

$$(\langle n \rangle, s, t)$$

where s and t contain the basepoint and partition the non-basepoint elements:

$$s \cup t = \langle n \rangle, \quad s \cap t = \{*\}$$

A morphism to $(\langle n' \rangle, s', t')$ is a pointed map $f : \langle n \rangle \to \langle n' \rangle$ satisfying $f(s) \subset s'$ and $f(t) \subset t'$. There are three evident functors

$$\pi_-(\langle n \rangle, s, t) = [s], \quad \pi(\langle n \rangle, s, t) = \langle n \rangle, \quad \pi_+(\langle n \rangle, s, t) = [t]$$

The pair (π_-, π_+) is an equivalence

$$\mathbf{Sub} \simeq \mathbf{Fin}_* \times \mathbf{Fin}_*$$

The extra projection π remembers how the two ordered lists of inputs are interleaved inside one list. This is the only additional information needed to combine two operads.

6.3.2 The Box Product

Definition 6.3.5 (Box Product of ∞-Operads). Let $\mathscr{C}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ and $\mathscr{D}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ be ∞-operads. Define $\mathscr{C}^{\otimes} \boxplus \mathscr{D}^{\otimes}$ by the pullback square

$$\begin{tikzcd}
\mathscr{C}^{\otimes} \boxplus \mathscr{D}^{\otimes} \arrow[r] \arrow[d] & \mathscr{C}^{\otimes} \times \mathscr{D}^{\otimes} \arrow[d] \\
\mathrm{N}_{\bullet}(\mathbf{Sub}) \arrow[r, "\mathrm{N}_{\bullet}(\pi_- \times \pi_+)"'] & \mathrm{N}_{\bullet}(\mathbf{Fin}_*) \times \mathrm{N}_{\bullet}(\mathbf{Fin}_*)
\end{tikzcd}$$

Its structure map to $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ is induced by $\pi : \mathbf{Sub} \to \mathbf{Fin}_*$. We call it the *box product* of $\mathscr{C}^{\otimes}$ and $\mathscr{D}^{\otimes}$.

As an ∞-category, the box product is equivalent to $\mathscr{C}^{\otimes} \times \mathscr{D}^{\otimes}$. Its operadic projection is different, however: π merges the two input lists. In particular, the colors of the box product form the disjoint union of the colors of $\mathscr{C}$ and $\mathscr{D}$, not their Cartesian product.

Theorem 6.3.2 (Coproduct of ∞-Operads). The box product $\mathscr{C}^{\otimes} \boxplus \mathscr{D}^{\otimes}$ is an ∞-operad. Moreover, for every ∞-operad $\mathscr{E}^{\otimes}$, restriction induces an equivalence

$$\mathbf{Alg}_{\mathscr{C} \boxplus \mathscr{D}}(\mathscr{E}) \simeq \mathbf{Alg}_{\mathscr{C}}(\mathscr{E}) \times \mathbf{Alg}_{\mathscr{D}}(\mathscr{E})$$

Passing to maximal subgroupoids gives

$$\mathrm{Map}_{\mathbf{Op}_\infty}(\mathscr{C} \boxplus \mathscr{D}, \mathscr{E}) \simeq \mathrm{Map}_{\mathbf{Op}_\infty}(\mathscr{C}, \mathscr{E}) \times \mathrm{Map}_{\mathbf{Op}_\infty}(\mathscr{D}, \mathscr{E})$$

Hence the box product represents the coproduct of $\mathscr{C}$ and $\mathscr{D}$ in $\mathbf{Op}_\infty$.

Equivalently, an algebra over $\mathscr{C} \boxplus \mathscr{D}$ is a pair consisting of a $\mathscr{C}$-algebra and a $\mathscr{D}$-algebra on the same target; the coproduct imposes no further compatibility law between them.

Proof sketch. Split every input according to s and t, then apply the Segal decompositions in $\mathscr{C}^{\otimes}$ and $\mathscr{D}^{\otimes}$ separately. The extreme suboperads $t = \{*\}$ and $s = \{*\}$ are operadically equivalent to $\mathscr{C}$ and $\mathscr{D}$. A map out of the box product is therefore determined, up to a contractible choice, by one map from each factor. □

6.4 Monoidal Envelopes

The monoidal envelope is the universal construction which turns an ∞-operad into a monoidal ∞-category. It keeps the original operations but freely supplies tensor products and their coCartesian transport.

6.4.1 Construction by Active Morphisms

Let $\mathcal{O}^{\otimes}$ be an ∞-operad. Denote by

$$\mathbf{Act}(\mathcal{O}^{\otimes}) \subset \mathbf{Fun}(\Delta^1, \mathcal{O}^{\otimes})$$

the full subcategory spanned by the active morphisms, and write $\mathrm{ev}_0, \mathrm{ev}_1$ for evaluation at the source and target.

Definition 6.4.6 (Relative Monoidal Envelope). Let

$$p : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$$

be a fibration of ∞-operads. Its $\mathcal{O}$*-monoidal envelope* is the pullback

$$\mathbf{Env}_{\mathcal{O}}(\mathcal{C})^{\otimes} := \mathcal{C}^{\otimes} \times_{\mathcal{O}^{\otimes}} \mathbf{Act}(\mathcal{O}^{\otimes})$$

where the two maps to $\mathcal{O}^{\otimes}$ are p and ev_0. Its projection to $\mathcal{O}^{\otimes}$ is induced by ev_1. When $\mathcal{O} = \mathbf{Comm}$, we simply write $\mathbf{Env}(\mathcal{C})^{\otimes}$.

i.e. defined by the pullback square

$$\begin{tikzcd}
\mathbf{Env}_{\mathcal{O}}(\mathcal{C})^{\otimes} \arrow[r] \arrow[d] & \mathcal{C}^{\otimes} \arrow[d, "p"] \\
\mathbf{Act}(\mathcal{O}^{\otimes}) \arrow[r, "\mathrm{ev}_1"'] & \mathcal{O}^{\otimes}
\end{tikzcd}$$

Thus an object of $\mathbf{Env}_{\mathcal{O}}(\mathcal{C})^{\otimes}$ is a pair (c, α), where $c \in \mathcal{C}^{\otimes}$ and

$$\alpha : p(c) \to x$$

is active in $\mathcal{O}^{\otimes}$. A morphism is a commutative square between such active arrows together with a compatible morphism in $\mathcal{C}^{\otimes}$.

The diagonal choice $\alpha = \mathrm{id}_{p(c)}$ defines a canonical map of ∞-operads

$$i : \mathcal{C}^{\otimes} \to \mathbf{Env}_{\mathcal{O}}(\mathcal{C})^{\otimes}$$

This map is fully faithful: the envelope adds formal tensor transport but does not identify or discard any operation already present in $\mathcal{C}$.

6.4.2 The Induced Monoidal Structure

Theorem 6.4.3 (The Envelope is $\mathcal{O}$-Monoidal). Evaluation at the target induces a coCartesian fibration of ∞-operads

$$q : \mathbf{Env}_{\mathcal{O}}(\mathcal{C})^{\otimes} \to \mathcal{O}^{\otimes}$$

A morphism in the envelope is q-coCartesian if and only if its projection to $\mathscr{C}^{\otimes}$ is inert.

Proof sketch. Factor morphisms in $\mathscr{O}^{\otimes}$ into active and inert parts. The operadic lifting axioms supply the inert part, while composition of active arrows supplies the active part. This constructs the required coCartesian lifts. The Segal equivalence is inherited by applying the same factorization separately to every output slot. □

Corollary 6.4.4 (Symmetric Monoidal Envelope). Every ∞-operad $\mathscr{C}^{\otimes} \to \mathbf{Comm}^{\otimes}$ has a canonical symmetric monoidal envelope $\mathbf{Env}(\mathscr{C})^{\otimes}$.

Its underlying ∞-category has a particularly concrete description:

$$\mathbf{Env}(\mathscr{C}) \simeq \mathscr{C}^{\otimes}_{\mathrm{act}}$$

The right-hand side has all objects of $\mathscr{C}^{\otimes}$ but only active morphisms. Equivalently, its objects are finite lists $(c_1, \ldots, c_n)$ of colors of $\mathscr{C}$, its unit is the empty list, and its tensor product is concatenation

$$(c_1, \ldots, c_m) \otimes (d_1, \ldots, d_n) = (c_1, \ldots, c_m, d_1, \ldots, d_n)$$

In this structure, tensor products of active coCartesian morphisms are again active and coCartesian.

6.4.3 Universal Property

Theorem 6.4.4 (Universal Property of the Monoidal Envelope). Let $\mathscr{D}^{\otimes} \to \mathscr{O}^{\otimes}$ be an $\mathscr{O}$-monoidal ∞-category. Restriction along i induces an equivalence

$$i^* : \mathbf{Fun}^{\otimes}_{\mathscr{O}}(\mathbf{Env}_{\mathscr{O}}(\mathscr{C}), \mathscr{D}) \simeq \mathbf{Alg}_{\mathscr{C}/\mathscr{O}}(\mathscr{D})$$

In words, an operad map from $\mathscr{C}$ to an $\mathscr{O}$-monoidal category extends uniquely up to a contractible space of choices to an $\mathscr{O}$-monoidal functor from its envelope.

Proof sketch. Restriction gives the forward map. Conversely, for an operad map $f : \mathscr{C}^{\otimes} \to \mathscr{D}^{\otimes}$, send (c, α) to the coCartesian transport of $f(c)$ along α. CoCartesian lifts are unique up to a contractible space of choices and compose coherently, so this construction gives the inverse equivalence. □

Corollary 6.4.5 (The Envelope Adjunction). The forgetful functor

$$U : \mathbf{Cat}^{\otimes}_{\infty} \to \mathbf{Op}_{\infty}$$

admits the monoidal-envelope functor as a left adjoint

$$\mathbf{Env} : \mathbf{Op}_{\infty} \to \mathbf{Cat}^{\otimes}_{\infty}$$

Equivalently, for every symmetric monoidal ∞-category $\mathscr{D}^{\otimes}$,

$$\mathbf{Fun}^{\otimes}_{\mathbf{Comm}}(\mathbf{Env}(\mathscr{C}), \mathscr{D}) \simeq \mathbf{Alg}_{\mathscr{C}}(\mathscr{D})$$

6.5 Tensor Products of ∞-Operads

The tensor product of operads is different from both the coproduct above and the Cartesian product of their total ∞-categories. It is characterized by the principle that an algebra over $\mathscr{O} \otimes \mathscr{P}$ is an $\mathscr{O}$-algebra internal to $\mathscr{P}$-algebras.

6.5.1 Multiplying Arities

Define the smash-product functor

$$\wedge : \mathbf{Fin}_* \times \mathbf{Fin}_* \to \mathbf{Fin}_*$$

by

$$\langle m \rangle \wedge \langle n \rangle = \langle mn \rangle$$

Its non-basepoint elements are pairs (a, b), ordered lexicographically. For pointed maps f and g,

$$(f \wedge g)(a, b) = \begin{cases} * & \text{if } \ f(a) = * \ \text{ or } \ g(b) = * \\ (f(a), g(b)) & \text{otherwise} \end{cases}$$

The functor $\wedge$ is associative and preserves coproducts separately in each variable.

> **Remark (Why Symmetry is Subtle).** The chosen lexicographic order makes associativity strict, but the symmetry
>
> $$\langle m \rangle \wedge \langle n \rangle \simeq \langle n \rangle \wedge \langle m \rangle$$
>
> is not literally the identity. Hence this single functor does not by itself give a strict symmetric structure. The space of all smash-product functors $\mathbf{Fin}_*^i \to \mathbf{Fin}_*$ is nevertheless contractible; using this whole space supplies the required coherent symmetries.

6.5.2 Operadic Bifunctors

> **Definition 6.5.7 (Bifunctor of ∞-Operads).** Let $\mathscr{O}^\otimes$, $\mathscr{P}^\otimes$, and $\mathscr{Q}^\otimes$ be ∞-operads. An *operadic bifunctor* is a functor
>
> $$f : \mathscr{O}^\otimes \times \mathscr{P}^\otimes \to \mathscr{Q}^\otimes$$
>
> fitting into the commutative square
>
> $$\begin{tikzcd} \mathscr{O}^\otimes \times \mathscr{P}^\otimes \arrow[r, "f"] \arrow[d] & \mathscr{Q}^\otimes \arrow[d] \\ \mathrm{N}_\bullet(\mathbf{Fin}_*) \times \mathrm{N}_\bullet(\mathbf{Fin}_*) \arrow[r, "\mathrm{N}_\bullet(- \wedge -)"] & \mathrm{N}_\bullet(\mathbf{Fin}_*) \end{tikzcd}$$
>
> and such that $f(\alpha, \beta)$ is inert whenever both α and β are inert. Denote the resulting ∞-category by
>
> $$\mathbf{BiFun}(\mathscr{O}, \mathscr{P}; \mathscr{Q})$$

Remark (The Concrete Construction of BiFun). Give $\mathcal{O}^{\otimes} \times \mathcal{P}^{\otimes}$ its map to $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ by composing the two projections with $\mathrm{N}_{\bullet}(- \wedge -)$. Formally, $\mathbf{BiFun}(\mathcal{O}, \mathcal{P}; \mathcal{Q})$ is the full subcategory of

$$\mathbf{Fun}_{/\mathrm{N}_{\bullet}(\mathbf{Fin}_*)}\left(\mathcal{O}^{\otimes} \times \mathcal{P}^{\otimes}, \mathcal{Q}^{\otimes}\right)$$

spanned by the functors which carry every pair of inert morphisms to an inert morphism. Since it is a full subcategory, a morphism $f \to g$ in **BiFun** is simply a natural transformation over $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$; no extra condition is imposed on the transformation.

Concretely, choose colors $x_1, \ldots, x_m, x$ of $\mathcal{O}$ and $y_1, \ldots, y_n, y$ of $\mathcal{P}$, together with operations

$$\alpha \in \mathrm{Mul}_{\mathcal{O}}\left(\{x_i\}_{1 \le i \le m}; x\right), \quad \beta \in \mathrm{Mul}_{\mathcal{P}}\left(\{y_j\}_{1 \le j \le n}; y\right)$$

A bifunctor produces the mn-ary operation

$$f(\alpha, \beta) \in \mathrm{Mul}_{\mathcal{Q}}\left(\left\{f\left(x_i, y_j\right)\right\}_{1 \le i \le m, 1 \le j \le n}; f(x, y)\right)$$

The inert condition guarantees that these mn inputs really are the matrix of pairs $\left(x_i, y_j\right)$. Functoriality says that first composing along the rows and then along the columns agrees coherently with composing in the opposite order. This is the *interchange law*. Equivalently, f is operadic in each variable separately, with the two operadic actions made coherently compatible.

Definition 6.5.8 (Tensor Product of ∞-Operads). A bifunctor

$$\mathcal{O}^{\otimes} \times \mathcal{P}^{\otimes} \to (\mathcal{O} \otimes \mathcal{P})^{\otimes}$$

exhibits $\mathcal{O} \otimes \mathcal{P}$ as the *tensor product* if, for every ∞-operad $\mathcal{Q}$, composition induces an equivalence

$$\mathbf{Alg}_{\mathcal{O} \otimes \mathcal{P}}(\mathcal{Q}) \simeq \mathbf{BiFun}(\mathcal{O}, \mathcal{P}; \mathcal{Q})$$

Theorem 6.5.5 (Existence of the Operadic Tensor Product). Every pair of small ∞-operads admits a tensor product, unique up to a canonical equivalence.

Remark (Model-Categorical Construction). This existence statement is the technically difficult part. For marked preoperads $\overline{x} = (x, m)$ and $\overline{y} = (y, n)$, first form

$$\overline{x} \odot \overline{y} := (x \times y, m \times n)$$

over $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ using the smash-product map $\mathrm{N}_{\bullet}(\wedge)$. The functor $\odot : \mathbf{POp}_{\infty} \times \mathbf{POp}_{\infty} \to \mathbf{POp}_{\infty}$ is left Quillen in each variable, making $\mathbf{POp}_{\infty}$ a monoidal model category. A fibrant replacement of this external product represents $\mathcal{O} \otimes \mathcal{P}$. Thus the tensor product is derived rather than the naive product of the underlying simplicial sets.

6.5.3 Universal Consequences

Corollary 6.5.6 (Algebraic Exponential Law). If $\mathscr{C}^{\otimes}$ is symmetric monoidal, then there is a canonical equivalence

$$\mathbf{Alg}_{\mathscr{O}\otimes\mathscr{P}}(\mathscr{C}) \simeq \mathbf{Alg}_{\mathscr{O}}(\mathbf{Alg}_{\mathscr{P}}(\mathscr{C}))$$

Consequently, $\mathbf{Alg}_{\mathscr{O}}(\mathscr{C})$ inherits a symmetric monoidal structure whose tensor product is computed pointwise in $\mathscr{C}$.

Remark (Interchange, Not Mere Coexistence). Compare this with the coproduct $\mathscr{O} \boxplus \mathscr{P}$. An algebra over the coproduct merely carries independent $\mathscr{O}$- and $\mathscr{P}$-structures. An algebra over $\mathscr{O} \otimes \mathscr{P}$ carries both structures together with coherent interchange: every $\mathscr{O}$-operation is a morphism of $\mathscr{P}$-algebras, and conversely.

Theorem 6.5.6 (The Symmetric Monoidal ∞-Category of Operads). Operadic tensor products assemble into a symmetric monoidal ∞-category

$$\mathbf{Op}_{\infty}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_{*})$$

whose underlying ∞-category is $\mathbf{Op}_{\infty}$. Its tensor unit is the trivial operad $\mathbf{Triv}^{\otimes}$, and its multimorphism spaces classify multilinear operad maps.

Proof sketch. The contractible groupoids of smash-product functors provide all coherent choices of parenthesization and permutation. Their operadic nerves assemble the derived external products of preoperads into a symmetric monoidal structure. The universal property above identifies its binary operation with $\mathscr{O} \otimes \mathscr{P}$, while the one-color unary operad **Triv** supplies the unit. □

Corollary 6.5.7 (Unary Operads Recover Cartesian Products). Under the fully faithful embedding $\mathbf{Cat}_{\infty} \to \mathbf{Op}_{\infty}$ by unary operads, the operadic tensor product restricts to the Cartesian product:

$$\mathscr{C} \otimes \mathscr{D} \simeq \mathscr{C} \times \mathscr{D}$$

Equivalently, the symmetric monoidal subcategory of unary operads is symmetric monoidally equivalent to $\mathbf{Cat}_{\infty}^{\times}$.

6.6 Day Convolution

The pointwise tensor product on a functor category uses only the monoidal structure of the target and ignores that of the source. Day convolution does the opposite: it extends the tensor product of the source to functors by freely summing over every way an object can be assembled as a tensor product.

6.6.1 A First Example: Graded Objects

Example (The Cauchy Product). Regard $\mathbb{N}$ as a discrete symmetric monoidal category under addition, and let $\mathscr{D} = \mathbf{Ab}$. A functor $a : \mathbb{N} \to \mathbf{Ab}$ is a nonnegatively graded abelian group $\{a_n\}_{n\geq 0}$. Day convolution is

$$(a \circledast b)_n \simeq \oplus_{i+j=n}\, a_i \otimes b_j$$

Its unit is $\mathbb{Z}$ in degree 0 and 0 in every positive degree. Thus Day convolution is exactly the Cauchy product of graded objects, analogous to multiplication of generating series:

$$\left(\sum_{i\geq 0} a_i t^i\right)\left(\sum_{j\geq 0} b_j t^j\right) = \sum_{n\geq 0} (\oplus_{i+j=n}\, a_i \otimes b_j) t^n$$

The summands indexed by $i + j = n$ are the simplest instance of the general rule: sum over all morphisms $c_0 \otimes c_1 \to c$.

6.6.2 Definition and Basic Properties

Definition 6.6.9 (Day Convolution). Let $\mathscr{C}$ be a small monoidal ∞-category and let $\mathscr{D}$ admit small colimits. For $f, g : \mathscr{C} \to \mathscr{D}$, their *Day convolution* is

$$f \circledast g := \mathrm{Lan}_{\otimes_{\mathscr{C}}}(\otimes_{\mathscr{D}} \circ (f \times g))$$

whenever this left Kan extension exists. Pointwise,

$$(f \circledast g)(c) \simeq \operatorname*{colim}_{c_0\otimes c_1 \to c} f(c_0) \otimes g(c_1)$$

where the indexing ∞-category has objects $(c_0, c_1, u : c_0 \otimes c_1 \to c)$.

Theorem 6.6.7 (The Day Monoidal Structure). Suppose the tensor product of $\mathscr{D}$ preserves small colimits separately in each variable. Then Day convolution makes $\mathbf{Fun}(\mathscr{C}, \mathscr{D})$ monoidal. If $\mathscr{C}$ and $\mathscr{D}$ are symmetric monoidal, this structure is symmetric monoidal. It preserves colimits separately in both functor variables.

Its unit is the functor

$$1_{\mathrm{Day}}(c) \simeq \mathrm{Map}_{\mathscr{C}}(1_{\mathscr{C}}, c) \odot 1_{\mathscr{D}}$$

where $k \odot d$ denotes the tensoring, or copower, of an object d by a space k.

Proof sketch. Both parenthesizations are iterated colimits over threefold tensor factorizations. Fubini and the associators identify them coherently; the same argument gives the unit and symmetry. Colimit preservation lets the tensor product of $\mathscr{D}$ pass through the Kan extensions. □

Corollary 6.6.8 (Algebra Objects). There is a canonical equivalence

$$\mathbf{CAlg}\big(\mathbf{Fun}(\mathscr{C}, \mathscr{D})_{\mathrm{Day}}\big) \simeq \mathbf{Fun}^{\otimes}_{\mathrm{lax}}(\mathscr{C}, \mathscr{D})$$

Thus commutative algebras for Day convolution are lax symmetric monoidal functors. Replacing commutative by associative gives lax monoidal functors.

6.6.3 The Coend Viewpoint

Assume that $\mathscr{D}$ is tensored over spaces. The comma-category formula for the Kan extension can be rewritten as the coend

$$(f \circledast g)(c) \simeq \int^{c_0, c_1 \in \mathscr{C}} \mathrm{Map}_{\mathscr{C}}(c_0 \otimes c_1, c) \odot (f(c_0) \otimes g(c_1))$$

The mapping space supplies the *weight*: every map $c_0 \otimes c_1 \to c$ contributes a copy of $f(c_0) \otimes g(c_1)$, while the coend imposes functorial identifications in c_0 and c_1. This is the enriched form of the colimit in the definition.

Remark (Representables). Put

$$h^a(c) := \mathrm{Map}_{\mathscr{C}}(a, c) \odot 1_{\mathscr{D}}$$

The co-Yoneda lemma gives

$$h^a \circledast h^b \simeq h^{a \otimes b}$$

Thus the covariant Yoneda embedding $\mathscr{C}^{\mathrm{op}} \to \mathbf{Fun}(\mathscr{C}, \mathscr{D})$ is strong monoidal after extending scalars from spaces to $\mathscr{D}$. Conceptually, Day convolution is the colimit-preserving extension of the tensor product on representables.

Remark (Closed Structure). If $\mathscr{D}$ is closed symmetric monoidal and admits the relevant limits, Day convolution is closed. Its internal Hom can be written as the end

$$[f, g]_{\mathrm{Day}}(c) \simeq \int_{x \in \mathscr{C}} [f(x), g(c \otimes x)]_{\mathscr{D}}$$

The coend formula for $\circledast$ and this end formula are adjoint manifestations of the same Kan-extension construction.

6.6.4 Operadic Generalization

The preceding construction uses only one binary tensor product. Norms package the same idea for every operation of an arbitrary ∞-operad $\mathscr{O}^{\otimes}$ simultaneously.

Definition 6.6.10 (Norm). Let $p : \mathscr{C}^{\otimes} \to \mathscr{O}^{\otimes}$ be a coCartesian fibration of ∞-operads and let $\mathscr{E}^{\otimes} \to \mathscr{C}^{\otimes}$ be a fibration. A *norm of* $\mathscr{E}$ *along* p is a fibration $\mathscr{N}^{\otimes} \to \mathscr{O}^{\otimes}$ together with

$$\alpha : \mathscr{N}^{\otimes} \times_{\mathscr{O}^{\otimes}} \mathscr{C}^{\otimes} \to \mathscr{E}^{\otimes}$$

such that, for every $\mathscr{P}^{\otimes} \to \mathscr{O}^{\otimes}$, composition with α induces an equivalence

$$\mathbf{Alg}_{\mathscr{P}/\mathscr{O}}(\mathscr{N}) \simeq \mathbf{Alg}_{(\mathscr{P} \times_{\mathscr{O}} \mathscr{C})/\mathscr{C}}(\mathscr{E})$$

We write it as $\mathrm{Nm}_{\mathscr{C}/\mathscr{O}}(\mathscr{E})^{\otimes}$.

Theorem 6.6.8 (Existence of Norms). Norms exist and are unique up to a contractible space of choices. In other words, the norm is the operadic right adjoint to base change along p.

Remark (The Technical Point). A naive functor category over $\mathcal{O}^\otimes$ need not satisfy the operadic Segal condition. The norm is constructed as an internal Hom of marked preoperads and then restricted to simplices which preserve the required inert morphisms. Establishing fibrancy of this marked object is the role of the long horn-filling argument in the source.

Definition 6.6.11 (The Functor Operad). Given an $\mathcal{O}$-monoidal $\mathcal{C}^\otimes$ and a fibration $\mathcal{D}^\otimes \to \mathcal{O}^\otimes$, define

$$\mathbf{Fun}^{\mathcal{O}}(\mathcal{C},\mathcal{D})^\otimes := \mathrm{Nm}_{\mathcal{C}/\mathcal{O}}(\mathcal{C} \times_{\mathcal{O}} \mathcal{D})^\otimes$$

Its fiber at a color x is

$$\mathbf{Fun}^{\mathcal{O}}(\mathcal{C},\mathcal{D})_x \simeq \mathbf{Fun}(\mathcal{C}_x,\mathcal{D}_x)$$

and its algebra objects satisfy

$$\mathbf{Alg}_{\mathcal{O}}\left(\mathbf{Fun}^{\mathcal{O}}(\mathcal{C},\mathcal{D})\right) \simeq \mathbf{Alg}_{\mathcal{C}/\mathcal{O}}(\mathcal{D})$$

The last equivalence says that algebras in the functor operad are precisely maps $\mathcal{C}^\otimes \to \mathcal{D}^\otimes$ over $\mathcal{O}^\otimes$.

Theorem 6.6.9 (Operadic Day Convolution). Suppose every fiber $\mathcal{C}_x$ is essentially small, every $\mathcal{D}_y$ admits the required colimits, and all tensor operations of $\mathcal{D}$ preserve them separately. Then $\mathbf{Fun}^{\mathcal{O}}(\mathcal{C},\mathcal{D})^\otimes \to \mathcal{O}^\otimes$ is a coCartesian fibration of ∞-operads.

For $\varphi \in \mathrm{Mul}_{\mathcal{O}}\left(\{x_i\}_{i\in\iota}; y\right)$, the induced operation is the left Kan extension along $\otimes_\varphi^{\mathcal{C}}$. At $c \in \mathcal{C}_y$ it is

$$\left(\otimes_\varphi^{\mathrm{Day}}\left(\{f_i\}_{i\in\iota}\right)\right)(c) \simeq \operatorname*{colim}_{\otimes_\varphi^{\mathcal{C}}\left(\{c_i\}_{i\in\iota}\right)\to c} \otimes_\varphi^{\mathcal{D}}\left(\left\{f_{i(c_i)}\right\}_{i\in\iota}\right)$$

Remark (Multimorphisms before Kan Extension). Before choosing the universal left Kan extension, an operation from $\{f_i\}_{i\in\iota}$ to g is a natural transformation

$$\otimes_\varphi^{\mathcal{D}} \circ \prod_{i\in\iota} f_i \to g \circ \otimes_\varphi^{\mathcal{C}}$$

Thus the norm first constructs the correct space of lax comparison maps; Day convolution then selects its universal target by left Kan extension.

7 Disintegration and Assembly

Let A be an associative ring. An *involution* on A is a map $\sigma : A \to A$ such that

$$(a+b)^\sigma = a^\sigma + b^\sigma, \quad (ab)^\sigma = b^\sigma a^\sigma, \quad (a^\sigma)^\sigma = a$$

Write **Ring** for the category of associative rings and $\mathbf{Ring}^\sigma$ for the category of rings with involution. The functor

$$A \mapsto A^{\mathrm{op}}$$

defines a Σ_2-action on **Ring**, and an involution is a coherent identification $A \simeq A^{\mathrm{op}}$ whose square is the identity. Hence $\mathbf{Ring}^\sigma$ is the category of homotopy fixed points of this action.

This admits an operadic reformulation. The category **Ring** is the category of algebras over the associative operad $\mathcal{O}$ in **Ab**, while $\mathbf{Ring}^\sigma$ is the category of algebras over an enlarged operad $\mathcal{O}'$. The reversal of the order of the inputs defines a Σ_2-action on $\mathcal{O}$, and, up to the appropriate coherent interpretation,

$$\mathcal{O}' \simeq \mathcal{O} \rtimes \Sigma_2$$

Here $\mathcal{O}$ contains only the associative operations, while $\mathcal{O}'$ also remembers the involution. Disintegration separates these two layers; assembly reconstructs the enlarged operad from the operadic fibers and their symmetry.

We shall extend this idea using unitalization, reduced ∞-operads, and generalized ∞-operads. The main principle is that a unital ∞-operad whose underlying ∞-category is a Kan complex can be disintegrated into a parametrized family of reduced ∞-operads and then assembled back.

> **Remark (Assembly and Semidirect Products).** In the example above, assembly recovers the familiar semidirect-product operad $\mathcal{O} \rtimes \Sigma_2$. The general assembly construction should be viewed as its homotopy-coherent and parametrized analogue.

7.1 Unital ∞-Operads

For objects $\{x_i\}_{1\le i\le n}$ and y of an ∞-operad $\mathcal{O}^\otimes$, the space

$$\mathrm{Mul}_{\mathcal{O}}\big(\{x_i\}_{1\le i\le n}; y\big)$$

classifies n-ary operations with the indicated input and output colors. For $n = 1$ these are the ordinary mapping spaces of the underlying ∞-category. The case $n = 0$ instead records *constants*, so it is here that the presence of units becomes visible.

> **Definition 7.1.1 (Unital ∞-Operad).** An ∞-operad $\mathcal{O}^\otimes$ is *unital* if, for every color x,
>
> $$\mathrm{Mul}_{\mathcal{O}}(\varnothing; x)$$
>
> is contractible. Thus there is an essentially unique nullary operation with output x.

Remark (Two Uses of “Unital”). Classically, “unital operad” may mean either that a distinguished unary identity has been chosen or that there is a unique nullary operation. The first condition is already built into the definition of an ∞-operad; here *unital* always refers to the second.

Both $\mathbf{Comm}^{\otimes}$ and $\mathbb{E}_0^{\otimes}$ are unital. The trivial operad $\mathbf{Triv}^{\otimes}$ is not, since it contains no nullary operation.

Proposition 7.1.1 (Categorical Criterion). An ∞-operad $\mathcal{O}^{\otimes}$ is unital if and only if its total ∞-category $\mathcal{O}^{\otimes}$ is pointed.

Proof sketch. The fiber over $\langle 0\rangle$ contains a canonical final object 0. If x lies over $\langle n\rangle$ and has components $x_1, \dots, x_n$, the operadic Segal condition gives

$$\mathrm{Map}_{\mathcal{O}^{\otimes}}(0, x) \simeq \prod_{1 \leq i \leq n} \mathrm{Mul}_{\mathcal{O}}(\emptyset; x_i)$$

Hence 0 is also initial exactly when all nullary operation spaces are contractible. □

7.1.1 Localization at Nullary Operations

Proposition 7.1.2 (Unital Operads as Local Objects). The operad $\mathbb{E}_0^{\otimes}$ is idempotent for the operadic tensor product:

$$\mathbb{E}_0 \otimes \mathbb{E}_0 \simeq \mathbb{E}_0$$

Consequently

$$U(\mathcal{O}) := \mathcal{O} \otimes \mathbb{E}_0$$

defines an idempotent localization of $\mathbf{Op}_\infty$ whose essential image is the full subcategory $\mathbf{Op}_\infty^{\mathrm{un}}$ of unital ∞-operads. Equivalently, $\mathcal{O}$ is unital if and only if the canonical map

$$\mathcal{O} \to U(\mathcal{O})$$

is an equivalence.

Proof sketch. The tensor product with $\mathbb{E}_0$ freely makes the nullary operation spaces contractible. The equivalence $\mathbb{E}_0 \otimes \mathbb{E}_0 \simeq \mathbb{E}_0$ makes U idempotent, and the universal property of the operadic tensor product then identifies its local objects with the unital operads. □

7.1.2 Unitalization by Pointed Objects

Definition 7.1.2 (Unitalization). A map $u : \mathcal{O}^u \to \mathcal{O}$ *exhibits $\mathcal{O}^u$ as a unitalization* if $\mathcal{O}^u$ is unital and, for every unital ∞-operad $\mathcal{C}$, restriction along u induces an equivalence

$$\mathbf{Alg}_{\mathcal{C}}(\mathcal{O}^u) \simeq \mathbf{Alg}_{\mathcal{C}}(\mathcal{O})$$

Such an object is unique up to equivalence.

Proposition 7.1.3 (Explicit Unitalization). Let $\mathcal{O}_*^{\otimes}$ be the ∞-category of pointed objects of $\mathcal{O}^{\otimes}$. Then the forgetful map

$$p : \mathcal{O}_*^{\otimes} \to \mathcal{O}^{\otimes}$$

is a fibration of ∞-operads, $\mathcal{O}_*^{\otimes}$ is unital, and p exhibits it as a unitalization of $\mathcal{O}^{\otimes}$.

Proof sketch. Pointing is compatible with the operadic product decomposition, so the pointed objects again form an ∞-operad. Its zero object supplies the unique nullary operations. If $\mathcal{E}$ is unital, then $\mathcal{E}^{\otimes}$ is pointed by the categorical criterion; a map from it to $\mathcal{O}^{\otimes}$ therefore lifts to pointed objects through a contractible space of choices. This is exactly the stated universal property. □

Thus unital operads form both a localization and a colocalization of $\mathbf{Op}_{\infty}$: tensoring with $\mathbb{E}_0$ gives the former, while pointed objects give the latter.

7.1.3 Unitality in Monoidal Families

Proposition 7.1.4 (Fiberwise Criterion). Let

$$p : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$$

be a coCartesian fibration of ∞-operads, where $\mathcal{O}$ is unital. Then $\mathcal{C}^{\otimes}$ is unital if and only if, for every color $x \in \mathcal{O}$, the unit object 1_x is initial in the fiber $\mathcal{C}_x$.

Proof sketch. A nullary operation in the total operad is a choice of a nullary operation in the base together with a map from the corresponding fiberwise unit. Since the nullary operation in $\mathcal{O}$ is unique, contractibility reduces exactly to the initiality of each 1_x. □

Corollary 7.1.1 (Compatibility with Algebra Objects). If $q : \mathcal{C}^u \to \mathcal{C}$ is a unitalization over a unital operad $\mathcal{O}$, then the composite $\mathcal{C}^u \to \mathcal{O}$ is again a coCartesian fibration of ∞-operads. Moreover, for every map of unital operads $\mathcal{E} \to \mathcal{O}$, restriction induces an equivalence

$$\mathbf{Alg}_{\mathcal{E}/\mathcal{O}}(\mathcal{C}^u) \simeq \mathbf{Alg}_{\mathcal{E}/\mathcal{O}}(\mathcal{C})$$

Thus unitalization changes the ambient operad without changing algebra objects whose acting operad is already unital.

7.2 Generalized ∞-Operads

For an ordinary ∞-operad, the Segal equivalence gives

$$\mathcal{O}_{\langle n\rangle}^{\otimes} \simeq \mathcal{O}^n$$

and the fiber over $\langle 0\rangle$ is contractible. A generalized ∞-operad keeps the same active–inert calculus but allows this fiber to be an arbitrary ∞-category. Absolute powers are consequently replaced by fiber products over the zero-input fiber.

Definition 7.2.3 (Generalized ∞-Operad). A *generalized ∞-operad* is an ∞-category

$$q : \mathcal{O}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

satisfying the following relative Segal conditions.

1. Every inert morphism in $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ admits a q-coCartesian lift.
2. An inert square in $\mathbf{Fin}_*$ is called *exact* when the corresponding square of nonbasepoint sets is a pushout. Every exact inert square induces a pullback square between the fibers of q.
3. A square of q-coCartesian lifts of an exact inert square is a q-limit diagram.

In particular, if $\mathscr{B} := \mathcal{O}^{\otimes}_{\langle 0\rangle}$ and $\mathscr{A} := \mathcal{O}^{\otimes}_{\langle 1\rangle}$, then

$$\mathcal{O}^{\otimes}_{\langle n\rangle} \simeq \mathscr{A} \times_{\mathscr{B}} \dots \times_{\mathscr{B}} \mathscr{A}$$

where the right-hand side has n factors.

Remark (What Was Relaxed?). For an ordinary ∞-operad, $\mathscr{B}$ is contractible, so the relative product above reduces to $\mathscr{A}^n$. A generalized ∞-operad therefore has the same operations and substitution laws, but its colors may vary over the parameter ∞-category $\mathscr{B}$.

For $n = 2$, the relative Segal equivalence is the pullback square

$$\begin{tikzcd}
\mathcal{O}^{\otimes}_{\langle 2\rangle} \arrow[r, "\rho^1"] \arrow[d, "\rho^2"'] & \mathscr{A} \arrow[d, "\varepsilon"] \\
\mathscr{A} \arrow[r, "\varepsilon"'] & \mathscr{B}
\end{tikzcd}$$

Thus $\mathcal{O}^{\otimes}_{\langle 2\rangle} \simeq \mathscr{A} \times_{\mathscr{B}} \mathscr{A}$. When $\mathscr{B} \simeq *$, this becomes the ordinary product $\mathscr{A} \times \mathscr{A}$.

Definition 7.2.4 (Maps). A map of generalized ∞-operads is a functor over $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ which carries inert morphisms to inert morphisms. It is a *fibration of generalized ∞-operads* when its underlying functor is an isofibration. We denote the resulting ∞-category by $\mathbf{Op}^{\mathrm{gn}}_{\infty}$.

Proposition 7.2.5 (Recovering Ordinary ∞-Operads). A generalized ∞-operad $\mathcal{O}^{\otimes}$ is an ordinary ∞-operad if and only if

$$\mathcal{O}^{\otimes}_{\langle 0\rangle}$$

is a contractible Kan complex.

Proof sketch. This is exactly the missing ordinary Segal condition. Once the zero-input fiber is contractible, every relative fiber product over it becomes the corresponding absolute product. The converse is part of the usual definition of an ∞-operad. □

Remark (Generalized Operadic Model Structure). Marked preoperads admit a model structure whose cofibrations are monomorphisms and whose fibrant objects are generalized ∞-operads. The ordinary operadic model structure is a further localization of it. Hence there is a fully faithful inclusion

$$\mathbf{Op}_\infty \subset \mathbf{Op}_\infty^{\mathrm{gn}}$$

and ordinary operads are precisely the generalized operads satisfying the contractibility condition above.

7.2.1 The Parameter Category

Taking the zero-input fiber defines a forgetful functor

$$F : \mathbf{Op}_\infty^{\mathrm{gn}} \to \mathbf{Cat}_\infty, \quad F(\mathcal{O}) = \mathcal{O}^{\otimes}_{\langle 0 \rangle}$$

Proposition 7.2.6 (The Trivial Generalized Operad). The functor F has a fully faithful right adjoint

$$G : \mathbf{Cat}_\infty \to \mathbf{Op}_\infty^{\mathrm{gn}}, \quad G(\mathcal{C}) = \mathcal{C} \times \mathrm{N}_\bullet(\mathbf{Fin}_*)$$

In particular, every ∞-category occurs as the parameter category of a generalized ∞-operad.

Proof sketch. A map from a generalized operad to $\mathcal{C} \times \mathrm{N}_\bullet(\mathbf{Fin}_*)$ is forced along inert morphisms and is therefore determined by its restriction to the zero-input fiber. Conversely, every functor from that fiber to $\mathcal{C}$ extends by right Kan extension. The space of extensions is contractible. □

Remark (The Adjunction at a Glance). The two functors fit into

$$\mathbf{Cat}_\infty \xrightarrow{G} \mathbf{Op}_\infty^{\mathrm{gn}} \xrightarrow{F} \mathbf{Cat}_\infty$$

with $F \circ G \simeq \mathrm{id}_{\mathbf{Cat}_\infty}$. The middle term remembers all operadic fibers; F keeps only their parameter category, while G inserts the constant family over it.

7.2.2 Families of ∞-Operads

Definition 7.2.5 ($\mathcal{C}$-Family of ∞-Operads). Let $\mathcal{C}$ be an ∞-category. A *$\mathcal{C}$-family of ∞-operads* is an isofibration

$$p : \mathcal{P}^{\otimes} \to \mathcal{C} \times \mathrm{N}_\bullet(\mathbf{Fin}_*)$$

such that:

1. inert morphisms admit p-coCartesian lifts;
2. the component maps over $\rho^i : \langle n \rangle \to \langle 1 \rangle$ exhibit an object as the corresponding p-limit of its n components;
3. for every $c \in \mathcal{C}$, the fiber $\mathcal{P}^{\otimes}_c \to \mathrm{N}_\bullet(\mathbf{Fin}_*)$ is an ordinary ∞-operad.

Remark (Reading a Family Fiberwise). For every $c \in \mathscr{C}$, the ordinary operad over c is obtained from the pullback square

$$\begin{tikzcd} \mathscr{P}^{\otimes}_c \arrow[r] \arrow[d] & \mathscr{P}^{\otimes} \arrow[d, "p"] \\ \{c\} \times \mathrm{N}_\bullet(\mathbf{Fin}_*) \arrow[r] & \mathscr{C} \times \mathrm{N}_\bullet(\mathbf{Fin}_*) \end{tikzcd}$$

A morphism $c \to d$ in $\mathscr{C}$ supplies coherent transport from the operad $\mathscr{P}^{\otimes}_c$ to $\mathscr{P}^{\otimes}_d$. The generalized operad is the total object obtained by assembling all these fibers.

Theorem 7.2.1 (Generalized Operads Are Families). Let

$$p : \mathscr{P}^{\otimes} \to \mathscr{C} \times \mathrm{N}_\bullet(\mathbf{Fin}_*)$$

be an isofibration. The following are equivalent:

1. $\mathscr{P}^{\otimes}$ is a generalized ∞-operad and the induced map $\mathscr{P}^{\otimes}_{\langle 0 \rangle} \to \mathscr{C}$ is a trivial Kan fibration;
2. p exhibits $\mathscr{P}^{\otimes}$ as a $\mathscr{C}$-family of ordinary ∞-operads.

Consequently a generalized ∞-operad is equivalently a pair $\left(\mathscr{C}, \mathscr{P}^{\otimes}\right)$ consisting of an ∞-category $\mathscr{C}$ and a $\mathscr{C}$-family of ∞-operads. Canonically one may take

$$\mathscr{C} = \mathscr{P}^{\otimes}_{\langle 0 \rangle}$$

Proof sketch. The relative Segal conditions are exactly the ordinary Segal conditions in each fiber together with coherent transport along $\mathscr{C}$. The trivial Kan condition identifies $\mathscr{C}$ with the zero-input fiber, and the converse construction glues the transported fibers. □

7.3 Assembly and Approximations to ∞-Operads

The inclusion $i : \mathbf{Op}_\infty \hookrightarrow \mathbf{Op}^{\mathrm{gn}}_\infty$ is reflective: every generalized operad has a universal ordinary replacement.

Definition 7.3.6 (Assembly). A map from a generalized ∞-operad to an ordinary one

$$\gamma : \mathscr{O}^{\otimes} \to \mathscr{P}^{\otimes}$$

assembles $\mathscr{O}^{\otimes}$ into $\mathscr{P}^{\otimes}$ if, for every ordinary ∞-operad $\mathscr{Q}^{\otimes}$, precomposition with γ induces an equivalence

$$\mathbf{Alg}_{\mathscr{P}}(\mathscr{Q}) \simeq \mathbf{Alg}_{\mathscr{O}}(\mathscr{Q})$$

Proposition 7.3.7 (The Assembly Reflection). The inclusion $i : \mathbf{Op}_\infty \hookrightarrow \mathbf{Op}^{\mathrm{gn}}_\infty$ has a left adjoint $\mathrm{Assem} : \mathbf{Op}^{\mathrm{gn}}_\infty \to \mathbf{Op}_\infty$. Its unit $\mathscr{O}^{\otimes} \to \mathrm{Assem}(\mathscr{O})^{\otimes}$ is an assembly, unique up to a contractible space of choices. Equivalently, $\mathbf{Op}_\infty$ is a localization of $\mathbf{Op}^{\mathrm{gn}}_\infty$.

Remark (Assembly as a Colimit). If $\mathcal{O}^{\otimes}$ is presented by a $\mathscr{C}$-family $\{\mathcal{O}_c^{\otimes}\}_{c\in\mathscr{C}}$, then its assembly behaves as the operadic colimit of that family. Schematically,

$$\begin{tikzcd} \mathcal{O}^{\otimes} \arrow[rr, "g"] \arrow[dr, "\eta"'] & & \mathcal{Q}^{\otimes} \\ & \mathrm{Assem}(\mathcal{O})^{\otimes} \arrow[ur, "\tilde{g}"'] & \end{tikzcd}$$

Every map g from $\mathcal{O}^{\otimes}$ to an ordinary operad factors through the unit η, and the factor $\tilde{g}$ is unique up to contractible choice. This is the precise sense in which assembly forgets only the varying parameter category.

7.3.1 Approximation and Weak Approximation

Assembly is an abstract localization. An approximation gives a concrete category of operations from which the same operad can be reconstructed.

Definition 7.3.7 (Approximation). Let $p : \mathcal{O}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ be an ∞-operad and let $f : \mathscr{E} \to \mathcal{O}^{\otimes}$ be an isofibration. We call f an *approximation* if:

1. for every $e \in \mathscr{E}$ over $\langle n\rangle$ and every inert projection $\rho^i : \langle n\rangle \to \langle 1\rangle$, there is a locally f-coCartesian lift $e \to e_i$ whose image is inert;
2. every active morphism $\alpha : x \to f(e)$ admits an f-Cartesian lift $\overline{\alpha} : \overline{x} \to e$.

More generally, f is a *weak approximation* if the first condition holds and, for every morphism $\alpha : x \to f(e)$, the ∞-category of factorizations

$$x \xrightarrow{\gamma} f(e') \xrightarrow{f(\beta)} f(e)$$

with γ inert is weakly contractible. An arbitrary functor is called a (weak) approximation when it becomes one after factoring it as an equivalence followed by an isofibration.

Remark (The Two Lifting Rules). The definition separates the two roles of morphisms in an operad:

$$\begin{tikzcd} \overline{x} \arrow[r, "\overline{\alpha}"] \arrow[d, "f"'] & e \arrow[d, "f"] \\ x \arrow[r, "\alpha\ \text{active}"'] & f(e) \end{tikzcd}$$

Inert arrows decompose an object into its inputs and lift coCartesianly; active arrows encode operations and lift Cartesianly. A weak approximation replaces the chosen Cartesian lift by a contractible space of inert–active factorizations.

Proposition 7.3.8 (Active-Fiber Criterion). Suppose the inert lifting condition above holds. Then f is a weak approximation if and only if, for every $e \in \mathscr{E}$ and every active $\alpha : x \to f(e)$, the ∞-category

$$\mathscr{E}_{/e} \times_{\mathscr{O}^{\otimes}_{/f(e)}} \{\alpha\}$$

of lifts of α is weakly contractible. Hence only active arrows need to be tested.

Proof sketch. Factor an arbitrary arrow into an inert arrow followed by an active arrow. The inert lift is already supplied by hypothesis, so the remaining choice is precisely a point of the displayed active lifting category. The active–inert factorization system and cofinality identify its contractibility with the definition above. □

Proposition 7.3.9 (Formal Properties). Approximations satisfy the following useful rules.

1. Every approximation is a weak approximation.
2. If f is a map of generalized ∞-operads, then f is an approximation exactly when its restrictions to the ordinary operadic parameter fibers are approximations.
3. Given composable maps of ∞-operads $\mathscr{O}^{\otimes} \xrightarrow{f} \mathscr{P}^{\otimes} \xrightarrow{g} \mathscr{Q}^{\otimes}$ whose underlying color categories are Kan complexes, if g is an approximation, then f is an approximation if and only if $g \circ f$ is.
4. For a map of ∞-operads whose underlying categories are Kan complexes, approximation and weak approximation are equivalent; equivalently, the induced map on active subcategories is an equivalence followed by a right fibration.

7.3.2 The Approximation Theorem

To formulate the central conclusion, let $f : \mathscr{E} \to \mathscr{O}^{\otimes}$ be a weak approximation. An $\mathscr{E}$-algebra in an ∞-operad $\mathscr{P}^{\otimes}$ is a functor $a : \mathscr{E} \to \mathscr{P}^{\otimes}$ over $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ which sends the chosen inert lifts in $\mathscr{E}$ to inert morphisms. It is *locally constant* if it sends every arrow over $\mathrm{id}_{\langle 1\rangle}$ to an equivalence. Write $\mathbf{Alg}_{\mathscr{E}}(\mathscr{P})$ and $\mathbf{Alg}^{\mathrm{loc}}_{\mathscr{E}}(\mathscr{P})$ for the resulting ∞-categories.

Theorem 7.3.2 (Approximation Theorem). Let $f : \mathscr{E} \to \mathscr{O}^{\otimes}$ be a weak approximation and let $\mathscr{P}^{\otimes}$ be any ∞-operad. Precomposition defines

$$\theta : \mathbf{Alg}_{\mathscr{O}}(\mathscr{P}) \to \mathbf{Alg}_{\mathscr{E}}(\mathscr{P})$$

1. If f induces an equivalence $\mathscr{E}_{\langle 1\rangle} \to \mathscr{O}$ on colors, then θ is an equivalence.
2. If $\mathscr{O}$ is a Kan complex and the map on colors is only a weak homotopy equivalence, then θ induces an equivalence

$$\mathbf{Alg}_{\mathscr{O}}(\mathscr{P}) \simeq \mathbf{Alg}^{\mathrm{loc}}_{\mathscr{E}}(\mathscr{P})$$

Proof sketch. Replace f by an equivalent isofibration and extend an $\mathscr{E}$-algebra by right Kan extension. The inert lifting condition makes the extension preserve input decompositions, while the active-fiber criterion says that every indexing category used to define an operation is contractible. The extension therefore exists and is unique up to contractible choice. If the color map is only a weak equivalence, precisely the locally constant algebras descend. □

Remark (What the Theorem Says). An approximation may contain many more objects and morphisms than $\mathcal{O}^{\otimes}$, but it contains exactly one coherent way to represent each operation. Thus it presents the same algebra theory:

$$\begin{tikzcd} \mathscr{E} \arrow[rr, "a \circ f"] \arrow[dr, "f"'] & & \mathscr{P}^{\otimes} \\ & \mathcal{O}^{\otimes} \arrow[ur, "a"'] \arrow[d, dash, dashed] & \\ & \mathbf{Alg}_{\mathscr{E}}(\mathscr{P}) \simeq \mathbf{Alg}_{\mathcal{O}}(\mathscr{P}) & \end{tikzcd}$$

The upper triangle is the concrete presentation; the lower node records that restriction along f loses no algebraic structure.

Corollary 7.3.2 (Detecting Equivalences of Operads). Let $f : \mathscr{E}^{\otimes} \to \mathcal{O}^{\otimes}$ be a weak approximation of ordinary ∞-operads. If f induces an equivalence on their underlying color ∞-categories, then f is an equivalence of ∞-operads.

Remark (Model-Categorical Form). Under the same color-equivalence hypothesis, an approximation induces a left Quillen equivalence between the corresponding slice model categories of ∞-preoperads. The technical fibrization construction in the full proof replaces an arbitrary approximation by a fibration with the same homotopy type; the contractibility criteria above guarantee that this replacement does not change the represented operad.

7.4 Disintegration of ∞-Operads

Unitality trivializes nullary operations. Reducedness also trivializes colors and unary operations, but not the higher arities.

Definition 7.4.8 (Reduced ∞-Operad). An ordinary ∞-operad $\mathcal{O}^{\otimes}$ is *reduced* if it is unital and its color category $\mathcal{O}$ is a contractible Kan complex.

Remark (What "Reduced" Removes). For a reduced operad, both the nullary and unary operation spaces are contractible:

$$\mathrm{Mul}_{\mathcal{O}}(\varnothing; x) \simeq *, \quad \mathrm{Mul}_{\mathcal{O}}(\{x\}; y) = \mathrm{Map}_{\mathcal{O}}(x, y) \simeq *$$

Higher-arity operation spaces may still be nontrivial.

Definition 7.4.9 (Reduced Generalized ∞-Operad). Regard a generalized operad $\mathscr{P}^{\otimes}$ as a family of ordinary operads over

$$\mathscr{B} = \mathscr{P}^{\otimes}_{\langle 0 \rangle}$$

It is *reduced* if $\mathscr{B}$ is a Kan complex and every fiber $\mathscr{P}^{\otimes}_b$ is a reduced ordinary ∞-operad. We write $\mathbf{Op}_{\infty}^{\mathrm{gn,rd}}$ for the full subcategory of reduced generalized ∞-operads.

If $\mathscr{B}$ is a Kan complex, the fiber over b may equivalently be described by the slice operad $\mathscr{P}^{\otimes}_{/b}$. Hence reducedness is invariant under replacing the family by an equivalent generalized operad.

Theorem 7.4.3 (Disintegration–Assembly Equivalence). Let $\mathbf{Op}_{\infty}^{\mathrm{u,gpd}} \subset \mathbf{Op}_{\infty}$ denote the full subcategory spanned by the unital ∞-operads whose underlying color categories are Kan complexes. Assembly restricts to an equivalence

$$\mathrm{Assem} : \mathbf{Op}_{\infty}^{\mathrm{gn,rd}} \simeq \mathbf{Op}_{\infty}^{\mathrm{u,gpd}}$$

Consequently every unital ∞-operad with groupoidal color category is, in an essentially unique way, the assembly of a family of reduced ∞-operads.

Remark (Where the Equivalence Lives). The theorem identifies the indicated full subcategories inside the assembly reflection:

$$\begin{tikzcd}
\mathbf{Op}_{\infty}^{\mathrm{gn,rd}} \arrow[r, "\mathrm{Assem}"] \arrow[d, "i"'] & \mathbf{Op}_{\infty}^{\mathrm{u,gpd}} \arrow[d, "i"] \\
\mathbf{Op}_{\infty}^{\mathrm{gn}} \arrow[r, "\mathrm{Assem}"'] & \mathbf{Op}_{\infty}
\end{tikzcd}$$

The upper arrow is an equivalence. The lower arrow is only a localization: reducedness is exactly the extra hypothesis that makes assembly reversible.

Proof sketch. For a unital operad $\mathscr{O}^{\otimes}$ with $\mathscr{O}$ a Kan complex, disintegrate it over its color groupoid. The fiber at x is the slice operad $\mathscr{O}^{\otimes}_{/x}$. Its color category is contractible and unitality supplies the unique nullary operations, so the fiber is reduced. Assembly glues these slices back to $\mathscr{O}^{\otimes}$.

Conversely, a reduced family has contractible unary data in each fiber. The approximation theorem therefore shows that maps out of its assembly are detected fiberwise. This proves full faithfulness, while the preceding slice construction proves essential surjectivity. □

7.4.1 Recognizing an Assembly

Proposition 7.4.10 (Assembly Criterion). Let

$$f : \mathscr{P}^{\otimes} \to \mathscr{O}^{\otimes}$$

be a map of generalized ∞-operads, and suppose the parameter and color categories involved are Kan complexes.

1. If f is a weak approximation and induces a weak homotopy equivalence on colors, then f exhibits $\mathscr{O}^{\otimes}$ as the assembly of $\mathscr{P}^{\otimes}$.
2. Conversely, suppose every fiber of $\mathscr{P}^{\otimes}$ is unital. If f is an assembly, then f is an approximation and induces a weak homotopy equivalence on colors.

Proof sketch. The forward implication is the approximation theorem: restriction along f induces equivalences on all algebra categories, which is precisely the universal property of assembly. For the converse, apply the universal property to free algebras. Fiberwise unitality makes the relevant restriction maps equivalences; the active-fiber criterion then recovers both the approximation property and the equivalence on colors. □

Remark (The Homotopy-Theoretic Input). A weak homotopy equivalence $u : x \to y$ with y a Kan complex is left cofinal. Consequently, if a diagram sends every edge of y to an equivalence, replacing its indexing space along u does not change its colimit. This elementary observation is what allows the proof to pass freely between an operad, an equivalent color space, and the corresponding assembled family.

7.4.2 The Free-Algebra Test

Suppose $\mathscr{C}^{\otimes}$ is a presentably symmetric monoidal ∞-category whose tensor product preserves small colimits separately in each variable. For a map $f : \mathscr{P}^{\otimes} \to \mathscr{O}^{\otimes}$, restriction gives a square

$$\begin{tikzcd}
\mathbf{Alg}_{\mathscr{O}}(\mathscr{C}) \arrow[r, "\theta"] \arrow[d, "U_{\mathscr{O}}"'] & \mathbf{Alg}_{\mathscr{P}}(\mathscr{C}) \arrow[d, "U_{\mathscr{P}}"] \\
\mathbf{Fun}(\mathscr{O}, \mathscr{C}) \arrow[r, "\theta'"'] & \mathbf{Fun}(\mathscr{P}, \mathscr{C})
\end{tikzcd}$$

Let $F_{\mathscr{O}}$ and $F_{\mathscr{P}}$ denote the respective free-algebra functors. The square determines a comparison transformation

$$F_{\mathscr{P}} \circ \theta' \to \theta \circ F_{\mathscr{O}}$$

Proposition 7.4.11 (Free-Algebra Criterion). If f is a weak approximation, the comparison transformation above is an equivalence. Conversely, it is enough to test this statement in spaces on the constant terminal diagram: if that comparison is an equivalence, then f is an approximation.

8 (co)Cartesian Symmetric Monoidal Categories

8.1 Cartesian Monoidal Categories

Let $\mathscr{C}$ be an ∞-category with finite products. The ordinary product already suggests a symmetric monoidal operation: its unit is the final object, and its structural maps are the projections. The point of this section is that these elementary facts determine all higher coherence automatically.

8.1.1 Intrinsic Characterization

Definition 8.1.1 (Cartesian Symmetric Monoidal Structure). Let $\mathscr{C}^{\otimes} \to \mathrm{N}_\bullet(\mathbf{Fin}_*)$ be a symmetric monoidal ∞-category, with underlying ∞-category $\mathscr{C}$. Its monoidal structure is *Cartesian* if

1. the unit $1_{\mathscr{C}}$ is final in $\mathscr{C}$
2. for every $c, d \in \mathscr{C}$, the canonical projections

$$c \otimes d \to c, \quad c \otimes d \to d$$

   exhibit $c \otimes d$ as $c \times d$

Remark (The Defining Picture). The tensor product is not merely equivalent to a product as an object: the equivalence is detected by the two inert projections.

$$\begin{tikzcd} & c \otimes d \arrow[dl, "\mathrm{pr}_1"'] \arrow[dr, "\mathrm{pr}_2"] & \\ c & & d \end{tikzcd}$$

Thus maps into the tensor product split as

$$\mathrm{Map}_{\mathscr{C}}(x, c \otimes d) \simeq \mathrm{Map}_{\mathscr{C}}(x, c) \times \mathrm{Map}_{\mathscr{C}}(x, d)$$

Definition 8.1.2 (Lax Cartesian Structure). Let $p : \mathscr{O}^{\otimes} \to \mathrm{N}_\bullet(\mathbf{Fin}_*)$ be an ∞-operad and let $\mathscr{D}$ admit finite products. A functor $\pi : \mathscr{O}^{\otimes} \to \mathscr{D}$ is a *lax Cartesian structure* if, whenever $x \in \mathscr{O}^{\otimes}_{\langle n \rangle}$ has inert components x_i, the canonical maps exhibit

$$\pi(x) \simeq \prod_{i=1}^{n} \pi(x_i)$$

If $\mathscr{O}^{\otimes}$ is symmetric monoidal, the structure is *weak Cartesian* when π also carries every coCartesian arrow over an active map to an equivalence. A weak Cartesian structure is *Cartesian* when its restriction $\mathscr{O} \to \mathscr{D}$ is an equivalence.

The first condition says that a list of inputs is evaluated componentwise. The second says that an active tensor operation introduces no information beyond the product already present in $\mathscr{D}$.

8.1.2 The Canonical Construction

The bookkeeping category behind the construction is $\Gamma^{\times}$. Its objects are pairs $(\langle n\rangle, S)$ with $S \subseteq \langle n\rangle^{\circ}$, and a morphism

$$(\langle n\rangle, S) \xrightarrow{\alpha} (\langle m\rangle, T)$$

is a pointed map $\alpha : \langle n\rangle \to \langle m\rangle$ such that $\alpha^{-1}(T) \subseteq S$. The subset S records which inputs remain visible. In particular, singleton subsets recover the individual factors.

Remark (Why Subsets Appear). Over $\langle 3\rangle$, the relevant part of a product diagram has the form

$$\begin{tikzcd}
& c_1 \times c_2 \times c_3 \arrow[dl] \arrow[d] \arrow[dr] & \\
c_1 \times c_2 \arrow[d] \arrow[dr] & c_1 \times c_3 \arrow[dl] \arrow[dr] & c_2 \times c_3 \arrow[dl] \arrow[d] \\
c_1 & c_2 & c_3
\end{tikzcd}$$

The arrows forget entries. The entire diagram is forced by the three singleton values: each value at S is their product over $i \in S$.

For an ∞-category $\mathscr{C}$, one first forms an ∞-category over $\mathrm{N}_\bullet(\mathbf{Fin}_*)$ whose fiber over $\langle n\rangle$ consists of diagrams

$$F : \mathscr{P}\big(\langle n\rangle^{\circ}\big)^{\mathrm{op}} \to \mathscr{C}$$

Here $\mathscr{P}$ denotes the poset of subsets. Let $\mathscr{C}^{\times}$ be the full subcategory on the diagrams satisfying

$$F(S) \simeq \prod_{i \in S} F(\{i\})$$

Equivalently, F is the right Kan extension of its values on the singleton subsets. This is precisely the condition that turns the formal list $(c_1, \ldots, c_n)$ into its coherent system of partial products.

Proposition 8.1.1 (Structure of $\mathscr{C}^{\times}$). The projection $\mathscr{C}^{\times} \to \mathrm{N}_\bullet(\mathbf{Fin}_*)$ has the following properties.

1. Evaluation on singleton subsets induces an equivalence

$$\mathscr{C}^{\times}_{\langle n\rangle} \simeq \mathscr{C}^{n}$$

2. The value at the empty subset is the terminal object, while the value at S is the product of the entries indexed by S.
3. A morphism of subset diagrams over $\alpha : \langle n\rangle \to \langle m\rangle$ is coCartesian exactly when, for every $T \subseteq \langle m\rangle^{\circ}$, its comparison map

$$F\big(\alpha^{-1}(T)\big) \to F'(T)$$

is an equivalence.

4. If $\mathscr{C}$ admits finite products, the active fold $\langle n\rangle \to \langle 1\rangle$ induces the ordinary n-fold product functor $\mathscr{C}^n \to \mathscr{C}$.

Remark (Nullary and Binary Cases). For $n = 0$, the construction selects the empty product $1_{\mathscr{C}}$. For $n = 2$, the inert projections select c and d, while the active fold selects $c \times d$. Thus the unit, projections, and multiplication of the monoidal structure all come from the same subset diagram.

Theorem 8.1.1 (Existence and Uniqueness of the Cartesian Structure). An ∞-category $\mathscr{C}$ admits a Cartesian symmetric monoidal structure if and only if it admits finite products. In that case

$$\mathscr{C}^{\times} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$$

is a Cartesian symmetric monoidal ∞-category, and every other Cartesian symmetric monoidal structure on $\mathscr{C}$ is symmetrically monoidally equivalent to it through an equivalence homotopic to the identity on $\mathscr{C}$.

Proof sketch. Finite products extend every tuple of objects uniquely, up to contractible choice, to the subset diagram above. Reindexing subsets supplies the coCartesian transport over $\mathbf{Fin}_*$, while the singleton decomposition gives the operadic Segal equivalences. Conversely, the unit and the two inert projections in any Cartesian structure recover the final object and binary products. Since every higher product is their iterated universal product, no additional coherent choices remain. □

Example (Spaces and Additive Categories). The ∞-category $\mathbf{Ani}$ carries its Cartesian monoidal structure by ordinary products of spaces. A commutative algebra object in $\mathbf{Ani}^{\times}$ is therefore an E_∞-space and is genuine additional structure.

In an additive ∞-category, finite products and finite coproducts agree as biproducts. Its Cartesian and coCartesian monoidal structures therefore have the same underlying tensor object, although their universal descriptions use projections and inclusions respectively.

8.1.3 Universal Property

Write $\mathbf{Fun}^{\mathrm{fp}}(\mathscr{C}, \mathscr{D})$ for the ∞-category of finite-product- preserving functors.

Theorem 8.1.2 (Functorial Characterization). If $\mathscr{C}$ and $\mathscr{D}$ admit finite products, restriction to the underlying ∞-categories induces an equivalence

$$\mathbf{Fun}^{\otimes}(\mathscr{C}^{\times}, \mathscr{D}^{\times}) \simeq \mathbf{Fun}^{\mathrm{fp}}(\mathscr{C}, \mathscr{D})$$

Consequently, the ∞-category of Cartesian symmetric monoidal ∞-categories and symmetric monoidal functors is equivalent to the ∞-category of ∞-categories with finite products and functors which preserve them.

Proof sketch. A symmetric monoidal functor preserves the tensor unit and the tensors encoded by active fold maps; in the Cartesian models these are precisely the final object and finite products. Conversely, a product-preserving functor can be applied pointwise to every subset diagram. It preserves the right Kan extension condition and hence defines a symmetric monoidal functor $\mathscr{C}^\times \to \mathscr{D}^\times$. The space of such extensions is contractible, which gives full faithfulness as well as essential surjectivity. □

Remark (Coherence Is a Consequence of Products). A product-preserving functor automatically preserves the unit, every n-fold tensor product, permutations, and all associativity coherences. Thus "symmetric monoidal" is not extra structure in the Cartesian setting; it is a property of the underlying functor.

Proposition 8.1.2 (Algebras as Cartesian Structures). Let $\mathscr{O}^\otimes$ be an ∞-operad and let $\mathscr{D}$ admit finite products. Composition with the Cartesian structure $\pi : \mathscr{D}^\times \to \mathscr{D}$ induces a trivial Kan fibration

$$\mathbf{Alg}_{\mathscr{O}}(\mathscr{D}) \to \mathbf{Fun}^{\text{lax-Cart}}(\mathscr{O}^\otimes, \mathscr{D})$$

Hence a lax Cartesian structure lifts, through a contractible space of choices, to an $\mathscr{O}$-algebra in $\mathscr{D}^\times$. If $\mathscr{O}^\otimes$ is symmetric monoidal, symmetric monoidal functors into $\mathscr{D}^\times$ correspond in the same way to weak Cartesian structures.

Proof sketch. A map of ∞-operads into $\mathscr{D}^\times$ is determined by its singleton components. After composing with π, the operadic Segal condition becomes exactly the assertion that the value of a tuple is the product of those components. Thus lax Cartesian structures are precisely the diagrams which admit the required lift. Since products are universal, the space of lifts and of all their coherences is contractible. □

Remark (What Is Being Lifted). The product diagram in $\mathscr{D}$ already contains all target-side coherence. An algebra structure is therefore the unique operadic lift of the lower functor in the triangle

$$\begin{tikzcd}
\mathscr{O}^\otimes \arrow[rr, "a^\otimes"] \arrow[dr, "\pi \circ a^\otimes"'] & & \mathscr{D}^\times \arrow[dl, "\pi"] \\
& \mathscr{D} &
\end{tikzcd}$$

8.2 Monoid Objects

For an ordinary category with finite products, a commutative monoid is an object equipped with a unit and a multiplication. In an ∞-category the same description hides infinitely many coherences. A more robust definition packages all powers of the object into a single product-preserving functor.

Definition 8.2.3 ($\mathscr{O}$-Monoid). Let $\mathscr{C}$ be an ∞-category with finite products and let $p : \mathscr{O}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ be an ∞-operad. An *$\mathscr{O}$-monoid* in $\mathscr{C}$ is a functor

$$m : \mathscr{O}^{\otimes} \to \mathscr{C}$$

such that, for every $x \in \mathscr{O}^{\otimes}_{\langle n \rangle}$ with inert components $x_1, \ldots, x_n$, the canonical map

$$m(x) \to \prod_{i=1}^{n} m(x_i)$$

is an equivalence. Write $\mathbf{Mon}_{\mathscr{O}}(\mathscr{C})$ for the full subcategory of $\mathbf{Fun}(\mathscr{O}^{\otimes}, \mathscr{C})$ spanned by these functors. When $\mathscr{O}^{\otimes} = \mathbf{Comm}^{\otimes}$, write

$$\mathbf{CMon}(\mathscr{C}) := \mathbf{Mon}_{\mathbf{Comm}}(\mathscr{C})$$

Remark (Recovering the Classical Operations). Let $m \in \mathbf{CMon}(\mathscr{C})$ and put $a = m(\langle 1 \rangle)$. The Segal equivalences give

$$m(\langle n \rangle) \simeq a^n$$

The unique active fold $\mu_2 : \langle 2 \rangle \to \langle 1 \rangle$ and the nullary active map therefore give

$$a \times a \simeq m(\langle 2 \rangle) \to a, \quad 1_{\mathscr{C}} \simeq m(\langle 0 \rangle) \to a$$

Functoriality of m packages commutativity, associativity, unitality, and every higher homotopy between them. Thus a commutative monoid object is the coherent version of the familiar data $(a, 1 \to a, a \times a \to a)$.

Proposition 8.2.3 (Monoids versus Algebras). Let $\mathscr{C}^{\otimes}$ be a symmetric monoidal ∞-category and let $\pi : \mathscr{C}^{\otimes} \to \mathscr{D}$ be a Cartesian structure. For every ∞-operad $\mathscr{O}^{\otimes}$, composition with π induces an equivalence

$$\mathbf{Alg}_{\mathscr{O}}(\mathscr{C}) \simeq \mathbf{Mon}_{\mathscr{O}}(\mathscr{D})$$

In particular, for the canonical Cartesian structure on an ∞-category with finite products,

$$\mathbf{Alg}_{\mathscr{O}}(\mathscr{C}) \simeq \mathbf{Mon}_{\mathscr{O}}(\mathscr{C})$$

Proof sketch. This is the preceding lifting theorem with the terminology changed. The Segal condition for an algebra map into $\mathscr{C}^{\times}$ says exactly that its composite with π carries an n-tuple to the product of its singleton values. Conversely, those product diagrams have a unique coherent lift to $\mathscr{C}^{\times}$. □

Example (Monoidal ∞-Categories as Monoid Objects). Taking $\mathscr{C} = \mathbf{Cat}_{\infty}$, a commutative monoid object is the same thing as a symmetric monoidal ∞-category. More generally, $\mathbf{Mon}_{\mathscr{O}}(\mathbf{Cat}_{\infty})$ is the ∞-category of $\mathscr{O}$-monoidal ∞-categories and $\mathscr{O}$-monoidal functors.

Proposition 8.2.4 (Approximation Invariance). Let $f : \mathscr{E}^{\otimes} \to \mathscr{O}^{\otimes}$ be a weak approximation which is an equivalence on colors. If $\mathscr{C}$ has finite products, restriction induces

$$\mathbf{Mon}_{\mathcal{O}}(\mathscr{C}) \simeq \mathbf{Mon}_{\mathscr{E}}(\mathscr{C})$$

Proof sketch. Extend an $\mathscr{E}$-monoid by right Kan extension. Approximation makes every indexing category of active lifts contractible, while the product condition supplies the inert extensions. Hence the extension exists and is unique up to contractible choice. □

Remark (Passing to Opposites). Although a symmetric monoidal structure is not manifestly self-dual, the involution $\mathscr{C} \mapsto \mathscr{C}^{\mathrm{op}}$ on $\mathbf{Cat}_\infty$ carries commutative monoid objects to commutative monoid objects. Consequently a symmetric monoidal structure on $\mathscr{C}$ canonically determines one on $\mathscr{C}^{\mathrm{op}}$, up to a contractible space of choices. This is the formal bridge from Cartesian to coCartesian structures.

8.3 coCartesian Monoidal Categories

The dual theory starts from finite coproducts.

Definition 8.3.4 (coCartesian Symmetric Monoidal Structure). A symmetric monoidal structure on $\mathscr{C}$ is *coCartesian* if its unit $0_{\mathscr{C}}$ is initial and the canonical inclusions

$$c \to c \otimes d \leftarrow d$$

exhibit $c \otimes d$ as the coproduct $c \sqcup d$.

Remark (Product versus Coproduct). The two universal cones point in opposite directions.

$$\begin{tikzcd} & c \otimes d \arrow[dl, "\mathrm{pr}_1"'] \arrow[dr, "\mathrm{pr}_2"] & \\ c & & d \end{tikzcd} \qquad \begin{tikzcd} c \arrow[dr, "i_1"'] & & d \arrow[dl, "i_2"] \\ & c \otimes d & \end{tikzcd}$$

The left cone is Cartesian; the right cone is coCartesian. Formally the latter is obtained by applying the Cartesian construction to $\mathscr{C}^{\mathrm{op}}$ and then taking opposites.

8.3.1 An Explicit coCartesian Operad

The duality argument proves existence, but a direct model makes the multimorphisms transparent.

Definition 8.3.5 (The Category $\Gamma^{\sqcup}$). An object of $\Gamma^{\sqcup}$ is a pair $(\langle n \rangle, i)$ with $i \in \langle n \rangle^{\circ}$. A morphism

$$(\langle m \rangle, i) \xrightarrow{\alpha} (\langle n \rangle, j)$$

is a pointed map $\alpha : \langle m \rangle \to \langle n \rangle$ satisfying $\alpha(i) = j$.

For any ∞-category $\mathscr{C}$, define $\mathscr{C}^{\sqcup}$ over $\mathrm{N}_\bullet(\mathbf{Fin}_*)$ as follows. An object over $\langle n\rangle$ is a tuple $(c_1, \dots, c_n)$. A morphism from $(c_1, \dots, c_m)$ to $(d_1, \dots, d_n)$ over $\alpha : \langle m\rangle \to \langle n\rangle$ is a collection

$$\left\{f_i : c_i \to d_{\alpha(i)}\right\}_{\alpha(i)\neq *}$$

No map is required from an input which α sends to the base point.

Remark (The Wiring Encoded by a Morphism). If $\alpha(1) = \alpha(2) = 1$ and $\alpha(3) = 2$, a morphism over α is exactly the displayed collection of ordinary arrows.

$$\begin{tikzcd}
c_1 \arrow[d, "f_1"'] & c_2 \arrow[dl, "f_2"] & c_3 \arrow[d, "f_3"] \\
d_1 & & d_2
\end{tikzcd}$$

The coCartesian lift replaces all inputs in each fiber $\alpha^{-1}(\{j\})$ by their coproduct.

Proposition 8.3.5 (Structure of $\mathscr{C}^{\sqcup}$). The projection $p : \mathscr{C}^{\sqcup} \to \mathrm{N}_\bullet(\mathbf{Fin}_*)$ has the following properties.

1. It is an ∞-operad for every ∞-category $\mathscr{C}$.
2. Its fiber over $\langle n\rangle$ is canonically equivalent to $\mathscr{C}^n$.
3. A morphism over $\alpha : \langle m\rangle \to \langle n\rangle$ is p-coCartesian exactly when, for every $1 \le j \le n$, its component maps
$$\{c_i \to d_j\}_{i\in\alpha^{-1}(\{j\})}$$
exhibit d_j as $\sqcup_{i\in\alpha^{-1}(\{j\})} c_i$.
4. The projection p is a coCartesian fibration, hence a symmetric monoidal ∞-category, if and only if $\mathscr{C}$ admits finite coproducts.

Proof sketch. The fiber decomposition is immediate from the description by tuples. The mapping-space criterion for coCartesian arrows reduces independently at every output j to the universal property of a coproduct. Such lifts exist for all pointed maps precisely when all finite coproducts exist; the empty fiber produces the initial object. □

Remark (Multimorphisms). The active fold gives

$$\mathrm{Mul}_{\mathscr{C}^{\sqcup}}\big(\{c_i\}_{1\le i\le n}, d\big) \simeq \prod_{i=1}^{n} \mathrm{Map}_{\mathscr{C}}(c_i, d)$$

If $\mathscr{C}$ has finite coproducts, this is

$$\mathrm{Map}_{\mathscr{C}}(\sqcup_{i=1}^{n} c_i, d)$$

Thus the higher operadic composition law contains no new operation: it is composition of ordinary arrows after applying the coproduct universal property.

Definition 8.3.6 (coCartesian ∞-Operad). An ∞-operad is *coCartesian* if it is equivalent, as an ∞-operad, to $\mathscr{C}^{\sqcup}$ for some ∞-category $\mathscr{C}$. A symmetric monoidal ∞-category is coCartesian in this sense if and only if it is coCartesian according to the intrinsic definition above.

8.3.2 Universal Properties

Corollary 8.3.1 (The coCartesian Dual). If $\mathscr{C}$ admits finite coproducts, it carries an essentially unique coCartesian symmetric monoidal structure $\mathscr{C}^{\sqcup}$. Symmetric monoidal functors

$$\mathscr{C}^{\sqcup} \to \mathscr{D}^{\sqcup}$$

are exactly the functors $\mathscr{C} \to \mathscr{D}$ which preserve finite coproducts.

Example (Disjoint Union). The category of sets, or the ∞-category of spaces, is coCartesian symmetric monoidal under disjoint union. For a space x, the fold map

$$x \sqcup x \to x$$

is the multiplication of its canonical commutative algebra structure. This should not be confused with a commutative monoid structure on x under the Cartesian product.

Remark (Canonical Algebra and Coalgebra Structures). In a coCartesian monoidal ∞-category every object has a canonical, essentially unique commutative algebra structure: its multiplication and unit are the fold maps

$$c \sqcup c \to c, \quad 0_{\mathscr{C}} \to c$$

Dually, in a Cartesian monoidal ∞-category every object has a canonical cocommutative coalgebra structure given by the diagonal and terminal map

$$c \to c \times c, \quad c \to 1_{\mathscr{C}}$$

This is why algebra theory in a coCartesian monoidal category is formally trivial, whereas commutative monoid objects in a Cartesian category can carry genuine extra structure.

Corollary 8.3.2 (Triviality of coCartesian Algebra Theory). Evaluation on the underlying object is an equivalence

$$\mathbf{CAlg}(\mathscr{C}) \simeq \mathscr{C}$$

Its inverse equips c with all iterated fold maps $\sqcup_{i=1}^{n} c \to c$. The commutative algebra axioms follow from the universal property of coproducts, so the entire coherent structure is unique up to contractible choice.

Theorem 8.3.3 (Algebras in a coCartesian Target). Let $\mathscr{P}^{\otimes}$ be a unital ∞-operad and let $\mathscr{C}^{\sqcup}$ be coCartesian. Restriction to colors induces an equivalence

$$\mathbf{Alg}_{\mathscr{P}}(\mathscr{C}) \simeq \mathbf{Fun}(\mathscr{P}, \mathscr{C})$$

More generally, maps between coCartesian ∞-operads are determined by their underlying functors. In particular, the forgetful functor from coCartesian symmetric monoidal ∞-categories to ∞-categories with finite coproducts is an equivalence onto the subcategory of finite-coproduct-preserving functors.

Proof sketch. Unitality determines all nullary operations. Every higher operation in the target is a coproduct followed by an ordinary morphism, so a functor on colors forces the images of all operations. The coproduct universal property makes the space of coherent extensions contractible. □

Theorem 8.3.4 (The Universal Family Property). Let $\mathscr{B}$ be an ∞-category and let $\mathscr{O}^{\otimes}$ and $\mathscr{D}^{\otimes}$ be ∞-operads. Put

$$\mathscr{P}^{\otimes} := \mathscr{B}^{\sqcup} \times_{\mathrm{N}_{\bullet}(\mathbf{Fin}_*)} \mathscr{O}^{\otimes}$$

Then restriction induces an equivalence

$$\mathbf{Alg}_{\mathscr{P}}(\mathscr{D}) \simeq \mathbf{Fun}(\mathscr{B}, \mathbf{Alg}_{\mathscr{O}}(\mathscr{D}))$$

Taking $\mathscr{O}^{\otimes} = \mathbf{Comm}^{\otimes}$ gives

$$\mathbf{Alg}_{\mathscr{B}^{\sqcup}}(\mathscr{D}) \simeq \mathbf{Fun}(\mathscr{B}, \mathbf{CAlg}(\mathscr{D}))$$

Remark (Meaning of the Family Property). A map of operads out of $\mathscr{B}^{\sqcup}$ is a coherently $\mathscr{B}$-indexed family of commutative algebra objects. Notice that maps of operads preserve inert morphisms but need not preserve every coCartesian morphism; the result concerns lax monoidal data, not only strong symmetric monoidal functors.

8.3.3 Recognition by Fold Maps

Proposition 8.3.6 (Recognition Criterion). Let $\mathscr{C}^{\otimes}$ be a symmetric monoidal ∞-category. The following conditions are equivalent.

1. The symmetric monoidal structure on $\mathscr{C}$ is coCartesian.
2. The induced symmetric monoidal structure on $h\mathscr{C}$ is coCartesian.
3. The unit $1_{\mathscr{C}}$ is initial and there is a natural family of fold maps $\delta_c : c \otimes c \to c$ satisfying

$$\delta_c \circ (\mathrm{id}_c \otimes u_c) \simeq \mathrm{id}_c$$

$$f \circ \delta_c \simeq \delta_d \circ (f \otimes f)$$

and

$$\delta_{c \otimes d} \circ \beta_{c,d} \simeq \delta_c \otimes \delta_d$$

Here $u_c : 1_{\mathscr{C}} \to c$ is the unique map and $\beta_{c,d} : (c \otimes c) \otimes (d \otimes d) \simeq (c \otimes d) \otimes (c \otimes d)$ is the canonical symmetry.

Remark (Naturality of the Fold). The middle identity is the homotopy commutativity of

$$\begin{tikzcd} c \otimes c \arrow[r, "f \otimes f"] \arrow[d, "\delta_c"'] & d \otimes d \arrow[d, "\delta_d"] \\ c \arrow[r, "f"'] & d \end{tikzcd}$$

Proof sketch. Only the last implication needs proof. The canonical maps $c \to c \otimes d \leftarrow d$ are induced by the initial unit. For every a, define

$$\mathrm{Map}_{\mathscr{C}}(c, a) \times \mathrm{Map}_{\mathscr{C}}(d, a) \to \mathrm{Map}_{\mathscr{C}}(c \otimes d, a)$$

by sending (f, g) to

$$c \otimes d \overset{f \otimes g}{\to} a \otimes a \overset{\delta_a}{\to} a$$

Unitality, naturality, and multiplicativity of δ show that this map is inverse to restriction along the two inclusions. Hence $c \otimes d$ is a coproduct. The same argument at the unit proves that the entire symmetric monoidal structure is coCartesian. □

8.4 Wreath Products

The tensor product of two ∞-operads is characterized by a universal property, but the marked preoperad used to construct it must usually be replaced by a fibrant object. The *wreath product* is a more explicit intermediate model: it records one operad acting on blocks and a second operad acting inside each block. It is generally not itself an ∞-operad, but after marking its inert arrows it represents the same derived object as the operadic tensor product.

8.4.1 Flattening Lists of Arities

For an ∞-category $\mathscr{J}$, let $\mathscr{J}^{\sqcup}$ denote the simplicial set whose objects are finite lists

$$(j_1, \ldots, j_m)$$

and whose morphisms from $(j_1, \ldots, j_m)$ to $(j'_1, \ldots, j'_n)$ consist of a pointed map $\alpha : \langle m \rangle \to \langle n \rangle$ together with maps

$$j_i \to j'_{\alpha(i)} \quad (\alpha(i) \neq *)$$

When $\mathscr{J} = \mathrm{N}_\bullet(\mathscr{I})$, this is the nerve of the ordinary category with exactly this description. Applying the construction to $\mathrm{N}_\bullet(\mathbf{Fin}_*)$ gives a canonical *flattening functor*

$$\Phi : \mathrm{N}_\bullet(\mathbf{Fin}_*)^{\sqcup} \to \mathrm{N}_\bullet(\mathbf{Fin}_*)$$

which concatenates a list of pointed finite sets. On objects,

$$\Phi(\langle k_1 \rangle, \ldots, \langle k_m \rangle) = \langle k_1 + \ldots + k_m \rangle$$

Remark (Two Levels of Arity). The outer arity counts blocks, while the inner arities count inputs inside those blocks. Flattening forgets the subdivision and retains the total collection of inputs.

$$\begin{tikzcd} \langle m \rangle \arrow[d, "\text{block arities}"] \\ (\langle k_1 \rangle, \ldots, \langle k_m \rangle) \arrow[d, "\Phi"] \\ \langle k_1 + \ldots + k_m \rangle \end{tikzcd}$$

8.4.2 The Explicit Model

Definition 8.4.7 (Wreath Product). Let $\mathscr{C}^{\otimes}$ and $\mathscr{D}^{\otimes}$ be ∞-operads. Their wreath product is the ∞-category

$$\mathscr{C}^{\otimes} \wr \mathscr{D}^{\otimes} := \mathscr{C}^{\otimes} \times_{\mathrm{N}_{\bullet}(\mathbf{Fin}_*)} \left(\mathscr{D}^{\otimes}\right)^{\sqcup}$$

equipped with the structure map to $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ obtained from Φ.

Thus an object can be written

$$(c; d_1, \ldots, d_m)$$

where c lies over $\langle m \rangle$ and d_i lies over $\langle k_i \rangle$. Its total arity is $\langle k_1 + \ldots + k_m \rangle$. A morphism has two layers: a morphism in $\mathscr{C}^{\otimes}$ rearranges and combines the blocks, while compatible morphisms in $\mathscr{D}^{\otimes}$ act inside the blocks.

Remark (The Inert Marking). Let $\mathscr{M}$ be the class of morphisms in $\mathscr{C}^{\otimes} \wr \mathscr{D}^{\otimes}$ for which the outer morphism in $\mathscr{C}^{\otimes}$ is inert and every induced inner morphism in $\mathscr{D}^{\otimes}$ is inert. The pair

$$\left(\mathscr{C}^{\otimes} \wr \mathscr{D}^{\otimes}, \mathscr{M}\right)$$

is an ∞-preoperad. The marking is essential: without it the wreath product remembers the two-level combinatorics but not which arrows play the role of inert projections.

There is a canonical monomorphism

$$i : \mathscr{C}^{\otimes} \times \mathscr{D}^{\otimes} \to \mathscr{C}^{\otimes} \wr \mathscr{D}^{\otimes}$$

It sends a pair (c, d), of arities m and n, to $(c; d, \ldots, d)$ with m copies of d. Its total arity is mn, in agreement with the smash-product arity used in the operadic tensor product.

8.4.3 Comparison with the Operadic Tensor Product

Lemma 8.4.1 (Extension over a Finite coCartesian Cone). Let s be a finite discrete simplicial set and suppose $x \to s^{\triangleleft}$ and $y \to s^{\triangleleft}$ are coCartesian fibrations whose fibers at the cone point are products of the fibers over s. Restriction from functors preserving coCartesian morphisms to their values over s is a trivial Kan fibration.

Proof sketch. A value at the cone point must be the product of the prescribed values over s, so it exists and is unique up to a contractible space of choices. The coCartesian universal property then supplies the structure maps and all higher coherences. This is the extension principle used repeatedly in the comparison theorem. □

Theorem 8.4.5 (Wreath Product Comparison). The canonical inclusion induces a weak equivalence of ∞-preoperads

$$\mathscr{C}^{\otimes,\natural} \odot \mathscr{D}^{\otimes,\natural} \to (\mathscr{C}^{\otimes} \wr \mathscr{D}^{\otimes}, \mathscr{M})$$

Consequently, every fibrant replacement of the marked wreath product represents the operadic tensor product $\mathscr{C}^{\otimes} \otimes \mathscr{D}^{\otimes}$.

Proof sketch. Restriction along i is a trivial Kan fibration on mapping spaces into every fibrant ∞-preoperad. The extension lemma handles the choices over each finite family of blocks. To pass from boundary data to arbitrary simplices, filter the missing simplices by dimension and combinatorial complexity. Each new simplex is attached together with a distinguished quasi-degenerate face, so the required extension reduces to an inner-horn lifting problem or to the finite-cone lemma. Hence i induces the same derived mapping spaces against all fibrant targets. □

Remark (Three Models of the Same Tensor Product). The comparison may be summarized by

$$\begin{tikzcd}
\mathscr{C}^{\otimes,\natural} \odot \mathscr{D}^{\otimes,\natural} \arrow[d, "\text{weak equivalence}"] \\
(\mathscr{C}^{\otimes} \wr \mathscr{D}^{\otimes}, \mathscr{M}) \arrow[d, "\text{fibrant replacement}"] \\
\mathscr{C}^{\otimes} \otimes \mathscr{D}^{\otimes}
\end{tikzcd}$$

The first line is convenient for the model structure, the second exposes the two-level operadic combinatorics, and the last is the intrinsic ∞-operadic tensor product. The long simplex filtration in the complete proof establishes only the first arrow; it adds no further structure to the final object.

Part III

Algebras and Modules

9 Free Algebras

We have already introduced the notion of $\mathcal{O}$-algebra objects of a symmetric $\mathcal{O}$-monoidal ∞-category $\mathscr{C}$, which is defined by a map of ∞-operads $\mathcal{O}^{\otimes} \to \mathscr{C}^{\otimes}$, and the ∞-category of algebras $\mathbf{Alg}_{\mathcal{O}}(\mathscr{C})$. The aim of this part is to study this notion in more detail.

9.1 From Symmetric Powers to Free Algebras

Let $\mathscr{C}^{\otimes}$ be a symmetric monoidal ∞-category. Restriction along the inclusion of operads

$$\mathbf{Triv}^{\otimes} \hookrightarrow \mathbf{Comm}^{\otimes}$$

is the forgetful functor

$$\theta : \mathbf{CAlg}(\mathscr{C}) \to \mathscr{C}$$

In an ordinary cocomplete symmetric monoidal category, if tensor product preserves colimits separately in each variable, the left adjoint is the symmetric algebra functor

$$\mathrm{Sym}^*(c) = \sqcup_{n\geq 0}\, c^{\otimes n}/\Sigma_n$$

In the ∞-categorical setting the quotient is replaced by homotopy orbits, so the expected formula is

$$\mathrm{Sym}^{*(c)} = \sqcup_{n\geq 0} \left(c^{\otimes n}\right)_{h\Sigma_n}$$

This formula is the model example, but the natural problem is more general. A map of ∞-operads

$$f : \mathcal{O}'^{\otimes} \to \mathcal{O}^{\otimes}$$

induces restriction of algebra structures

$$f^* : \mathbf{Alg}_{\mathcal{O}}(\mathscr{C}) \to \mathbf{Alg}_{\mathcal{O}'}(\mathscr{C})$$

We want to construct its left adjoint $f_!$. The correct construction is an *operadic left Kan extension*. Its local building blocks are operadic colimit diagrams.

> **Remark (Why Ordinary Colimits Are Insufficient).** A colimit of underlying objects need not inherit an algebra structure. The obstruction is that every multimorphism must remain compatible with the colimit. For example, if $a = \operatorname*{colim}_i a_i$, then the multiplication on a requires the comparison
>
> $$\left(\operatorname*{colim}_i a_i\right) \otimes \left(\operatorname*{colim}_j a_j\right) \to \operatorname*{colim}_{i,j}(a_i \otimes a_j)$$
>
> to be an equivalence. Operadic colimits encode exactly this compatibility, and do so one tensor variable at a time.

9.2 Operadic Colimit Diagrams

Let

$$q : \mathscr{C}^{\otimes} \to \mathscr{O}^{\otimes}$$

be a fibration of ∞-operads. Write $\mathscr{C}^{\otimes}_{\mathrm{act}}$ and $\mathscr{O}^{\otimes}_{\mathrm{act}}$ for the subcategories containing all objects and only active morphisms.

9.2.1 Weak and Genuine Operadic Colimits

Let $\overline{p} : k^{\triangleright} \to \mathscr{C}^{\otimes}_{\mathrm{act}}$ be a cone and let $p = \overline{p}|_k$.

Definition 9.2.1 (The Active Cocone Category). For a diagram $d : s \to \mathscr{C}^{\otimes}_{\mathrm{act}}$, define

$$\mathbf{Cocone}^{\mathrm{act}}_{\mathscr{C}}(d) := \mathbf{Fun}\left(s^{\triangleright}, \mathscr{C}^{\otimes}_{\mathrm{act}}\right) \times_{\mathbf{Fun}\left(s, \mathscr{C}^{\otimes}_{\mathrm{act}}\right)} \{d\}$$

Thus $\mathbf{Cocone}^{\mathrm{act}}_{\mathscr{C}}(d)$ is the homotopy fiber over d of the restriction functor from $s^{\triangleright}$-diagrams to s-diagrams.

1. An object is an extension

$$\overline{d} : s^{\triangleright} \to \mathscr{C}^{\otimes}_{\mathrm{act}}$$

   together with an identification $\overline{d}|_s \simeq d$.
2. A morphism is a homotopy-coherent natural transformation between two such extensions which is fixed on s.

The superscript act restricts only the morphisms: all objects of $\mathscr{C}^{\otimes}$ remain, but every arrow in the cocone is required to be active. Equivalently,

$$\mathbf{Cocone}^{\mathrm{act}}_{\mathscr{C}}(d) \simeq \left(\mathscr{C}^{\otimes}_{\mathrm{act}}\right)_{d/}$$

Remark (A Two-Object Cocone). If $s = \{0, 1\}$ is discrete and $d(0) = x_0$, $d(1) = x_1$, an object of the active cocone category has the form

$$\begin{tikzcd} x_0 \arrow[dr, "\text{active}"'] & & x_1 \arrow[dl, "\text{active}"] \\ & c & \end{tikzcd}$$

If the x_i are tuples of colors and c lies over $\langle 1 \rangle$, these arrows are multimorphisms with common output c.

There is now a canonical comparison map

$$\psi_{\overline{p}} : \mathbf{Cocone}^{\mathrm{act}}_{\mathscr{C}}(\overline{p}) \to \mathbf{Cocone}^{\mathrm{act}}_{\mathscr{C}}(p) \times_{\mathbf{Cocone}^{\mathrm{act}}_{\mathscr{O}}(qp)} \mathbf{Cocone}^{\mathrm{act}}_{\mathscr{O}}(q\overline{p})$$

Remark (Meaning of the Comparison Map). The left-hand side is the category of further active cocones on the already chosen cone $\overline{p}$. A point on the right consists of an active cocone on p in $\mathscr{C}^{\otimes}$ and a compatible further cocone on $q\overline{p}$ in the base. The map $\psi_{\overline{p}}$ forgets the actual active lift from the cone point of $\overline{p}$. Hence it is a trivial Kan fibration precisely when every compatible base cocone has a contractibly unique lift.

Definition 9.2.2 (Weak Operadic Colimit). The cone $\overline{p}$ is a *weak operadic q-colimit diagram* if $\psi_{\overline{p}}$ is a trivial Kan fibration.

Equivalently, every active cocone compatible with the prescribed cocone in $\mathcal{O}^{\otimes}$ lifts through q, and the space of such lifts is contractible.

Definition 9.2.3 (Operadic Colimit). An active cone $\overline{p} : k^{\triangleright} \to \mathcal{C}^{\otimes}_{\mathrm{act}}$ is an *operadic q-colimit diagram* if, for every object $c \in \mathcal{C}$, the cone obtained by adjoining c

$$\overline{p} \oplus c : k^{\triangleright} \to \mathcal{C}^{\otimes}_{\mathrm{act}}$$

is a weak operadic q-colimit diagram.

Here $\oplus$ denotes concatenation of operadic inputs. The second definition is stronger because it asks the universal property to survive after an arbitrary extra input is inserted.

Remark (Warning). An ordinary q-colimit need not be an operadic q-colimit, and an operadic q-colimit need not be an ordinary q-colimit. They coincide only under additional compatibility hypotheses.

Remark (The Lifting Picture). Weak operadic universality can be tested by filling the dotted arrow in every diagram of the following form, with the cone point landing in the underlying category of colors.

$$\begin{tikzcd}
k \star \partial\Delta^n \arrow[r] \arrow[d] & \mathcal{C}^{\otimes}_{\mathrm{act}} \arrow[d, "q"] \\
k \star \Delta^n \arrow[r] \arrow[ur, dashed] & \mathcal{O}^{\otimes}_{\mathrm{act}}
\end{tikzcd}$$

9.2.2 Formal Properties

Proposition 9.2.1 (Basic Stability Properties). Operadic colimit diagrams satisfy the following rules.

1. A weak operadic colimit remains one after left-cofinal reindexing.
2. A finite product of operadic q-colimit diagrams is again an operadic q-colimit diagram.
3. If the indexing simplicial sets in such a finite product are weakly contractible, the resulting cone is also an ordinary q-colimit.
4. Suppose $\overline{p}_0$ and $\overline{p}_1$ are joined by a natural transformation whose components on the original diagram are q-coCartesian and whose component at the cone point is an equivalence. Then one is an operadic q-colimit if and only if the other is.

Proof sketch. The comparison maps defining weak operadic colimits are left fibrations, so it suffices to prove that their fibers are contractible. Cofinal changes do not alter these fibers. Finite products are handled one variable at a time, using the contractible choice of extensions over a coCartesian cone. The natural-transformation statement follows by comparing the corresponding active cocone categories and applying two-out-of-three. □

Proposition 9.2.2 (Detecting coCartesian Fibrations). The fibration q is a coCartesian fibration of ∞-operads if and only if every active arrow in $\mathcal{O}^{\otimes}$ admits a lift which, regarded as a cone on one object, is an operadic q-colimit diagram.

Thus a coCartesian lift is precisely a unary operadic colimit. This is the first indication that operadic colimits are the correct relative analogue of ordinary colimits.

9.2.3 The Fiberwise Criterion

Assume now that q is a coCartesian fibration. An active operation

$$m : x \oplus y \to z$$

in $\mathcal{O}^{\otimes}$ induces a pushforward functor between the corresponding fibers. If $\overline{p} : k^{\triangleright} \to \mathcal{C}_x$ is a cone, then adjoining an object $d \in \mathcal{C}_y$ and pushing forward along m gives

$$k^{\triangleright} \xrightarrow{\overline{p}} \mathcal{C}_x \xrightarrow{-\oplus d} \mathcal{C}_{x \oplus y} \xrightarrow{m_!} \mathcal{C}_z$$

Theorem 9.2.1 (Fiberwise Criterion). The cone $\overline{p}$ is an operadic q-colimit diagram if and only if every composite above is an ordinary colimit diagram in $\mathcal{C}_z$, for every choice of y, z, d, and active operation $m : x \oplus y \to z$.

Remark (Symmetric Monoidal Case). If $\mathcal{C}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ is a symmetric monoidal ∞-category, the criterion reduces to a familiar statement: a cone

$$\overline{p} : k^{\triangleright} \to \mathcal{C}$$

is an operadic colimit exactly when, for every $c \in \mathcal{C}$,

$$k^{\triangleright} \xrightarrow{\overline{p}} \mathcal{C} \xrightarrow{c \otimes -} \mathcal{C}$$

is an ordinary colimit diagram. In short,

> “An operadic colimit is a colimit preserved by tensoring with every object.”

Proof sketch. The Segal equivalence decomposes an object over a finite pointed set into its input components. The fibers of the comparison map $\psi_{\overline{p}}$ are therefore the fibers of the ordinary colimit comparison maps obtained after adjoining the remaining inputs and applying an active operation. Contractibility of all these fibers is exactly the stated fiberwise condition. □

9.2.4 Existence from Compatible Colimits

Definition 9.2.4 (Compatibility with Indexed Colimits). Let k be a simplicial set. A coCartesian fibration of ∞-operads

$$q : \mathscr{C}^{\otimes} \to \mathscr{O}^{\otimes}$$

is *compatible with k-indexed colimits* if

1. every fiber $\mathscr{C}_x$ admits k-indexed colimits
2. every operadic tensor functor

$$\otimes_{\varphi} : \prod_{1 \le i \le n} \mathscr{C}_{x_i} \to \mathscr{C}_y$$

preserves k-indexed colimits separately in each variable

Theorem 9.2.2 (Existence of Operadic Colimits). If q is compatible with k-indexed colimits, then every k-diagram in $\mathscr{C}^{\otimes}_{\mathrm{act}}$, together with a prescribed cone over its image in $\mathscr{O}^{\otimes}_{\mathrm{act}}$, extends to an operadic q-colimit diagram.

Proof sketch. First form the required ordinary colimit in the fiber containing the cone point. The preservation hypothesis implies that adjoining any other inputs and applying any active operation still produces a colimit cone. The fiberwise criterion upgrades this ordinary cone to an operadic one, while coCartesian lifting supplies the required coherent map over the prescribed base cone. □

Corollary 9.2.1 (Small-Colimit Criterion). For a regular cardinal κ, the map q is a coCartesian fibration compatible with κ-small colimits if and only if every κ-small diagram equipped with a cone in the base admits an operadic q-colimit lift.

Remark (Role in the Construction of Free Algebras). Under the compatibility hypotheses, operadic left Kan extension along f is left adjoint to restriction. The free algebra functor is the special case which freely adds the operations encoded by $\mathscr{O}$.

9.3 Operadic Left Kan Extensions

For an ordinary functor $u : \mathscr{A} \to \mathscr{B}$, the value of a left Kan extension at $b \in \mathscr{B}$ is computed from all arrows $u(a) \to b$. The operadic version has the same shape, except that one must allow an arbitrary finite tuple of inputs and use active morphisms in place of ordinary arrows.

9.3.1 Correspondences of Operads

Definition 9.3.5 (Correspondence of Operads). A *correspondence of ∞-operads* is a Δ^1-family of ∞-operads

$$r : \mathscr{M}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*) \times \Delta^1$$

Its fibers over 0 and 1 will be denoted by $\mathscr{A}^{\otimes}$ and $\mathscr{B}^{\otimes}$. Thus $\mathscr{M}^{\otimes}$ interpolates from $\mathscr{A}^{\otimes}$ to $\mathscr{B}^{\otimes}$ and records the operations which may carry inputs from the first fiber to outputs in the second.

$$
\begin{tikzcd}
\mathscr{A}^{\otimes} \arrow[r, hook] \arrow[d] & \mathscr{M}^{\otimes} \arrow[d] & \mathscr{B}^{\otimes} \arrow[l, hook'] \arrow[d] \\
\{0\} \arrow[r, hook] & \Delta^1 & \{1\} \arrow[l, hook']
\end{tikzcd}
$$

Let $q : \mathscr{C}^{\otimes} \to \mathscr{O}^{\otimes}$ be a fibration of ∞-operads and let $F : \mathscr{M}^{\otimes} \to \mathscr{C}^{\otimes}$ be a map of generalized ∞ -operads. For $b \in \mathscr{B}^{\otimes}$, define the active indexing category

$$\mathscr{K}_b := \left(\mathscr{M}^{\otimes}_{\mathrm{act}}\right)_{/b} \times_{\mathscr{M}^{\otimes}} \mathscr{A}^{\otimes}$$

Here the map from the active slice to $\mathscr{M}^{\otimes}$ remembers the source. Consequently, an object of $\mathscr{K}_b$ is an active operation $a \to b$ whose input object a lies in the initial fiber $\mathscr{A}^{\otimes}$. The identity of b supplies a canonical cone

$$\mathscr{K}_b^{\triangleright} \to \left(\mathscr{M}^{\otimes}\right)_{/b} \to \mathscr{M}^{\otimes} \xrightarrow{F} \mathscr{C}^{\otimes}$$

Definition 9.3.6 (Operadic Left Kan Extension). The map F is an *operadic q-left Kan extension* of $F|_{\mathscr{A}^{\otimes}}$ if the cone above is an operadic q-colimit diagram for every $b \in \mathscr{B}^{\otimes}$.

Remark (What the Definition Says). The restriction of F to $\mathscr{A}^{\otimes}$ supplies the input data. For an output b, the category $\mathscr{K}_b$ lists every active way of feeding that data into b. The operadic colimit condition says that $F(b)$ is universal among objects receiving all these operations, and that this universality survives after any further operadic input is adjoined. In particular, $F(b)$ is determined up to a contractible space of choices.

9.3.2 Existence and Coherent Extension

Theorem 9.3.3 (Local-to-Global Extension Principle). Consider a lifting problem of Δ^n-families of operads over the fibration q.

1. If $n = 1$, an operadic left Kan extension exists if and only if, for every object b in the terminal fiber, the diagram indexed by $\mathscr{K}_b$ can be completed to an operadic q-colimit cone lifting the prescribed cone in $\mathscr{O}^{\otimes}$.
2. If $n > 1$ and the restriction to the edge $\{0, 1\}$ is already an operadic left Kan extension, then the compatible partial family extends over the whole n-simplex.

Proof sketch. Factor morphisms in $\mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ into inert and active parts and attach the missing simplices dimension by dimension. At a new vertex, the required filler is exactly an operadic colimit cone. Higher-dimensional attachments are inner-horn lifting problems and are solved by the fibration q. The active–inert factorization makes these attachments compatible with the operadic Segal condition. Thus the only genuinely new choices occur at objects, and their spaces are contractible by the operadic colimit property. □

Remark (Why the Higher-Simplex Statement Matters). Pointwise existence alone would not produce a functor. The second part of the theorem supplies all naturality and higher coherence simultaneously. It is the operadic counterpart of the fact that pointwise ordinary Kan extensions assemble into a homotopy-coherent functor.

9.3.3 The Pointwise Formula

Suppose now that

$$f : \mathcal{O}'^{\otimes} \to \mathcal{O}^{\otimes}$$

is a map of ∞-operads, a is an $\mathcal{O}'$-algebra in $\mathcal{C}^{\otimes}$, and the relevant operadic colimits exist. For a color $y \in \mathcal{O}$, set

$$\mathcal{I}_y := \left(\mathcal{O}^{\otimes}_{\mathrm{act}}\right)_{/y} \times_{\mathcal{O}^{\otimes}} \mathcal{O}'^{\otimes}$$

An object of $\mathcal{I}_y$ is a tuple of colors $x_1, \ldots, x_n$ of $\mathcal{O}'$ together with an active operation

$$\alpha : f(x_1) \oplus \ldots \oplus f(x_n) \to y$$

Let $\otimes_\alpha$ denote the pushforward functor classified by this operation. The value of the operadic left Kan extension is

$$(f_! a)(y) \simeq \operatorname*{colim}_{(x_1,\ldots,x_n;\alpha)\in\mathcal{I}_y} \otimes_\alpha (a(x_1), \ldots, a(x_n))$$

Remark (Symmetric Monoidal Form). In a symmetric monoidal ∞-category there is a single ambient fiber, and $\otimes_\alpha$ is the corresponding iterated tensor product. Hence the formula becomes

$$(f_! a)(y) \simeq \operatorname*{colim}_{(x_1,\ldots,x_n;\alpha)\in\mathcal{I}_y} a(x_1) \otimes \ldots \otimes a(x_n)$$

The indexing category retains the operation α and all of its homotopies; replacing it by a mere set of operations would lose the required coherent symmetry.

Remark (Free Algebras). Take $f : \mathbf{Triv}^{\otimes} \to \mathcal{O}^{\otimes}$. For a one-colored operad and an object c, the preceding formula specializes to

$$\mathrm{Free}_{\mathcal{O}}(c) \simeq \sqcup_{n\geq 0} \left(\mathcal{O}(n) \otimes c^{\otimes n}\right)_{h\Sigma_n}$$

The $n = 0$ term inserts the constants of $\mathcal{O}$. For $\mathcal{O} = \mathbf{Comm}$, every $\mathcal{O}(n)$ is contractible and this recovers

$$\mathrm{Sym}^*(c) \simeq \sqcup_{n\geq 0} \left(c^{\otimes n}\right)_{h\Sigma_n}$$

9.3.4 Computational Summary

“ Operadic left Kan extension is a colimit over active operations, computed in the output fiber and tested after adjoining every extra input. ”

The comparison with the ordinary pointwise formula is

$$(\mathrm{Lan}_u F)(b) \simeq \operatorname*{colim}_{(a,u(a)\to b)} F(a)$$

$$(f_! a)(y) \simeq \operatorname*{colim}_{(x_1,\dots,x_n;\alpha: f(x_1)\oplus\dots\oplus f(x_n)\to y)} \otimes_\alpha (a(x_1),\dots,a(x_n))$$

The practical procedure is therefore:

1. form the active slice $\mathcal{I}_y$ of all operations with output y
2. evaluate the input algebra and apply the operation-specific tensor functor $\otimes_\alpha$
3. take the ordinary colimit in the fiber $\mathscr{C}_y$
4. verify, using the fiberwise criterion, that every operadic tensor functor preserves this colimit separately in each additional variable

Under compatibility with the required indexed colimits, the final condition is automatic. The resulting objects assemble coherently by the local-to-global extension theorem and define the left adjoint

$$f_! : \mathbf{Alg}_{\mathcal{O}'}(\mathscr{C}) \to \mathbf{Alg}_{\mathcal{O}}(\mathscr{C})$$

> **Remark (Colimits, Not a Formal Duality).** The construction in this section is the operadic *colimit* and left Kan extension theory. Its pointwise formulas resemble ordinary colimits, but the active-operation index and the stability under extra inputs are essential. A formal dual theory is not obtained merely by reversing every arrow, because the definition of an operad is intrinsically asymmetric in its many inputs and single output.

9.3.5 The Adjunction $f_! \dashv f^*$

> **Definition 9.3.7 (Restriction of Algebra Structures).** Let $q : \mathscr{C}^\otimes \to \mathcal{O}^\otimes$ be an $\mathcal{O}$-monoidal ∞-category and let
>
> $$f : \mathcal{O}'^\otimes \to \mathcal{O}^\otimes$$
>
> be a map of ∞-operads. Its base change is the $\mathcal{O}'$-monoidal ∞-category
>
> $$\mathscr{C}_f^\otimes := \mathscr{C}^\otimes \times_{\mathcal{O}^\otimes} \mathcal{O}'^\otimes \to \mathcal{O}'^\otimes$$
>
> By convention, $\mathbf{Alg}_{\mathcal{O}'}(\mathscr{C})$ below means $\mathbf{Alg}_{\mathcal{O}'}\left(\mathscr{C}_f\right)$. An $\mathcal{O}$-algebra is an inert-preserving section
>
> $$b : \mathcal{O}^\otimes \to \mathscr{C}^\otimes$$
>
> Restriction along f is precomposition:
>
> $$f^* b := b \circ f : \mathcal{O}'^\otimes \to \mathscr{C}^\otimes$$
>
> Since $q \circ b \circ f = f$, this map is equivalently a section of $\mathscr{C}_f^\otimes \to \mathcal{O}'^\otimes$, and hence an $\mathcal{O}'$-algebra. On a morphism $\theta : b \to b'$, the functor is defined by the same precomposition rule $f^{*(\theta)} = \theta \circ f$.

Remark (Meaning of the Two Sides). The two algebra categories record different amounts of operadic structure.

1. $\mathbf{Alg}_{\mathcal{O}'}(\mathcal{C})$ consists of objects carrying the operations and coherences prescribed by $\mathcal{O}'$.
2. $\mathbf{Alg}_{\mathcal{O}}(\mathcal{C})$ consists of objects carrying the operations and coherences prescribed by $\mathcal{O}$.

The functor f^* views an $\mathcal{O}$-algebra only through the operations coming from $\mathcal{O}'$. If f is an inclusion, it literally forgets the operations not lying in its image; for a general f, it reindexes colors and operations along f. In the opposite direction, $f_!$ freely supplies the missing $\mathcal{O}$-operations by the operadic colimits of the preceding subsection, subject only to the original $\mathcal{O}'$-relations.

Theorem 9.3.4 (Operadic Kan Extension Adjunction). Let

$$f : \mathcal{O}'^{\otimes} \to \mathcal{O}^{\otimes}$$

be a map of ∞-operads, and let $\mathcal{C}^{\otimes}$ be an $\mathcal{O}$-monoidal ∞-category compatible with all operadic colimits required by the pointwise formula. Restriction along f admits the operadic left Kan extension as a left adjoint:

$$f_! : \mathbf{Alg}_{\mathcal{O}'}(\mathcal{C}) \rightleftarrows \mathbf{Alg}_{\mathcal{O}}(\mathcal{C}) : f^*$$

In other words, $f_! \dashv f^*$, where f^* is restriction of the algebra structure and $f_!$ is computed by the operadic left Kan extension formula above.

For $a \in \mathbf{Alg}_{\mathcal{O}'}(\mathcal{C})$ and $b \in \mathbf{Alg}_{\mathcal{O}}(\mathcal{C})$, the adjunction is characterized by a natural equivalence of mapping anima

$$\mathrm{Map}_{\mathbf{Alg}_{\mathcal{O}}(\mathcal{C})}(f_! a, b) \simeq \mathrm{Map}_{\mathbf{Alg}_{\mathcal{O}'}(\mathcal{C})}(a, f^* b)$$

The unit and counit are natural transformations

$$\eta : \mathrm{id}_{\mathbf{Alg}_{\mathcal{O}'}(\mathcal{C})} \to f^* f_!, \quad \varepsilon : f_! f^* \to \mathrm{id}_{\mathbf{Alg}_{\mathcal{O}}(\mathcal{C})}$$

Their components are induced by the canonical inclusions into, and evaluation maps out of, the active-operation colimits. They satisfy the triangle identities

$$\varepsilon_{f_! a} \circ f_!(\eta_a) \simeq \mathrm{id}_{f_! a}, \quad f^{*(\varepsilon_b)} \circ \eta_{f^* b} \simeq \mathrm{id}_{f^* b}$$

Proof sketch. An $\mathcal{O}$-algebra map $f_! a \to b$ is determined by compatible maps from every term

$$\otimes_{\alpha} (a(x_1), \dots, a(x_n))$$

in the active-operation colimit defining $f_! a$. By the universal property of this operadic colimit, such a compatible family is exactly an $\mathcal{O}'$-algebra map $a \to f^* b$. The equivalence is natural in both variables, so it determines the adjunction together with its unit, counit, and triangle identities. □

9.4 Construction of Free Algebras

Definition 9.4.8 (Relative Free Algebra). Let $f : \mathcal{O}'^{\otimes} \to \mathcal{O}^{\otimes}$ be a map of ∞-operads and suppose that $f_! \dashv f^*$ exists. For $a \in \mathbf{Alg}_{\mathcal{O}'}(\mathcal{C})$, the *relative free $\mathcal{O}$-algebra generated by a* is simply $f_! a$; equivalently, its unit $a \to f^* f_! a$ induces, naturally in every $b \in \mathbf{Alg}_{\mathcal{O}}(\mathcal{C})$, the equivalence

$$\mathrm{Map}_{\mathbf{Alg}_{\mathcal{O}}(\mathcal{C})}(f_! a, b) \simeq \mathrm{Map}_{\mathbf{Alg}_{\mathcal{O}'}(\mathcal{C})}(a, f^* b)$$

We now express the preceding adjunction directly in terms of generators and operations. Let $q : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$ be a fibration of ∞-operads and suppose that we are given maps

$$\mathcal{A}^{\otimes} \xrightarrow{i} \mathcal{B}^{\otimes} \xrightarrow{j} \mathcal{O}^{\otimes}$$

Restriction along i gives

$$\theta = i^* : \mathbf{Alg}_{\mathcal{B}/\mathcal{O}}(\mathcal{C}) \to \mathbf{Alg}_{\mathcal{A}/\mathcal{O}}(\mathcal{C})$$

9.4.1 Free Relative Algebras

Let $a \in \mathbf{Alg}_{\mathcal{A}/\mathcal{O}}(\mathcal{C})$. For a color $y \in \mathcal{B}$, define

$$\mathcal{K}_y := \mathcal{A}^{\otimes} \times_{\mathcal{B}^{\otimes}} \left(\mathcal{B}^{\otimes}_{\mathrm{act}}\right)_{/y}$$

Thus $\mathcal{K}_y$ parametrizes active operations in $\mathcal{B}$ with output y and with every input supplied by $\mathcal{A}$. Evaluation of a produces a diagram $\mathcal{K}_y \to \mathcal{C}^{\otimes}_{\mathrm{act}}$ over the corresponding diagram in $\mathcal{O}^{\otimes}_{\mathrm{act}}$.

Definition 9.4.9 (Free Algebra Generated by a Relative Algebra). Let

$$b \in \mathbf{Alg}_{\mathcal{B}/\mathcal{O}}(\mathcal{C}), \quad \eta : a \to i^* b$$

The map η *exhibits b as a q-free $\mathcal{B}$-algebra generated by a* if, for every $y \in \mathcal{B}$, the induced cone

$$\mathcal{K}_y^{\triangleright} \to \mathcal{C}^{\otimes}_{\mathrm{act}}$$

is an operadic q-colimit diagram.

Proposition 9.4.3 (Universal Property of a Free Algebra). If $\eta : a \to i^* b$ exhibits b as a q-free algebra generated by a, then for every $b' \in \mathbf{Alg}_{\mathcal{B}/\mathcal{O}}(\mathcal{C})$, composition with η induces a natural equivalence

$$\mathrm{Map}_{\mathbf{Alg}_{\mathcal{B}/\mathcal{O}}(\mathcal{C})}(b, b') \simeq \mathrm{Map}_{\mathbf{Alg}_{\mathcal{A}/\mathcal{O}}(\mathcal{C})}(a, i^* b')$$

Consequently, b is unique up to a contractible space of choices and is the value $i_! a$ whenever the left adjoint $i_!$ exists.

Proof sketch. A map $b \to b'$ is determined at each output y by a compatible cocone from the active-operation diagram indexed by $\mathcal{K}_y$. The operadic colimit property identifies such cocones with maps from the original $\mathcal{A}$-algebra a to $i^* b'$. Compatibility with inert morphisms assembles these pointwise equivalences into the stated equivalence of mapping anima. □

9.4.2 Existence

Theorem 9.4.5 (Pointwise Existence Criterion). For $a \in \mathbf{Alg}_{\mathscr{A}/\mathscr{O}}(\mathscr{C})$, the following conditions are equivalent.

1. There exist a $\mathscr{B}$-algebra b and a map $a \to i^*b$ which exhibit b as a q-free algebra generated by a.
2. For every color $y \in \mathscr{B}$, the diagram indexed by $\mathscr{K}_y$ extends to an operadic q-colimit cone lifting the canonical cone in $\mathscr{O}^{\otimes}_{\mathrm{act}}$.

Corollary 9.4.2 (Existence under Small-Colimit Hypotheses). Let κ be an uncountable regular cardinal. Suppose that $\mathscr{A}^{\otimes}$ and $\mathscr{B}^{\otimes}$ are essentially κ-small and that q is a coCartesian fibration compatible with κ-small colimits. Then restriction admits a left adjoint

$$i_! : \mathbf{Alg}_{\mathscr{A}/\mathscr{O}}(\mathscr{C}) \rightleftarrows \mathbf{Alg}_{\mathscr{B}/\mathscr{O}}(\mathscr{C}) : i^*$$

and $i_!a$ is the q-free $\mathscr{B}$-algebra generated by a.

Proof sketch. The active indexing categories $\mathscr{K}_y$ are κ-small. Compatibility supplies the required operadic colimits, so the pointwise existence criterion constructs $i_!a$. The universal property above then identifies the resulting functor as the left adjoint of restriction. □

Remark (The Unital Improvement). If the operads are unital, the relevant active indexing categories are weakly contractible. It is then enough to assume the existence and separate preservation of weakly contractible κ-small colimits. This weaker form is often the useful one in practice.

Remark (Naturality). Let $T : \mathscr{C}^{\otimes} \to \mathscr{D}^{\otimes}$ be an $\mathscr{O}$-monoidal functor which preserves the colimits appearing above. Then T preserves free algebras:

$$T(i_!a) \simeq i_!(T(a))$$

9.4.3 Operadic Symmetric Powers

We specialize to the inclusion

$$\mathbf{Triv}^{\otimes} \to \mathscr{O}^{\otimes}$$

which selects a color $x \in \mathscr{O}$. Let $c \in \mathscr{C}_x$ and let $y \in \mathscr{O}$. For $n \geq 0$, denote by $\mathscr{P}_{x,y}(n)$ the anima of active operations with n inputs of color x and output y, including their permutations and higher homotopies. Its fiber over a chosen ordering is

$$\mathrm{Mul}_{\mathscr{O}}(\{x\}_{1\leq i\leq n}; y)$$

CoCartesian transport along an operation $\alpha \in \mathscr{P}_{x,y}(n)$ gives an object

$$\otimes_{\alpha} (c, \dots, c) \in \mathscr{C}_y$$

Definition 9.4.10 (Operadic Symmetric Power). The nth operadic symmetric power of c from x to y is

$$\mathrm{Sym}^n_{\mathcal{O};x,y}(c) := \operatorname*{colim}_{\alpha\in\mathcal{P}_{x,y}(n)} \otimes_\alpha (c, \dots, c)$$

Theorem 9.4.6 (Explicit Free-Algebra Formula). Under the small-colimit hypotheses above, the free $\mathcal{O}$-algebra on $c \in \mathcal{C}_x$ exists. Its value at a color y is

$$\mathrm{Free}_{\mathcal{O}}(c)(y) \simeq \sqcup_{n\geq 0} \mathrm{Sym}^n_{\mathcal{O};x,y}(c)$$

A map $c \to a(x)$ exhibits an $\mathcal{O}$-algebra a as free on c if and only if these canonical maps are equivalences for every y.

Proof sketch. Decompose the active indexing category $\mathcal{I}_y$ by the number n of inputs. Its nth component is $\mathcal{P}_{x,y}(n)$, so the pointwise operadic Kan-extension formula becomes the coproduct of the corresponding colimits. The universal property follows from the adjunction $\mathrm{Free}_{\mathcal{O}} \dashv U$.□

Example (Commutative Algebras). For $\mathcal{O} = \mathbf{Comm}$ there is one color and $\mathcal{P}_{x,x}(n) \simeq B\Sigma_n$. Therefore

$$\mathrm{Sym}^*(c) \simeq \sqcup_{n\geq 0} \left(c^{\otimes n}\right)_{h\Sigma_n}$$

which is the homotopy-coherent version of the classical symmetric algebra.

Remark (Relation with Operadic Left Kan Extension). Free-algebra construction introduces no new mechanism: it is precisely the operadic left Kan extension along i. The factorization through a correspondence of operads is used only to organize its coherent construction. The long technical proof attaches simplices one at a time; active faces impose the universal colimit condition, while inert faces impose the operadic Segal compatibility.

9.5 Transitivity of Operadic Left Kan Extensions

Suppose that we have composable maps of ∞-operads

$$\mathcal{O}_0^{\otimes} \xrightarrow{f} \mathcal{O}_1^{\otimes} \xrightarrow{g} \mathcal{O}_2^{\otimes}$$

and an $\mathcal{O}_2$-monoidal ∞-category $\mathcal{C}^{\otimes}$. Assume that the operadic left Kan extensions below exist; it is enough, for example, that the relevant active indexing categories are small and that the operadic tensor products in $\mathcal{C}$ preserve their colimits separately in each variable.

Theorem 9.5.7 (Transitivity). For every $a \in \mathbf{Alg}_{\mathcal{O}_0}(\mathcal{C})$, there is a canonical natural equivalence

$$(g \circ f)_! a \simeq g_!(f_! a)$$

Thus extending an algebra structure from $\mathcal{O}_0$ directly to $\mathcal{O}_2$ agrees with first extending it to $\mathcal{O}_1$ and then to $\mathcal{O}_2$.

$$\begin{tikzcd}
\mathbf{Alg}_{\mathcal{O}_0}(\mathscr{C}) \arrow[rr, "f_!"] \arrow[dr, "(g\circ f)_!"'] & & \mathbf{Alg}_{\mathcal{O}_1}(\mathscr{C}) \arrow[dl, "g_!"] \\
& \mathbf{Alg}_{\mathcal{O}_2}(\mathscr{C}) &
\end{tikzcd}$$

Remark (Pointwise Meaning). Let $\mathscr{K}_{f(y)}$ denote the active operations from $\mathcal{O}_0$ whose image under f has output $y \in \mathcal{O}_1$, and define $\mathscr{K}_{g(z)}$ similarly. The iterated construction of $g_! f_! a$ is indexed by

$$\int_{\beta \in \mathscr{K}_{g(z)}} \prod_i \mathscr{K}_{f(y_i)}$$

where the inputs of β are $y_1, \ldots, y_n$. Operadic substitution composes these operations and gives the active indexing category for $(g \circ f)_! a$.

Proof sketch. Substitute the colimit formula for each value $(f_! a)(y_i)$ into the pointwise formula for $g_!$. The Fubini rule for colimits rewrites the resulting iterated colimit as a single colimit over the displayed Grothendieck construction. Operadic composition identifies this index cofinally with the active slice defining $(g \circ f)_! a$, producing the claimed equivalence. Equivalently, once the adjunctions exist,

$$(g \circ f)^* = f^* \circ g^*$$

and both $(g \circ f)_!$ and $g_! \circ f_!$ are left adjoint to this restriction functor, hence are canonically equivalent. □

Remark (Why a Flatness Hypothesis Appears). In the family-valued formulation, one assumes that the family over Δ^2 is flat. Its role is exactly to ensure that the two-stage active indexing category above computes the direct one without introducing extra homotopy. For ordinary composable operad maps under the preceding colimit-compatibility assumptions, the practical conclusion is simply the transitivity formula above.

10 (co)Limits of Algebras

Let $\mathscr{C}^{\otimes}$ be a symmetric monoidal ∞-category and let $\mathscr{O}^{\otimes}$ be an ∞-operad. This chapter studies limits and colimits in $\mathbf{Alg}_{\mathscr{O}}(\mathscr{C})$.

> “Limits of algebras are usually detected on the underlying objects; colimits must also be compatible with every operadic multiplication.”

More precisely, limits are generally computed in the underlying fibers whenever the required limits exist there; no tensor-preservation hypothesis is needed. Colimits are subtler. Sifted colimits are often still computed underneath when tensor products preserve them separately in each variable, whereas arbitrary coproducts usually are not. The free-algebra construction of the preceding chapter provides the standard method: resolve a diagram by free algebras, compute there, and then recover its colimit geometrically.

> **Remark (Two Important Extremes).** For commutative algebras, finite coproducts are tensor products in the underlying symmetric monoidal category. At the opposite extreme, the empty coproduct is the initial algebra. We begin with the latter because it is controlled entirely by nullary operations and unit objects.

10.1 Unit Objects and Trivial Algebras

Let

$$p : \mathscr{C}^{\otimes} \to \mathscr{O}^{\otimes}$$

be a fibration of ∞-operads, and suppose that $\mathscr{O}^{\otimes}$ is unital. Write $\mathscr{C}$ and $\mathscr{O}$ for the fibers over $\langle 1 \rangle$. The fiber over $\langle 0 \rangle$ records the nullary input.

10.1.1 Relative Unit Objects

> **Definition 10.1.1 (x-Unit Object).** Let $x \in \mathscr{O}^{\otimes}$ and let $1_x \in \mathscr{C}^{\otimes}_x$. A morphism
>
> $$u : c_0 \to 1_x$$
>
> *exhibits* 1_x *as an x-unit object* if $c_0 \in \mathscr{C}^{\otimes}_{\langle 0 \rangle}$ and the edge u, regarded as a cone $\Delta^1 \simeq (\Delta^0)^{\triangleright} \to \mathscr{C}^{\otimes}$, is an operadic p-colimit diagram.

Thus 1_x is not merely a chosen object: it is the universal value of the empty tensor product in the fiber over x. Its universal property makes it unique up to a contractible space of choices.

> **Remark (A Unit Relative to a Color).** The object 1_x is the unit *relative to the color x* of the base operad. Different colors may therefore have units in different fibers $\mathscr{C}_x$; for a one-colored operad this reduces to the usual tensor unit $1 \in \mathscr{C}$.

Definition 10.1.2 (Unit Object in a Tuple). A morphism $u : c_0 \to c$ in $\mathscr{C}^{\otimes}$ *exhibits c as a unit object* if, for every inert projection $c \to c_i$ to a component over $\langle 1 \rangle$, the composite

$$c_0 \to c \to c_i$$

exhibits c_i as a $p(c_i)$-unit object.

We say that p *has unit objects* if such a lift exists for every object of $\mathcal{O}^{\otimes}$.

Remark (How to Recognize Them). Every morphism exhibiting a unit is p-coCartesian. Conversely, a p-coCartesian lift out of a nullary object exhibits a unit as soon as each unary component does. Consequently it is enough to construct units one color at a time; units for tuples are obtained by concatenating the corresponding unary units.

Proposition 10.1.1 (Units in a Monoidal Family). If p is a coCartesian fibration of ∞-operads and $\mathcal{O}^{\otimes}$ is unital, then p has unit objects. Equivalently, for every color $x \in \mathcal{O}$, the fiber $\mathscr{C}_x$ contains an essentially unique object 1_x representing the empty tensor product.

Proof sketch. The unique nullary operation of the unital operad $\mathcal{O}^{\otimes}$ has a coCartesian lift. The operadic colimit criterion shows that its target is an x-unit. CoCartesian composition constructs the units of arbitrary tuples, and the same universal property makes all choices contractible. □

10.1.2 Trivial Algebras

Definition 10.1.3 (Trivial Algebra). An algebra $a \in \mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathscr{C})$ is *trivial* if, for every $x \in \mathcal{O}^{\otimes}$, the canonical morphism

$$a(0) \to a(x)$$

exhibits $a(x)$ as a unit object. Here 0 denotes the nullary object of $\mathcal{O}^{\otimes}$.

In other words, a trivial algebra interprets every color by its corresponding unit and every operation by the uniquely induced map between units. It has no algebraic data beyond the units forced by the operad.

Theorem 10.1.1 (Initial Algebra Criterion). Assume that p has unit objects. For $a \in \mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathscr{C})$, the following conditions are equivalent.

1. The algebra a is initial in $\mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathscr{C})$.
2. The section a is the operadic p-left Kan extension of its restriction to the nullary fiber $\mathcal{O}^{\otimes}_{\langle 0 \rangle}$.
3. The algebra a is trivial.

Proof sketch. Extending from the nullary fiber forces every value of a to be the appropriate unit object, so the second and third conditions agree. A map from units into the values of any algebra is determined by the unit maps of that algebra and has a contractible space of coherent choices. Hence a trivial algebra admits an essentially unique map to every algebra and is initial.

Conversely, an initial algebra agrees with the nullary left Kan extension, which exists because p has unit objects. □

Proposition 10.1.2 (Existence). For a fibration of ∞-operads over a unital $\mathcal{O}^{\otimes}$, the following conditions are equivalent.

1. The fibration p has unit objects.
2. The ∞-category $\mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathscr{C})$ contains a trivial algebra.

Whenever they hold, the trivial algebra is the initial algebra.

Corollary 10.1.1 (The Initial Commutative Algebra). Let $\mathscr{C}^{\otimes}$ be a symmetric monoidal ∞-category. Then $\mathbf{CAlg}(\mathscr{C})$ has an initial object. A commutative algebra a is initial if and only if its unit map

$$1 \to a$$

is an equivalence in $\mathscr{C}$. Thus the initial commutative algebra is the tensor unit equipped with its canonical commutative multiplication.

10.2 Limits of Algebras

The basic phenomenon already appears for ordinary commutative algebras. Let $a, b \in \mathbf{CAlg}(\mathscr{C})$ and suppose that the product $a \times b$ exists in $\mathscr{C}$. Its multiplication is the composite

$$(a \times b) \otimes (a \times b) \to (a \otimes a) \times (b \otimes b) \to a \times b$$

obtained from the projections and the multiplications of a and b. Hence the product in $\mathbf{CAlg}(\mathscr{C})$ is the product of the underlying objects. The same argument works for arbitrary operads, arbitrary diagram shapes, and relative limits.

10.2.1 Relative Limits

Let $\mathcal{O}^{\otimes}$ be an ∞-operad and let

$$p : \mathscr{C}^{\otimes} \to \mathscr{D}^{\otimes}$$

be a fibration of ∞-operads over $\mathcal{O}^{\otimes}$. It induces a functor

$$q : \mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathscr{C}) \to \mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathscr{D})$$

For each color $x \in \mathcal{O}$, write $p_x : \mathscr{C}_x \to \mathscr{D}_x$ for the induced functor between fibers.

Definition 10.2.4 (Relative Limit). A cone in $\mathscr{C}_x$ lying over a fixed cone in $\mathscr{D}_x$ is a *p_x-limit cone* if it is terminal among the cones with that fixed image in $\mathscr{D}_x$. Thus a relative limit is computed while the cone in the base is held fixed.

Theorem 10.2.2 (Fiberwise Criterion for Relative Limits). Let k be a simplicial set, let

$$d : k \to \mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathscr{C})$$

be a diagram, and fix a cone

$$e : k^{\triangleleft} \to \mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathcal{D})$$

extending $q \circ d$. Suppose that, for every color $x \in \mathcal{O}$, the evaluated diagram $d_x : k \to \mathcal{C}_x$ admits a p_x-limit cone lying over e_x. Then:

1. these fiberwise cones assemble into a cone $\overline{d} : k^{\triangleleft} \to \mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathcal{C})$ lying over e
2. a cone over d is a q-limit cone if and only if its evaluation at every color x is a p_x-limit cone

Remark (Relative versus Ordinary Limits). A p_x-limit is generally stronger than an ordinary limit in $\mathcal{C}_x$: its universal property is relative to the prescribed cone in $\mathcal{D}_x$. The two notions coincide when the base is terminal, which is the absolute situation below.

Proof sketch. Choose the relative limit object separately in each fiber. For an operation $\alpha : (x_1, \ldots, x_n) \to y$, the projections of the cones give compatible maps from the tuple of limit objects to every stage of the diagram in $\mathcal{C}_y$. The universal property of the p_y-limit produces the required structure map at y. Its coherence is forced because it can be checked after all cone projections. The same argument shows that the global cone is universal precisely when each evaluated cone is universal. □

10.2.2 Absolute Limits Are Computed Fiberwise

Now let

$$r : \mathcal{C}^{\otimes} \to \mathcal{O}^{\otimes}$$

be a coCartesian fibration of ∞-operads. For every color x, evaluation defines a functor

$$\mathrm{ev}_x : \mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathcal{C}) \to \mathcal{C}_x$$

Corollary 10.2.2 (Limits of Algebras). Let k be a simplicial set and suppose that every fiber $\mathcal{C}_x$ admits k-indexed limits. Then $\mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathcal{C})$ admits k-indexed limits, and a cone of algebras is a limit cone if and only if its evaluation in every $\mathcal{C}_x$ is a limit cone. In particular, every evaluation functor ev_x preserves these limits.

Remark (No Tensor-Preservation Hypothesis). Unlike colimits, this statement does not require the operadic tensor products to preserve limits. To define an operation on a limit, one maps from the tensor product of the limit objects to every stage and then uses the universal property of the target limit. For colimits the arrows point in the opposite direction, which is exactly why separate preservation by tensor products becomes necessary there.

Corollary 10.2.3 (Equivalences Are Detected Colorwise). A morphism $\eta : a \to b$ of $\mathcal{O}$-algebras is an equivalence if and only if

$$\eta_x : a(x) \to b(x)$$

> is an equivalence in $\mathscr{C}_x$ for every color $x \in \mathcal{O}$.

“To compute a limit of operadic algebras, forget to every color, take the limit in the corresponding fiber, and recover all operations from the universal property of those limits.”

10.3 Colimits of Algebras

For colimits the preceding argument cannot simply be reversed: an operation must carry the colimits of its inputs to the colimit of its outputs. We first treat sifted diagrams, where this compatibility gives a pointwise answer, and then obtain general colimits from free-algebra resolutions.

10.3.1 Sifted Colimits Are Computed Fiberwise

Let

$$p : \mathscr{C}^{\otimes} \to \mathcal{O}^{\otimes}$$

be a coCartesian fibration of ∞-operads, let k be a sifted simplicial set, and assume that p is compatible with k-indexed colimits: every fiber $\mathscr{C}_x$ admits them and every operadic tensor functor preserves them separately in each variable.

> **Theorem 10.3.3 (Pointwise Sifted Colimits).** The ∞-category $\mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathscr{C})$ admits k-indexed colimits. A cocone
>
> $$\overline{d} : k^{\triangleright} \to \mathbf{Alg}_{\mathcal{O}/\mathcal{O}}(\mathscr{C})$$
>
> is a colimit cocone if and only if, for every color $x \in \mathcal{O}$, its evaluation
>
> $$\overline{d}_x : k^{\triangleright} \to \mathscr{C}_x$$
>
> is a colimit cocone. Consequently the evaluation functors create these colimits.

> **Remark (Why Siftedness Is the Right Hypothesis).** For an n-ary operation, separate preservation gives colimits in one input at a time. Siftedness says that the diagonal $k \to k^n$ is cofinal, so these iterated colimits agree with the colimit of the original single k-diagram. This is precisely what lets the pointwise colimit inherit all n-ary operations.

Proof sketch. Form $c_x \coloneqq \operatorname*{colim}_{i \in k} d_i(x)$ in every fiber. For an operation $\alpha : (x_1, \ldots, x_n) \to y$, compatibility with colimits and cofinality of the diagonal give

$$\otimes_\alpha \left(c_{x_1}, \ldots, c_{x_n}\right) \simeq \operatorname*{colim}_{i \in k} \otimes_\alpha \left(d_i(x_1), \ldots, d_i(x_n)\right)$$

The algebra structures of the d_i therefore induce a structure map to c_y. Operadic coherence is inherited stagewise, and the same calculation identifies maps out of c with compatible cocones, proving universality. □

Corollary 10.3.4 (Sifted Colimits of Commutative Algebras). Let $\mathscr{C}^{\otimes}$ be symmetric monoidal and compatible with k-indexed sifted colimits. Then $\mathbf{CAlg}(\mathscr{C})$ admits these colimits, and the forgetful functor

$$U : \mathbf{CAlg}(\mathscr{C}) \to \mathscr{C}$$

creates them.

Remark (A Unital Improvement). Suppose that the structure map

$$\mathscr{O}^{\otimes} \to \mathbf{Comm}^{\otimes}$$

factors through $\mathbb{E}_0^{\otimes}$. Then every operation has arity at most one. In this case the same pointwise theorem holds for any weakly contractible indexing simplicial set k, provided the fiber transports preserve the relevant k-colimits. There are no genuinely multivariable operations, so siftedness is no longer needed; weak contractibility deals with the nullary operation.

10.3.2 General Colimits from Free Algebras

Theorem 10.3.4 (Existence of Small Colimits). Let κ be an uncountable regular cardinal. Suppose that $\mathscr{O}^{\otimes}$ is essentially κ-small and that $p : \mathscr{C}^{\otimes} \to \mathscr{O}^{\otimes}$ is compatible with κ-small colimits. Then $\mathbf{Alg}_{\mathscr{O}/\mathscr{O}}(\mathscr{C})$ admits all κ-small colimits.

Proof sketch. The forgetful functor from algebras with only their underlying colored objects has the free-algebra functor as a left adjoint. Every algebra is the geometric realization of its simplicial free-algebra resolution. Sifted colimits and geometric realizations are computed fiberwise by the preceding theorem, while coproducts of free algebras exist because the free functor preserves colimits. Combining geometric realizations, sifted colimits, and finite coproducts constructs every κ-small colimit. □

Remark (Why the General Formula Is Less Explicit). The theorem gives existence but usually not a short pointwise formula for an arbitrary coproduct of algebras. The free resolution replaces the original diagram by one whose coproduct is easy, then imposes the algebraic relations through a geometric realization.

10.3.3 Accessibility and Presentability

Theorem 10.3.5 (Presentability Criterion). Assume that $\mathscr{O}^{\otimes}$ is essentially small.

1. If every fiber $\mathscr{C}_x$ is accessible and all transport functors induced by morphisms of $\mathscr{O}^{\otimes}$ are accessible, then $\mathbf{Alg}_{\mathscr{O}/\mathscr{O}}(\mathscr{C})$ is accessible.
2. If every fiber $\mathscr{C}_x$ is presentable and p is compatible with all small colimits, then $\mathbf{Alg}_{\mathscr{O}/\mathscr{O}}(\mathscr{C})$ is presentable.

Corollary 10.3.5 (The Standard Symmetric Monoidal Case). If $\mathscr{C}$ is a presentable ∞-category and its symmetric monoidal tensor product preserves small colimits separately in each variable, then $\mathbf{CAlg}(\mathscr{C})$ is presentable. More generally, the same holds for $\mathscr{O}$-algebras whenever $\mathscr{O}^{\otimes}$ is essentially small.

> Sifted colimits of algebras are computed colorwise; arbitrary colimits are built by resolving into free algebras; presentability follows when these constructions are accessible.

10.4 Tensor Products of Commutative Algebras

The preceding existence theorem constructs coproducts indirectly. This is unavoidable for associative algebras: their coproduct is a free product and has no simple formula in terms of the underlying objects. For commutative algebras, however, the tensor product already has the coproduct universal property. We first record the operadic construction which produces this tensor product.

10.4.1 A Family of Algebra Categories

Let

$$f : \mathscr{O}^{\otimes} \times \mathscr{P}^{\otimes} \to \mathscr{Q}^{\otimes}$$

be an operadic bifunctor and let

$$q : \mathscr{C}^{\otimes} \to \mathscr{Q}^{\otimes}$$

be a fibration of ∞-operads. For a color $y \in \mathscr{P}$, write $f_y : \mathscr{O}^{\otimes} \to \mathscr{Q}^{\otimes}$ for the restriction of f to $\mathscr{O}^{\otimes} \times \{y\}$.

Definition 10.4.5 (The Parametrized Algebra Operad). The ∞-operad

$$\mathbf{Alg}_f(\mathscr{C})^{\otimes} \to \mathscr{P}^{\otimes}$$

is the family whose fiber over y is the ∞-category of lifts a in the diagram

$$\begin{tikzcd} \mathscr{O}^{\otimes} \arrow[r, "a"] \arrow[d, "f_y"'] & \mathscr{C}^{\otimes} \arrow[d, "q"] \\ \mathscr{Q}^{\otimes} \arrow[r, "\mathrm{id}"] & \mathscr{Q}^{\otimes} \end{tikzcd}$$

Thus its objects over y are $\mathscr{O}$-algebra objects in $\mathscr{C}$ whose structural map to $\mathscr{Q}^{\otimes}$ is prescribed by f_y. Transport in the $\mathscr{P}$-direction varies this algebra structure functorially.

Proposition 10.4.3 (Operadic Structure of the Family). The projection $\mathbf{Alg}_f(\mathscr{C})^{\otimes} \to \mathscr{P}^{\otimes}$ is a fibration of ∞-operads. Moreover:

1. a morphism is inert precisely when its image in $\mathscr{P}^\otimes$ is inert and its value at every color of $\mathcal{O}$ is inert in $\mathscr{C}^\otimes$;
2. if q is a coCartesian fibration, then so is $\mathbf{Alg}_f(\mathscr{C})^\otimes \to \mathscr{P}^\otimes$;
3. in that case, a morphism is coCartesian precisely when it is q-coCartesian after evaluation at every color of $\mathcal{O}$.

Proof sketch. The Segal decomposition reduces an object and a morphism over $\langle n \rangle$ to their values at the n inert projections. Consequently inertness and the coCartesian mapping-space criterion can be tested color by color. CoCartesian lifts are obtained by lifting the corresponding component maps in $\mathscr{C}^\otimes$ and then reassembling them by the same Segal equivalence. □

10.4.2 Pointwise Tensor Products of Algebras

There is a unique operadic bifunctor

$$\mathbf{Comm}^\otimes \times \mathcal{O}^\otimes \to \mathbf{Comm}^\otimes$$

for every ∞-operad $\mathcal{O}^\otimes$. Applying the preceding construction to a symmetric monoidal ∞-category $\mathscr{C}^\otimes$ gives an $\mathcal{O}$-monoidal structure on $\mathbf{Alg}_{\mathcal{O}}(\mathscr{C})$.

Proposition 10.4.4 (Pointwise Tensor Product). The induced tensor product is evaluated colorwise. For $a, b \in \mathbf{Alg}_{\mathcal{O}}(\mathscr{C})$ and every color $x \in \mathcal{O}$,

$$(a \otimes b)(x) \simeq a(x) \otimes b(x)$$

The unit is the constant algebra determined by $1_{\mathscr{C}}$, and every evaluation functor

$$\mathrm{ev}_x : \mathbf{Alg}_{\mathcal{O}}(\mathscr{C}) \to \mathscr{C}$$

is symmetric monoidal.

Remark (Why Commutativity Matters). Pointwise multiplication uses the interchange map which lets an $\mathcal{O}$-operation pass through the tensor product. For associative algebras, two maps into a target need not commute, so their coproduct is a free product. In a commutative target the required interchange is automatic; this is what turns the pointwise tensor product into a coproduct.

10.4.3 Tensor Product Is Coproduct

Theorem 10.4.6 (Tensor Products of Commutative Algebras). Let $\mathscr{C}^\otimes$ be a symmetric monoidal ∞-category. The pointwise symmetric monoidal structure on $\mathbf{CAlg}(\mathscr{C})$ is coCartesian. Hence $\mathbf{CAlg}(\mathscr{C})$ admits finite coproducts, its initial object is the unit algebra $1_{\mathscr{C}}$, and

$$a \sqcup_{\mathbf{CAlg}(\mathscr{C})} b \simeq a \otimes b$$

for all commutative algebra objects a, b.

Proof sketch. The multiplication of a commutative algebra a gives a natural fold map

$$\delta_a : a \otimes a \to a$$

and its unit gives $1_{\mathscr{C}} \to a$. The exponential law

$$\mathbf{CAlg}(\mathbf{CAlg}(\mathscr{C})) \simeq \mathbf{CAlg}(\mathscr{C})$$

makes these fold maps coherent. Commutativity, associativity, and unitality then verify the intrinsic criterion for a coCartesian symmetric monoidal structure. Consequently

$$\mathrm{Map}_{\mathbf{CAlg}(\mathscr{C})}(a \otimes b, c) \simeq \mathrm{Map}_{\mathbf{CAlg}(\mathscr{C})}(a, c) \times \mathrm{Map}_{\mathbf{CAlg}(\mathscr{C})}(b, c)$$

which is exactly the coproduct universal property. □

Corollary 10.4.6 (A Concrete Coproduct Formula). Whenever the underlying tensor product is familiar, it directly computes coproducts of commutative algebras. For example, for commutative rings,

$$a \sqcup b \simeq a \otimes_{\mathbb{Z}} b$$

while for commutative algebra spectra it is the smash product $a \wedge_{\mathbb{S}} b$ over the sphere spectrum.

Remark (Maps out of a coCartesian Monoidal Category). Let $\mathscr{B}$ admit finite coproducts. A functor $g : \mathscr{B} \to \mathscr{D}$ defines a symmetric monoidal functor from $\mathscr{B}^{\sqcup}$ precisely when it preserves finite coproducts. Thus, once $\mathbf{CAlg}(\mathscr{C})$ is equipped with the tensor product above, a symmetric monoidal functor out of it is the same thing as a functor preserving its initial object and binary coproducts.

“ In commutative algebra, tensor product is not merely compatible with coproducts: it is the coproduct. ”

11 Modules

Let $\mathscr{C}^{\otimes}$ be a symmetric monoidal category and let $a \in \mathbf{Alg}(\mathscr{C})$ be an associative unital algebra object, with multiplication $\mu : a \otimes a \to a$ and unit $\eta : 1 \to a$. A *left a-module* is an object $m \in \mathscr{C}$ equipped with an action

$$\lambda : a \otimes m \to m$$

which is associative and unital. Concretely, the following diagrams commute.

$$\begin{tikzcd} a \otimes a \otimes m \arrow[r, "\mu \otimes \mathrm{id}_m"] \arrow[d, "\mathrm{id}_a \otimes \lambda"'] & a \otimes m \arrow[d, "\lambda"] \\ a \otimes m \arrow[r, "\lambda"'] & m \end{tikzcd} \qquad \begin{tikzcd} 1 \otimes m \arrow[r, "\eta \otimes \mathrm{id}_m"] \arrow[dr, "\simeq"'] & a \otimes m \arrow[d, "\lambda"] \\ & m \end{tikzcd}$$

Dually, a right a-module has an action $\rho : m \otimes a \to m$. We write $\mathbf{LMod}_a(\mathscr{C})$ and $\mathbf{RMod}_a(\mathscr{C})$ for the corresponding categories. If a is commutative, the symmetry of $\mathscr{C}^{\otimes}$ canonically converts left actions into right actions and conversely; in this case we simply write $\mathbf{Mod}_a(\mathscr{C})$.

The basic operation on modules is the *relative tensor product*. Given a right a-module m and a left a-module n, it is classically defined by the balanced coequalizer

$$m \otimes a \otimes n \rightrightarrows m \otimes n \to m \times_a n$$

where the two parallel arrows are $\rho \otimes \mathrm{id}_n$ and $\mathrm{id}_m \otimes \lambda$. When a is commutative, this operation often makes $\mathbf{Mod}_a(\mathscr{C})$ into a symmetric monoidal category whose unit is a.

> **Remark (The Homotopy-Correct Construction).** The displayed coequalizer is only the classical shadow. In an ∞-category the relative tensor product is computed by the geometric realization of the two-sided bar construction
>
> $$m \times_a n \simeq |\mathbf{Bar}_\bullet(m, a, n)|, \quad \mathbf{Bar}_k(m, a, n) = m \otimes a^{\otimes k} \otimes n$$
>
> This requires the relevant geometric realizations to exist and tensoring to preserve them separately in each variable. Without these hypotheses, $m \times_a n$ need not exist, and $\mathbf{Mod}_a(\mathscr{C})$ need not inherit the expected monoidal structure.

11.1 Coherent ∞-Operads

The module construction above needs more than associativity of the operad. To form relative tensor products and compose module actions, one must be able to insert an extra input into an operation and make these insertions compatible with composition. A *coherent ∞-operad* is precisely a unital operad for which these spaces of extensions glue by homotopy pushouts.

11.1.1 Semi-Inert Morphisms

Definition 11.1.1 (Semi-Inert and Null Morphisms). A morphism $\alpha : \langle m \rangle \to \langle n \rangle$ in $\mathbf{Fin}_*$ is *semi-inert* if every non-basepoint $j \in \langle n \rangle^{\circ}$ has at most one inverse image. It is *null* if every element of $\langle m \rangle$ is sent to the basepoint.

Let $p : \mathcal{O}^{\otimes} \to \mathrm{N}_{\bullet}(\mathbf{Fin}_*)$ be an ∞-operad. A morphism $u : x \to y$ of $\mathcal{O}^{\otimes}$ is semi-inert if $p(u)$ is semi-inert and, for every inert $v : y \to z$, the composite $v \circ u$ is inert whenever its image under p is inert. It is null if $p(u)$ is null.

Remark (Input–Output Meaning). Write $x = (x_1, \ldots, x_m)$ and $y = (y_1, \ldots, y_n)$. A morphism over $\alpha : \langle m \rangle \to \langle n \rangle$ is specified by operations

$$\varphi_j \in \mathrm{Mul}_{\mathcal{O}}\left(\{x_i\}_{\alpha(i)=j}, y_j\right)$$

It is semi-inert exactly when, for every output y_j, either no input is assigned to it, or a unique input x_i is assigned and $\varphi_j : x_i \to y_j$ is an equivalence. Thus semi-inert maps may *insert empty input slots* inert maps are those with no empty output slot, while null maps have only empty slots. In a unital operad the operations belonging to the empty slots form contractible spaces.

Semi-inertness is insensitive to an inert change of coordinates: in a commutative triangle $h = v \circ u$, if u is inert, then v is semi-inert if and only if h is semi-inert.

11.1.2 Extension Spaces

Definition 11.1.2 (Extension of an Active Morphism). Let $f : x \to y$ be active. An *extension of f* is a commutative square

$$\begin{tikzcd} x \arrow[r, "f"] \arrow[d, "i"'] & y \arrow[d, "u"] \\ x' \arrow[r, "f'"'] & y' \end{tikzcd}$$

in which i is semi-inert and inserts exactly one new input, u is an equivalence, and f' is active. The full ∞-category of such squares and their equivalences is denoted $\mathbf{Ext}(f)$.

Remark (Meaning of an Extension). Since u is an equivalence, we may identify y' with y. If x_0 is the new (single) input omitted by i, then we may similarly read x' as $x \oplus x_0$. The defining square becomes

$$\begin{tikzcd} x \arrow[r, "f"] \arrow[d, "i"'] & y \arrow[d, "\mathrm{id}_y"] \\ x \oplus x_0 \arrow[r, "f'"'] & y \end{tikzcd}$$

Thus an extension is precisely a coherent way to insert one extra input into f. More generally, for a chain σ of active morphisms and a downward-closed subset s of its vertices, i.e. a morphism

$$\sigma : \Delta^n \to \mathcal{O}^{\otimes}_{\mathrm{act}}$$

$\mathbf{Ext}(\sigma, s)$ records compatible extension squares in which the new input is inserted at the vertices in s and transported through the remaining active arrows. i.e.

$$\begin{tikzcd} \sigma : & x_0 \arrow[r, "f_1"] \arrow[d, "g_0"', "\simeq"] & x_1 \arrow[r, "f_2"] \arrow[d, "g_1"', "\simeq"] & \dots \arrow[r, "f_r"] & x_r \arrow[r, "f_{r+1}"] \arrow[d, "g_r"'] & \dots \arrow[r, "f_n"] & x_n \arrow[d, "g_n", "\simeq"'] \\ & x_0' \arrow[r, "f_1'"'] & x_1' \arrow[r, "f_2'"'] & \dots \arrow[r, "f_r'"'] & x_r \oplus x_{r,0} \arrow[r, "f_{r+1}'"'] & \dots \arrow[r, "f_n'"'] & x_n' \end{tikzcd}$$

$$(r \in s)$$

These categories supply the restriction maps needed for higher coherence. If the color category $\mathcal{O}$ is Kan, every morphism in $\mathbf{Ext}(\sigma, s)$ is an equivalence, so it is a Kan complex.

Definition 11.1.3 (Coherent ∞-Operad). An ∞-operad $\mathcal{O}^{\otimes}$ is *coherent* if:

1. $\mathcal{O}^{\otimes}$ is unital
2. its underlying ∞-category of colors $\mathcal{O}$ is a Kan complex
3. for every composable pair of active morphisms $x \xrightarrow{f} y \xrightarrow{g} z$, the homotopy-coherent extension diagram

$$\begin{tikzcd} \mathbf{Ext}(\mathrm{id}_y) \arrow[r] \arrow[d] & \mathbf{Ext}(g) \arrow[d] \\ \mathbf{Ext}(f) \arrow[r] & \mathbf{Ext}(g \circ f) \end{tikzcd}$$

   is a homotopy pushout square of Kan complexes.

Remark (What the Pushout Says). An extension of the composite $g \circ f$ can be produced in two ways: insert the extra input at the f-stage, or insert it at the g-stage. Their common degenerate case is an extension of id_y. Coherence says that these are all the possibilities and that their identifications contain no hidden higher ambiguity.

Restriction along a morphism of simplices gives maps between extension spaces. In particular, extensions of a longer active chain restrict to extensions of any initial segment; when $\mathcal{O}$ is Kan and the segment has a largest selected vertex, the corresponding restriction is a trivial fibration. This is the mechanism that makes iterated module constructions independent of parenthesization.

11.1.3 Examples and Invariance

Proposition 11.1.1 (Basic Coherent Operads). The commutative operad $\mathbf{Comm}^{\otimes}$ and the operad $\mathbb{E}_0^{\otimes}$ are coherent. More generally, the little-cubes operads $\mathbb{E}_k^{\otimes}$ are coherent for every $k \geq 0$.

Proof sketch. For $\mathbf{Comm}^{\otimes}$, every operation is determined by its underlying map of finite pointed sets, so each extension space is equivalent to the discrete set of possible new input positions; the coherence square becomes an elementary pushout of finite sets. The same computation for $\mathbb{E}_0^{\otimes}$ allows only nullary and unary active operations. The little-cubes case is a geometric refinement: configuration spaces replace finite sets, and the required square is obtained by gluing configurations according to where the new cube is inserted. □

Proposition 11.1.2 (Invariance under Approximation). Let $F : \mathcal{O}^{\otimes} \to \mathcal{O}'^{\otimes}$ be a map of unital ∞-operads whose color categories are Kan complexes, and suppose that F is an approximation. If $\mathcal{O}'^{\otimes}$ is coherent, then $\mathcal{O}^{\otimes}$ is coherent. Conversely, coherence descends from $\mathcal{O}^{\otimes}$ to $\mathcal{O}'^{\otimes}$ when the induced map $\pi_0\mathcal{O} \to \pi_0\mathcal{O}'$ is surjective.

Proof sketch. An approximation induces equivalences on the extension spaces $\mathbf{Ext}(\sigma, s)$. It therefore transports the defining homotopy-pushout square. Surjectivity on connected components ensures that every square in the target is represented by one in the source. □

Remark (Why Coherence Matters for Modules). If a unital operad with Kan color space is presented as the assembly of a family of reduced operads $\left(\mathcal{O}_x^{\otimes}\right)_{x\in\mathcal{O}}$, then it is coherent exactly when every fiber $\mathcal{O}_x^{\otimes}$ is coherent. Hence an $\mathcal{O}$-algebra $a = (a_x)$ in an $\mathcal{O}$-monoidal ∞-category gives a compatible family of module categories

$$\left(\mathbf{Mod}_{a_x}^{\mathcal{O}_x}(\mathcal{C}_x)\right)_{x\in\mathcal{O}}$$

Coherence is the condition that lets these fiberwise module theories glue and that makes their relative tensor products associative up to all higher homotopies. It is therefore the operadic hypothesis behind a usable theory of modules over $\mathcal{O}$-algebras.

11.1.4 A Coherence Criterion

The definition by extension spaces is conceptual, but it appears to require checking a homotopy-pushout square for every pair of active operations. There is a more economical criterion involving only the source of a semi-inert morphism.

Definition 11.1.4 (The Semi-Inert Arrow Category). Let $\mathcal{O}^{\otimes}$ be a unital ∞-operad. Define

$$\mathcal{K}_{\mathcal{O}} \subseteq \mathbf{Fun}(\Delta^1, \mathcal{O}^{\otimes})$$

to be the full subcategory spanned by the semi-inert morphisms, and let

$$e_0, e_1 : \mathcal{K}_{\mathcal{O}} \to \mathcal{O}^{\otimes}$$

be evaluation at the source and target. A morphism of $\mathcal{K}_{\mathcal{O}}$ is called inert if both of its endpoint evaluations are inert.

Remark (Flat Isofibrations). An isofibration $r : x \to s$ is *flat* if pullback along r preserves categorical equivalences over s. Thus flatness says that the fibers of r vary homotopy-exactly under every base change; it is the ∞-categorical replacement for a well-behaved family of categories.

Theorem 11.1.1 (Coherence Criterion). Let $\mathcal{O}^{\otimes}$ be unital and suppose that its color category $\mathcal{O}$ is a Kan complex. Then the following are equivalent:

1. $\mathcal{O}^{\otimes}$ is coherent
2. the source-evaluation map

$$e_0 : \mathcal{K}_{\mathcal{O}} \to \mathcal{O}^{\otimes}$$

   is a flat isofibration

This reformulates coherence as a base-change property of a single functor. Informally, once the source of a semi-inert arrow is moved through an active operation, flatness guarantees that the space of compatible targets remains homotopically correct.

11.1.4.1 Reduction to One Missing Input

Definition 11.1.5 (m-Semi-Inert Morphism). A semi-inert morphism $u : x \to x'$ is *m-semi-inert* if the underlying pointed map

$$p(u) : \langle n \rangle \to \langle n' \rangle$$

leaves at most m non-basepoint elements of $\langle n' \rangle$ outside its image. We say that $\mathcal{O}^{\otimes}$ is *m-coherent* when the flatness condition for e_0 is required only over such arrows.

Thus 0-semi-inert arrows are inert up to equivalence, while a 1-semi-inert arrow inserts at most one genuinely new input. Although flatness initially asks for every m, the unital structure lets us add the missing inputs one at a time.

Proposition 11.1.3 (The One-Input Test). For a unital ∞-operad, the following conditions are equivalent:

1. $e_0 : \mathcal{K}_{\mathcal{O}} \to \mathcal{O}^{\otimes}$ is a flat isofibration
2. $\mathcal{O}^{\otimes}$ is m-coherent for every $m \geq 0$
3. $\mathcal{O}^{\otimes}$ is 1-coherent

Proof sketch. Factor an m-semi-inert morphism into a chain in which each step inserts one missing output. The relevant extension categories for successive steps form flat families. Weak contractibility for the one-input fibers therefore propagates inductively from $m-1$ to m. The case $m=0$ is automatic from inert lifting. □

11.1.4.2 A Local Contractibility Test

Fix an active cocone to a color z and a diagram

$$\begin{tikzcd} x \arrow[r] \arrow[d, "u"'] \arrow[drr, bend right, "\text{active}"] & y \arrow[dr, "\text{active}"] & \\ x' \arrow[rr, "\text{active}"'] & & z \end{tikzcd}$$

where u is semi-inert. Let $\mathscr{B}[\sigma, z]$ be the ∞-category of factorizations obtained by adjoining a square

$$\begin{tikzcd} x \arrow[r] \arrow[d, "u"'] & y \arrow[d, "v"] & \\ x' \arrow[r] & y' \arrow[r, "\text{active}"] & z \end{tikzcd}$$

such that v is semi-inert and the induced pointed map

$$p(x') \sqcup_{p(x)} p(y) \to p(y')$$

is surjective. Objects of $\mathscr{B}[\sigma, z]$ are precisely the compatible ways to transport the newly inserted inputs across the active map $x \to y$.

Proposition 11.1.4 (Local Form of Flatness). The source evaluation e_0 is flat if and only if every category $\mathscr{B}[\sigma, z]$ above is weakly contractible. It is enough to check this when u is 1-semi-inert.

Remark (Why Surjectivity Appears). Semi-inertness controls how many old inputs reach each output, while the surjectivity condition says that every output of y' is accounted for by either x' or y. Together they rule out an additional, invisible input appearing during transport.

Lemma 11.1.1 (Flat Families with Contractible Fibers). Let $r : x \to s$ be a flat inner fibration. Assume that every fiber x_t is weakly contractible and that, for every vertex $a \in x$, the induced map of relative overcategories

$$x_{a/} \to s_{r(a)/}$$

has weakly contractible fibers. Then every base change $x \times_s s' \to s'$ is a weak homotopy equivalence. In particular, r is a weak homotopy equivalence.

Proof sketch. Build s' simplex by simplex. Flatness identifies the new pullback as a homotopy pushout. In dimensions greater than one the horn attachment is a categorical equivalence;

dimensions zero and one reduce respectively to the contractibility of the fibers and of the relative overcategories. □

Proof sketch. Proof of the coherence criterion. The formal coherence square of extension spaces is a square of Kan fibrations. Taking a fiber at a chosen extension converts its homotopy pushout into one of the local categories $\mathscr{B}[\sigma, z]$. The comparison maps are trivial Kan fibrations, so the square is a homotopy pushout exactly when this local category is weakly contractible. The one-input test reduces the latter condition to 1-semi-inert arrows, while the local flatness proposition identifies it with flatness of e_0. □

> “Coherence is a global compatibility law, but it is detected locally by transporting one new input through one active operation.”

11.2 Module Objects

Let a be an ordinary associative algebra. A left a-module is an object m equipped with an action $a \otimes m \to m$. The essential point is not the binary map itself, but what happens under iteration: in every composite there is exactly one input belonging to m, while all remaining inputs belong to a. The algebra inputs may branch and multiply, but the module input follows one distinguished path through the whole operation tree.

The same idea works for an arbitrary operad. Let $p : \mathscr{C}^{\otimes} \to \mathscr{O}^{\otimes}$ be an $\mathscr{O}$-monoidal ∞-category and let $a \in \mathbf{Alg}_{\mathscr{O}}(\mathscr{C})$. For an operation

$$\alpha \in \mathrm{Mul}_{\mathscr{O}}\Big(\{x_j\}_{1 \le j \le n}; y\Big)$$

and a chosen input i, an $\mathscr{O}$-module m over a should supply an action of the form

$$a(x_1) \otimes \ldots \otimes m(x_i) \otimes \ldots \otimes a(x_n) \xrightarrow{\lambda_{\alpha,i}} m(y)$$

These maps must be compatible with equivalences, units and operadic substitution. Under substitution, the distinguished input of the outer operation is replaced by the distinguished input of the inner operation; all other branches are evaluated by the algebra a.

> “An $\mathscr{O}$-module is an $\mathscr{O}$-algebra operation with exactly one input declared to be the module variable; composition follows that marked input and evaluates every side branch in the algebra.”

> **Definition 11.2.6 ($\mathscr{O}$-Module Object over an Algebra).** Assume that $p : \mathscr{C}^{\otimes} \to \mathscr{O}^{\otimes}$ is coCartesian, and let $a \in \mathbf{Alg}_{\mathscr{O}}(\mathscr{C})$. An *$\mathscr{O}$-module object over a* consists of the following data.
>
> 1. For every color $x \in \mathscr{O}$, an object
>
> $$m(x) \in \mathscr{C}_x$$
>
> 2. For every $n \ge 1$, every operation
>
> $$\alpha \in \mathrm{Mul}_{\mathscr{O}}\Big(\{x_j\}_{1 \le j \le n}; y\Big)$$

and every distinguished input $1 \le i \le n$, an action map in $\mathscr{C}_y$

$$\lambda_{\alpha,i} : \otimes_\alpha \left(a(x_1), \ldots, a(x_{i-1}), m(x_i), a(x_{i+1}), \ldots, a(x_n)\right) \to m(y)$$

Here

$$\otimes_\alpha : \prod_{1 \le j \le n} \mathscr{C}_{x_j} \to \mathscr{C}_y$$

is the operadic tensor functor classified by the coCartesian lift of α. Thus every input except the ith is evaluated in the algebra a, while the ith input and the output are evaluated in the module m.

3. These maps are equivariant under permutations of the inputs. For an identity operation id_x, the map $\lambda_{\mathrm{id}_x,1}$ is equivalent to $\mathrm{id}_{m(x)}$.
4. They are coherently compatible with substitution: after substituting operations into α, the unique inner operation containing the distinguished input is evaluated by a module action λ, while all other inner operations are evaluated by the algebra structure of a. The resulting composite agrees coherently with the action associated to the substituted operation.

There is no marked action for a nullary operation, since it has no input to distinguish. Nullary operations still enter through the algebra a when they occur on an unmarked side branch.

In the one-colored case, writing a for the algebra and m for the module, an n-ary operation α with marked input i therefore determines

$$\lambda_{\alpha,i} : a^{\otimes(i-1)} \otimes m \otimes a^{\otimes(n-i)} \to m$$

We denote the resulting ∞-category of module objects and compatible module maps by $\mathbf{Mod}_a^{\mathscr{O}}(\mathscr{C})$.

Remark (First Examples). For the commutative operad, the choice of the marked slot is immaterial and this recovers the usual notion of a module over a commutative algebra. For an associative operad, the order of the marked slot records the appropriate one-sided or two-sided action data prescribed by that operad. At the other extreme, $\mathbb{E}_0$ has no nontrivial multi-input operation, so an $\mathbb{E}_0$-module is only an object of the ambient category.

Remark (Why Semi-Inert Arrows Appear). The preceding action formula is the intuition, but by itself it does not record all higher coherences. A semi-inert arrow is precisely the combinatorial device that moves one marked input without copying it. Its occupied slots are evaluated by a, while its unique unoccupied slot is filled by m. Thus semi-inert arrows package all maps $\lambda_{\alpha,i}$ and all compatibilities among them into a single operadic object.

11.2.1 From Semi-Inert Arrows to Partial Algebras

Retain the semi-inert arrow category $\mathscr{K}_{\mathscr{O}}$ and its evaluations

$$e_0, e_1 : \mathscr{K}_{\mathscr{O}} \to \mathscr{O}^{\otimes}$$

Let $\mathcal{K}^0_{\mathcal{O}}$ be its full subcategory spanned by the null arrows. A null arrow forgets every ordinary input; unitality gives the basic contractibility statement

Lemma 11.2.2 (Null Arrows). If $\mathcal{O}^{\otimes}$ is unital, then

$$(e_0, e_1) : \mathcal{K}^0_{\mathcal{O}} \to \mathcal{O}^{\otimes} \times \mathcal{O}^{\otimes}$$

is a trivial Kan fibration. In particular, a null arrow with prescribed source and target is unique up to a contractible space of choices.

A *partial $\mathcal{O}$-algebra* in $\mathcal{C}$ records the algebra operations seen along these null arrows. Denote its ∞-category by $\mathbf{PAlg}_{\mathcal{O}}(\mathcal{C})$. Restricting a genuine algebra to null arrows gives a canonical functor

$$\mathbf{Alg}_{\mathcal{O}}(\mathcal{C}) \to \mathbf{PAlg}_{\mathcal{O}}(\mathcal{C})$$

The semi-inert construction produces an ∞-category $\overline{\mathbf{Mod}}^{\mathcal{O}}(\mathcal{C})^{\otimes}$ over $\mathcal{O}^{\otimes}$: its points are compatible lifts to $\mathcal{C}^{\otimes}$ of the semi-inert arrows in $\mathcal{K}_{\mathcal{O}}$. We retain only those points that carry inert morphisms of $\mathcal{K}_{\mathcal{O}}$ to inert morphisms of $\mathcal{C}^{\otimes}$.

Remark (What Is Being Marked?). Over an object $(x_1, \dots, x_n)$, a semi-inert arrow may leave one output without an old input. That empty position is the future module variable; all occupied positions are algebra variables. Transport along $\mathcal{K}_{\mathcal{O}}$ remembers coherently which slot is the module slot.

11.2.2 The Module Operad over an Algebra

Definition 11.2.7 (Module Operad). Let $a \in \mathbf{Alg}_{\mathcal{O}}(\mathcal{C})$. The *$\mathcal{O}$-module operad over a*, denoted $\mathbf{Mod}^{\mathcal{O}}_a(\mathcal{C})^{\otimes}$, is the ∞-operad whose objects are $\mathcal{O}$-module objects over a and whose active morphisms are the coherently compatible marked action maps $\lambda_{\alpha,i}$.

Precisely, it is obtained by the fiber product

$$\mathbf{Mod}^{\mathcal{O}}_a(\mathcal{C})^{\otimes} := \overline{\mathbf{Mod}}^{\mathcal{O}}(\mathcal{C})^{\otimes} \times_{\mathbf{PAlg}_{\mathcal{O}}(\mathcal{C})} \{a\}$$

The map to $\mathbf{PAlg}_{\mathcal{O}}(\mathcal{C})$ forgets the marked module input and remembers the algebra inputs. Pulling back along a therefore fixes every unmarked input to be a. Its fiber over a color $x \in \mathcal{O}$ is denoted $\mathbf{Mod}^{\mathcal{O}}_{a,x}(\mathcal{C})$. Its objects are the possible module objects of color x, and its active operations encode the action of the algebra inputs on the distinguished module input.

Thus the construction separates two kinds of data

$$\begin{tikzcd}
a \arrow[r, "\text{algebra operations}"] & \mathbf{PAlg}_{\mathcal{O}}(\mathcal{C}) \\
\mathbf{Mod}^{\mathcal{O}}_a(\mathcal{C})^{\otimes} \arrow[u] \arrow[r] & \overline{\mathbf{Mod}}^{\mathcal{O}}(\mathcal{C})^{\otimes} \arrow[u, "\text{restrict to null arrows}"']
\end{tikzcd}$$

The square is a pullback: fixing a fixes the ordinary algebra inputs, while the remaining variable is the module object.

Remark (How to Read the Construction). The fiber product can be read in four steps.

1. A semi-inert arrow chooses a single *hole* among the inputs of an $\mathcal{O}$-operation. The hole is never duplicated, so it can be followed through a composite operation.
2. A lift to $\mathscr{C}^{\otimes}$ decorates every occupied input by algebra data and decorates the hole by a variable object m. Before choosing a, this is the space $\overline{\mathbf{Mod}}^{\mathcal{O}}(\mathscr{C})^{\otimes}$ of all such decorated marked operations.
3. Restriction to null arrows forgets the moving hole and reads only the algebra decoration. The fiber product with $\{a\}$ imposes the boundary condition that this decoration is the fixed algebra a. Thus the fiber product does not create an action: it selects exactly those marked operations whose unmarked branches already use the multiplication of a.
4. What remains free is the object occupying the hole. An active marked operation therefore has the form

$$\begin{tikzcd}
a(x_1) \arrow[dr] & m(x_i) \arrow[d, "\text{marked}"] & a(x_n) \arrow[dl] \\
& (\alpha, i) \arrow[d, "\lambda_{\alpha,i}"'] & \\
& m(y) &
\end{tikzcd}$$

When marked operations are substituted, the marked edge is joined to the next marked edge, while every side branch is evaluated by a. Coherence of $\mathcal{O}^{\otimes}$ says that all ways of performing these substitutions give the same result up to a contractible space of choices. This is exactly why the selected diagrams assemble into an ∞-operad.

Remark (A Binary Example). For a binary operation μ, choosing the second input gives

$$a \otimes m \xrightarrow{\lambda_{\mu,2}} m$$

Substituting multiplication into the algebra input and substituting another marked action into the module input produce the two sides of the familiar associativity law. The module axiom is therefore not added separately: it is the image of operadic substitution inside $\mathbf{Mod}_a^{\mathcal{O}}(\mathscr{C})^{\otimes}$.

Theorem 11.2.2 (Existence of the Module Operad). Suppose that $\mathcal{O}^{\otimes}$ is coherent and that $p : \mathscr{C}^{\otimes} \to \mathcal{O}^{\otimes}$ is a fibration of ∞-operads. For every algebra $a \in \mathbf{Alg}_{\mathcal{O}}(\mathscr{C})$, the projection

$$\mathbf{Mod}_a^{\mathcal{O}}(\mathscr{C})^{\otimes} \to \mathcal{O}^{\otimes}$$

is a fibration of ∞-operads. Hence modules over a, together with all their operadic action maps, form an honest ∞-operad rather than merely a simplicial set of diagrams.

The role of coherence is now concrete: by the coherence criterion, $e_0 : \mathscr{K}_{\mathcal{O}} \to \mathcal{O}^{\otimes}$ is flat. Consequently semi-inert lifts survive base change and compose with the homotopy-exactness required by the operad axioms.

11.2.3 Detecting Inert Morphisms

Proposition 11.2.5 (Inertness in the Module Operad). A morphism f in $\mathbf{Mod}_a^{\mathcal{O}}(\mathscr{C})^{\otimes}$ is inert if and only if:

1. its underlying morphism f_0 in $\mathcal{O}^{\otimes}$ is inert
2. for every e_0-coCartesian lift $\tilde{f}$ of f_0 in $\mathscr{K}_{\mathcal{O}}$, the endpoint morphism in $\mathscr{C}^{\otimes}$ classified by f and evaluated at $\tilde{f}$ is inert

Remark (Meaning of the Second Test). The first condition checks the ordinary algebra slots. The second moves the distinguished module slot through every compatible semi-inert lift and checks it there. Thus an inert module morphism is inert in both the algebra directions and the marked module direction.

Proof sketch. The null-arrow lemma first makes the partial-algebra construction an ∞-category over $\mathcal{O}^{\otimes}$. Coherence makes e_0 a flat isofibration, so pullback along semi-inert arrows preserves the categorical equivalences used in the operadic Segal conditions. The displayed inertness criterion is precisely the condition ensuring that these pullbacks preserve inert edges. Restricting to the fiber over a then gives the stated fibration of ∞-operads. □

11.2.4 The $\mathbb{E}_0$ Sanity Check

Proposition 11.2.6 ($\mathbb{E}_0$-Modules). Let $q : \mathscr{C}^{\otimes} \to \mathbb{E}_0^{\otimes}$ be a fibration of ∞-operads. For every $\mathbb{E}_0$-algebra a, evaluation at the distinguished module input induces an equivalence

$$\mathbf{Mod}_a^{\mathbb{E}_0}(\mathscr{C}) \to \mathscr{C}$$

Indeed, $\mathbb{E}_0$ has no nontrivial multi-input operation that could impose an additional action law. Once the $\mathbb{E}_0$-algebra a is fixed, an a-module is therefore just an arbitrary object of $\mathscr{C}$. This extreme case confirms that the module operad adds exactly the operations prescribed by the base operad, and no hidden structure.

11.3 Algebra Objects in Module Categories

The preceding construction does more than produce a category of modules: it also remembers enough operadic structure to form algebras *inside* that category. Classically, if a is a commutative algebra, then a commutative algebra object of $\mathbf{Mod}_a(\mathscr{C})$ is simply a commutative algebra b of $\mathscr{C}$ together with a map $a \to b$. The ∞-categorical statement is the same, but the map $a \to b$ and all of its compatibilities are encoded by the semi-inert construction.

Write $\mathbf{Alg}_{\mathcal{O}}(\mathscr{C})^{a/}$ for the undercategory whose objects are morphisms $a \to b$ of $\mathcal{O}$-algebras.

11.3.1 The Universal Arrow Description

Proposition 11.3.7 (Algebras in the Universal Module Operad). Suppose that $\mathcal{O}^{\otimes}$ is coherent and that $\mathscr{C}^{\otimes} \to \mathcal{O}^{\otimes}$ is a fibration of ∞-operads. There is a canonical equivalence

$$\mathbf{Alg}_{\mathcal{O}}\left(\overline{\mathbf{Mod}}^{\mathcal{O}}(\mathscr{C})\right) \simeq \mathbf{Fun}\left(\Delta^1, \mathbf{Alg}_{\mathcal{O}}(\mathscr{C})\right)$$

Thus an algebra in the universal module construction is exactly an arrow $a \to b$ between $\mathcal{O}$-algebras.

Remark (Why an Arrow Appears). The source of a semi-inert arrow records the algebra inputs, while its distinguished empty slot records the future module variable. An algebra object supplies compatible operations at both endpoints. Evaluation at the two endpoints therefore produces two algebras a, b, and transport along the marked slot produces the algebra map $a \to b$. Conversely, such a map lets every operation of a act on b through the corresponding operation of b.

$$\begin{tikzcd}
a \arrow[rr, "\text{algebra map}"] & & b \\
& \mathbf{Alg}_{\mathcal{O}}\left(\overline{\mathbf{Mod}}^{\mathcal{O}}(\mathscr{C})\right) \arrow[ul, "e_0"] \arrow[ur, "e_1"'] &
\end{tikzcd}$$

11.3.2 Algebras over a Fixed Algebra

Theorem 11.3.3 (Algebras in a-Modules). Let $a \in \mathbf{Alg}_{\mathcal{O}}(\mathscr{C})$. Restricting the preceding equivalence to the fiber over a gives a canonical equivalence

$$\mathbf{Alg}_{\mathcal{O}}\left(\mathbf{Mod}_a^{\mathcal{O}}(\mathscr{C})\right) \simeq \mathbf{Alg}_{\mathcal{O}}(\mathscr{C})^{a/}$$

In other words, an $\mathcal{O}$-algebra object in a-modules is precisely an $\mathcal{O}$-algebra b in $\mathscr{C}$ equipped with a morphism $a \to b$.

In the commutative case this reads

$$\mathbf{CAlg}\left(\mathbf{Mod}_a(\mathscr{C})\right) \simeq \mathbf{CAlg}\left(\mathscr{C}\right)^{a/}$$

The familiar phrase "a commutative a-algebra" therefore has no additional meaning hidden in it: it is exactly a commutative algebra under a.

Proof sketch. The universal module operad maps to the arrow category of $\mathbf{Alg}_{\mathcal{O}}(\mathscr{C})$ by endpoint evaluation. Coherence makes the source evaluation on semi-inert arrows flat, so restriction and the required Kan extensions preserve the operadic Segal conditions. The resulting endpoint functor is an equivalence. Pulling it back along the vertex $\{a\}$ identifies its fiber with the under-category of arrows beginning at a. □

11.3.3 Iterating the Module Construction

Let $\mathbf{Mod}^{\mathcal{O}}(\mathscr{C})$ denote the total ∞-category of pairs (b, m) consisting of an $\mathcal{O}$-algebra b and a b-module m. Its projection remembers b. The preceding theorem upgrades to the pullback formula

$$\mathbf{Mod}^{\mathcal{O}}\left(\mathbf{Mod}_a^{\mathcal{O}}(\mathscr{C})\right) \simeq \mathbf{Mod}^{\mathcal{O}}(\mathscr{C}) \times_{\mathbf{Alg}_{\mathcal{O}}(\mathscr{C})} \mathbf{Alg}_{\mathcal{O}}(\mathscr{C})^{a/}$$

An object on either side is therefore the same triple

$$a \to b, \quad m \in \mathbf{Mod}_b^{\mathcal{O}}(\mathscr{C})$$

Proposition 11.3.8 (Modules in Modules). Let b be an $\mathcal{O}$-algebra object of $\mathbf{Mod}_a^{\mathcal{O}}(\mathscr{C})$, and let $|b|$ be the target algebra in $\mathscr{C}$ corresponding to the map $a \to |b|$. Then there is a canonical equivalence

$$\mathbf{Mod}_b^{\mathcal{O}}\left(\mathbf{Mod}_a^{\mathcal{O}}(\mathscr{C})\right) \simeq \mathbf{Mod}_{|b|}^{\mathcal{O}}(\mathscr{C})$$

Thus a b-module internal to a-modules is just a module over the underlying algebra $|b|$ in $\mathscr{C}$.

Remark (Classical Reading). For commutative rings, if $a \to b$ is a ring map, then a b -module in the category of a-modules carries no extra structure beyond its ordinary b-module structure: the a-action is already obtained by restriction of scalars along $a \to b$. The proposition is the fully coherent operadic form of this elementary fact.

Proof sketch. Both sides are obtained by fixing the same algebra arrow $a \to |b|$ in the universal category of marked operations. The defining squares are homotopy pullbacks, and endpoint evaluation is an equivalence; taking the relevant fibers therefore gives the displayed equivalence of module ∞-categories. □

“ Algebra objects in a-modules are algebras under a, and modules over such an algebra are the same whether computed inside a-modules or in the original ambient category. ”